Proximal-to-Distal Energy Transfer in the Golf Swing

Mechanisms, Evidence, and Counterfactual Tests of Distal-Handoff Timing

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Abstract

Proximal-to-distal (P→D) sequencing — the ordered peaking of pelvis, thorax, arm, and club rotational speeds — is the most widely reported organizing principle of the golf downswing, and it is usually read as evidence that mechanical energy should be passed outward along the kinetic chain as early and as completely as possible. This study reviews the mechanics and the empirical literature behind that reading and asks the sharper question: *when* during the downswing should the P→D transfer occur? Across double-pendulum mechanics, delayed-release simulation, optimal control, and the segmental kinetic-energy literature, the evidence favors a strategy that *retains* energy in the proximal system early in the downswing and *accelerates* the handoff late. We formalize this retain-early/release-late hypothesis in the language of zero-torque counterfactual (ZTCF) and zero-velocity counterfactual (ZVCF) decompositions, and test it in a reproducible two-segment golf model using an exact interaction-force decomposition, a matched-state torque-killswitch comparison, a wrist-torque timing sweep (92 torque programs), pointwise drift/control counterfactual decompositions, and Robertson-Winter style wrist-interface power accounting, all generated by committed, reproducible scripts. In this model, driving the wrist from the top produced 15% less clubhead speed than a fully passive wrist, while delaying the wrist drive to the optimal onset produced 12% more, and a program that actively restrains the wrist early and drives it late is best of all at both shoulder-torque levels tested. The energy accounting shows the winning programs move *less* energy into the club early and far more late, with the best program performing *negative* wrist work early — active retention. We interpret these results against the literature, state their limitations (a planar 2-DOF model with a fixed hub and rigid shaft). A one-at-a-time sensitivity study spanning 13 parameter cases preserves the strategy ordering but shifts the optimal onset, showing that the modeled mechanism is more robust than any single timing prescription. The force analysis resolves the geometric sine and cosine projections that determine whether distal centripetal and tangential forces transfer power. A separate audit of the author's prior two-hand WSCG model reconstructs all 2,801 stored samples and shows that a -19.63 N m pointwise zero-command midpoint couple is generated entirely by separated contact forces. It also distinguishes couple sign from power sign and verifies the predicted linear grip-separation and relative-orientation geometry. A matched rigid/flexible three-coordinate surrogate then separates control, momentum, gravity, joint damping, shaft elasticity, and shaft damping with closed work–energy accounting. In its reference case, flex changes delivery speed by +0.108 m/s, while gravity and joint-damping ablations are substantially larger; broad stiffness, damping, torque-cut, endpoint-window, and timestep sweeps bound the result. A forward constrained two-arm model then branches an exact same-state zero-command trajectory at 0.200 s. Its contact-force couple remains negative for the full 50 ms branch and reaches -7.43 N m; removing both grip moment arms collapses that couple exactly to zero, while timestep and projection studies bound the numerical result. A coupled extension then admits a finite-mass translating base, two constrained arms, and a two-segment compliant club in one forward KKT system. Base motion and shaft flex are endogenous; a same-state zero-command branch retains a negative force-generated couple for 50 ms, the coincident-grip control removes it exactly, and work–energy and contact-power identities close under timestep refinement. These are planar model-feasibility results, not physiological or human evidence. A separate distributed-shaft study then reuses the shared Euler–Bernoulli finite-element model to compare a first-mode reduction with a six-mode reference. Synthetic modal identification recovers its declared modulus and tip mass, mesh and work–energy gates close, and a short load pulse exposes sixty times more RMS response discrepancy than a slow pulse. This is structural verification, not equipment calibration. A subsequent planar experiment couples three transported shaft modes into the moving-base, two-hand forward solve. The declared baseline stays within a 5% deflection screen; the zero-command grip couple remains negative for 30 ms; coincident and reversed moment-arm controls close exactly; and work–energy error decreases under timestep refinement. The result supports only the declared synthetic planar mechanism. A common-observable model ladder then carries reference-explicit force, couple, velocity, angular velocity, and power through a third coordinate, prescribed hub motion, two-hand constraint geometry, and proper 3-D frame transformations. A reduced full-body extension executes nonplanar common-state inverse dynamics through MuJoCo and an independent Lagrange–Christoffel formulation. Across 20 generalized coordinates and 61 states, the maximum relative discrepancy is 2.14×10^{-11}; reversing the hand moment arm reverses a -4.32 N m force-generated couple, and coincident hands remove it exactly. Because contact loads are prescribed, passive spatial contact remains inconclusive at that tier. A subsequent reduced forward-contact experiment independently integrates MuJoCo and Pinocchio with two finite-mass hand carriages, paired compliant interfaces, and no direct club actuation. Its exact same-state driver-killswitch branch retains a negative swing-normal contact couple for 37.5 ms; coincident and reversed-moment-arm controls close exactly, trajectory and wrench gates pass, and work–energy residual decreases under timestep refinement. This supports a reduced spatial mechanism, not anatomical arms, muscle coordination, human strategy, or equipment calibration. A coupled uncertainty and control experiment then varies 12 inputs simultaneously, audits structural and practical identifiability, and compares eight preselected delayed-command programs on separate training and held-out ensembles. The net planar wrench leaves one individual-hand-force mode unidentified, the six summary observables cannot identify the full 12-parameter vector, and all eight programs are nondominated on held-out samples. An early opposing command improves lower-tail speed relative to late drive but worsens the planar face/path proxy, rejecting a universal optimum and any direct physiological interpretation. A preregistered experimental protocol then fixes synchronized motion, bilateral hand-wrench, club, ground-reaction, launch, and activation measurements; participant-level holdout; filtering; calibration; residuals; identity protection; and four outcome falsifiers. Its synthetic dry run qualifies only the fail-closed pipeline. No human outcome is analyzed, and every experimental prediction remains untested. A separate hand-path attribution study evaluates the MacKenzie force-work-per-path-length estimand in an impact-truncated double pendulum, a forward-simulated one-arm model, and a mechanically closed two-arm local sweep. Drift supplies 101.2% and 103.5% of signed hand-path force work in the two forward cases because the contemporaneous control contribution is slightly opposing; in the two-arm sweep, drift and control instead cancel with an index of 0.962. Joint- and time-resolved impulse, power, work, force-vector, and common/differential-mode results show why a large net hand-path force is a mechanical transfer observable rather than a direct measure of biological effort. A final matched-task allocation experiment separates direct wrist moments from the moment of bilateral hand forces at identical club states. Internal demands vary even when club-moment closure is exact. A separate dead-zone transmission model predicts that preserving each channel's loaded direction avoids a reversal gap, whereas reversing both modeled channels crosses zero transmission. These conditional predictions do not identify scapular action or establish a preferred human technique. A constrained-contact extension then defines frame-explicit configuration, velocity, control, and external-load contributions to reaction force and fails closed when contact allocation is nonunique. In the fixed-support double-pendulum benchmark, pointwise ZTCF predicts the modeled support-reaction waveform with componentwise R-squared values of 0.871 and 0.814 but retains large amplitude error; opposing control also makes the vertical ZTCF impulse exceed the total. This benchmark is not a bilateral human force-plate validation, and the required synchronized force-plate and whole-body kinematic experiment is specified as a falsification protocol. References, analysis code, machine-readable results, and provenance are available with the document for independent inspection and reuse.

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Scope and Evidence Categories

This document distinguishes three kinds of statement. **Empirical findings** summarize published measurements of golfers. **Model-derived findings** report deterministic simulations generated by the archived analysis code and parameter sets. **Hypotheses and interpretations** state claims that remain to be tested in higher-fidelity models or human data. The distinction matters because agreement between a simplified model and prior literature is not independent empirical confirmation.

Chapter 1

Introduction

1.1 Motivation

Clubhead speed at impact is the dominant controllable determinant of drive distance, and it correlates strongly with playing standard (Fradkin, Sherman, and Finch 2004; Hellström 2009; Hume, Keogh, and Reid 2005). The golf downswing is among the most-studied examples of a *sequenced* multi-segment rotation in sport biomechanics (McPhee 2022; Dillman and Lange 1994). Across measurement systems and populations, the downswing of a skilled golfer shows a characteristic order: the pelvis reaches its peak angular speed first, then the thorax, then the lead arm, and finally the club, with each successive segment peaking faster than the one before (Cheetham et al. 2008; Tinmark et al. 2010). This pattern — the *kinematic sequence*, an instance of the general proximal-to-distal (P→D) sequencing observed in throwing and striking skills (Putnam 1993) — is routinely interpreted as the visible signature of mechanical energy generated in the large proximal segments and transferred outward to the small, fast distal ones, and it has been translated into coaching doctrine accordingly (Smith et al. 2015).

That interpretation leaves a first-order question unanswered:

> **Should a golfer attempt to transfer energy from proximal to distal segments *early* in the downswing, or should energy be *retained* in the proximal segments early and the transfer deliberately *accelerated* late?**

The two strategies imply opposite coaching cues ("fire the club from the top" versus "hold the lag and let it release"), opposite torque programs at the wrists and torso, and different predictions for what counterfactual decompositions of a measured swing should show. The question also connects to a long-standing thread in the modeling literature: double-pendulum analyses have argued since the 1960s that the highest clubhead speeds arise when the distal joint is *not* driven early — when release is passively or actively delayed (Cochran and Stobbs 1968; T. Jorgensen 1970; Milburn 1982; Pickering and Vickers 1999) — while the coaching translation of the kinematic sequence often emphasizes initiating distal motion promptly after transition.

This study approaches the question with *counterfactual dynamics*. Two decompositions are evaluated: the **zero-torque counterfactual (ZTCF)** — the acceleration the system would exhibit at the current instant if all applied joint torques were removed, isolating what accumulated motion, gravity, and inter-segment coupling produce on their own — and the **zero-velocity counterfactual (ZVCF)** — the acceleration the applied torques, together with gravity, would produce at the current pose from rest, removing all motion-dependent effects. Preliminary torque-removal analyses motivated the hypothesis examined throughout this document: that *restraining* the P→D handoff early in the downswing and accelerating it late may be the more effective strategy.

1.2 Research Questions

The document is organized around four questions:

- **RQ1 — Mechanism.** By what mechanical pathways does energy generated proximally reach the club, and what does each pathway imply about the *timing* of the handoff? (Section 2)
- **RQ2 — Evidence.** Where does the P→D transfer occur in measured swings, and how do skilled and less-skilled players differ — in sequence order, in magnitudes, or in transfer efficiency? (1)
- **RQ3 — Formalization.** How can "transfer early" versus "retain-early/release-late" be stated as falsifiable claims about measurable quantities, and what do the ZTCF/ZVCF counterfactuals predict under each? (Section 21)
- **RQ4 — Model test.** What does a two-link model show when the timing of the distal handoff is manipulated directly, with the energy accounting and counterfactual decompositions computed along every trajectory? (Section 23, Section 24, Section 25)

1.3 Approach and Contributions

The document makes six contributions:

1. a synthesis of the mechanics of sequenced swings, from the summation-of-speed principle through segment-interaction dynamics, parametric energy transfer, and optimal control (Section 2);
2. a methodological account of what "energy transfer" means operationally — joint power, joint-force power, segmental energy accounting, induced accelerations — and where the popular kinematic-sequence proxy fails (Section 2.5);
3. a critical review of the empirical evidence, including the directly relevant segmental kinetic-energy sequencing studies (Kenny et al. 2008; Anderson, Wright, and Stefanyshyn 2006) and the skill-level literature (1);
4. a formalization of the timing question and of the ZTCF/ZVCF counterfactual framework, including the distinction between pointwise and re-simulation counterfactuals (Section 21);
5. a **reproducible model test** on a double-pendulum golf model — 92 torque programs swept over wrist-drive onset time and profile, scored at a consistently defined impact, with pointwise drift/control decompositions and Robertson-Winter wrist-interface power accounting computed along every representative trajectory (Section 23, Section 24); and
6. a bridge from these 2-DOF results to broader tests: what the simplified model can and cannot establish, and the concrete, steps that extend the analysis to re-simulation parity, full-body engines, and motion-capture replication (Section 25, Section 26).

1.4 Reading Guide

Readers interested only in the new quantitative material can read Section 21 for the hypothesis, Section 23 for the setup, and Section 24 and Section 25 for the outcome. Readers using this document as a literature resource will find the review chapters (Section 2, 1) and the verified bibliography self-contained. Appendices document software and data availability (Section 27), reproducibility (Section 28), collect validation evidence (Section 29), and tabulate model parameters (Section 30).

Chapter 2

Mechanics of Sequenced Swings

This chapter assembles the mechanical foundations the rest of the document builds on: the dynamics formalism (Section 2.1), the summation-of-speed principle and its dynamical correction (Section 2.2), the double-pendulum tradition that made release timing an explicit variable (Section 2.3), the optimal-control literature (Section 2.4), and the operational meanings of "energy transfer" (Section 2.5).

2.1 Dynamics Formalism

Write the equation of motion of an articulated chain in standard manipulator form (Featherstone 2008):

$$M(q)\,\ddot{q} \;=\; \tau \;-\; b(q,\dot{q}), \qquad b(q,\dot{q}) \;=\; C(q,\dot{q})\,\dot{q} \;+\; g(q) \;+\; d(\dot{q}), \tag{2.1}$$

with configuration q, inertia matrix M, applied generalized torques τ, Coriolis/centripetal terms $C\dot{q}$, gravity g, and velocity-dependent dissipation d. Two structural facts drive everything that follows. First, the system is *coupled*: M is not diagonal, so torque at any joint induces acceleration at every joint — the basis of induced-acceleration analysis (Zajac and Gordon 1989) and of the counterfactual decompositions of Section 21. Second, the equation is *affine in* τ: the acceleration splits exactly into a drift part (what accumulated motion and gravity produce alone) and a control part ($M^{-1}\tau$), a property enforced to numerical tolerance by the accompanying tests (Section 29).

2.2 The Summation-of-Speed Principle and Its Limits

The oldest formalization of P→D sequencing is the *summation of speed* principle: in skills demanding high end-point speed, each segment should begin its motion at the moment of peak speed of the segment proximal to it, so that distal speed is built on top of proximal speed (Bunn 1972). Milburn applied the principle to golf with a double-pendulum analysis of the downswing and found the characteristic pattern of an initially dominant proximal (arm) rotation followed by rapid distal (club) rotation, noting that *"an initial delay in the uncocking of the wrist"* allows the arm to reach greater acceleration and the club's acceleration to be summed with the proximal segment's maximum (Milburn 1982). Note the structure of that finding: the summation principle, taken naively, prescribes *sequential initiation*; Milburn's analysis showed the benefit comes specifically from *delaying* the distal contribution.

Putnam's segment-interaction analyses put the principle on a dynamical footing and exposed its limits (Putnam 1991, 1993). In a linked chain, each segment's angular acceleration depends not only on joint torques but on *motion-dependent interaction torques* — centripetal and Coriolis terms proportional to squares and products of segment angular velocities (the $C\dot{q}$ term of Equation 2.1). Putnam showed the P→D pattern in striking and throwing arises substantially from these interaction torques: proximal deceleration and distal acceleration late in the motion are coupled through the joint force and occur even without a distal joint torque. Feltner and Dapena's classic analysis of the baseball pitch quantified the same interplay at the shoulder and elbow (Feltner and Dapena 1986), and Herring and Chapman showed in simulated throws that the *timing* of torque onsets and torque reversals — including deliberately reversed (restraining) torques — materially changes end-point speed (Herring and Chapman 1992). Two implications matter for golf. First, kinematic order does not reveal *which* torques produced it: an identical peak sequence can arise from different mixes of muscular and interaction torques. Second, the proximal deceleration observed in experts need not be an active "braking" strategy — it can be the dynamical *consequence* of the distal segment accelerating and back-reacting on the chain.

Marshall and Elliott added a measurement caution: planar peak-ordering analyses miss long-axis rotations, which peak very late in racquet skills and contribute substantially to end-point speed (Marshall and Elliott 2000); the golf analogue is forearm rotation and wrist action in the final phase of the downswing.

2.3 Double-Pendulum Mechanics of the Golf Downswing

The two-lever, one-hinge model of the downswing — an "arm" link rotating about a hub, connected by a wrist hinge to a "club" link — has been the workhorse of golf mechanics since Cochran and Stobbs' *Search for the Perfect Swing* (Cochran and Stobbs 1968) and Jorgensen's dynamical analysis fitted to stroboscopic photographs of a professional swing (T. Jorgensen 1970; T. P. Jorgensen 1999); Penner's review surveys the tradition (Penner 2003). Its enduring value here is that it makes the *timing* of the P→D handoff an explicit variable (the "release": the point at which the wrist angle is allowed, or driven, to open) and exposes the mechanism by which delay pays. Three results anchor the analysis:

Delayed release increases clubhead speed. Pickering and Vickers, sweeping release angles in a driven double-pendulum model, found clubhead velocity at impact increases as release is delayed — the classical "late hit" (Pickering and Vickers 1999). Jorgensen's fitted model reproduced the expert swing only with a substantially delayed opening of the wrist angle (T. Jorgensen 1970; T. P. Jorgensen 1999). In these models an early release (early P→D handoff) is strictly inferior: club kinetic energy purchased early is paid for with arm speed that never develops.

Late wrist torque helps; early wrist torque is the wrong tool. Sprigings and Neal, using a torque-driven three-segment model with physiologically bounded actuators, found that an active wrist torque applied *during the latter stages of the downswing* increased clubhead speed by approximately 4% (Sprigings and Neal 2000); Sprigings and MacKenzie confirmed the delayed-release benefit in a follow-up simulation focused on release timing (Sprigings and MacKenzie 2002). The point is not that distal effort is useless, but that its productive window is late.

Energy transfer to the club can be largely passive — governed by the changing geometry. White analyzed an *undriven* double pendulum and showed that the transfer of kinetic energy from arms to club late in the downswing is a parametric-transfer phenomenon: as the club swings out, the effective moment of inertia of the system changes, and conservation of energy and angular momentum move kinetic energy distally without any wrist torque at all; the wrist-cock angle maintained before release is the principal efficiency-determining parameter (White 2006). Miura's analysis of the "inward pull" complements this: an expert's hands moving toward the body near impact — shrinking the hub radius when centrifugal loading is fully developed — performs work against the centrifugal force that appears as additional clubhead kinetic energy (Miura 2001). Nesbit and McGinnis' hub-path analyses generalize the same idea: the non-circular hand path of skilled golfers, including its late-downswing tightening, is a mechanism for shaping kinetic transfer from golfer to club (Nesbit and McGinnis 2009; Nesbit and McGinnis 2014).

Together these results describe a machine whose highest-output mode is: *load the proximal rotation first, hold the distal joint folded (retaining energy in a compact, fast-rotating configuration), then let geometry and — in the last phase — modest distal torque move the energy outward rapidly.* The double-pendulum literature is already an argument for **retain-early / release-late**. Section 24 tests exactly this proposition in the numerical implementation used here.

2.4 Optimal Control and Torque Sequencing

If the timing question is posed as an optimization — what torque program maximizes clubhead speed subject to actuator limits? — the answer has been studied repeatedly. Lampsa's early optimal-control study of a driven double pendulum computed torque histories maximizing drive distance, and notably found the *optimal* torques differed in structure from those measured (via inverse dynamics) in his own swing (Lampsa 1975) — early evidence that human swings need not be optimal and that the optimum is a meaningful target rather than a redescription of expert behavior. Sharp, fitting shoulder-arm-club models to expert swing data and then optimizing torque programs, established a pattern for the best use of the available driving torques across the downswing phases, with the distal contribution structured late rather than distributed evenly (Sharp 2009). Three-dimensional forward-dynamics models extended the approach with anatomically meaningful actuators and flexible shafts: MacKenzie and Sprigings' 3D model (MacKenzie and Sprigings 2009), the carry-optimized model of Balzerson, Banerjee and McPhee (Balzerson, Banerjee, and McPhee 2016), and the six-degree-of-freedom dynamic optimization of McNally and McPhee (McNally and McPhee 2018) all generate expert-like clubhead speeds with torque programs whose distal elements act late in the downswing. McPhee's review surveys this modeling lineage (McPhee 2022).

A caution cuts the other way: optimal control optimizes *within a model*, and the sensitivity of optimal timing to model structure (hub mobility, shaft flexibility (Betzler et al. 2012; Osis and Stefanyshyn 2012), actuator rate limits) is why cross-engine validation (Section 22) is valuable when revisiting the question.

2.5 What "Energy Transfer" Means Operationally

Claims about energy transfer require an accounting framework. Four levels of analysis appear in the literature, and they are not interchangeable.

2.5.1 Joint Power and Work

For a joint spanning segments i and $i+1$ with net joint torque τ and relative angular velocity ω, the *joint (moment) power* is $P_m = \tau\omega$; its time integral over a phase is the mechanical work done by the net moment. Positive work indicates generation, negative work absorption. This is the quantity most golf studies report, and it is the basis of the work-power budget of Nesbit and Serrano (Section 20.1) (Nesbit and Serrano 2005).

2.5.2 Joint-Force Power and Segmental Energy Accounting

Joint moments are not the only conduit of mechanical energy: the joint reaction force $\mathbf{F}$ acting at a joint moving with velocity $\mathbf{v}$ transmits *joint-force power* $P_f = \mathbf{F} \cdot \mathbf{v}$ from one segment to the next. Robertson and Winter's classic formulation balances, for each segment, the rate of change of segment mechanical energy against the sum of joint-force powers and moment powers at its proximal and distal ends, distinguishing energy *transferred between* segments from energy *generated or absorbed* at joints (Robertson and Winter 1980; Winter 2009). This is the correct level of analysis for the timing question, because "retain-early / release-late" is precisely a claim about the *time course of inter-segment transfer terms*, not about peak angular velocities. Section 23 implements exactly this accounting at the model's wrist interface, including the energy-balance residual check.

2.5.3 Induced Accelerations and Dynamic Coupling

In a coupled chain, any single torque induces accelerations at *every* joint — including joints it does not span — through the inverse inertia matrix (Zajac and Gordon 1989). Induced-acceleration and power-transfer analyses in locomotion research (Zajac, Neptune, and Kautz 2002) show that intuitions of the form "muscle X accelerates joint Y" routinely fail in coupled systems. For golf this matters because a torso torque early in the downswing partly accelerates the *club* (through coupling), and a wrist torque late in the downswing back-reacts on the torso; the net energy routing is an emergent property of the whole chain. The ZTCF/ZVCF decompositions of Section 21 are the counterfactual cousins of this analysis.

2.5.4 Kinematic Sequence as a Proxy — And Its Failure Modes

Peak-ordering of segment angular speeds is a *proxy* for energy flow, not a measurement of it. Marsan and colleagues showed that the identified kinematic sequence in golf depends materially on which angular-velocity component (or norm, or projection) is analyzed: thirteen golfers analyzed with seven component-selection methods yielded different sequences (Marsan et al. 2019). Marshall and Elliott's long-axis-rotation caution (Marshall and Elliott 2000) is a special case of the same problem, and the near-planarity of the downswing is itself only an approximation (Coleman and Rankin 2005; Kwon et al. 2012). Any serious test of the timing hypothesis must therefore run at the level of energy and power accounting (Section 2.5.2), with the kinematic sequence reported as corroborating description only — which is how Section 24 uses it.

Chapter 3

Interaction-Force Mechanisms in the Double Pendulum

The phrase *energy transfer* is often used as if energy were a substance that waits in one link and is then released into the next. The mechanics are more specific. A distal segment gains mechanical energy when forces acting at its moving joint and moments acting on it do positive power. Those forces can grow without a large distal actuator torque because the segment must satisfy the acceleration constraints imposed by the moving chain. Their capacity to do work depends on orientation: a large force perpendicular to joint velocity transmits no instantaneous power, while a smaller force aligned with that velocity can be effective. This chapter derives those statements exactly for the double pendulum, tests them numerically, and identifies which conclusions survive when the commanded torques are removed.

3.1 The Question in Operational Form

Consider the club segment alone. It is acted on by gravity, a force at the wrist, and a net wrist moment. Its kinetic-energy rate is

$$\dot{T}_2 = \underbrace{\mathbf{F}_W \cdot \mathbf{v}_W}_{\text{joint-force power}} + \underbrace{M_W \omega_2^{\text{abs}}}_{\text{joint-moment power}} + \underbrace{m_2 \mathbf{g} \cdot \mathbf{v}_{C2}}_{\text{gravity power}}, \tag{3.1}$$

where $\mathbf{F}_W$ is the wrist force acting on the club, $\mathbf{v}_W$ is the velocity of the wrist point, M_W is the net moment acting on the club at the wrist, and $\omega_2^{\text{abs}} = \dot{\theta}_1 + \dot{\theta}_2$ is the club's absolute angular velocity. Equation Equation 3.1 separates three questions:

1. What creates a large wrist force?
2. What orients that force so it has a useful projection onto wrist velocity?
3. What part of the resulting club-energy gain requires commanded wrist torque?

The first two questions concern dynamic coupling and geometry. The third is a counterfactual question. None can be answered from angular-speed peak order alone (Putnam 1991; Robertson and Winter 1980; Winter 2009).

3.2 Coordinate and Sign Conventions

The arm angle θ_1 and relative wrist angle θ_2 are measured from the downward vertical and from the arm, respectively. The club absolute angle is $\phi = \theta_1 + \theta_2$. In swing-plane Cartesian coordinates, x points toward the target side and y points upward. Define

$$\mathbf{u}(\theta) = \begin{bmatrix} \sin\theta \\ -\cos\theta \end{bmatrix}, \qquad \mathbf{u}_\perp(\theta) = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}. \tag{3.2}$$

$\mathbf{u}$ points along a link and $\mathbf{u}_\perp$ points in its positive angular-velocity direction. Figure Figure 3.1 declares the free-body convention used in every calculation. $\mathbf{F}_W$ is the force *on the club*; the equal-and-opposite force acts on the arm. Positive $\mathbf{F}_W \cdot \mathbf{v}_W$ therefore means energy crosses the wrist interface into the club.

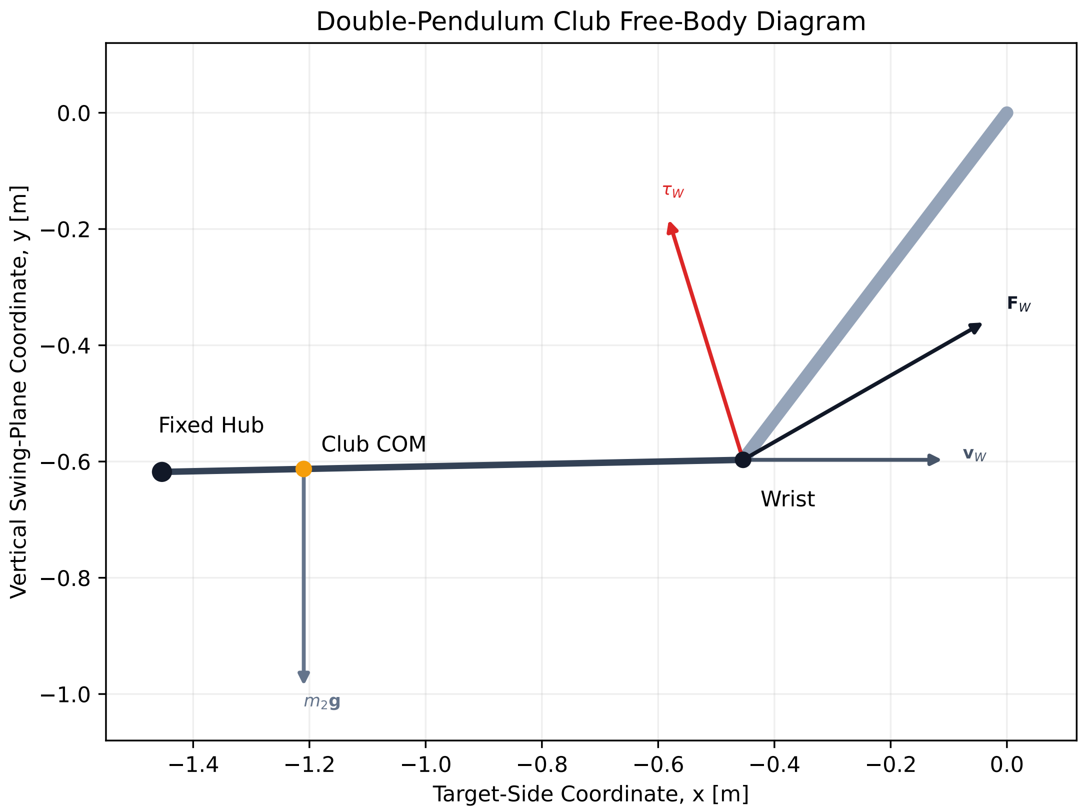

Figure 3.1: Double-Pendulum Club Free-Body Diagram

The force and moment are related but not interchangeable. Translation of the club centre of mass depends on the net force. Rotation about the centre of mass depends on both the wrist-force moment arm and M_W:

$$I_{2,C}\ddot{\phi} = (\mathbf{r}_W - \mathbf{r}_C) \times \mathbf{F}_W + M_W. \tag{3.3}$$

The executable validation checks Equation 3.3 at machine precision. This distinction becomes central in a two-hand model, where two hand forces can have a modest resultant force but a large equivalent couple.

3.3 Exact Decomposition of the Wrist Reaction Force

The wrist and club centre-of-mass positions are

$$\mathbf{r}_W = l_1\mathbf{u}(\theta_1), \qquad \mathbf{r}_{C2} = \mathbf{r}_W + l_{c2}\mathbf{u}(\phi). \tag{3.4}$$

Differentiating twice gives four acceleration terms:

$$\begin{aligned}\mathbf{a}_{C2} = \quad & \underbrace{l_1\ddot{\theta}_1\mathbf{u}_\perp(\theta_1)}_{\text{proximal tangential}} - \underbrace{l_1\dot{\theta}_1^2\mathbf{u}(\theta_1)}_{\text{proximal centripetal}} \\ & + \underbrace{l_{c2}\ddot{\phi}\mathbf{u}_\perp(\phi)}_{\text{distal tangential}} - \underbrace{l_{c2}\dot{\phi}^2\mathbf{u}(\phi)}_{\text{distal centripetal}} .\end{aligned} \tag{3.5}$$

Newton's second law for the club yields

$$\mathbf{F}_W = m_2(\mathbf{a}_{C2} - \mathbf{g}). \tag{3.6}$$

Combining Equation 3.5 and Equation 3.6 produces five exact force components. They are not fitted regressors and do not depend on numerical differentiation.

Table 3.1: Exact Wrist-Force Components in the Planar Double Pendulum

Component	Vector Expression	Physical Meaning
Proximal tangential	$m_2 l_1\ddot{\theta}_1\mathbf{u}_\perp(\theta_1)$	Force needed because the wrist has tangential acceleration
Proximal centripetal	$-m_2 l_1\dot{\theta}_1^2\mathbf{u}(\theta_1)$	Force needed because the wrist follows a curved path
Distal tangential	$m_2 l_{c2}\ddot{\phi}\mathbf{u}_\perp(\phi)$	Force associated with club angular acceleration about the wrist
Distal centripetal	$-m_2 l_{c2}\dot{\phi}^2\mathbf{u}(\phi)$	Force associated with the club COM curving about the wrist
Gravity reaction	$-m_2\mathbf{g}$	Wrist support needed in addition to gravity

Two cautions prevent common misreadings. First, *centripetal* describes an acceleration direction, not a separate applied agent. The joint reaction force is the physical constraint force; the named terms explain why that force is required. Second, gravity appears twice in a causal discussion: it contributes to the system accelerations through the equations of motion, and the explicit $-m_2\mathbf{g}$ term is required when the club free-body equation is solved for the wrist force. These roles must not be double-counted.

3.4 Why Zero Commanded Torque Does Not Imply Zero Force

For generalized coordinates $q = [\theta_1, \theta_2]^T$, the dynamics are

$$\ddot{q} = M(q)^{-1}\{\tau - b(q, \dot{q})\}. \tag{3.7}$$

Setting $\tau = 0$ leaves $\ddot{q}_0 = -M^{-1}b$, which is generally nonzero whenever gravity, velocity-dependent coupling, or damping is present. Substitution of $\ddot{q}_0$ into Equation 3.5 yields a nonzero centre-of-mass acceleration and therefore a nonzero wrist reaction force. This is the precise mechanical basis for the WSCG presentation's statement that zero joint torque does not mean zero interaction force (Olson 2024). It also follows directly from standard linked-segment dynamics (Putnam 1991, 1993; Featherstone 2008).

Because the equations are affine in torque, the *instantaneous* acceleration has an exact split:

$$\ddot{q}_{\text{total}} = \underbrace{-M^{-1}b}_{\ddot{q}_{\text{drift}}} + \underbrace{M^{-1}\tau}_{\ddot{q}_{\text{control}}}. \tag{3.8}$$

At a fixed state, all velocity-squared terms in Equation 3.5 are shared between the two cases. The force split is therefore

$$\mathbf{F}_{\text{total}} = \mathbf{F}_{\text{drift}} + \mathbf{F}_{\text{control}}, \tag{3.9}$$

where $\mathbf{F}_{\text{drift}}$ is evaluated at zero commanded torque and $\mathbf{F}_{\text{control}}$ is the change induced by the commanded torque. This is a local superposition at one state. Once two futures are integrated, their states diverge and the equality cannot be applied across time by subtracting unrelated samples.

Figure Figure 3.2 shows this decomposition at six phases of the reference swing. The force scale is fixed across panels. The pointwise zero-torque force becomes large late even though the commanded torque is set to zero in that calculation; accumulated angular velocity and changing geometry continue to demand a substantial wrist constraint force.

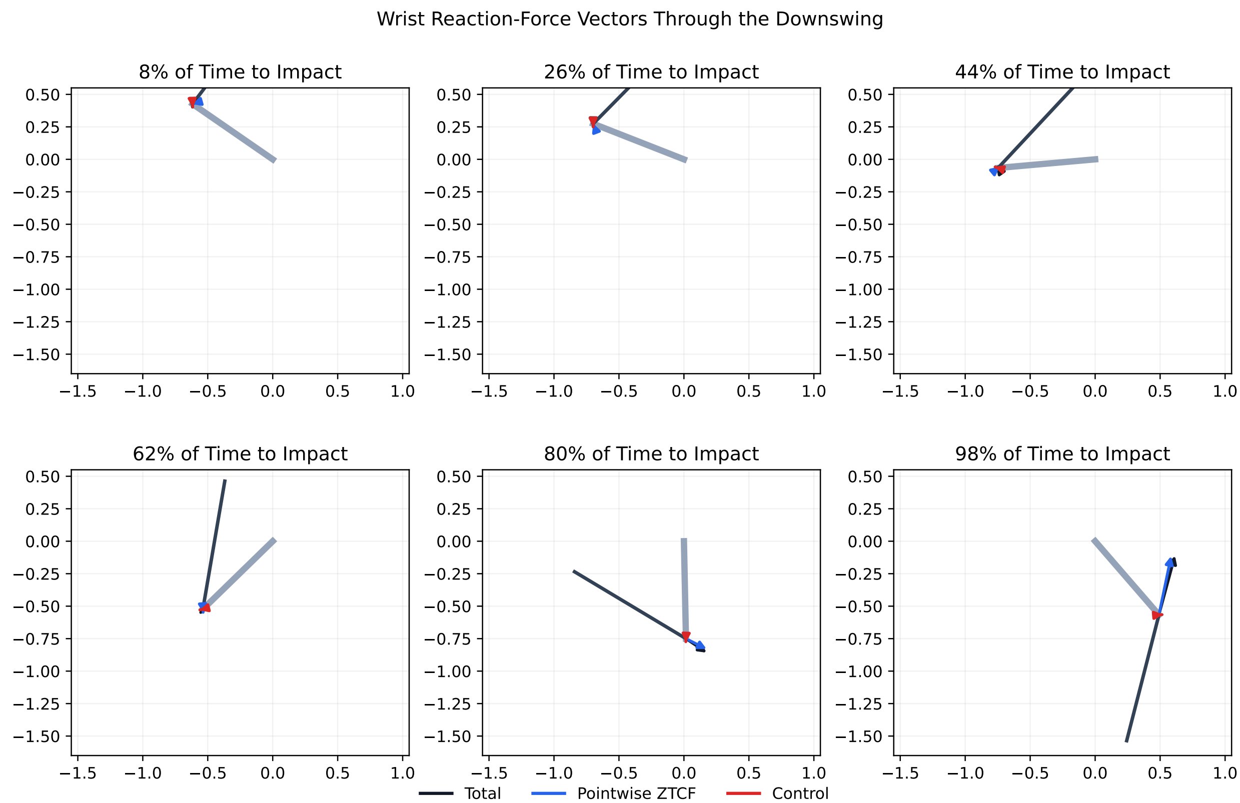

Figure 3.2: Wrist Reaction-Force Vectors Through the Downswing

3.5 Force Magnitude Is Not Transfer

The wrist velocity is

$$\mathbf{v}_W = l_1 \dot{\theta}_1 \mathbf{u}_\perp(\theta_1). \tag{3.10}$$

For any component $\mathbf{F}_k$, its contribution to club energy is $P_k = \mathbf{F}_k \cdot \mathbf{v}_W$. The dot product, not force magnitude, decides transfer. Substitution of the five components produces

$$
\begin{aligned}
P_{\text{prox,tan}} &= m_2 l_1^2 \ddot{\theta}_1 \dot{\theta}_1, \\
P_{\text{prox,cen}} &= 0, \\
P_{\text{dist,tan}} &= m_2 l_1 l_{c2} \ddot{\phi} \dot{\theta}_1 \cos\theta_2, \\
P_{\text{dist,cen}} &= -m_2 l_1 l_{c2} \dot{\phi}^2 \dot{\theta}_1 \sin\theta_2, \\
P_{g\text{-reaction}} &= m_2 g_{\text{proj}} l_1 \dot{\theta}_1 \sin\theta_1.
\end{aligned} \tag{3.11}
$$

Equation Equation 3.11 gives several strong results.

The proximal centripetal term does no work at a fixed circular hub path. It is radial to the arm while wrist velocity is tangential. It can be large in a force plot and still contribute exactly zero joint-force power. A mobile or noncircular hub breaks that orthogonality, which is why hand-path shaping is a separate mechanism in more complete models (Miura 2001; Nesbit and McGinnis 2009; Nesbit and McGinnis 2014).

The distal centripetal term is geometry gated. Its projection coefficient is $-\sin\theta_2$. With the club approximately 90° behind the arm ($\theta_2 \approx -90°$), the coefficient is near +1: a force directed toward the club's instantaneous centre can project strongly along hand motion. As the links align, this coefficient falls to zero.

The distal tangential term has the complementary gate. Its coefficient is $\cos\theta_2$: near a 90° wrist cock its projection is small, while it becomes large as the links align. The changing wrist angle therefore rotates the dominant transfer pathway from one inertial component toward another.

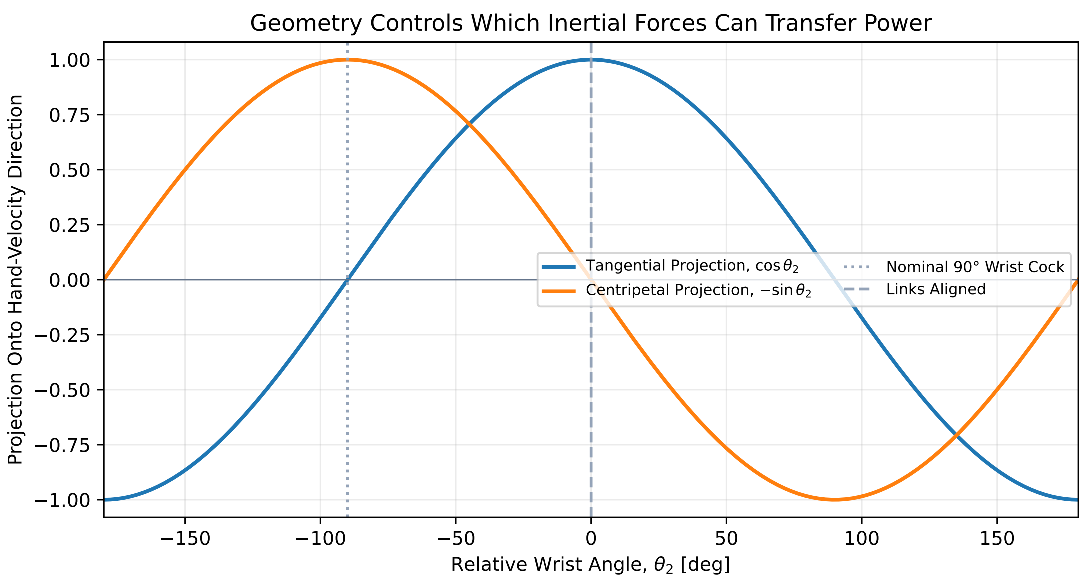

Figure 3.3: Geometry Controls Which Inertial Forces Can Transfer Power

This complementarity is the geometric core of the handoff. Early in the downswing, a compact configuration permits high proximal angular acceleration without forcing the club to acquire its final angular velocity immediately. As the wrist opens, the projection factors change while $\dot{\phi}^2$ grows. The force can therefore rise rapidly and rotate into a direction that performs positive work at the wrist. White's parametric-transfer description and Miura's hand-path analysis are consistent with this power-level account, but they emphasize different model freedoms (White 2006; Miura 2001).

3.6 Numerical Mechanism Audit

The numerical audit uses the existing reference program: 60 N · m at the shoulder, -10 N · m at the wrist until 0.10 s, and +15 N · m thereafter. Impact is the first valid upward crossing of club vertical. The program reaches that event at 0.3493 s with 38.85 m/s clubhead speed. These values describe the declared model, not a human normative target.

The exact force reconstruction reaches a peak wrist-force magnitude of 315.4 N before impact. The pointwise zero-torque case reaches 310.6 N, while the root-mean-square magnitude of the control-force increment is 16.3% of the root-mean-square total-force magnitude. The similarity is not evidence that torque is unimportant: prior torque shaped the state, and the present torque still changes the future. It does show that the late force demand at the observed state is dominated by the moving system's inertial and gravitational drift rather than by the instantaneous control increment.

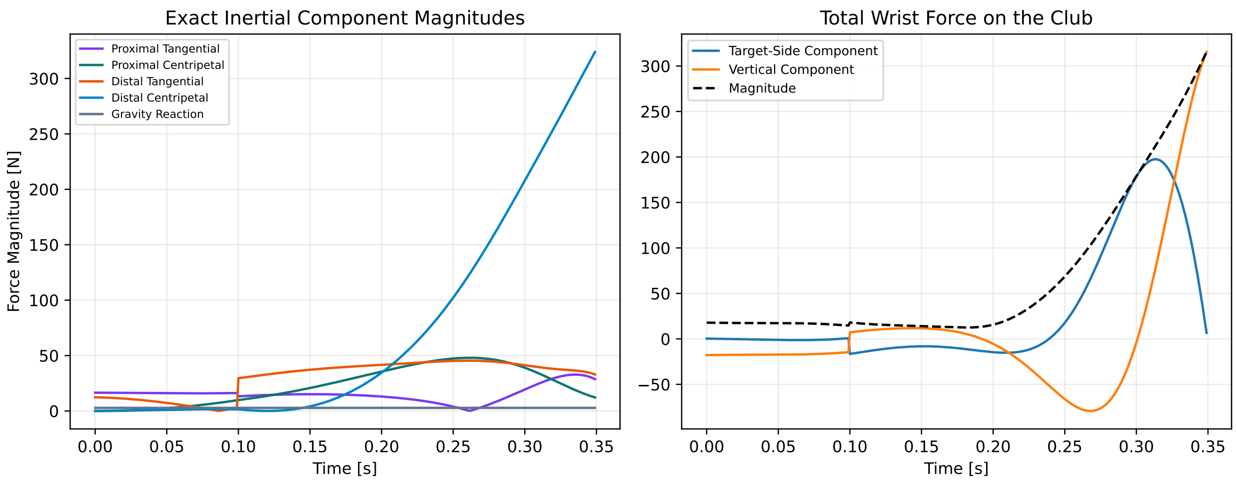

Figure 3.4: Exact Inertial Components and Total Wrist Force

Figure Figure 3.4 exposes the mechanism behind the late rise. The distal centripetal term grows approximately with $\dot{\phi}^2$ and is the largest component near impact. Its effect on transfer is then determined by the wrist-angle projection in Equation 3.11. The discontinuity in tangential terms at 0.10 s is expected: the prescribed wrist torque switches from restraint to drive, changing angular acceleration instantaneously while position and velocity remain continuous.

The integrated wrist-force work to impact is 130.7 J. The late half contributes 131.7 J, because the early half contains small positive and negative exchanges that nearly cancel. Across the full interval, positive force work is 136.8 J and negative force work is -6.1 J. The pointwise zero-torque force integrates to 132.2 J along the *commanded states*. That last quantity is an attribution diagnostic, not the energy of a realizable zero-torque future.

The force-power traces clarify why peak-force comparisons alone are insufficient. The proximal centripetal magnitude is visible in Figure 3.4 but its power trace remains zero by geometric orthogonality. The distal centripetal pathway grows into the dominant positive term late. The distal tangential pathway is negative through much of the middle downswing and turns positive near impact. Total transfer is their signed sum with proximal tangential and gravity-reaction contributions.

3.7 Pointwise and Forward Counterfactuals

Two zero-torque questions are useful, but they must have different names.

Pointwise ZTCF asks: *At this observed state, what acceleration and reaction force would the equations require if commanded torques were zero right now?* It holds $(q, \dot{q})$ fixed independently at every recorded sample. This is an instantaneous attribution and gives the exact affine split in Equation 3.9.

Matched-state torque killswitch asks: *Starting from this state, what future trajectory follows if torque remains zero?* It initializes commanded and zero-torque rollouts at the same $(q, \dot{q}, t)$ and integrates both.

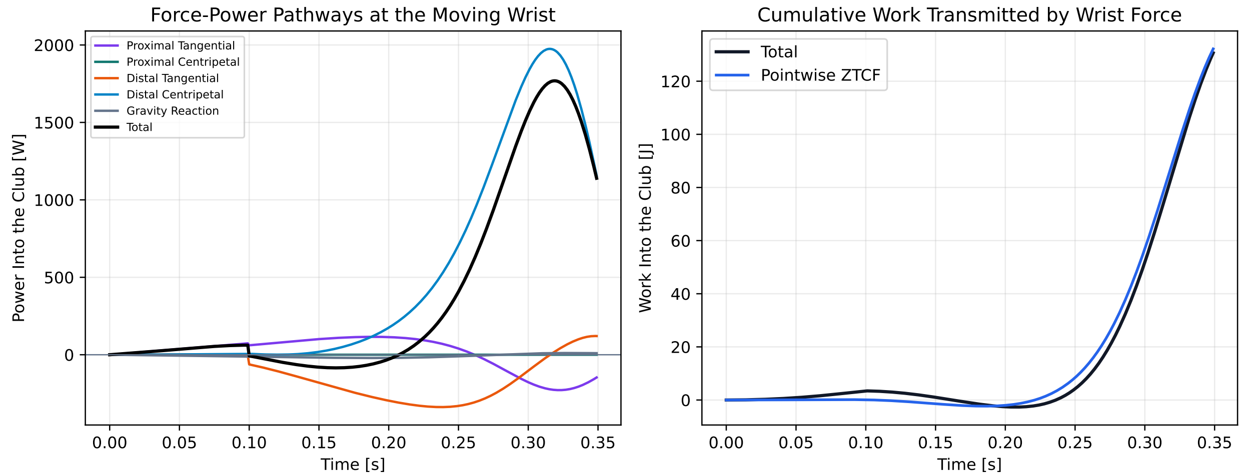

Figure 3.5: Force-Power Pathways and Cumulative Wrist-Force Work

Only the initial state is matched. As soon as accelerations differ, velocity, geometry, force, and power histories diverge.

The WSCG presentation constructs a trajectory-level counterfactual by applying kill switches at successive states and compiling the resulting values (Olson 2024). The present implementation makes the distinction executable. At a 0.251 s cut, the two rollouts begin with identical position and velocity, and the initial zero-torque acceleration exactly matches the pointwise drift calculation. After 0.120 s, their generalized positions differ by 0.540 rad and their generalized velocities by 8.49 rad/s. Figure Figure 3.6 shows why a pointwise ZTCF trace cannot be interpreted as a torque-free replay of the measured motion.

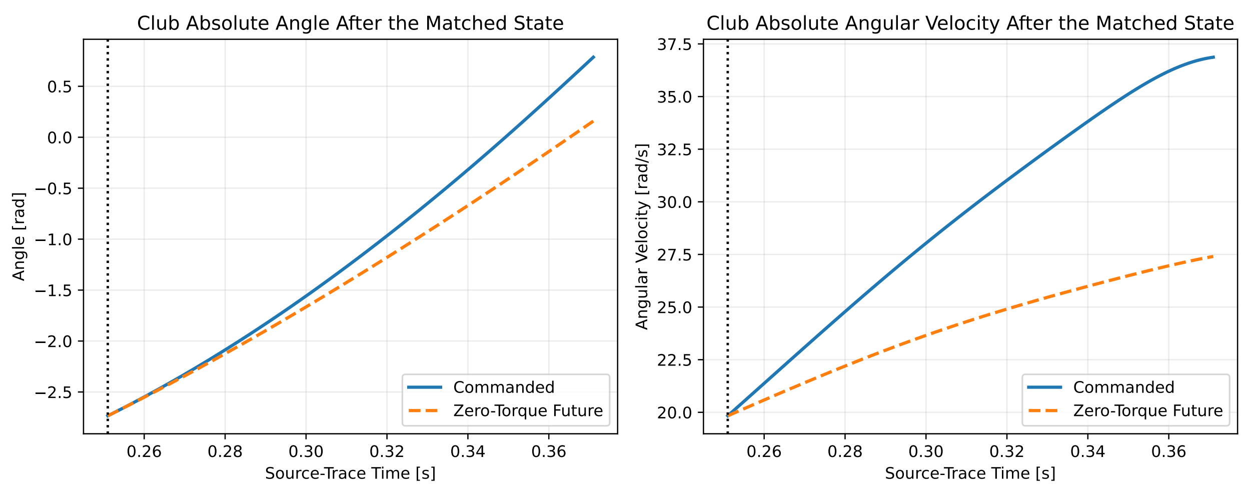

Figure 3.6: Commanded and Zero-Torque Futures From a Matched Late-Downswing State

Both counterfactuals are informative. The pointwise version isolates the force required by the state already achieved. The killswitch version tests whether that state can carry the club forward without continued control. A robust claim of passive late-swing transfer should be supported by both: large favorable pointwise drift power and persistence of the predicted behavior over a stated killswitch horizon.

3.8 Implications for Clubhead-Speed Strategy

Within this model class, the mechanism suggests a conditional strategy rather than a universal timing command.

1. **Build proximal motion while the system is compact.** Early shoulder work increases $\dot{\theta}_1$ and establishes the state on which later inertial forces depend. Premature distal drive can open the wrist before the favorable velocity-squared terms have grown.
2. **Preserve a useful relative angle long enough for inertial loading to develop.** Near a 90° lag angle, the distal centripetal force has a strong projection onto hand velocity. This is a mechanical condition, not a claim that a fixed angle should be copied across golfers.
3. **Allow geometry to redirect the transfer pathway.** As the wrist opens, centripetal projection falls and tangential projection rises. A productive release coordinates this rotation with increasing club angular speed.
4. **Use late wrist moment as a supplement, not as the sole explanation.** The tested late drive improves speed, yet most wrist-force work at the achieved states remains in the zero-torque drift term. Active torque helps create and steer the state; it need not directly account for most late transfer power.
5. **Judge a strategy at impact under constraints.** Maximizing an intermediate force, angular acceleration, or lag angle can delay delivery or create an invalid second-pass solution. Clubhead speed, orientation, actuator bounds, and robustness must be evaluated together (Pickering and Vickers 1999; Sprigings and Neal 2000).

These conclusions are model-conditional. A mobile hub adds radial hand velocity, a flexible shaft adds stored elastic energy, three-dimensional motion adds long-axis rotations, and a biological controller adds torque-rate and activation constraints. Each extension can change the best timing while leaving the power identity $P = \mathbf{F} \cdot \mathbf{v}$ intact.

3.9 Bridge to Two-Hand Negative Coupling

A single wrist point cannot produce an independent force couple: force and wrist moment are separate inputs. Two hands change the wrench geometry. If the lead and trail hands apply $\mathbf{F}_L$ and $\mathbf{F}_T$ at positions $\mathbf{r}_L$ and $\mathbf{r}_T$, their equivalent wrench about a reference point O is

$$\mathbf{F}_{\text{net}} = \mathbf{F}_L + \mathbf{F}_T, \qquad M_O = (\mathbf{r}_L - \mathbf{r}_O) \times \mathbf{F}_L + (\mathbf{r}_T - \mathbf{r}_O) \times \mathbf{F}_T. \tag{3.12}$$

Opposing hand-force components can cancel in $\mathbf{F}_{\text{net}}$ while adding in M_O. Consequently, a negative equivalent couple near impact does not by itself prove an actively commanded negative wrist torque. It can arise as the pair of constraint forces needed to enforce two moving hand contacts on the club. The WSCG two-hand planar simulation reported precisely this pattern: similar BASE and zero-torque net club forces, substantial opposing local hand forces, and a late reversal of the equivalent couple (Olson 2024).

The registered source series are redrawn in Figure 3.7. They preserve the reported BASE and counterfactual hand-force histories; they are not claimed as a reproduction of the underlying Simscape simulation.

This evidence motivates the next model test. The two-hand study must compute the equivalent wrench from local forces, verify sign conventions through power and moment balance, compare pointwise and killswitch counterfactuals, and separate momentum, gravity, shaft flex, damping, and numerical effects. The passive-negative-couple hypothesis survives only if the sign reversal persists under those controls and across reasonable contact spacing, hub paths, and shaft parameters.

3.10 Claims, Nonclaims, and Falsification Tests

The double-pendulum evidence supports four narrow claims:

- substantial wrist reaction force can exist at zero commanded torque;

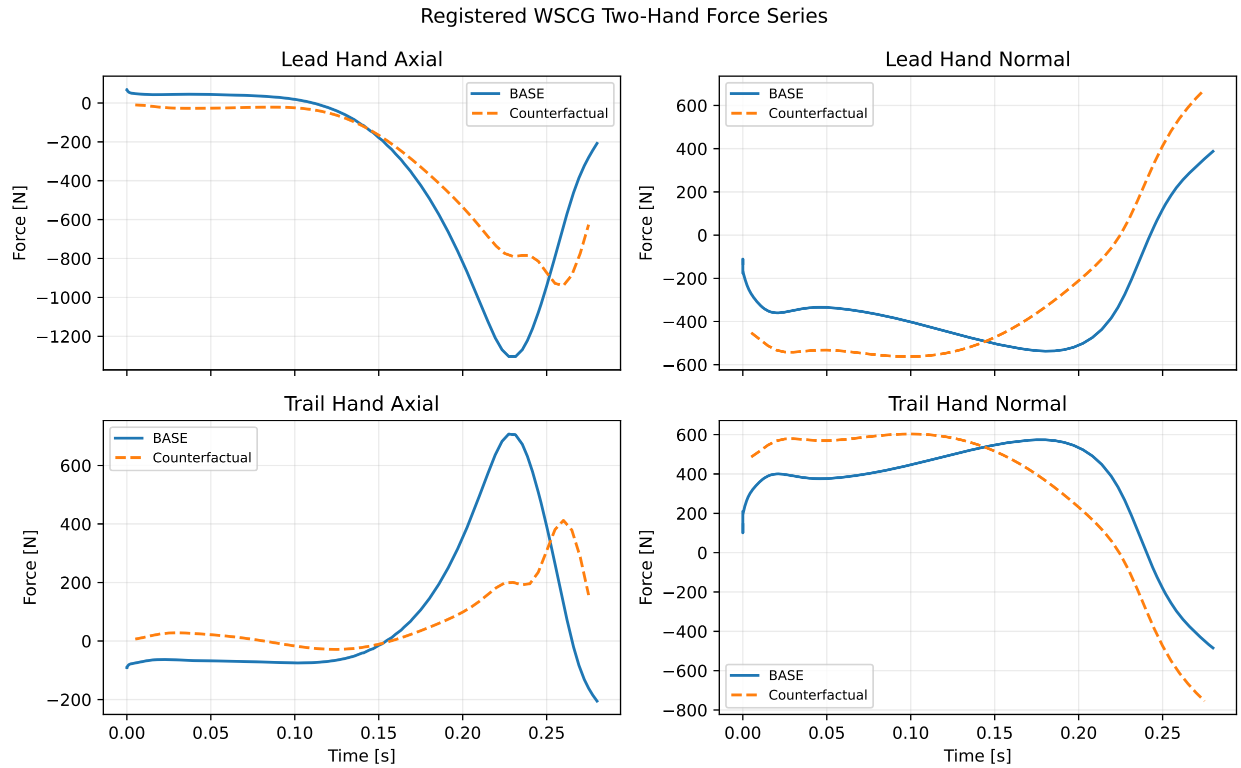

Figure 3.7: Registered WSCG Two-Hand BASE and Counterfactual Hand-Force Series

- force transfer is governed by a signed velocity projection, not force magnitude;
- wrist geometry produces exact sine and cosine gates on two major inertial power terms; and
- in the reference program, most positive wrist-force work occurs late and the pointwise drift force dominates the total-force magnitude at achieved states.

It does **not** establish that human golfers use the same torque history, that the two-hand negative couple is always passive, that more lag is always better, or that the reported parameter values are population estimates. The following tests could overturn or materially qualify the framework:

- a matched-state killswitch ensemble showing that favorable pointwise drift does not persist long enough to affect delivery;
- a two-hand wrench audit in which the negative couple disappears after gravity, shaft elasticity, damping, and numerical artifacts are separated;
- higher-order models in which hub translation or three-dimensional rotation reverses the dominant power pathway; or
- synchronized human motion and force estimates inconsistent with the modeled timing and sign structure.

These failure conditions bound the inference. The double-pendulum evidence identifies a reproducible mechanism; it does not convert behavior in one model into a population estimate or coaching prescription.

Chapter 4

Hand-Path Force Attribution Across the Model Ladder

4.1 The Observation and the Inference It Does Not Support

MacKenzie, McCourt, and Champoux analyzed drives from 76 right-handed amateur golfers with handicaps below 15 and represented the golfer's action on the driver as a resultant force at the mid-grip point plus a resultant couple (MacKenzie, McCourt, and Champoux 2020). Their path-averaged force was strongly associated with clubhead speed ($r = 0.96$); its square accounts for approximately 92% of the between-player variance in that sample. Mean linear work was 174 J, mean angular work was 39 J, mean hand-path length was 1.35 m, and mean force along the path was 129 N.

That result is important, but the observable must be named precisely. It is a signed mechanical line integral divided by path length,

$$\overline{F}_{\parallel} = \frac{\int_{t_a}^{t_b} \mathbf{F}_H \cdot \mathbf{v}_H \, dt}{\int_{t_a}^{t_b} \|\mathbf{v}_H\| \, dt} = \frac{W_{H,F}}{L_H}, \tag{4.1}$$

not a time average, not a force-magnitude average, and not an average of two individual hand-force magnitudes. It is also not a physiological effort measure. In a linked system, the force transmitted at the grip is the reaction required to satisfy the equations of motion and the joint constraints. Its value can contain a large contribution from gravity, velocity-dependent interaction dynamics, passive elements, prescribed-base motion, and earlier work stored in the achieved state. None of those contributions is identified by a net grip-force trace alone (Putnam 1991, 1993; Zajac and Gordon 1989). Even a specified endpoint force can admit wide ranges of muscle activation in a redundant musculoskeletal model (Sohn, McKay, and Ting 2013).

The correct conclusion from the 2020 study is therefore narrow and useful: golfers who produced more signed linear work per metre of hand travel also produced greater clubhead speed in that sample. The result does not establish that they experienced proportionally greater perceived effort, generated proportionally greater instantaneous muscle activation, or could obtain the same change by simply trying to pull harder along the observed path. This chapter asks the model-based question left open by that evidence: how much of the transmitted force, impulse, power, and work is present in the achieved state without the instantaneous command, and how does that attribution change with joint, phase, geometry, and model structure?

4.2 Four Distinct Counterfactual Quantities

For a control-affine model,

$$M(q)\ddot{q} + h(q, \dot{q}) = B(q)u, \tag{4.2}$$

the same-state acceleration is separated exactly as

$$\ddot{q}_{\text{total}} = \underbrace{-M^{-1}h}_{\ddot{q}_{\text{drift}}} + \underbrace{M^{-1}Bu}_{\ddot{q}_{\text{control}}}. \tag{4.3}$$

The **pointwise ZTCF sample** is the first term evaluated at the achieved state after the declared applied controls are set to zero. Repeating this operation along the achieved history produces a **stitched pointwise ZTCF trace**, not a single forward trajectory. In the analyses here, its inventory includes gravity and motion-dependent terms and includes a passive element only when that element belongs to the declared model. The **control increment** is total minus ZTCF at that identical state. The two close exactly because the state, constraints, parameters, and numerical protocol are held fixed.

The **ZVCF** is a different counterfactual: velocities are set to zero while the configuration and the declared commands are retained. Velocity-dependent terms disappear, but gravity, configuration-dependent stiffness, and the direct command generally remain. ZVCF is therefore not another name for the control increment. It answers which reactions and accelerations would remain at this configuration if the generalized velocity vanished under its stated protocol.

Finally, a **forward** or **branched ZTCF** starts a new trajectory after removing a command. It ceases to share the achieved state as soon as it is integrated. A pointwise ZTCF sample can attribute an instantaneous reaction; only a forward experiment can test persistence. Results from one experiment are not silently promoted into claims about the other.

For a constrained two-hand model, the same distinction applies to the complete Karush–Kuhn–Tucker solve,

$$\begin{bmatrix} M & -J^T \\ J & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} Bu - h \\ -\dot{J}\dot{q} \end{bmatrix}. \tag{4.4}$$

The contact reactions λ must be recomputed in each counterfactual. Subtracting unconstrained accelerations, assigning a residual by least squares, or comparing reactions from diverged trajectories does not provide a valid reaction-force attribution.

4.3 From Force Vectors to Transfer Metrics

At every moving interface, the path tangent is

$$\mathbf{e}_t = \frac{\mathbf{v}}{\|\mathbf{v}\|}, \qquad \|\mathbf{v}\| > v_\epsilon, \tag{4.5}$$

and the signed along-path components are

$$F_{\parallel,k} = \mathbf{F}_k \cdot \mathbf{e}_t, \qquad k \in \{\text{total}, \text{drift}, \text{control}, \text{ZVCF}\}. \tag{4.6}$$

The tangent is taken from the achieved trajectory for all pointwise additive components. Samples below the declared speed threshold are undefined rather than being assigned a convenient direction. This matters near the top of the backswing and at fixed joints.

Four integrals answer different questions:

$$\mathbf{J}_k = \int \mathbf{F}_k \, dt, \qquad J_{\parallel,k} = \int F_{\parallel,k} \, dt, \tag{4.7}$$

$$P_{F,k} = \mathbf{F}_k \cdot \mathbf{v}, \qquad W_{F,k} = \int P_{F,k}\, dt. \tag{4.8}$$

The vector impulse $\mathbf{J}$ retains direction in the global frame. The signed tangent impulse $J_\parallel$ integrates projections onto a tangent that rotates with the path, so it is not $\|\mathbf{J}\|$ and is not an energy measure. Force power weights the same signed projection by path speed; its integral is linear work. A component can dominate absolute impulse while doing little work, or do substantial work over a short high-speed interval without dominating time-integrated force.

At a joint that also transmits a couple, the complete planar wrench power is

$$P_k = \mathbf{F}_k \cdot \mathbf{v}_O + C_{O,k}\omega, \tag{4.9}$$

where the force, couple, translational velocity, and angular velocity use the same body, sign, frame, and reference point. Reporting force power and couple power separately reveals whether energy crossed an interface through translation, rotation, or opposing contributions whose net is small.

4.4 Fractions, Cancellation, and Undefined Ratios

For any additive signed quantity $X = X_d + X_c$, the familiar ratio X_d/X is useful only when X is safely separated from zero. Drift and control can oppose, so the ratio can be negative or exceed 100%; those values are physical sign information, not errors. Near cancellation, however, an arbitrarily large signed percentage is a poor summary. Each table therefore retains the raw values and reports both

$$s_d = \frac{X_d}{X}, \qquad s_c = \frac{X_c}{X}, \tag{4.10}$$

when the denominator is valid, and the cancellation-stable magnitude shares

$$m_d = \frac{|X_d|}{|X_d| + |X_c|}, \qquad m_c = \frac{|X_c|}{|X_d| + |X_c|}. \tag{4.11}$$

The companion cancellation index

$$\chi = 1 - \frac{|X|}{|X_d| + |X_c|} \tag{4.12}$$

is zero when the components reinforce and approaches one when large opposing components leave a small resultant. An attribution atlas without this guard would systematically exaggerate unstable percentages at exactly the joints and phases where the mechanics are most counterintuitive.

4.5 Model Ladder and Common Observables

The same observable contract is evaluated at three levels. Increasing model complexity is used as a stress test, not as a claim that the highest tier is a validated representation of an individual golfer.

4.5.1 Double Pendulum: Exact Minimal Mechanism

The two-link arm–club model has a fixed shoulder, a distributed-inertia arm, a distributed-inertia club, and two generalized controls. Exact equations of motion yield total and zero-command accelerations at every achieved state. The wrist reaction on the club is independently reconstructed from the club free-body equation, and the shoulder reaction is checked against the arm subsystem balance. This tier isolates the central mechanism: a large wrist reaction and positive wrist-force work can exist with zero instantaneous wrist command because the moving proximal segment, gravity, and joint geometry still constrain the club.

The fixed shoulder has no linear path and therefore no shoulder path-force estimand. It does have a transmitted couple and corresponding couple power. The wrist has both a translational path and a rotational interface. Keeping those observables separate prevents the phrase *force along the hand path* from being applied indiscriminately to every joint.

4.5.2 One-Arm Three-Link Tier: Joint-by-Joint Redistribution

The next tier introduces upper-arm, forearm, and club coordinates and reports shoulder, elbow, and wrist actions. The repository implementation uses distal point masses and is therefore a mechanism model rather than a calibrated anatomical limb. Its value is structural: adding an elbow allows the analysis to test whether drift dominance at the wrist implies drift dominance everywhere else. It does not. Each joint has a different distal subsystem, velocity, moment arm, and phase history, so force, couple, impulse, and work shares need not agree either across joints or within one joint.

4.5.3 Two-Arm Closed Loop: Constraint Reactions and Redundancy

The two-arm tier contains seven planar generalized coordinates: two relative joint pairs for the arms and a freely translating and rotating club. Four independent hand-position constraints close both hands on separated grip points. The constraint Jacobian must have row rank four, the mass matrix must remain positive definite, and both the acceleration-constraint and KKT residuals must satisfy declared tolerances. Singular poses fail closed.

This construction avoids a common but consequential shortcut in which the club translation is implicitly anchored to one hand and the second contact is added as if it supplied two further independent constraints. Such a model can return numbers from a least-squares solve while failing to identify the individual contact reactions required for the scientific claim.

4.6 Reference-Case Attribution Results

The three reference cases apply one observable contract but do not pretend to be three measurements of the same swing. The double pendulum is a forward simulation truncated at its first valid club-vertical impact (0.3493 s). The one-arm model is a forward simulation over a declared 0.400 s window. The two-arm closed loop is a 0.400 s, constraint-consistent **local kinematic sweep** with only 0.0425 m of mid-grip travel. Its line integrals diagnose the prescribed path; they are not realized forward-simulation work and are not scaled up to a full downswing.

Table Table 4.1 reports the MacKenzie-compatible primary estimand at the wrist for the open-chain models and at the mid-grip resultant for the closed loop. Drift is the stitched pointwise ZTCF contribution. Total equals drift plus control at the achieved state. ZVCF is displayed in a separate column because it is not an additive component of that equality; its force is projected on the achieved velocity only to make the diagnostic comparable in units.

Table 4.1: Primary Hand-Path Force-Work Attribution in the Declared Reference Cases

Model Tier	Path (m)	Total Work (J)	Drift Work (J)	Control Work (J)	ZVCF Projected Diagnostic (J)	$\overline{F}_{\parallel}$ Total / Drift / Control (N)
Double pendulum	2.2241	130.794	132.375	-1.580	-1.588	58.807 / 59.517 / -0.711
One-arm three-link	2.7336	27.802	28.781	-0.979	-1.827	10.170 / 10.528 / -0.358
Two-arm closed loop	0.0425	-0.00381	-0.05236	+0.04855	+0.00188	-0.0898 / -1.2330 / +1.1432

The first two rows answer the central inference question directly. In the double pendulum, the drift contribution is 101.2% of signed force work and the control increment is -1.2%; in cancellation-stable magnitude terms the split is 98.8% drift and 1.2% control. In the one-arm case, the signed split is 103.5% drift and -3.5% control, while the magnitude split is 96.7% and 3.3%. Thus a positive net force-work-per-path-length value can be generated mostly by the state-dependent reaction even while the current command contribution opposes the hand path. The result is about mechanical attribution within the declared trajectories. It does not imply that the achieved states were obtained without earlier control or biological effort.

The two-arm row exposes the opposite interpretive hazard. Drift and control work are individually about an order of magnitude larger than their net and oppose one another. Their magnitude shares are 51.9% and 48.1%, but the cancellation index is 0.962. A signed percentage of the near-zero resultant would exceed 1,000% and obscure the mechanics. Raw work, magnitude shares, and the cancellation index are therefore the defensible summary.

Figure Figure 4.1 shows why the integrated results cannot be inferred from a force peak alone. In the double pendulum, the late positive rise is almost entirely drift. In the one-arm model, drift changes sign and then becomes strongly positive while control remains opposing late. In the two-arm local sweep, much larger opposing drift and control components leave a small net resultant. The dashed ZVCF trace stays distinct from the control increment even when the two happen to be close over part of a history.

Figure Figure 4.2 retains vector direction rather than only the along-path scalar. All four splits use one force-arrow scale within each model row, and both two-hand contacts are shown. The achieved path tangent is a geometric reference, not an additional force. A control vector can therefore look counterintuitive in global coordinates while its tangent projection, normal component, and future geometric effect remain separately identifiable.

Figure Figure 4.3 separates the target-direction component of global vector impulse from force work. For example, the two-arm sweep has a positive 5.57 N s target-direction resultant assembled from -0.40 N s drift and +5.97 N s control, yet its net force work is nearly zero because work also depends on the direction and speed of the mid-grip path. ZVCF bars are again same-configuration projections, not a third additive share.

4.6.1 Every Joint and Time Window

The joint atlas uses total-wrench work, including both force and couple power. Each horizontal bar in Figure Figure 4.4 is $|W_d|/(|W_d| + |W_c|)$ for one joint and one normalized time quartile. The cross is plotted at the corresponding cancellation index when that index is at least 0.25. This presentation keeps all 36 model–joint–window combinations visible without converting cancellation into unstable signed percentages.

The atlas rejects a system-wide label such as "drift dominated." In the double pendulum, shoulder total-wrench work remains control dominated in all four windows, while wrist drift share rises from less than 1%

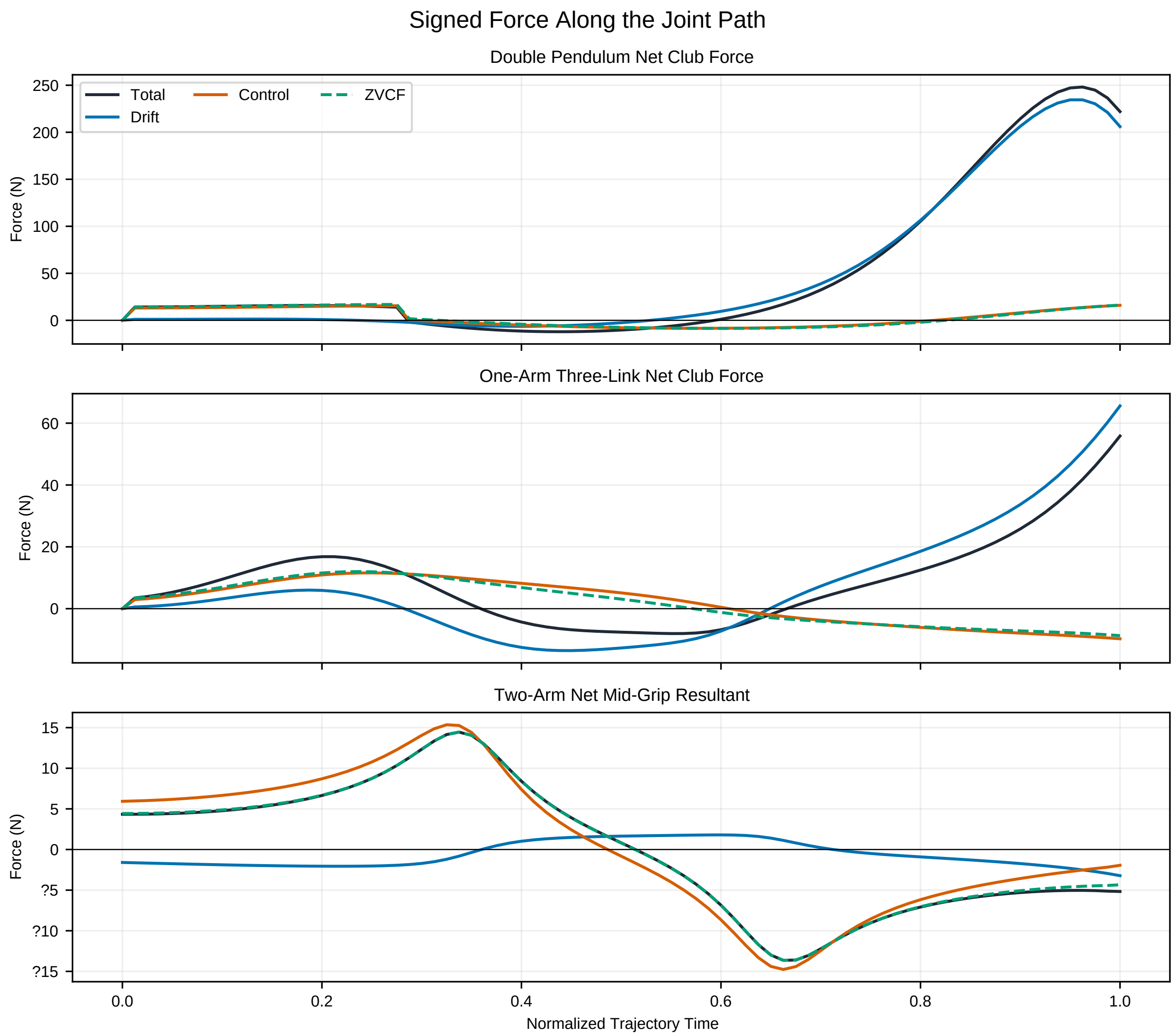

Figure 4.1: Signed Force Along the Achieved Hand Path for the Three Reference Cases

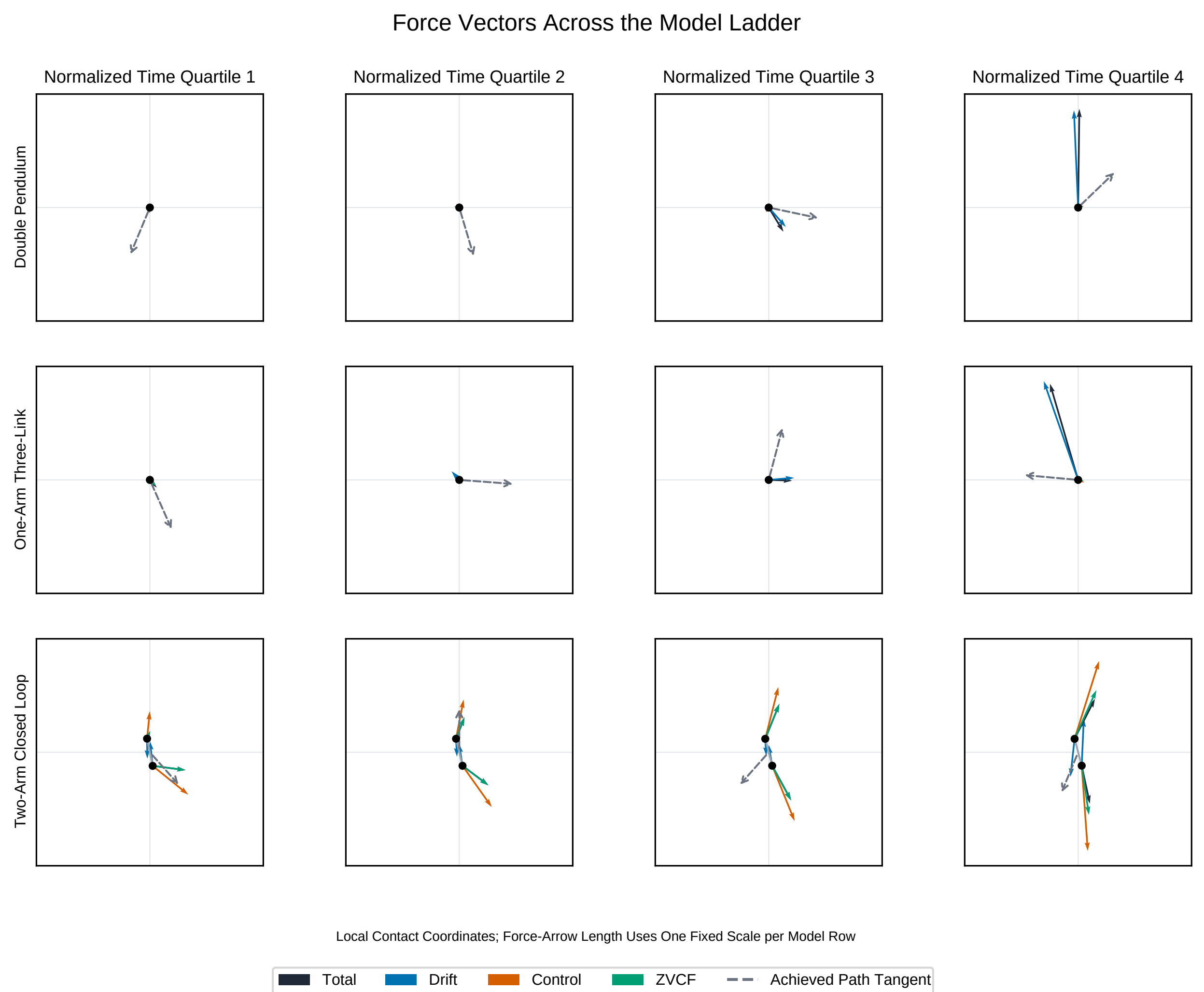

Figure 4.2: Total, Drift, Control, and ZVCF Force Vectors at Normalized Time Quartiles

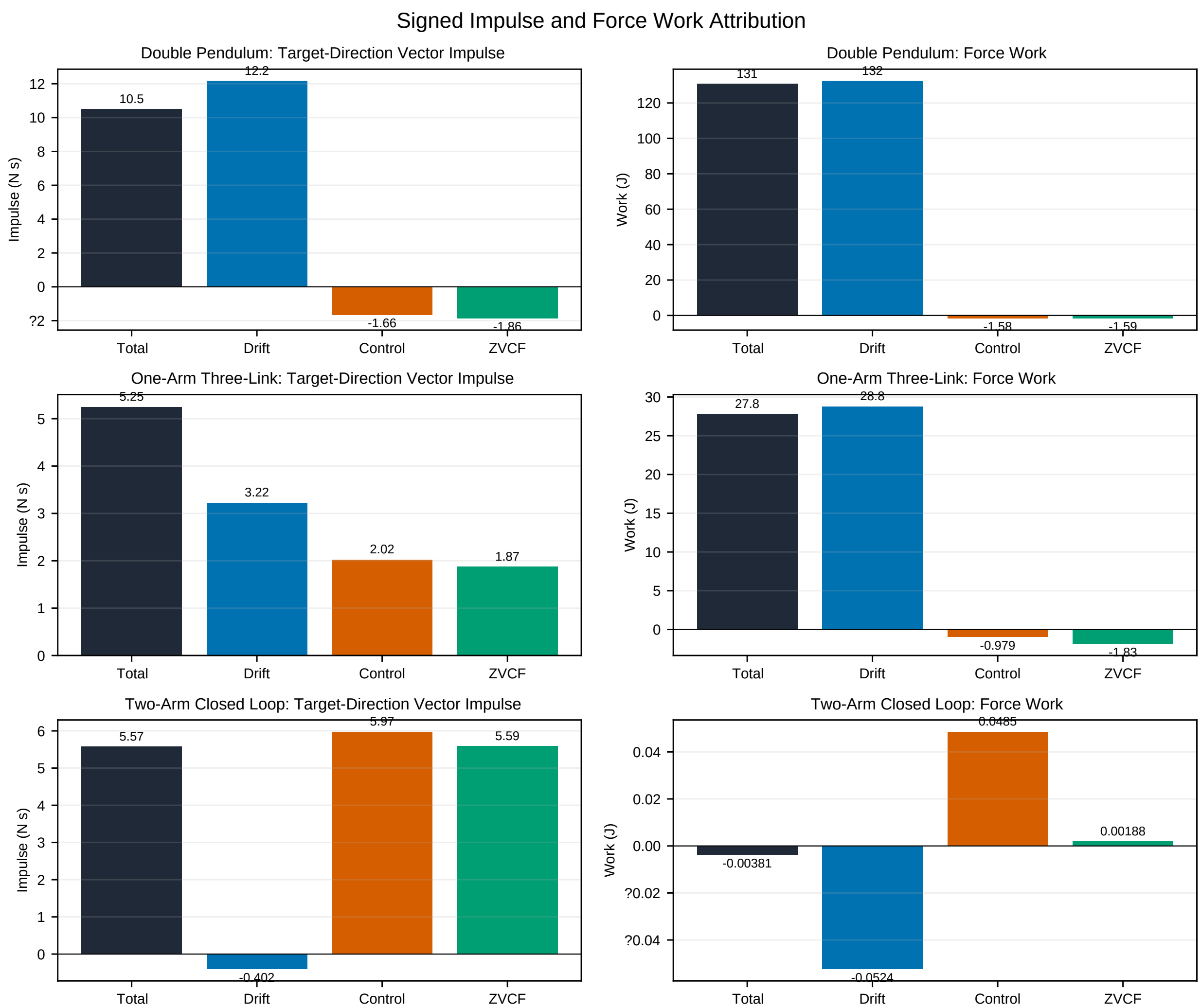

Figure 4.3: Signed Target-Direction Impulse and Force-Work Attribution

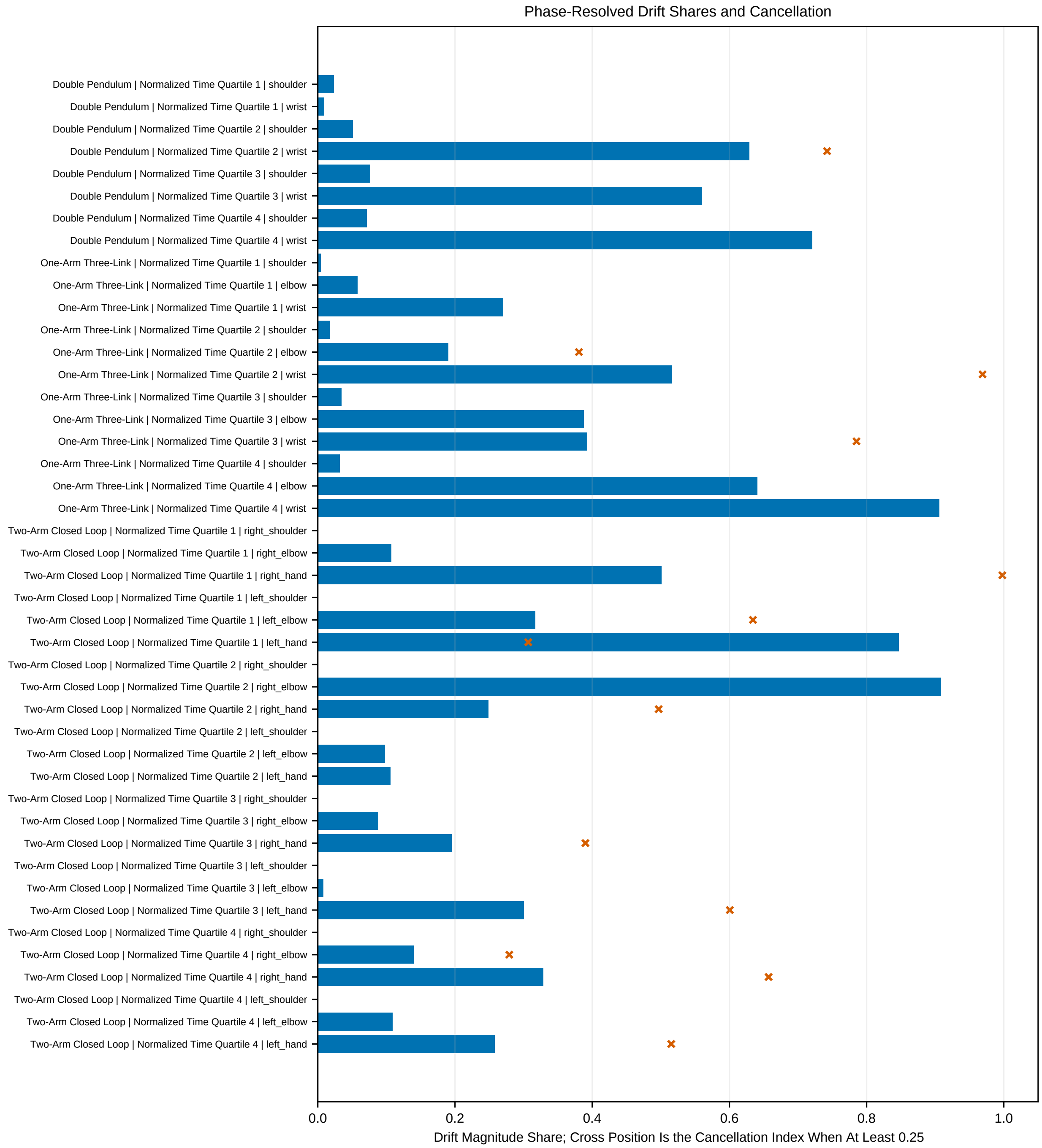

Figure 4.4: Phase-Resolved Drift Work Shares and Cancellation for Every Joint

in the first window to roughly 63%, 56%, and 72% thereafter. In the one-arm case, the shoulder is control dominated throughout, whereas the elbow and wrist become strongly drift influenced late. The two-arm model adds left/right asymmetry and contact redundancy: a drift-dominant hand contact can coexist with a control-dominant shoulder or elbow during the same interval. The exact raw work values and signed shares remain in the JSON evidence rather than being recoverable only from bar lengths.

Power requires an instantaneous guard analogous to the work guard:

$$m_{P,d}(t) = \frac{|P_d(t)|}{|P_d(t)| + |P_c(t)|}, \qquad \chi_P(t) = 1 - \frac{|P_d(t) + P_c(t)|}{|P_d(t)| + |P_c(t)|}. \tag{4.13}$$

Near-zero denominators are undefined and masked. Figure Figure 4.5 plots $m_{P,d}$ for every joint; crosses identify samples with $\chi_P \geq 0.25$. The rapid excursions are a substantive result: an integrated work share can be stable even though the instantaneous power allocation changes quickly or passes through near cancellation. This is why a single peak-power fraction is not substituted for the time history.

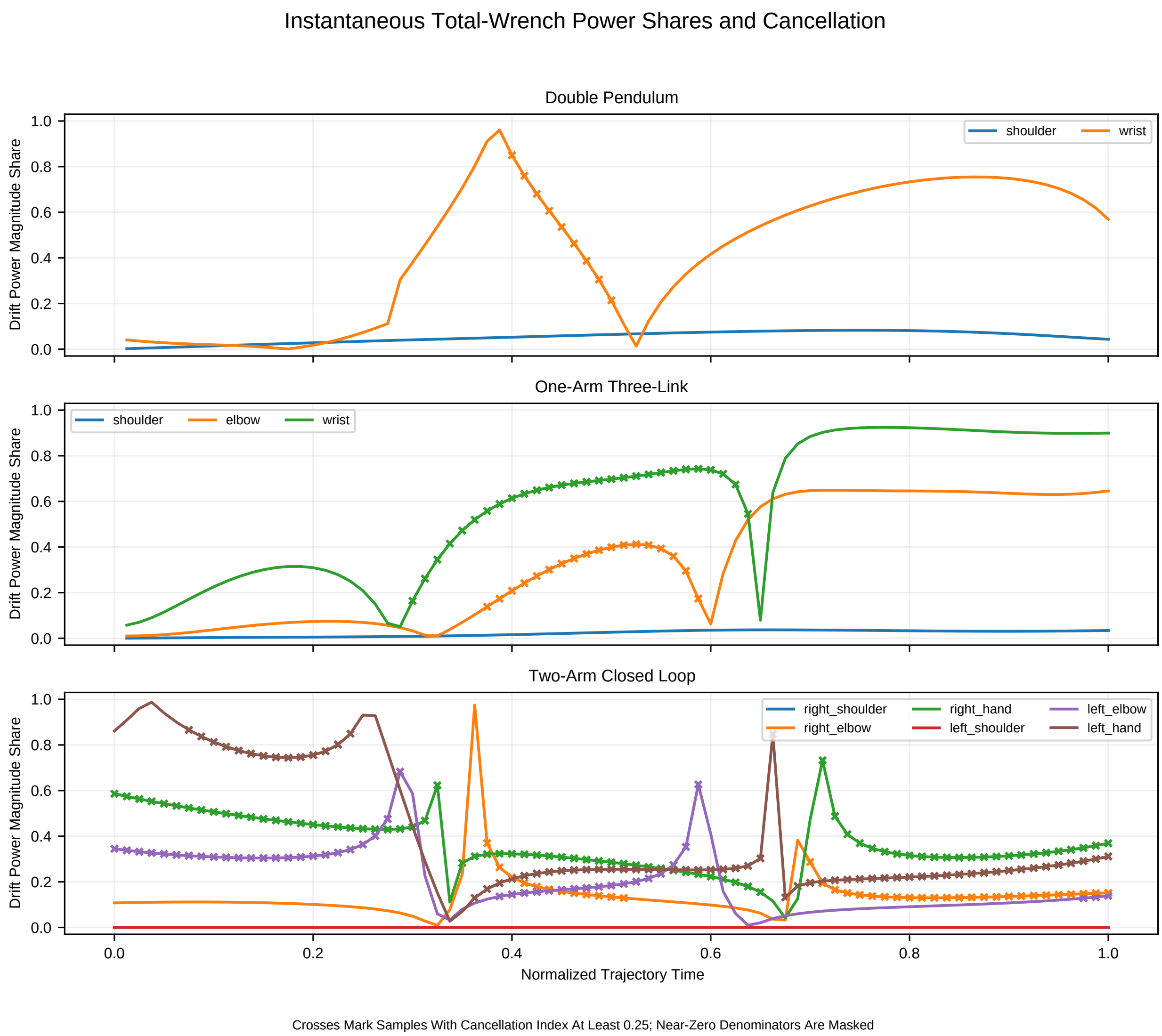

Figure 4.5: Instantaneous Drift Power Shares and Cancellation for Every Joint

4.6.2 Numerical Closure and Resolution Sensitivity

For all three tiers, the maximum additive force and couple residual is below 1.8×10^{-15} in SI units, the maximum power residual is below 2.3×10^{-13} W, and the maximum cumulative-work residual is below 5.7×10^{-14} J. The four windows cover every integration interval once and only once. Coarsening the stored histories from 81 to 41 samples changes primary force work by 0.044 J in the double pendulum, 0.059 J in the one-arm case, and less than 0.001 J in the two-arm sweep. At 21 samples, the respective changes are 1.11 J, 0.29 J, and less than 0.001 J. These are quadrature and sampling checks, not parameter uncertainty or biological validation.

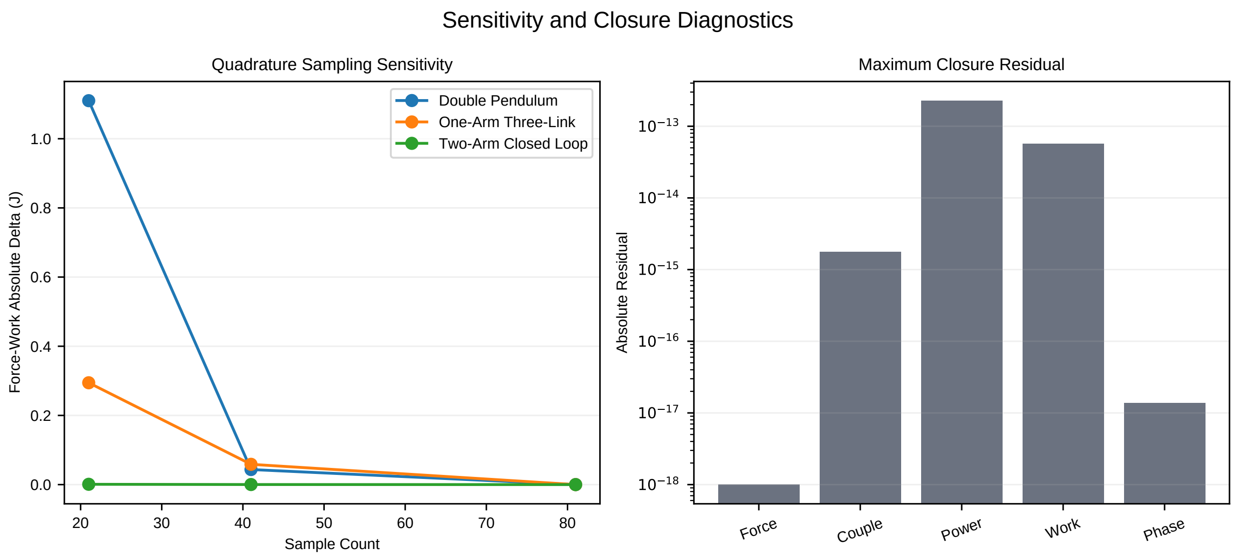

Figure 4.6: Quadrature Sensitivity and Additive Closure Diagnostics

4.7 Two Hands Create Observable and Internal Modes

Let $\mathbf{F}_R$ and $\mathbf{F}_L$ be the right- and left-hand forces on the club. The evidence package uses the explicitly declared convention

$$\mathbf{R} = \mathbf{F}_R + \mathbf{F}_L, \qquad \mathbf{F}_d = \frac{\mathbf{F}_R - \mathbf{F}_L}{2}. \tag{4.14}$$

The common mode is the resultant $\mathbf{R}$. About the grip midpoint, the differential mode creates the force couple

$$\mathbf{M}_{F,M} = -\mathbf{d} \times \mathbf{F}_d, \tag{4.15}$$

where $\mathbf{d}$ points from the right-hand contact to the left-hand contact. Two large opposing forces can therefore produce a small net force and a large club moment. Conversely, a large internal squeeze can increase contact loading while contributing no net rigid-club wrench. Neither the net hand force nor the club's rigid-body work identifies that internal demand (Zatsiorsky and Latash 2008).

The redundancy extends to control allocation. A desired equivalent club moment can be produced by the separated hand-force couple, by applied wrist free torques, or by combinations in which those pathways oppose. Equal net club action does not imply equal joint loading, actuator work, co-contraction, or robustness to perturbation. The constrained model can expose these mechanical alternatives, but without muscles, activation dynamics, and a physiological objective it cannot rank them as human effort strategies. Human

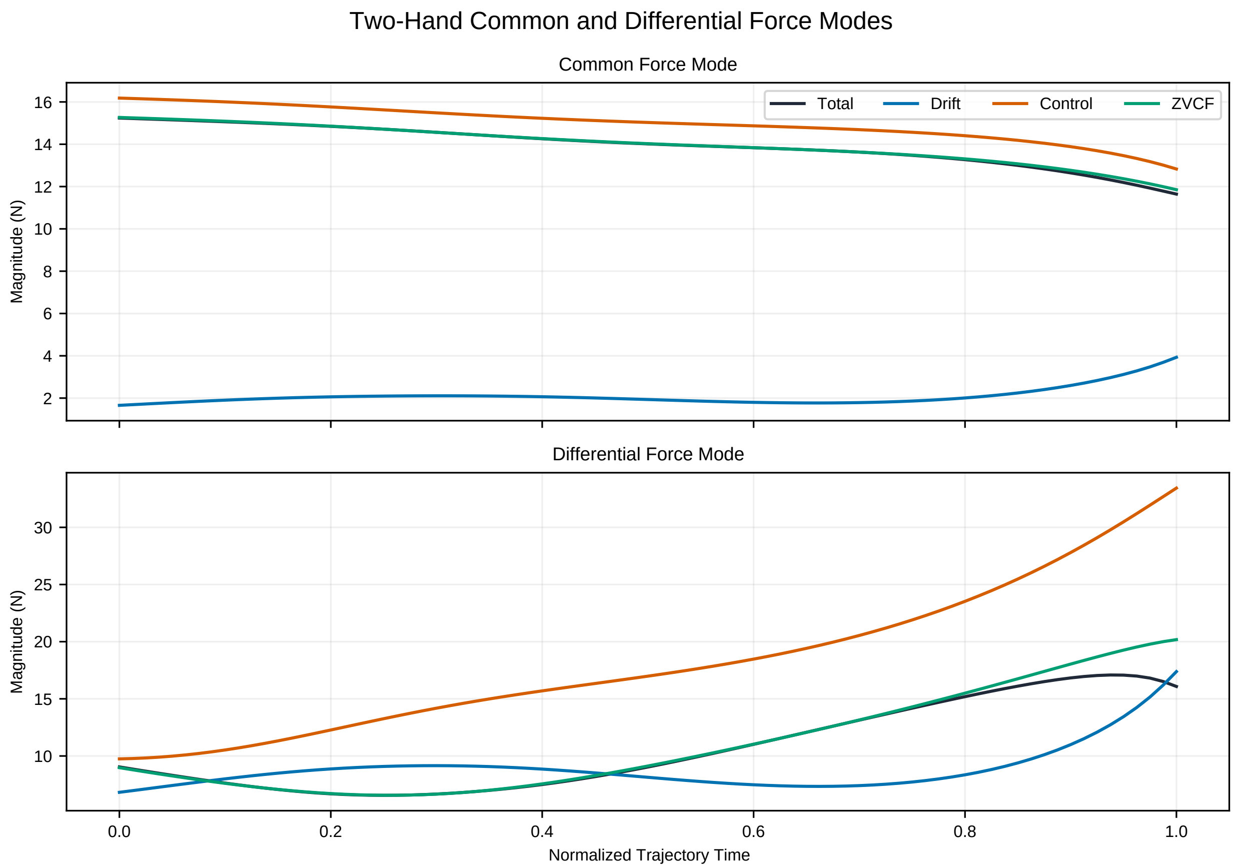

Figure 4.7: Common-Resultant and Differential Two-Hand Force Modes

two-hand experiments likewise show that similar object kinematics can coexist with substantially different hand torques and internal squeeze (Reinkensmeyer, Lum, and Lehman 1992).

Figure Figure 4.7 makes the hidden mode visible in the declared local sweep. The common resultant is largely control supplied while the drift contribution remains smaller. The differential-mode magnitudes are larger and diverge late, so the same interval can support a modest net mid-grip force and a substantial separated-contact couple. Magnitudes alone do not identify the sign of that couple; the signed contact vectors and the declared right-minus-left convention are required for that calculation.

4.8 Why an Apparently Counterproductive Control Can Be Useful

Optimal control need not point in the same direction as the instantaneous motion or the net reaction. A control increment may oppose the drift reaction while still improving a later objective through at least four mechanisms:

1. it can retain angular separation and delay distal acceleration;
2. it can rotate the future force vector toward a more useful path projection;
3. it can trade a transient negative actuator work interval for greater later positive force work; and
4. in a closed loop, it can alter the common and differential contact modes without the same change in net club wrench.

Consequently, a negative control force along the path is not automatically a mistake, and a positive drift force is not free propulsion. The achieved state embodies earlier control and energy history. A defensible strategy statement must specify the optimized objective, actuator limits, event definition, and cost of producing that state.

This interpretation is consistent with multijoint reaching evidence in which people exploited or compensated interaction torque differently as speed and load changed; reproducing the behavior required mixed kinematic, energetic, and dynamic costs rather than a single force-maximization rule (Vu, Isableu, and Berret 2016). It is supporting context for the mechanism, not golf-specific validation.

4.9 Late Reversal and the Preactivation Hypothesis

The archived WSCG result in Section 7 shows a late sign reversal of the force-generated two-hand couple under pointwise zero command. The present closed-loop model makes a related hypothesis testable: an opposed contact mode or wrist command established before that reversal may reduce the size or rate of the command change required when the drift contribution turns favorable. This is a mechanical **prepositioning** or **preloading** hypothesis.

It is not yet evidence of physiological preactivation. That stronger claim requires activation dynamics, electromechanical delay, torque-rate and torque–velocity limits, passive tissue impedance, and a matched physiological cost. It also requires experimental observables capable of separating net club action from individual-hand force and muscle activation. The model can show whether a prior internal-force state changes later wrench attribution; it cannot show that golfers use a neural preactivation strategy or that such a strategy improves performance in humans.

The hypothesis is falsifiable. It would be weakened if the effect disappears under timestep, event-boundary, geometry, and parameter perturbations; if it depends on a rank-deficient contact configuration; if equal-effort actuator models remove the apparent benefit; or if measured two-hand forces and muscle activity do not exhibit the predicted phase relation.

4.9.1 A Bounded Residual-Couple Preview Test

The archived WSCG traces permit one limited part of the hypothesis to be tested without inventing a muscle model. Treat the archived BASE equivalent midpoint couple as a reference, the stitched pointwise ZTCF couple as the reaction available at each achieved state, and their registered DELTA as the required control residual. At the pointwise ZTCF minimum (0.2148 s), the reaction is −19.63 N m while BASE is −22.78 N m. Thus 86.2% of the negative BASE couple at that instant is present in the declared zero-applied-control solve; the required residual is only −3.15 N m. Across the 0.16–0.25 s analysis window, the residual's peak magnitude is 3.18 N m compared with 23.17 N m for BASE.

Those values suggest a control-management rule within this archived model: control the residual, not the net couple. Commanding the entire BASE couple on top of the already-present drift reaction double-counts the mechanical action. To examine timing, a unit-gain first-order actuator was used solely as a signal model,

$$T_a \dot{C}_a + C_a = C_{\text{cmd}}, \tag{4.16}$$

with the archived BASE and pointwise ZTCF traces held fixed. For the reference $T_a = 30$ ms case, commanding the instantaneous residual produced a late-window tracking RMSE of 1.252 N m. Previewing that same residual by 24 ms reduced the RMSE to 0.531 N m, a 57.6% reduction. By contrast, sending the net BASE couple through the actuator and then adding the drift reaction produced 19.06 N m RMSE. Across assumed time constants from 10 to 50 ms, the best preview ranged from 9 to 35 ms and reduced residual-tracking error by 46.1% to 79.4%.

Figure Figure 4.8 establishes a narrow signal- tracking result: anticipation can compensate a declared first-order delay when the command targets BASE minus drift. It does not evaluate clubhead speed, muscle activation, metabolic cost, or a forward coupled golfer trajectory. The best preview is conditional on the assumed time constant and known future residual; it is not a recommended physiological timing value. The study turns preactivation into a reproducible hypothesis with explicit failure conditions rather than presenting it as an observed human mechanism.

4.10 Phase-Resolved Reporting Protocol

Every reference history is partitioned into four nonoverlapping, equal-duration windows whose exact time and sample boundaries are stored with the data. These are named **normalized time quartiles**, not anatomical swing phases. They make the bookkeeping and cross-tier comparisons reproducible, but equal fractions of elapsed time are not assumed to represent equivalent mechanics. When a model or experiment provides independently defined events, a second analysis should replace or supplement these neutral windows with intervals such as:

1. early acceleration and retention;
2. transition toward release;
3. rapid distal acceleration;
4. late drift-reaction reversal; and
5. the final delivery interval.

For every joint and time window, the evidence record stores vector impulse, signed tangent impulse, force work, couple work, total-wrench work, robust shares, cancellation indices, and closure residuals. Dense arrays retain the positive, negative, and absolute tangent-impulse accumulations and the pointwise force, couple, and total-power traces, so peak and sign-specific questions can be recomputed without reconstructing the dynamics. Path length and path-averaged force are reported separately. Total, drift, control, and ZVCF force traces are retained; ZVCF power is not called a realized quantity because its defining velocity is zero. A conclusion is called common across tiers only when its sign and interpretation survive the declared differences in coordinates, mass representation, constraints, and control allocation.

4.11 What the Models Can Establish

The model ladder can establish that a declared fraction of a mechanical reaction, impulse, power, or work is attributable to drift or control within a specified model at a specified state. It can show that drift dominance varies by joint and phase, that a large force may have a weak power projection, and that two-hand internal

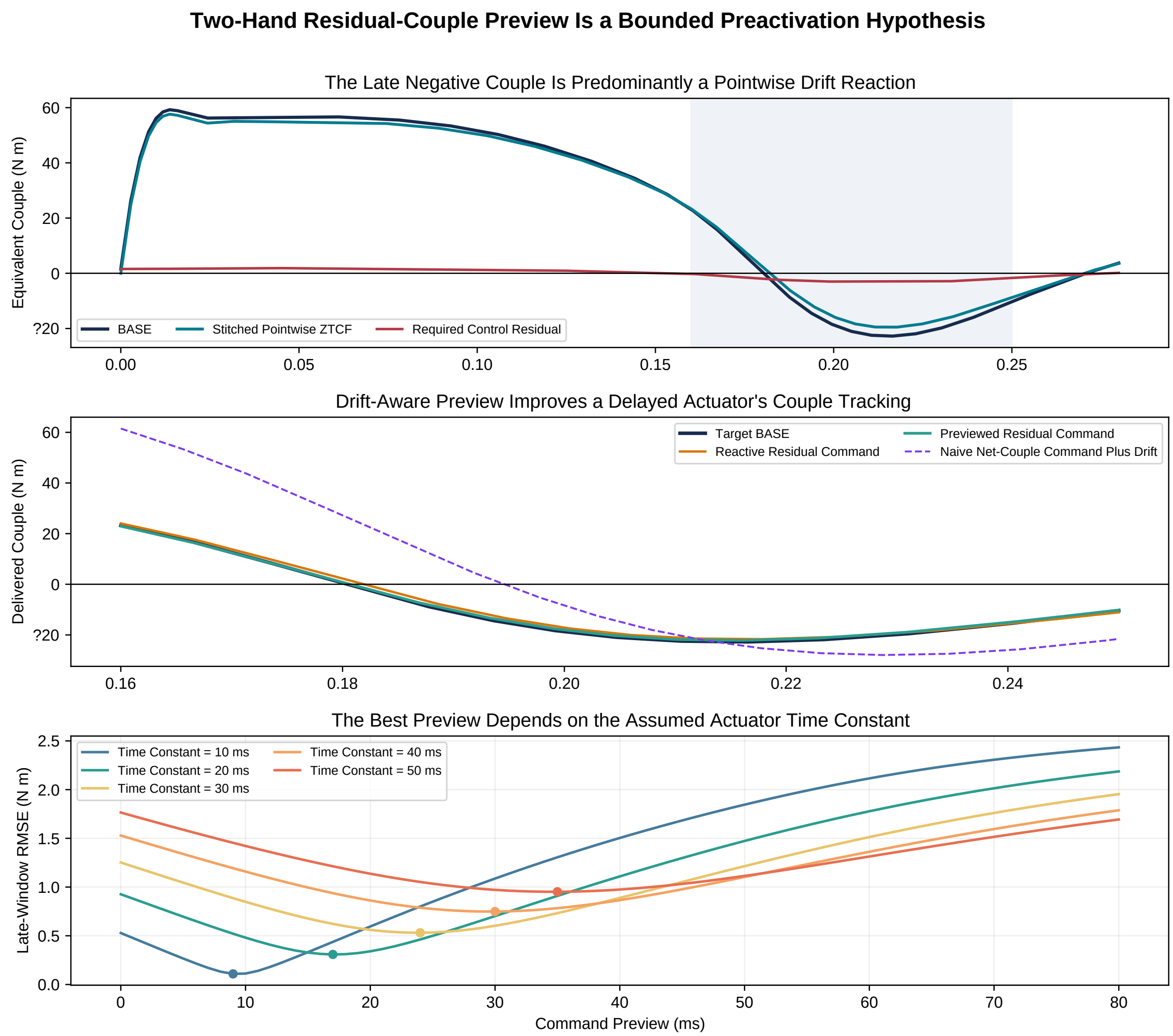

Figure 4.8: Residual-Couple Preview Test for the Archived Two-Hand Model

modes are hidden by the net grip wrench. It can also identify geometries and command allocations that are consistent with late positive transfer.

It cannot convert net force into perceived exertion, muscle activation, metabolic cost, injury risk, or a universal coaching instruction. Those require additional measurements and model layers. The neutral practical implication is therefore diagnostic rather than prescriptive: evaluate how the complete force vector, couple, and interface velocity evolve together, and distinguish the state-dependent reaction from the current command before interpreting a large hand-path force as greater effort.

Chapter 5

Ground-Reaction Drift Attribution

Ground-reaction force (GRF) is the external contact wrench by which the golfer and ground exchange linear and angular momentum. It is measured downstream of the neuromuscular commands, segment interactions, gravity, contact geometry, and inertial motion that jointly determine it. A force-plate trace is therefore neither a direct control signal nor a direct measure of energy transferred to the club. It is a constrained-dynamics observable. This chapter asks the narrower, falsifiable question: **given a declared multibody model and a measured state, how much of its predicted ground-reaction wrench is retained when controllable generalized torques are set to zero?**

The current evidence has two levels. First, a generic constrained-contact solver establishes the algebra and its identifiability conditions. Second, the reference double pendulum provides a deterministic fixed-support benchmark. The latter is a support-reaction analogue, not a human force-plate validation. No bilateral golf force-plate dataset synchronized to whole-body and club kinematics is present in the evidence package, so the human drift fraction remains an open empirical quantity.

5.1 Measurement and Model Boundaries

A six-axis force plate reports an equivalent contact wrench: three force components and three moments about a declared plate origin. Under a planar, rigid-ground convention, the center of pressure (COP) is inferred from the two horizontal moments and vertical force; the remaining vertical moment is the free moment. These derived quantities become unstable when vertical force is small and are frame- and sign-convention dependent. Pressure insoles directly measure normal pressure rather than the full six-axis wrench; recovering shear force and free moment requires an additional model. Joo and colleagues, for example, treated the nonvertical components as predictions learned against force-plate data rather than direct pressure measurements (Joo, Oh, and Mun 2016).

The golf literature also does not support one universal COP waveform. Ball and Best found two recurring direction-of-hit COP strategies across professional through high-handicap golfers (Ball and Best 2007). Han and colleagues associated specific lead-foot and trail-foot moment features with clubhead speed in a skilled male sample (Han et al. 2019), while Worsfold and colleagues measured shoe–turf torque differences by handicap and footwear condition (Worsfold, Smith, and Dyson 2008). The recent systematic review emphasizes continued variation in terminology, normalization, events, and analysis choices (Watson et al. 2026). These are population associations and strategy descriptions, not evidence that a particular GRF feature uniquely identifies a biological command.

5.2 Constrained-Reaction Equation

At one state, write the equations of motion as

$$M(q)\ddot{q} + h_0(q) + h_v(q, \dot{q}) = B(q)u + Q_{\text{ext}} + J(q)^\mathsf{T}\lambda, \tag{5.1}$$

with acceleration constraint

$$J(q)\ddot{q} + \gamma(q, \dot{q}) = 0. \tag{5.2}$$

Here h_0 is velocity independent, h_v is velocity dependent, u is the declared controllable input, Q_{ext} contains other known applied loads, and λ is the reaction expressed in contact coordinates. Define $A = JM^{-1}J^\mathsf{T}$. When J has full row rank and A is well conditioned,

$$\lambda = A^{-1}\left[JM^{-1}(h_0 + h_v - Bu - Q_{\text{ext}}) - \gamma\right]. \tag{5.3}$$

The implementation uses linear solves, not an explicit inverse. It fails closed when the contact rows are dependent. A pseudoinverse could return one minimum-norm allocation, but that numerical choice would not make the physical left/right allocation identifiable.

Equation Equation 5.3 gives four additive terms:

$$\lambda = \lambda_0 + \lambda_v + \lambda_u + \lambda_e,$$

where

$$\begin{aligned}
\lambda_0 &= A^{-1}JM^{-1}h_0, \\
\lambda_v &= A^{-1}(JM^{-1}h_v - \gamma), \\
\lambda_u &= -A^{-1}JM^{-1}Bu, \\
\lambda_e &= -A^{-1}JM^{-1}Q_{\text{ext}}.
\end{aligned} \tag{5.4}$$

This is an attribution under a stated model, not a causal identification from force-plate data alone. Modeling errors in M, J, segment inertias, marker kinematics, filtering, or external loads appear in the same residual one might otherwise mislabel as control. Dynamic inconsistency between measured motion and measured external force is a known inverse-dynamics problem (Sturdy et al. 2022; Werling et al. 2023).

5.3 Ground-Reaction ZTCF and ZVCF

The ground-reaction zero-torque counterfactual (ZTCF) sets $u = 0$ at the achieved $(q, \dot{q})$ while retaining declared non-control external loads:

$$\lambda_{\text{ZTCF}} = \lambda_0 + \lambda_v + \lambda_e.$$

The ground-reaction zero-velocity counterfactual (ZVCF) evaluates the same configuration and applied inputs with velocity set to zero:

$$\lambda_{\text{ZVCF}} = \lambda_0 + \lambda_u + \lambda_e.$$

Both are **pointwise**. Neither advances a zero-torque or zero-velocity system through time. They also overlap in $\lambda_0 + \lambda_e$; consequently, $\lambda_{\text{ZTCF}} + \lambda_{\text{ZVCF}}$ is generally not the total reaction. The complementary additive split is instead configuration, velocity, control, and other-external reaction in Equation 5.4.

In force-plate patterns, λ_0 supplies the configuration-dependent support baseline, λ_v can create phase-dependent shear and vertical features through centripetal, Coriolis, and moving-constraint terms, and λ_u

captures how applied generalized torques alter the constraint reaction. Similar-looking peaks can arise from different combinations. This is why timing coincidence between a GRF peak and a distal-speed increase does not by itself establish energy transfer through that force peak.

5.4 Fixed-Support Double-Pendulum Benchmark

The first executable case evaluates the shoulder support reaction of the same impact-truncated planar double pendulum used elsewhere in this report. At each of 161 achieved states, the analysis independently evaluates total, ZTCF, ZVCF, and zero-velocity/zero-control support forces. The fixed shoulder is an ideal support; it has no feet, COP, free moment, pelvis, or transverse-plane motion. Its purpose is to test the decomposition before introducing those degrees of freedom.

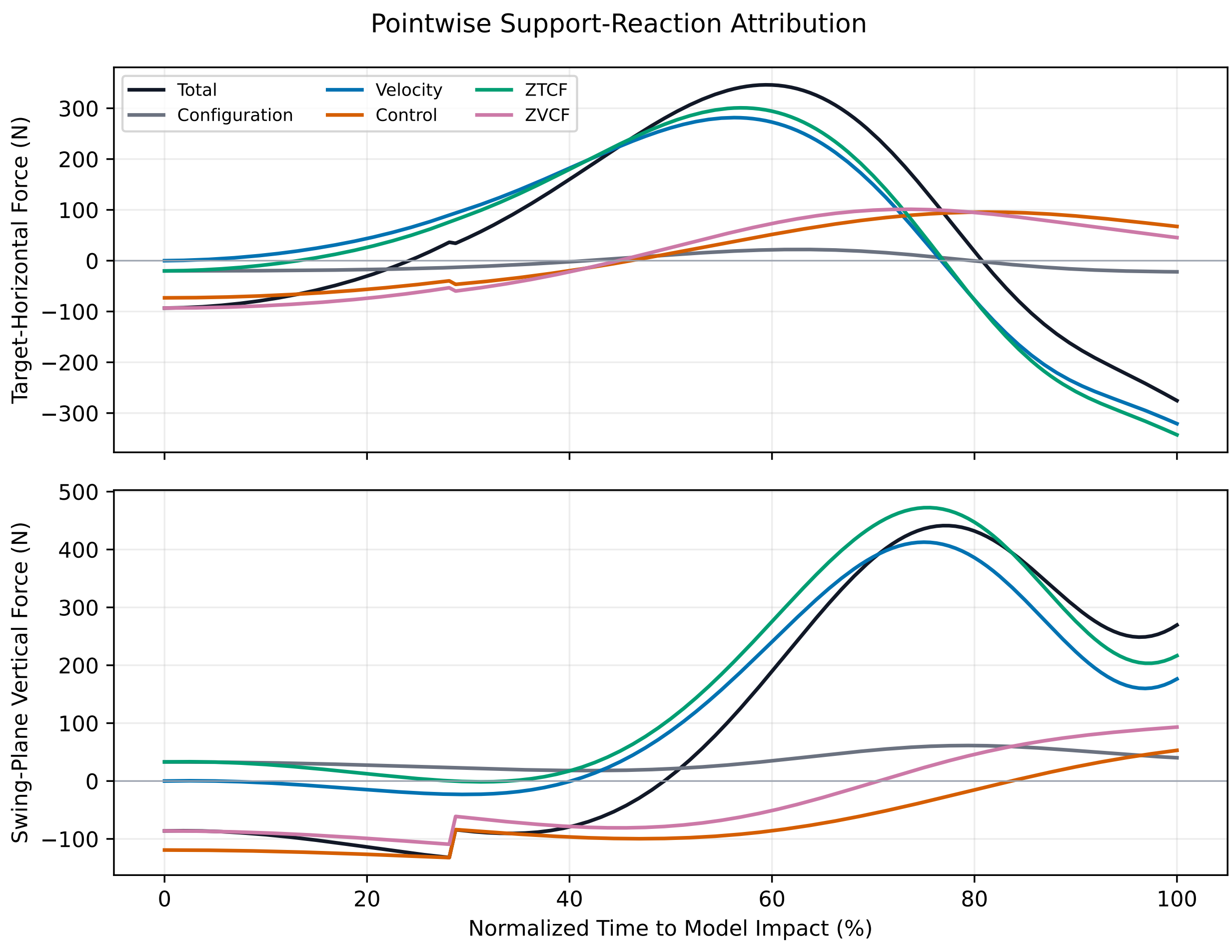

Figure 5.1: Pointwise Support-Reaction Attribution

The additive identities close to 5.7×10^{-14} N for the total split and 1.9×10^{-13} N for the independently evaluated ZVCF identity. Treating ZTCF as a drift-only predictor of the modeled total support reaction gives $R^2 = 0.871$ in the target-horizontal component and $R^2 = 0.814$ in the swing-plane vertical component. The corresponding RMSE values are 64.6 N and 89.8 N. They are 1.02 and 1.42 times the model's projected planar-weight scale, respectively. Thus the waveform association is strong while the pointwise amplitude error remains substantial; R^2 alone would overstate predictive adequacy.

The total vector impulse is $(22.92, 35.20)$ N s. ZTCF contributes $(17.91, 60.12)$ N s, while control contributes $(5.01, -24.93)$ N s. The vertical result is a clear cancellation case: the passive/configuration-plus- velocity impulse exceeds the total because control opposes it. A "percent of total" ratio would therefore exceed 100% and should not be interpreted as a fraction of biological effort.

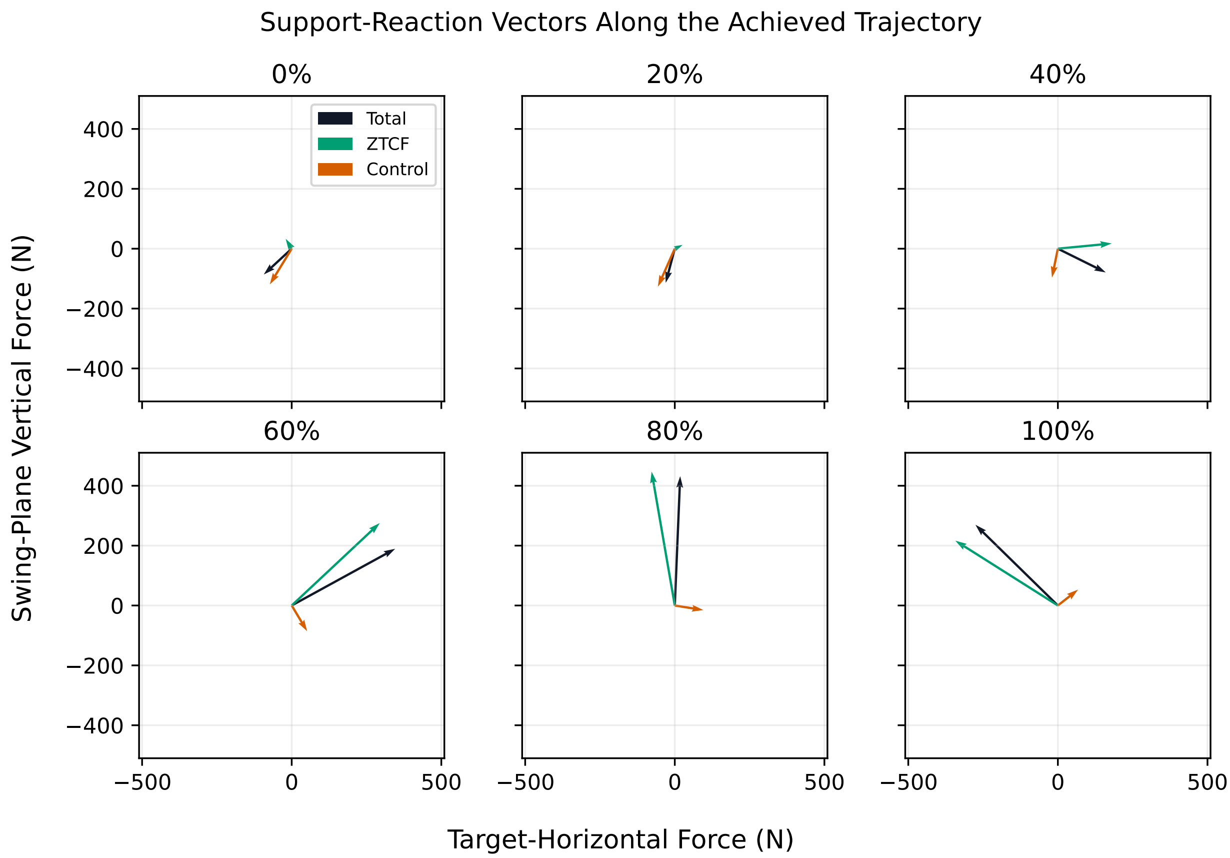

Figure 5.2: Support-Reaction Vectors Along the Achieved Trajectory

5.5 Relation to Common Golf GRF Patterns

The decomposition suggests testable, not prescriptive, interpretations:

1. **Vertical force peaks.** A peak can include configuration support, velocity-dependent vertical acceleration, and input-induced reaction. Zeroing velocity tests how much of the model peak requires motion; zeroing control tests how much persists passively at the achieved state.
2. **Anterior–posterior and medial–lateral shear.** These components are especially sensitive to frame definition, foot contact geometry, and moving-base terms. A resultant whole-body shear prediction does not specify its bilateral allocation.
3. **COP migration.** COP is a ratio derived from force and moment. It cannot be recovered from a planar point support and should not be decomposed by independently dividing component forces and moments. Each counterfactual wrench must first be reconstructed, then mapped to COP with low-force samples masked.
4. **Free moment.** The vertical free moment requires a three-dimensional distributed contact model. It is not present in the planar benchmark. The two-hand negative-couple mechanism elsewhere in this report is analogous in its force-couple geometry, but it is not evidence that a measured foot-ground free moment has the same source.
5. **Pressure transfer.** COP or plantar-pressure migration is not mass transfer and is not identical to energy transfer. Different COP strategies can be viable (Ball and Best 2007).

5.6 Human Falsification Protocol

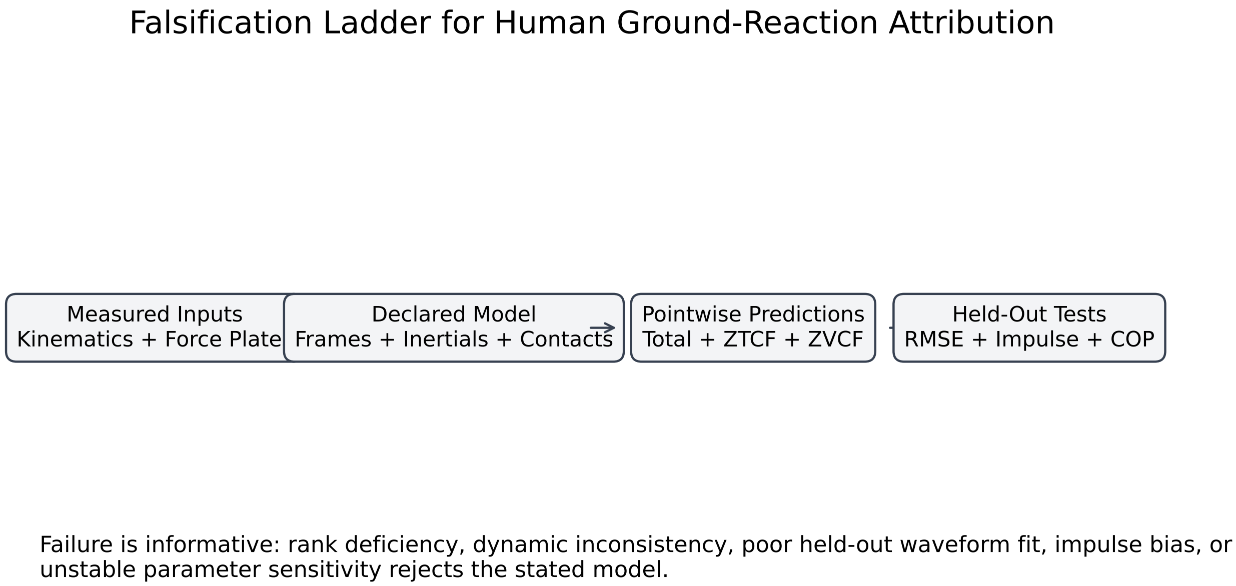

Figure 5.3: Falsification Ladder for Human Ground-Reaction Attribution

A human test requires synchronized bilateral six-axis force plates, whole-body and club kinematics, segment inertial estimates, and a declared contact/frame convention. The primary endpoint should be held-out waveform error for each force and moment component, supplemented by vector impulse error, peak timing error, COP error during adequately loaded samples, and free- moment error. Results should be stratified by participant rather than randomly splitting time samples from the same swing.

The following rejection criteria make the framework falsifiable:

- the full modeled wrench does not reproduce measured force and moment within preregistered tolerances;
- residual pelvis forces or moments remain large relative to measured GRF;
- ZTCF/ZVCF shares change qualitatively under plausible inertial, filtering, coordinate, or contact-model perturbations;
- a claimed bilateral attribution is not invariant across equally admissible contact allocations;
- a drift-only model fails held-out component, impulse, or event-timing tests; or
- a simpler baseline, such as center-of-mass acceleration plus gravity, predicts the same observables as well or better.

The machine-readable evidence records the declared frame, scale, component impulses, closure errors, prediction metrics, nonidentifiable quantities, and source hashes. This separates the demonstrated algebra and planar benchmark from the still-unmeasured human hypothesis.

Chapter 6

Matched-State Counterfactual Persistence

The previous chapter established an instantaneous result: at a fixed state, zero commanded torque can coexist with substantial wrist reaction force and positive joint-force power. That result does not show how long the behavior persists. This chapter tests persistence by releasing the commanded and zero-torque systems from identical states at multiple downswing phases and integrating both futures over controlled horizons.

6.1 Experimental Contract

Let the source trace provide a state $x_c = [q(t_c), \dot{q}(t_c)]$ at cut time t_c. Two simulations are initialized with exactly that state:

$$x_{\mathrm{cmd}}(0) = x_0(t_c) = x_c. \tag{6.1}$$

The commanded future continues the declared shoulder and wrist torque program; the killswitch future uses $\tau = 0$ at both joints. The initial acceleration difference is

$$\ddot{q}_{\mathrm{cmd}}(0) - \ddot{q}_0(0) = M(q_c)^{-1}\tau(t_c). \tag{6.2}$$

At the initial instant, the zero-torque acceleration, force, and force power are the pointwise ZTCF quantities. At every later instant, each future is evaluated on its own state. This construction preserves causal meaning:

- pointwise ZTCF attributes acceleration and force at the state already reached;
- the killswitch tests whether the state carries the passive mechanism forward; and
- commanded-minus-killswitch differences quantify the consequences of continued control over a declared horizon.

The ensemble spans eight cut times from 0.080 to 0.320 s, including 0.099, 0.100, and 0.101 s around the wrist-torque switch; four horizons from 20 to 120 ms; and integration steps of 0.5, 1, and 2 ms. This gives 96 baseline comparisons. Every row records position, velocity, wrist force, wrist-force power, integrated force work, and terminal clubhead-speed differences.

6.2 Why Cut Time and Horizon Matter

A counterfactual effect is not a property of a trajectory in the abstract. It depends on *when* the intervention occurs and *how long* the futures are allowed to diverge. A short early killswitch mostly tests the immediate acceleration change. A long late killswitch also changes geometry, club angular velocity, and the velocity-squared force terms. Figure Figure 6.1 shows this dependence directly.

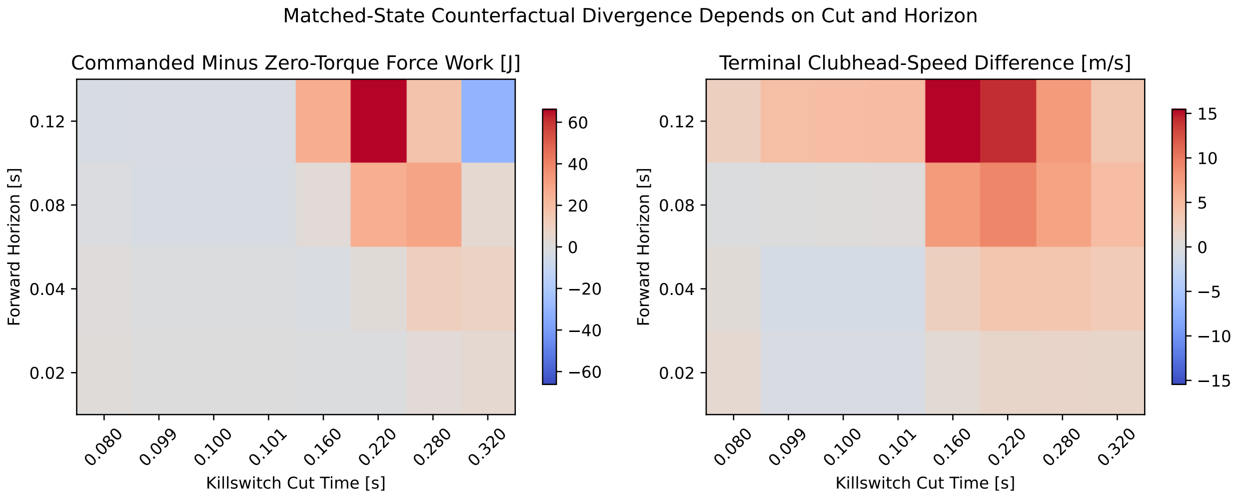

Figure 6.1: Matched-State Counterfactual Divergence by Cut Time and Horizon

At the 1 ms reference step, commanded-minus-zero-torque wrist-force work ranges from -30.88 J for a 0.320 s cut followed for 120 ms to +66.17 J for a 0.220 s cut over the same horizon. The corresponding terminal clubhead-speed differences are +3.42 and +14.20 m/s. A negative force-work difference does not mean the commanded future is globally slower: the wrist moment, gravity, and the state inherited at the cut also contribute to club energy. It means only that, over that horizon, continued control produces less *wrist-force work* than the zero-torque future.

The maps therefore reject a binary interpretation such as “the late swing is passive” or “continued torque is necessary.” Early and middle cuts generally show that continued control creates a rapidly different state and often more clubhead speed. Very late cuts can show less commanded force work because the two futures encounter different orientation and deceleration histories. The mechanism is phase dependent.

6.3 Separation of State, Force, and Power

Figure Figure 6.2 follows three representative cuts for 80 ms. The curves coincide at the vertical dotted line by construction. Their slopes then separate because acceleration differs. Force and power can diverge faster than angle because both depend directly on acceleration and velocity-squared terms.

The 0.12 s cut occurs soon after the restraint-to-drive transition. Continued wrist drive produces a more open club state, but commanded wrist-force power is initially more negative than in the zero-torque future. At 0.22 s, commanded force and power rise much faster than the killswitch values, producing a large positive work difference. At 0.30 s, both futures initially retain high positive power because both inherit a rapidly moving club. This last panel is the clearest persistence example: the zero-torque future does not lose its large interaction force when torque is removed; it loses ground gradually as the state evolves differently.

This behavior clarifies the meaning of passive contribution. A passive term can be large because earlier active work created high velocity. Removing torque late does not erase that history. Conversely, observing a large zero-torque force at the cut does not show that the same force would persist indefinitely. The matched-state horizon is part of every claim.

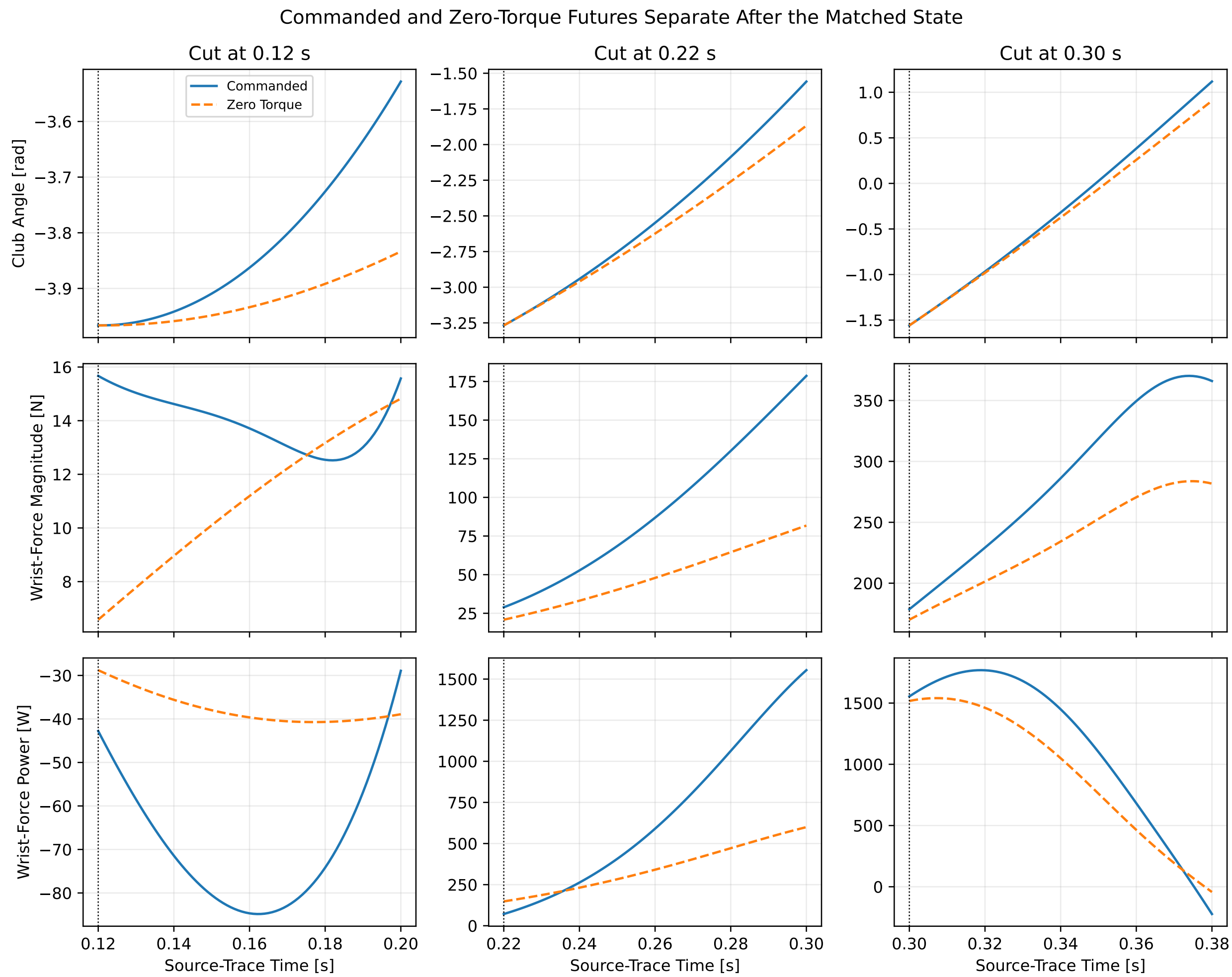

Figure 6.2: Commanded and Zero-Torque State, Force, and Power Futures

6.4 Control Discontinuity Audit

The prescribed wrist torque changes from -10 to +15 $N \cdot m$ at 0.100 s. A pointwise split changes discontinuously at that time because the control vector changes, while position and velocity remain continuous. Three cut times bracket the switch. Over an 80 ms future, commanded-minus-zero-torque force work is -2.83 J at 0.099 s, -2.87 J at 0.100 s, and -2.87 J at 0.101 s; terminal clubhead-speed differences are -0.03, +0.01, and +0.14 m/s. The force-distance metric changes smoothly from 19.98 to 20.85 N.

The close neighboring results show that the ensemble is not dominated by a single-sample numerical impulse. The acceleration changes discontinuously as the idealized step command requires, but RK4 integrates finite state changes, and the adjacent-cut futures remain continuous at the reported scale. A physiological torque-rate model remains preferable for human interpretation; the present audit establishes numerical behavior for the declared open-loop program only.

6.5 Timestep Convergence

The 0.5 ms result is used as the reference. Across all cut-time and horizon combinations, the maximum 1 ms difference is 3.9×10^{-10} rad in terminal angle distance, 1.1×10^{-8} rad/s in terminal velocity distance, 1.1×10^{-7} N in terminal force distance, 0.0174 J in force-work difference, and 4.5×10^{-9} m/s in terminal clubhead-speed difference. The 2 ms step produces larger but still bounded differences: up to 0.0091 rad, 0.0908 rad/s, 0.99 N, 0.356 J, and 0.190 m/s, respectively.

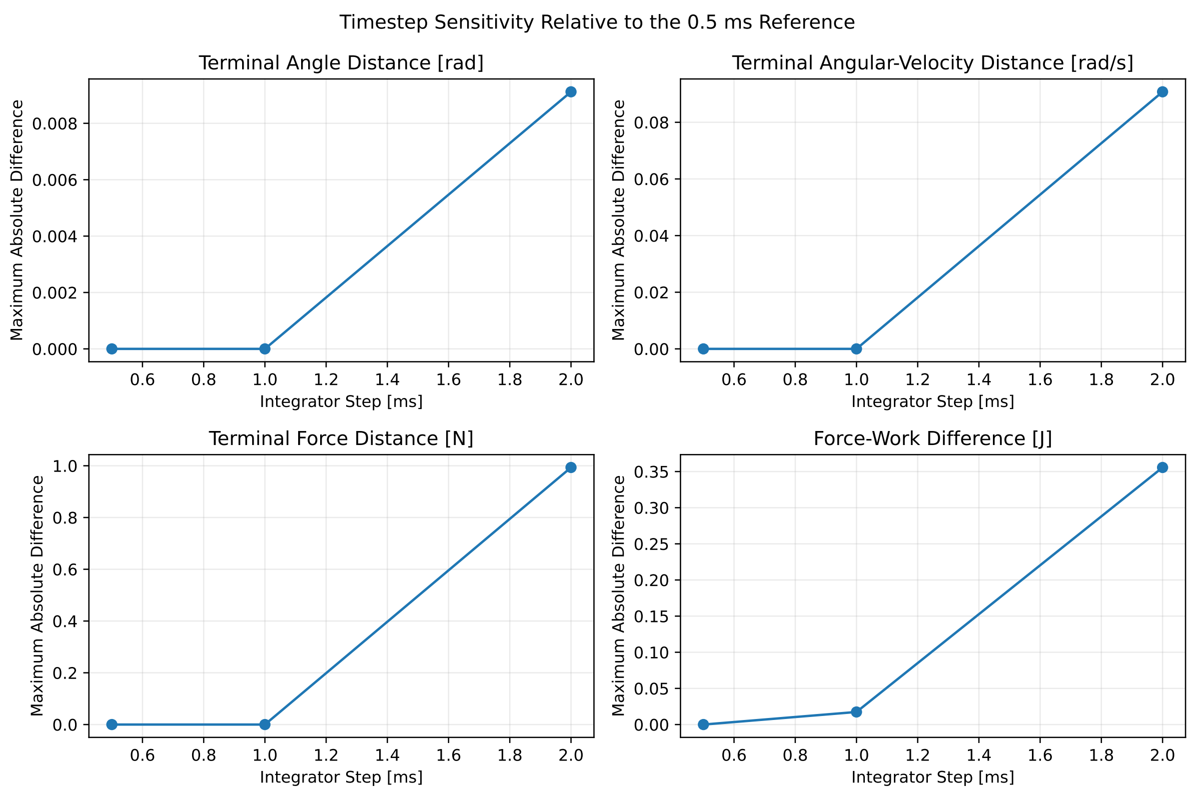

Figure 6.3: Timestep Sensitivity of Matched-State Divergence Metrics

The near-identity of the 0.5 and 1 ms state metrics supports use of the existing 1 ms article grid. Work is slightly more sensitive because it integrates a power difference and because some cases cross the idealized torque step. The 2 ms result is retained as a coarse-step warning rather than hidden by a single convergence number.

6.6 Gravity and Damping Ablations

Zero torque does not mean zero generalized forcing. Gravity and viscous damping remain inside the bias vector $b(q, \dot{q})$. To prevent the word *passive* from collapsing these mechanisms into one category, the 80 ms ensemble is repeated with projected gravity disabled and with both damping coefficients set to zero.

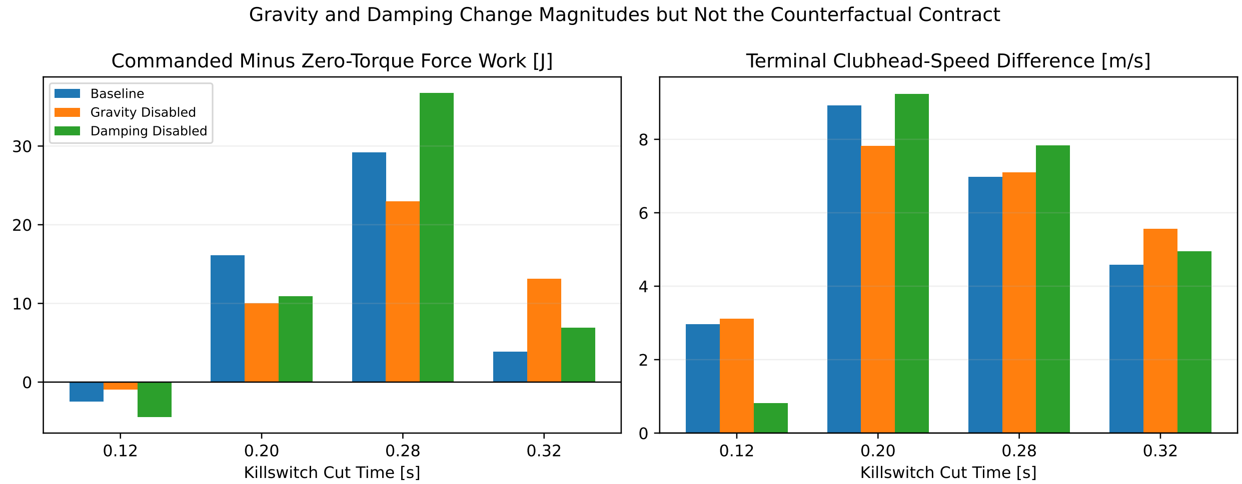

Figure 6.4: Gravity and Damping Ablations of the Killswitch Comparison

The ablations change magnitudes but do not eliminate phase dependence. At a 0.28 s cut, baseline commanded-minus-zero-torque force work is 29.17 J, versus 22.95 J without gravity and 36.73 J without damping. Terminal clubhead-speed differences remain positive in all three cases. At 0.12 s, force-work differences are negative in all three cases, while the damping-disabled speed difference is much smaller than baseline. These results show that gravity and damping materially shape the counterfactual; neither alone creates the late positive difference.

The ablations are not additive. Each altered model follows a different source trajectory before the cut, so subtracting variant numbers does not produce an orthogonal causal partition. A full mechanism partition would require matched source states with bias subterms selectively disabled inside the same dynamics evaluation. That finer separation is tracked in #8447.

6.7 WSCG BASE/ZTCF/DELTA Convention Check

The registered WSCG chart defines `DELTA = BASE - counterfactual`. Its four stored DELTA series can be reconstructed from the independently cached BASE and counterfactual series after time alignment. Maximum absolute residuals are 0.43 N for lead-hand axial, 1.48 N for lead-hand normal, 0.16 N for trail-hand axial, and 1.54 N for trail-hand normal. The residuals are small relative to the reported force ranges and arise from interpolating the dense BASE cache onto the 55-sample counterfactual grid.

This check establishes sign and subtraction convention, not model parity. The WSCG chart comes from a two-hand flexible-shaft model, while the present killswitch ensemble uses a rigid-shaft single-wrist double pendulum. Numerical agreement between their force magnitudes is neither expected nor claimed. The two-hand reproduction must compare like-for-like states, frames, hand contact points, and equivalent-wrench definitions.

6.8 Counterfactual Conclusions

The ensemble supports a stronger and more precise statement than the original pointwise analysis:

- pointwise ZTCF exactly equals the initial derivative of the matched-state zero-torque future;
- zero-torque interaction force and positive force power can persist for tens of milliseconds after a late cut;
- persistence and commanded-minus-zero-torque differences depend strongly on cut phase and horizon;
- the 1 ms integration step is converged for the reported state, force, work, and speed metrics relative to 0.5 ms; and
- gravity, damping, and the wrist-command discontinuity change quantitative results without explaining away the underlying distinction between instantaneous attribution and a realizable future.

These conclusions close the counterfactual-contract gap for the double pendulum. They do not establish the source of the two-hand negative equivalent couple. That question requires local hand forces and wrench geometry, addressed at the next fidelity level.

Chapter 7

Two-Hand Wrench Mechanics and Passive Couple Reversal

7.1 The Question Made Testable

The double-pendulum analysis established that a distal segment can receive a large reaction force even when the corresponding joint torque is zero. A two-hand grip adds a second mechanism: the two contact forces can form a substantial *couple* even when their vector sum is comparatively modest. The mechanical question is therefore not simply whether a wrist torque trace turns negative. It is whether the negative action on the club is attributable to commanded free torque, to the moment created by separated hand forces, or to a mixture of both.

The distinction matters because the three statements below are not interchangeable:

1. the resultant hand force is small;
2. the equivalent couple about the grip midpoint is negative; and
3. the golfer actively commands a negative wrist torque.

The first statement concerns translation, the second concerns rotation about a declared point, and only the third is an actuation claim. Two large and nearly opposing hand forces can satisfy the first two statements without satisfying the third. Conversely, a commanded free torque can be present even when the force-generated moment is zero.

This chapter tests the distinction against the archived output tables behind the WSCG presentation (Olson 2024). It does not digitize a picture. It reconstructs the force system from the stored contact positions, global hand forces, and free torques at all 2,801 time samples. The analysis is executable through the linked `run_two_hand_wscg_analysis.py`, and its mechanics are isolated in the backend-independent `two_hand_wrench.py`.

7.2 Source Cases and Evidentiary Boundary

The archived simulation supplies three synchronized tables:

- **BASE** is the commanded model solution;
- **ZTCF** evaluates the reactions after commanded joint torques and the designated damping terms are set to zero at each achieved BASE state; and
- **DELTA** is the stored operational difference, BASE minus ZTCF.

The ZTCF table is an instantaneous, state-matched attribution. It is not a single torque-free forward swing. This distinction is the same one developed in Section 6: a pointwise counterfactual can identify which part of the acceleration or reaction is present without current command, but it cannot by itself state how long that behavior would persist after commands were removed. Accordingly, the present result uses *passive* in the narrow actuation sense: the ZTCF contact moment exists while the evaluated command torques are exactly zero. It does not mean that the state was reached without earlier active work.

The binary input tables are registered by SHA-256 and exported to ordinary CSV by `export_two_hand_wscg_tables.m`. The exported cache makes the audit usable without MATLAB; the MATLAB function exists to prove where every column came from. The original MAT files and source presentations remain unchanged. Because these are project-originated model records, agreement with them is a reproduction of the author's prior simulation, not independent empirical validation.

7.3 Frames, Directions, and Signs

The archived right-handed model labels the left wrist as the lead contact and the right wrist as the trail contact. In this chapter:

- $\mathbf{r}_L$ and $\mathbf{r}_T$ are the lead- and trail-contact positions;
- $\mathbf{F}_L$ and $\mathbf{F}_T$ are forces exerted **by the wrists on the club**;
- τ_L and τ_T are free torques exerted on the club;
- O is the grip midpoint; and
- positive moment follows the right-hand rule about the model's $+z$ axis.

The local $+x$ axis points from the lead contact toward the trail contact. The fixed model-plane normal is

$$\mathbf{e}_z = \frac{1}{\sqrt{2}}(0, -1, 1), \tag{7.1}$$

and $\mathbf{e}_y = \mathbf{e}_z \times \mathbf{e}_x$. Projection of the stored global forces onto this basis reproduces the stored local axial and normal components. The maximum out-of-plane hand-force component is 1.78 N, small relative to in-plane peaks of several hundred newtons. Thus the archived result is numerically planar even though its vectors are stored in three coordinates.

Changing any one of force direction, hand labeling, axis direction, or moment sign reverses part of the interpretation. Those conventions are therefore data, not cosmetic plotting choices. They are written into the machine-readable provenance record and tested.

7.4 Reduction of Two Contacts to an Equivalent Wrench

About an arbitrary reference point O, the two-hand action reduces to

$$\mathbf{R} = \mathbf{F}_L + \mathbf{F}_T, \tag{7.2}$$

and

$$\mathbf{M}_O = (\mathbf{r}_L - \mathbf{r}_O) \times \mathbf{F}_L + (\mathbf{r}_T - \mathbf{r}_O) \times \mathbf{F}_T + \tau_L + \tau_T. \tag{7.3}$$

The scalar planar couple is $C_O = \mathbf{M}_O \cdot \mathbf{e}_z$. This is the quantity called the equivalent midpoint couple when O is the grip midpoint. The first two terms in Equation 7.3 are the **contact-force moment**; the last two are the **applied free torque**. Their sum, rather than either contribution alone, is the equivalent couple.

7.4.1 What Is and Is Not Reference Invariant

A useful correction is necessary here. The physical force system is invariant under a change of bookkeeping point, but the reported moment number generally is not. If the reference is moved from A to B,

$$\mathbf{M}_B = \mathbf{M}_A - (\mathbf{r}_B - \mathbf{r}_A) \times \mathbf{R}. \tag{7.4}$$

Only a pure couple ($\mathbf{R} = 0$) has the same moment about every point. A nonzero two-hand resultant requires the transport term in Equation 7.4. Thus an audit should demand *wrench equivalence under reference transport*, not invariance of the scalar midpoint moment. The test suite evaluates Equation 7.4 directly and also verifies invariance to a rigid translation and co-rotation of the complete force system.

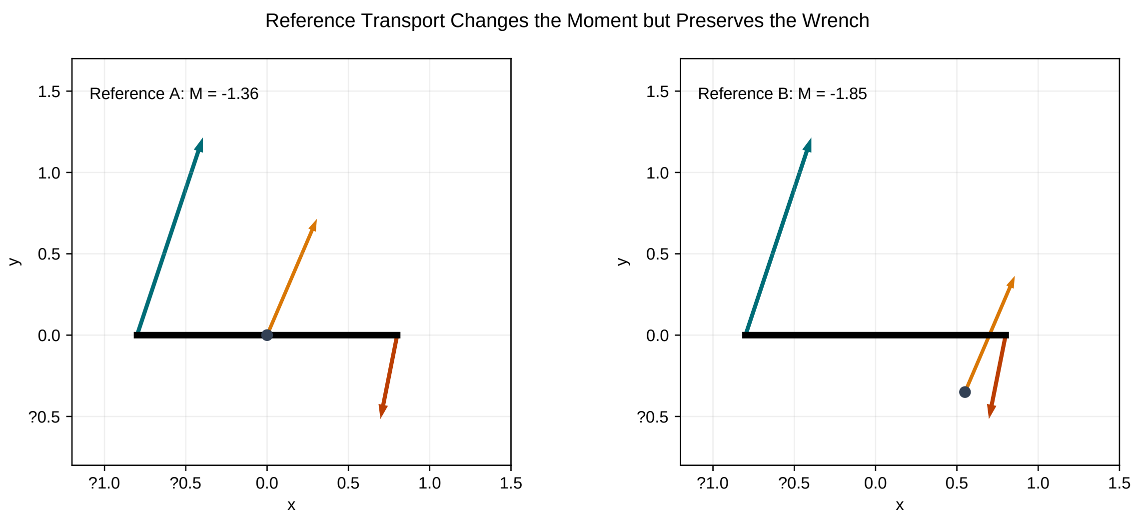

Figure 7.1: Reference Transport Changes the Moment but Preserves the Wrench

Figure 7.1 illustrates the distinction. The resultant arrow is unchanged, while its associated moment changes exactly enough to preserve the action of the original two forces. This prevents a common error: comparing couples calculated about different grip points as if they were the same physical observable.

7.5 Common and Opposed Force Modes

Let the grip midpoint be $\mathbf{r}_M = (\mathbf{r}_L + \mathbf{r}_T)/2$, define the contact-separation vector $\mathbf{d} = \mathbf{r}_T - \mathbf{r}_L$, and introduce common and differential force modes,

$$\mathbf{F}_c = \frac{\mathbf{F}_L + \mathbf{F}_T}{2}, \qquad \mathbf{F}_d = \frac{\mathbf{F}_L - \mathbf{F}_T}{2}. \tag{7.5}$$

Then $\mathbf{R} = 2\mathbf{F}_c$, while the force-generated midpoint moment is

$$\mathbf{M}_{F,M} = -\mathbf{d} \times \mathbf{F}_d. \tag{7.6}$$

Equation Equation 7.6 is the central two-hand mechanism. The common mode controls the resultant. The differential, or opposed, mode controls the couple. A large $\|\mathbf{F}_d\|$ may therefore be nearly invisible in $\|\mathbf{R}\|$.

If $\mathbf{d} = s\mathbf{e}_x$, only the differential normal component contributes to the planar force moment:

$$C_{F,M} = -sF_{d,y}. \tag{7.7}$$

Axial forces can be very large without directly producing a midpoint moment when their lines of action remain collinear with the grip. They still matter to constraint loading, deformation, and power. Normal-force imbalance, contact separation, and their signed relative orientation set the moment.

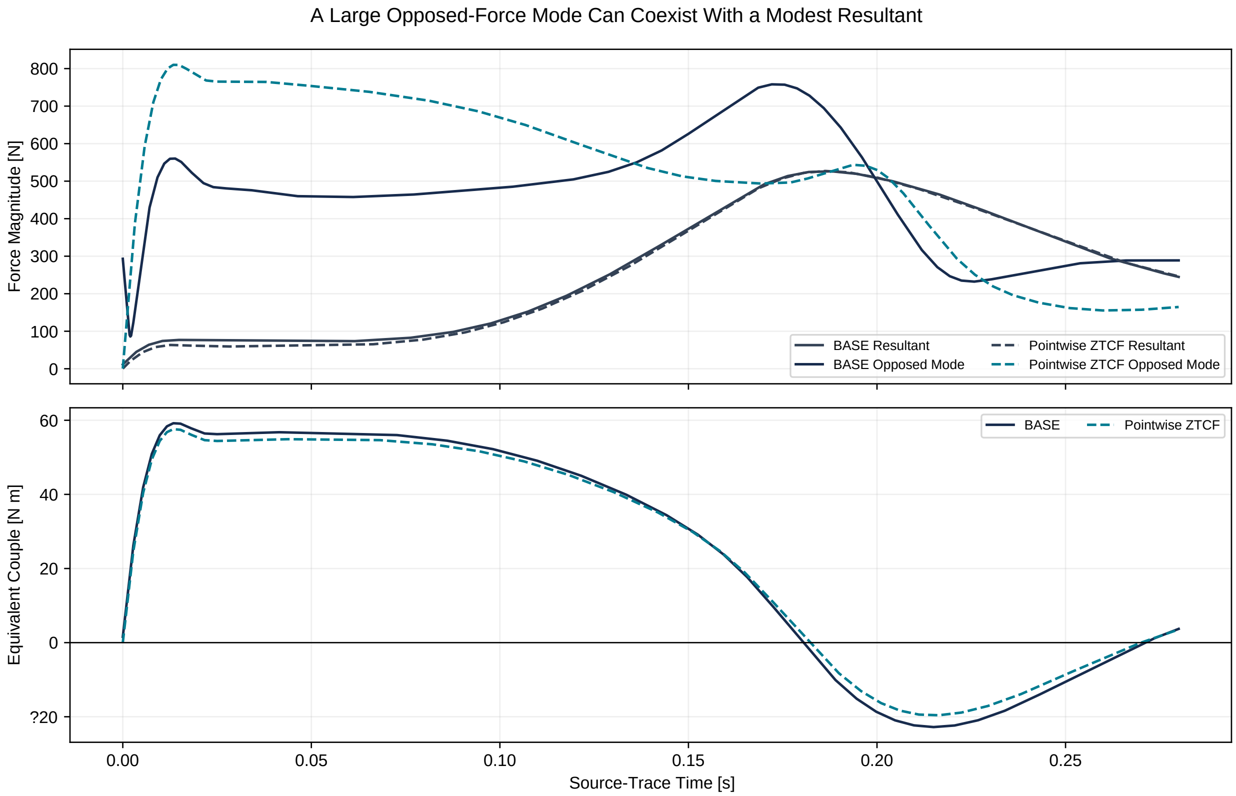

Figure 7.2: A Large Opposed-Force Mode Can Coexist With a Modest Resultant

In Figure 7.2, the opposed mode greatly exceeds the resultant over much of the early and middle trace. BASE and ZTCF resultants are nearly coincident, while their opposed modes differ more visibly. This is why a net-force plot alone cannot diagnose the rotational action of a two-hand grip.

7.6 Reconstruction and Verification Methods

Each of the 2,801 samples is evaluated by the same sequence:

1. read the two contact positions, forces, free torques, velocities, midpoint, and stored equivalent wrench;
2. compute the resultant with Equation 7.2;
3. compute the midpoint contact-force moment and add the free torques with Equation 7.3;
4. project global quantities onto the declared local axes;
5. locate every sign crossing by linear interpolation inside its adjacent 0.1 ms sample bracket; and
6. repeat the crossing estimate after downsampling by factors through 20, using every possible sample offset.

The reconstruction tolerances are absolute and deliberately much tighter than the signal magnitudes. Across BASE, ZTCF, and DELTA, the largest resultant residual is 1.1×10^{-12} N. The largest global couple residual is 0.057 N $\cdot$ m, or less than 0.1% of the approximately 59 N $\cdot$ m peak. BASE minus ZTCF minus DELTA closes to 1.2×10^{-13} N $\cdot$ m for the local equivalent couple.

The small nonzero couple reconstruction residual is consistent with archived table interpolation and coordinate-transform precision. It is not used to infer a physical torque. The test limit is 0.1 N $\cdot$ m, more than an order of magnitude above floating-point noise but far below the negative-couple peak.

7.7 Results: Where the Negative Couple Comes From

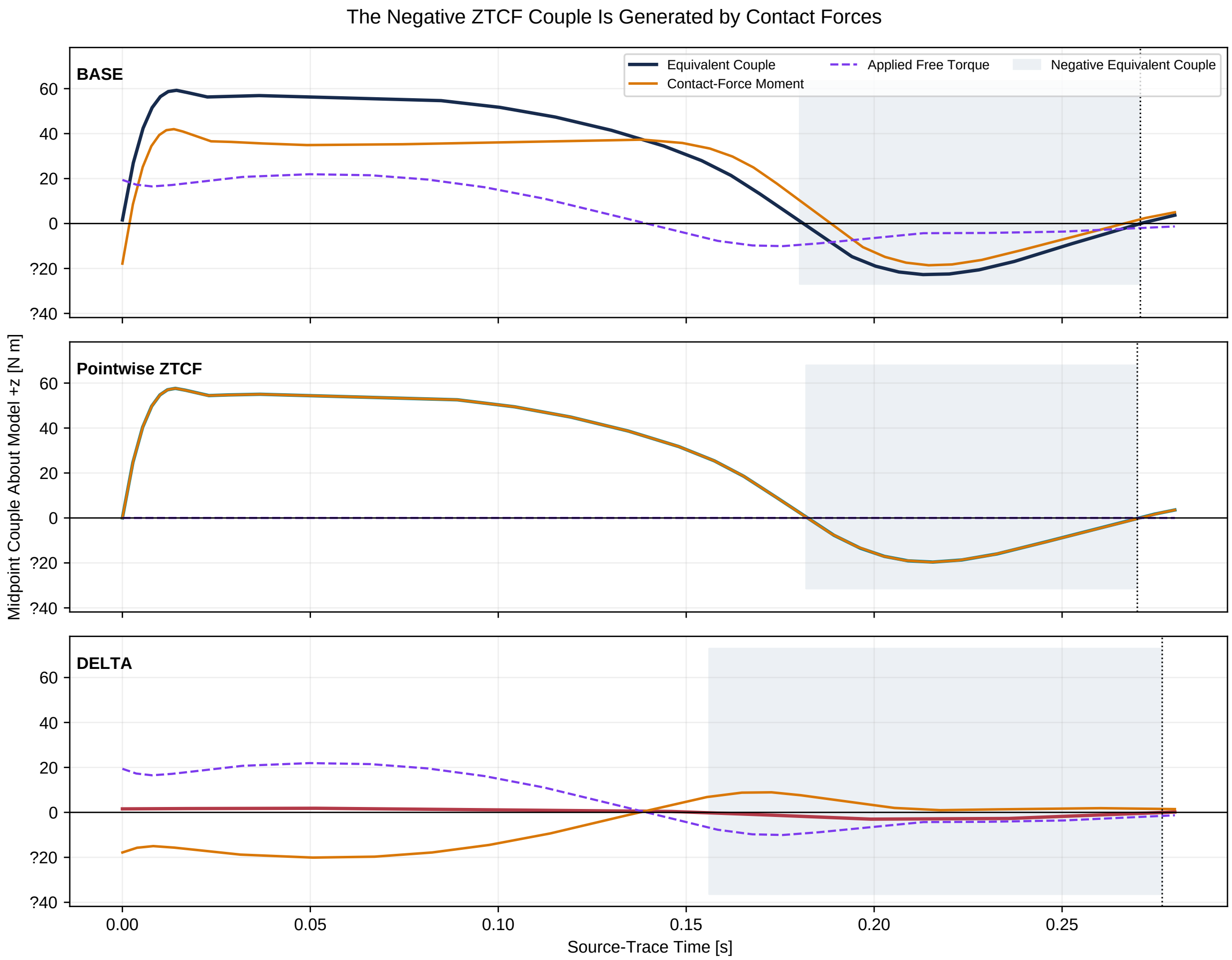

Figure 7.3: The Negative ZTCF Couple Is Generated by Contact Forces

Figure 7.3 provides the decisive decomposition. In BASE, the equivalent couple reaches a minimum of -22.79 N · m at 0.2151 s. At that instant, -18.60 N · m comes from separated contact forces and -4.22 N · m comes from applied free torque. In pointwise ZTCF, all command torque and kill damping are zero, yet the equivalent couple reaches -19.63 N · m at 0.2148 s. The reconstructed free-torque contribution there is below 3.1×10^{-13} N · m; the negative couple is therefore entirely a contact force moment to the resolution of the archive.

At the time of the BASE minimum, the ZTCF couple retains 86.1% of the BASE magnitude. DELTA is exact by construction, but its peak negative couple is only -3.18 N · m. These comparisons support a narrow conclusion: *within the achieved states of this model, most of the late negative midpoint couple does not require instantaneous commanded wrist torque.*

They do not support the stronger claim that wrist torque is irrelevant. Active torques helped create the state history on which ZTCF is evaluated, and BASE free torque contributes measurably to the total. Nor do they prove the same partition in a human swing.

7.7.1 Timing of the Two Sign Changes

All three cases contain an early positive-to-negative crossing and a later negative-to-positive crossing. For BASE, the negative interval begins at 0.180172 s and ends at 0.270824 s. For ZTCF, it begins at 0.181823 s and ends at 0.270016 s. Thus the late BASE and ZTCF reversals differ by 0.808 ms. Their respective negative intervals last approximately 90.7 and 88.2 ms.

The archived sample interval is 0.1 ms. After downsampling to effective steps as large as 2 ms and repeating every phase offset, the largest shift in the late crossing is 7.95 microseconds for BASE and 7.41 microseconds for ZTCF. This demonstrates interpolation and sampling stability for the stored smooth curves. It is not a new Simscape solver-tolerance study; changing solver, constraint stabilization, or shaft discretization remains a separate model sensitivity question.

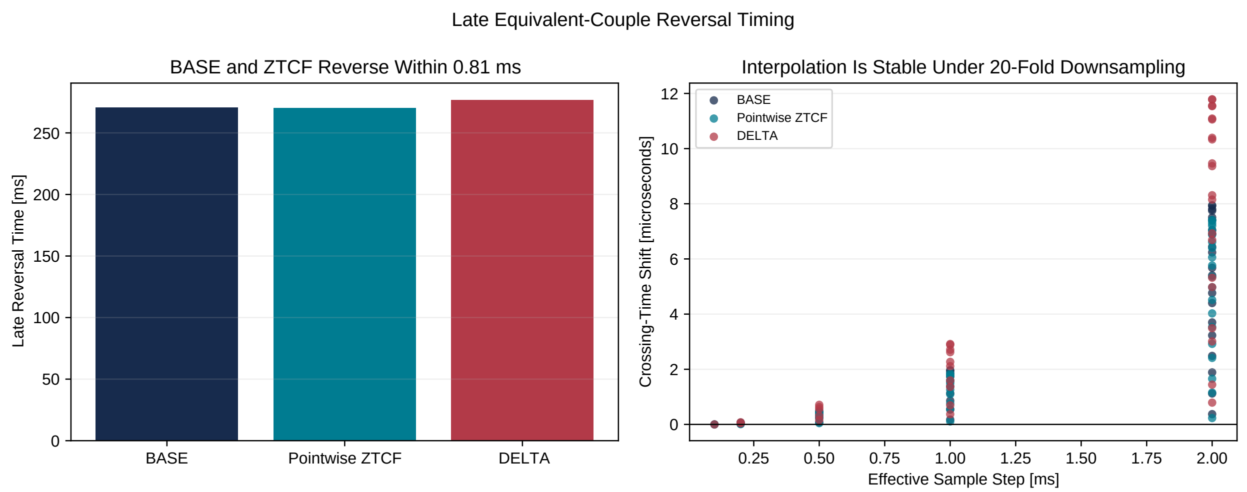

Figure 7.4: Late Equivalent-Couple Reversal Timing

The DELTA late crossing occurs later, at 0.276634 s. Treating the DELTA crossing as the onset of the passive mechanism would therefore be incorrect. DELTA asks how BASE differs from the pointwise zero-command reaction. The ZTCF curve itself is the relevant evidence for a moment that remains without current command.

7.8 Local Force Geometry Through the Downswing

The local force histories in Figure 7.5 expose the mechanism hidden by the resultant. Near the ZTCF minimum, the lead and trail normal forces are +249.8 N and -266.2 N. Their opposition makes the differential normal mode large and negative under Equation 7.7. At the same instant, the lead and trail axial forces are -497.5 N and +30.4 N. Those axial loads influence the resultant and internal loading, but their direct midpoint moment is suppressed by the small perpendicular lever arm.

The sign evolution is geometric. A force does not carry an intrinsic “accelerating” or “decelerating” label. Its moment depends on the cross product between its lever arm and direction, while its power depends on its dot product with contact velocity. During the downswing, both the grip direction and the contact-force directions rotate. The same anatomical hand can therefore move from one moment quadrant to another without an abrupt change in force magnitude.

Figure 7.6 shows six global poses. The view is three-dimensional because the archive stores global vectors in three coordinates, while the nearly equal global y and z coordinates reveal the declared oblique model plane. The force arrows change relative to the short grip segment: the visual transition from positive couple, through the negative minimum, and back through the late reversal is a cross-product change, not evidence of an invisible command.

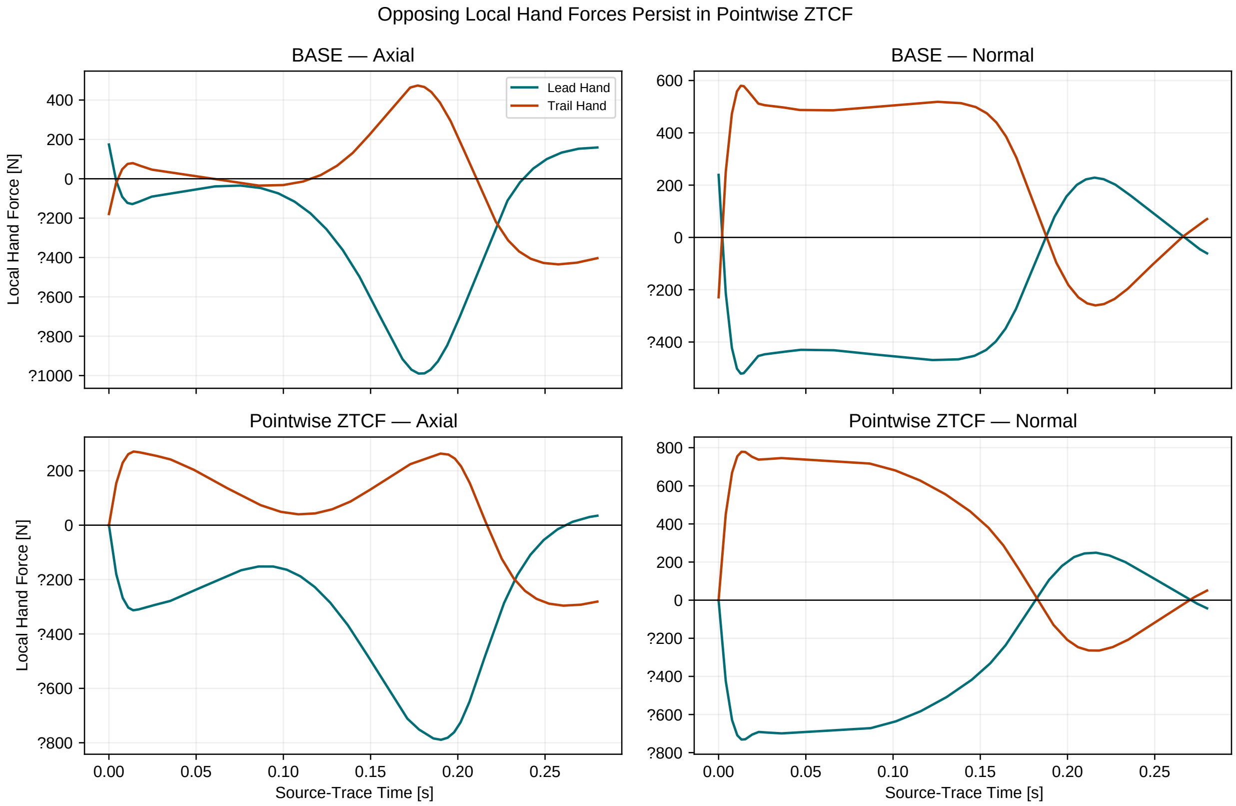

Figure 7.5: Opposing Local Hand Forces Persist in Pointwise ZTCF

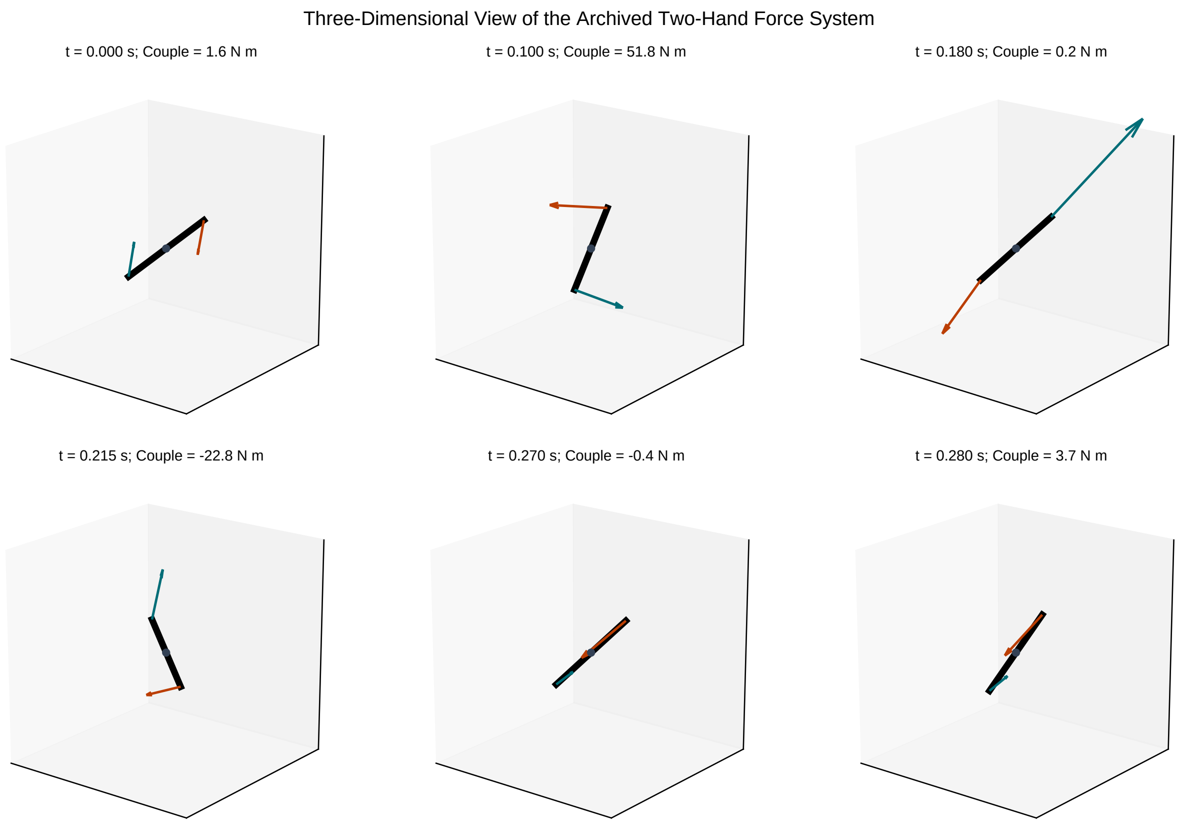

Figure 7.6: Three-Dimensional View of the Archived Two-Hand Force System

7.9 Geometry Counterfactuals

The force-generated couple can be perturbed without changing the force values. At the peak-negative ZTCF sample, two deterministic counterfactuals were applied:

1. scale both contact offsets from the midpoint while holding their directions and forces fixed; and
2. rotate the grip/contact offsets within the model plane while holding the global force directions fixed.

The first experiment produces an exactly linear response, as required by Equation 7.7. Doubling the separation doubles the contact-force moment; collapsing the contacts to one point removes it. This is a mechanics identity, not a recommendation to alter grip width without regard to anatomy or control.

The second experiment produces the expected sinusoidal orientation gate. At some relative orientations the same forces create a positive moment; at others they create a negative one. When the contacts and forces are rigidly co-rotated together, the moment is invariant to 1.5×10^{-14} N · m. Therefore global pose alone does not set the sign. The sign is controlled by the *relative* orientation of the contact-separation vector and differential force.

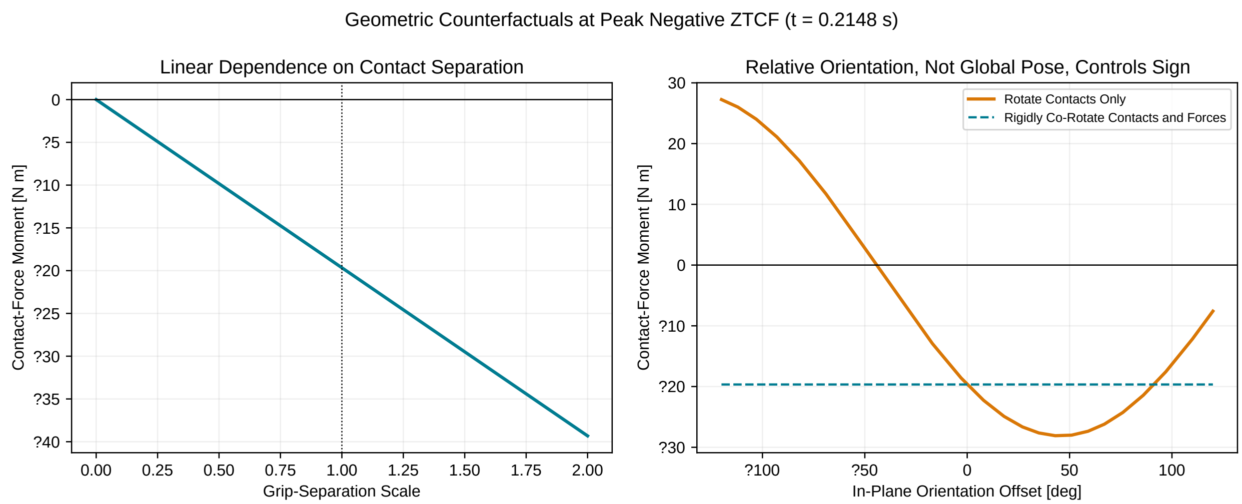

Figure 7.7: Geometric Counterfactuals at Peak Negative ZTCF

The practical implication is a constraint on strategy language. "Create more negative torque" is underspecified. A controller can change the moment by changing contact separation, force opposition, force direction relative to the grip, or a true applied free torque. Those routes have different effects on joint loading, net force, power, and robustness. They must not be collapsed into one scalar cue.

7.10 Couple Sign, Power, and Clubhead Speed

Moment sign is not energy-transfer sign. For a rigid body reduced at midpoint M, wrench power would be

$$P = \mathbf{R} \cdot \mathbf{v}_M + C_M \omega. \tag{7.8}$$

The archived model contains two distinct moving contacts. An exact two-point force-power identity is therefore more informative. With $\Delta \mathbf{v} = \mathbf{v}_T - \mathbf{v}_L$ and $\Delta \mathbf{F} = (\mathbf{F}_T - \mathbf{F}_L)/2$,

$$P_F = \mathbf{F}_L \cdot \mathbf{v}_L + \mathbf{F}_T \cdot \mathbf{v}_T = \mathbf{R} \cdot \mathbf{v}_M + \Delta \mathbf{F} \cdot \Delta \mathbf{v}. \tag{7.9}$$

The relative-velocity term can be separated into rigid rotation and a small contact-deformation remainder. The implemented identity closes within 2.0×10^{-11} W. In ZTCF, the maximum magnitude of the relative-contact remainder is only 0.188 W, compared with contact-force power peaks above 4 kW; the grip-contact kinematics are effectively rigid for this calculation.

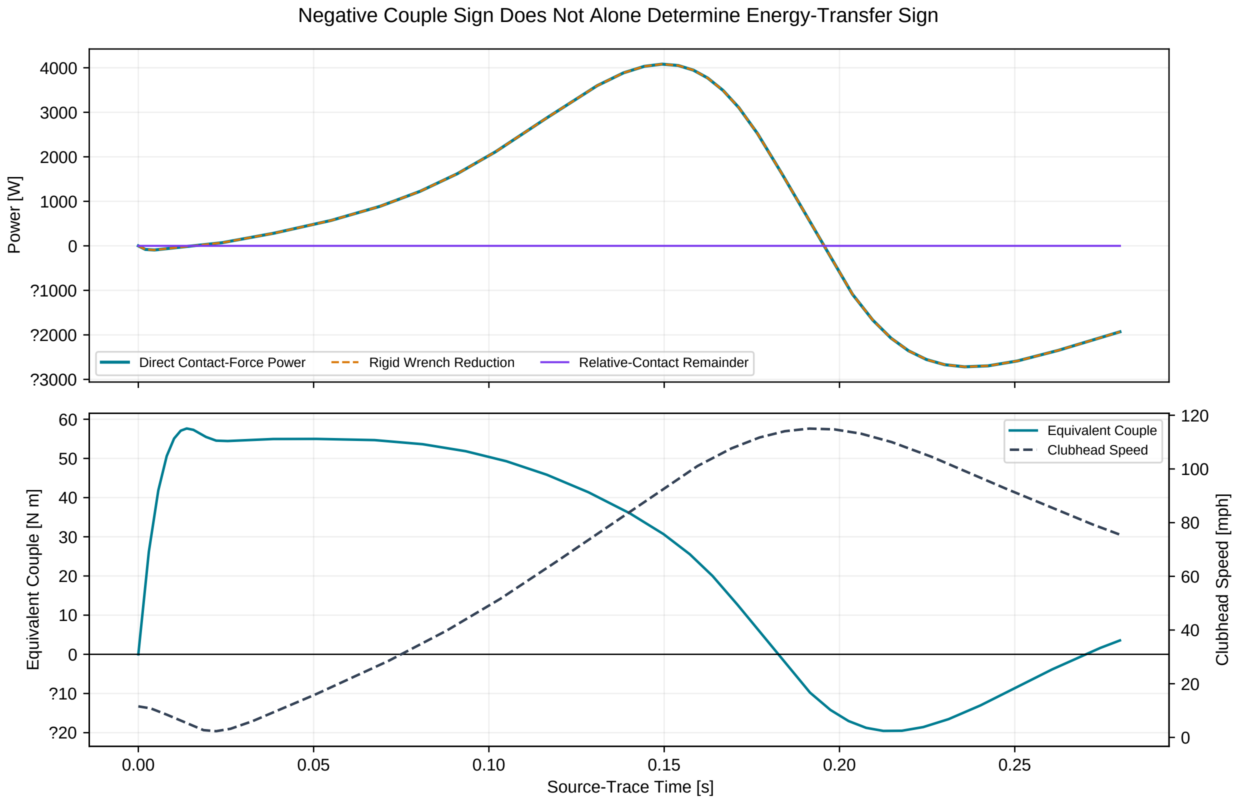

Figure 7.8: Negative Couple Sign Does Not Alone Determine Energy-Transfer Sign

At the ZTCF negative-couple minimum, the reconstructed grip angular velocity is 36.21 rad/s and the total contact-force power is -2.08 kW. Clubhead speed is already about 110.0 mph, close to the archived 115.1 mph maximum at 0.1925 s. Across the entire negative-couple interval, contact-force power is negative for approximately 85% of samples and integrates to about -149 J. In this particular trace, the passive negative couple is principally associated with late redirection and deceleration after peak speed, not with a direct positive-power boost to the club.

That finding sharpens, rather than rejects, the proposed mechanism. Passive negative torquing can be a consequence of momentum and constraint geometry without being the source of positive club work. Its strategic value, if any, would be indirect: allowing the system to reach a favorable high-speed state, controlling orientation, and avoiding unnecessary active opposition to the natural late reaction. Whether that improves impact delivery requires an optimization with impact timing, face orientation, actuator limits, and shaft stress—not inspection of the torque sign alone.

7.11 Mechanistic Interpretation by Phase

The archived trace can be read in four phases without assigning intent to the model:

Positive-Couple Formation (0–0.180 s). The two-hand force system initially creates a positive midpoint couple. Both BASE and ZTCF rise rapidly, showing that constraint reactions dominate even before the sign reversal.

Passive Negative-Couple Entry (approximately 0.180 s). BASE and ZTCF cross zero within 1.7 ms of one another. The similarity of their force-generated moments indicates that the sign transition is largely encoded in the achieved positions and velocities rather than the instantaneous wrist command.

Peak Negative Reaction (approximately 0.215 s). Opposed normal forces form the dominant negative couple while clubhead speed is near its maximum. BASE free torque augments the negative action by about $4.2\ \mathrm{N} \cdot \mathrm{m}$, but ZTCF demonstrates that the larger contact-force contribution remains with command removed.

Late Recovery and Reversal (approximately 0.270 s). Relative geometry and contact forces evolve until the couple returns positive. BASE and ZTCF reverse at nearly the same time; DELTA reverses later and should not be mistaken for the passive timing signal.

This phase description is mechanical, not anatomical. The model does not resolve muscle coordination, grip-pressure distributions, tendon elasticity, or neural control. It identifies a candidate force-system pathway that future measurements can attempt to confirm or falsify.

7.12 Implications for Speed-Creation Strategy

The present evidence supports constraints on a strategy rather than a universal instruction:

- **Build the state before judging the late reaction.** ZTCF is large because earlier motion has created large velocities and a particular geometry.
- **Track differential normal force, not net force alone.** The opposed mode is the direct geometric source of the two-hand couple.
- **Separate speed production from impact management.** Positive power earlier in the downswing creates speed; the late passive negative couple in this trace mostly accompanies energy removal and orientation control after peak speed.
- **Do not fight a passive reaction by default.** Adding positive command solely to cancel a negative measured couple could increase loading or disrupt impact orientation. The appropriate response is an optimization question.
- **Constrain the endpoint.** A credible optimization must evaluate clubhead speed at impact together with face/path orientation, grip loads, shaft stress, actuator work, and sensitivity to timing errors.

This framework rejects two simple slogans: neither “negative torque creates speed” nor “negative torque wastes speed” is generally valid. Power, timing, and endpoint constraints decide the effect.

7.13 Limitations and Falsification Tests

The strongest current result is internal: the archived model’s negative ZTCF couple reconstructs from separated contact forces with zero current command. Several limitations bound external interpretation:

- ZTCF is pointwise along the commanded state history, not a forward two-hand killswitch rollout.
- The source is one deterministic planar simulation and one parameter set.
- The model uses idealized wrist contacts and does not resolve pressure across the hands or grip.
- Shaft and grip compliance are present in the source architecture but have not yet been separated by matched rigid/flexible reruns.
- Downsampling verifies interpolation stability, not solver-tolerance or constraint-stabilization independence.
- No synchronized human hand-force and full-body kinematic data are used.

The mechanism would be materially weakened if a like-for-like rerun showed any of the following: the ZTCF negative couple disappears under solver refinement; the sign is caused by a frame or action-reaction error; rigid/flexible variants reverse the attribution; a forward killswitch leaves the negative region too quickly to matter; or measured human contact forces fail to exhibit the predicted differential-normal pattern.

The next model layer should therefore separate gravity, momentum-dependent terms, damping, and shaft elasticity while preserving the same wrench and power contracts. Higher-order and human-data studies should retain the distinction between resultant force, force-generated moment, free torque, and power. That shared vocabulary is the main reusable product of the present analysis.

Chapter 8

Forward Constrained Two-Hand Dynamics

8.1 Purpose and Evidential Scope

The preceding two-hand analysis established that separated contact forces can produce a club couple even when their resultant is small. It also reconstructed the archived WSCG zero-command solution point by point along a commanded state history (Olson 2024). That calculation answers an instantaneous question: what acceleration and constraint force would the model produce if the applied commands were removed at this state? It does not answer whether the resulting negative couple survives after the state begins to evolve under zero command.

This chapter supplies that missing forward test. Two planar two-link arms grasp a club with two independent point constraints. The club translates and rotates; the arm and club coordinates evolve under the constrained equations of motion. At a declared late-downswing state, all six applied joint torques are set to zero and a new trajectory is integrated from the *exact same* generalized position and velocity. The resulting branch is therefore a test of dynamic persistence, not a stitched collection of same-state counterfactuals.

The scope is deliberately narrow. The calculation establishes mechanical feasibility in a planar rigid-body model. It is not a fit to a golfer, contains no muscle model, does not identify neural intent, and does not prove that a human golfer uses the modeled strategy. Redundant two-hand tasks generally do not permit biological effort to be inferred uniquely from the net wrench (Reinkensmeyer, Lum, and Lehman 1992; Zatsiorsky and Latash 2008). The more defensible result is that a negative late force-generated club couple can persist without continued command in a nonsingular forward constrained model.

8.2 Coordinates, Geometry, and Closure

The generalized coordinate vector is

$$q = \begin{bmatrix}\theta_{Rs} & \theta_{Re} & \theta_{Ls} & \theta_{Le} & x_c & y_c & \phi_c\end{bmatrix}^{\mathsf{T}}, \tag{8.1}$$

where the four arm angles describe right and left shoulder–elbow chains and (x_c, y_c, ϕ_c) describe the floating club. Angles are measured in the declared planar frame, with x along the target-line coordinate and y upward. For a segment angle α, the direction and angular derivative are

$$d(\alpha) = \begin{bmatrix}\sin\alpha \\ -\cos\alpha\end{bmatrix}, \qquad d_{,\alpha}(\alpha) = \begin{bmatrix}\cos\alpha \\ \sin\alpha\end{bmatrix}. \tag{8.2}$$

The right and left hand positions follow from the corresponding two-link chains. The club grip points are

$$r_{g,R} = r_c + a_R d(\phi_c), \qquad r_{g,L} = r_c + a_L d(\phi_c), \tag{8.3}$$

with $a_R = +0.065$ m and $a_L = -0.065$ m in the reference case. The four holonomic constraints are

$$C(q) = \begin{bmatrix} r_{h,R}(q) - r_{g,R}(q) \\ r_{h,L}(q) - r_{g,L}(q) \end{bmatrix} = 0, \qquad J(q) = \frac{\partial C}{\partial q}. \tag{8.4}$$

The model has seven coordinates and four independent scalar constraints, leaving three admissible instantaneous degrees of freedom. Every accepted sample must retain rank $J = 4$. A rank-deficient Jacobian, a singular Schur complement, or a constraint residual above the declared limit causes the solver to fail closed; a least-squares fallback is not used. This is important because a numerically convenient force allocation at a singular grasp would not be evidence for a physically identified two-hand couple.

8.3 Constrained Equations of Motion

The mass matrix is assembled from center-of-mass translational Jacobians and segment angular Jacobians. Gravity and the velocity-dependent bias are kept explicit. This follows standard rigid-body dynamics accounting (Featherstone 2008) while retaining the small model's auditable analytical structure. At each state, the acceleration and constraint multiplier solve

$$\begin{bmatrix} M(q) & -J(q)^{\mathsf{T}} \\ J(q) & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} Q(u) - h(q, \dot{q}) \\ -\dot{J}(q, \dot{q})\dot{q} \end{bmatrix}. \tag{8.5}$$

Here $h = c + g$ contains Coriolis/centrifugal and gravitational terms. With the implemented sign convention, the force exerted by the hands on the club is $F_{h \to c} = -\lambda$ after reshaping the four multipliers into two planar forces. The residuals of both rows of Equation Equation 8.5 are stored for every sample.

The applied generalized force is not a six-entry vector copied directly into the seven coordinates. Wrist torque is an internal action–reaction pair: a positive wrist torque acts positively on the club and negatively on the two arm coordinates that contribute to the corresponding forearm angle. Thus,

$$Q(u) = \begin{bmatrix} u_{Rs} - u_{Rw} \\ u_{Re} - u_{Rw} \\ u_{Ls} - u_{Lw} \\ u_{Le} - u_{Lw} \\ 0 \\ 0 \\ u_{Rw} + u_{Lw} \end{bmatrix}. \tag{8.6}$$

This mapping prevents the directly applied wrist torque from being silently counted as a contact-force couple.

8.4 Forward Integration and Numerical Audit

The reference trajectory is integrated for 0.400 s with a 0.25 ms step. A velocity-Verlet update is paired with mass-metric position and velocity projection. For a position residual C, the local correction is

$$\delta q = M^{-1} J^{\mathsf{T}} \left(J M^{-1} J^{\mathsf{T}} \right)^{-1} C, \qquad q \leftarrow q - \delta q. \tag{8.7}$$

The projection is iterated to the declared tolerance. Velocity is projected onto $J\dot{q} = 0$ with the same mass metric. Projection correction and its numerical energy change are recorded separately. They are numerical stabilization diagnostics and are **not** reclassified as physical work.

The initial club center is $(0, -0.50)$ m, the club angle is 0.16 rad, and the arm angles are obtained by inverse kinematics on specified elbow branches. The initial generalized velocity is zero. The open-loop command is a smooth test input, not an inferred human torque history:

Applied Torque	Command Over $0 \le t \le 0.4$ s (N m)
Right Shoulder	$18 + 4t/0.4$
Right Elbow	$7 - 1.5t/0.4$
Right Wrist	$-3 + 2t/0.4$
Left Shoulder	$16 + 3t/0.4$
Left Elbow	$6 - t/0.4$
Left Wrist	$2 - t/0.4$

The commands create a repeatable state history through which mechanism interventions can be compared. They are not optimized for clubhead speed and are not presented as normative technique. More complete forward golf models have shown why joint-torque timing, body motion, and objective definition must be treated together (MacKenzie and Sprigings 2009; Balzerson, Banerjee, and McPhee 2016).

8.5 Registered Observables

The force-generated couple about the declared club center is

$$M_F = (r_{g,R} - r_c) \times F_R + (r_{g,L} - r_c) \times F_L. \tag{8.8}$$

The direct wrist contribution is stored separately as $M_w = u_{Rw} + u_{Lw}$. The total rotational action on the club is therefore $M_F + M_w$ before any other modeled external moment. This distinction is the central test: $M_F < 0$ cannot be attributed algebraically to a negative applied wrist torque because M_F is computed only from the two constraint forces.

The contact forces are also transformed into common and differential modes,

$$R = F_R + F_L, \qquad D = \tfrac{1}{2}(F_R - F_L). \tag{8.9}$$

The resultant R controls club translation. The differential mode D becomes rotationally effective through the signed grip moment arms. Large opposing forces can therefore produce a substantial couple with a modest resultant, as is familiar in multifinger prehension (Zatsiorsky and Latash 2008).

Finally, contact power is evaluated both pointwise and through the reduced wrench identity:

$$P_{contact} = F_R \cdot v_{g,R} + F_L \cdot v_{g,L} = R \cdot v_c + M_F \omega_c. \tag{8.10}$$

The two forms agree to 6.3×10^{-13} W in the reference run. Couple sign alone still does not determine energy transfer: rotational power also depends on club angular velocity, while the resultant contributes translational power. This is consistent with standard joint-force and segment-power bookkeeping (Winter 2009).

8.6 Counterfactual and Negative-Control Design

Four intervention families expose the result to contradiction.

1. **Matched-State Torque Killswitch.** At 0.200 s, the achieved q and $\dot{q}$ are copied bit for bit into a new forward trajectory and all six commands are set to zero for 50 ms. Additional branches start at 0.180, 0.190, 0.210, 0.220, and 0.240 s.
2. **Zero Contact Moment Arm.** Both grip offsets are set to zero and a new constraint-consistent initial state is constructed. If Equation Equation 8.8 is implemented correctly, the force-generated couple must vanish even though nonzero contact forces may remain.
3. **Timestep Refinement.** The 0.400 s baseline is repeated at 2.0, 1.0, and 0.5 ms. The sign-reversal time and minimum couple must remain stable while the work–energy residual contracts.
4. **Projection Sensitivity.** The position-projection tolerance is changed from 10^{-8} to 10^{-10} m at a fixed 1.0 ms step. A mechanism that depends materially on stabilization tolerance is rejected.

These are mechanistic interventions. They are stronger than visual similarity between two curves because each has a declared null consequence and an observable failure mode.

8.7 Results

8.7.1 Force-Generated Couple Reversal

The baseline force-generated couple first becomes negative at 0.19825 s and reaches −32.85 N m over the recorded interval. The direct wrist torque ranges only from −1.0 to 0 N m and is excluded from that force-generated series. The late negative value is therefore not a relabeled wrist command.

At the 0.200 s cut, the inherited force-generated couple is −2.57 N m. With all applied commands removed, it remains negative throughout the 50 ms branch and reaches −7.43 N m. The commanded continuation becomes more negative, reaching approximately −13 N m over the same horizon, but that difference is not the central claim. The decisive observation is that the negative force-generated couple does not collapse to zero when command is removed.

The branch ensemble also limits the interpretation. A branch at 0.180 s begins before reversal and crosses negative at 0.185 s. Branches initiated from 0.190 through 0.220 s remain negative for the full 50 ms window. The 0.240 s branch remains negative for 37.25 ms before changing sign. Passive persistence is therefore state dependent and finite; it is not an invariant property of the club or grip.

8.7.2 Geometric Mechanism

Figure Figure 8.2 displays the actual hand-on-club force vectors with one common scale. Between 0.180 and 0.240 s, the club and arm geometry change the signed moment arms and the tangential components of the two forces. The couple reversal is generated by that evolving vector geometry, not by force magnitude alone. At 0.300 s the force directions and club orientation have changed again, illustrating why a static instruction such as "pull harder" cannot specify the sign of M_F.

The zero-moment-arm intervention makes the geometric requirement explicit. When both grip points coincide with the club center, the maximum absolute force-generated couple is exactly 0 N m to machine representation. Contact force alone is insufficient; a nonzero signed lever arm and a transverse force component are jointly necessary.

8.7.3 Solver Closure and Convergence

The reference trajectory retains constraint rank four. The maximum position constraint residual is 9.51×10^{-11} m, the maximum velocity residual is 3.99×10^{-15} m/s, the KKT residual is below 3.77×10^{-13}, and the acceleration-constraint residual is below 3.43×10^{-13}. Over the 0.400 s interval, mechanical energy rises by 203.714 J and integrated applied-control work is 203.618 J, leaving a 0.096 J physical work–energy residual at the 0.25 ms reporting step.

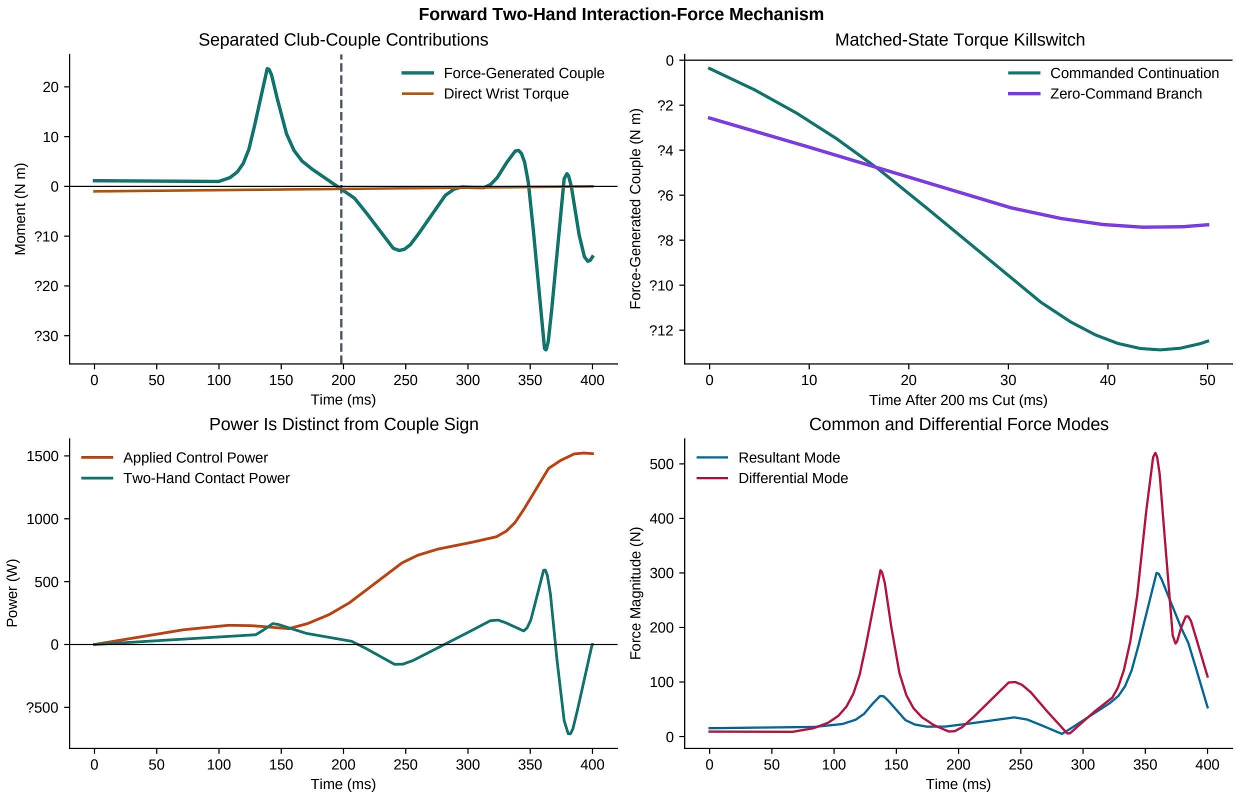

Figure 8.1: Forward Two-Hand Interaction-Force Mechanism. The force-generated couple is separated from direct wrist torque (upper left). A zero-command branch starts from the exact 200 ms state and retains a negative couple for the full 50 ms window (upper right). Contact power and common/differential force modes show why moment sign, power sign, and force magnitude are different observables.

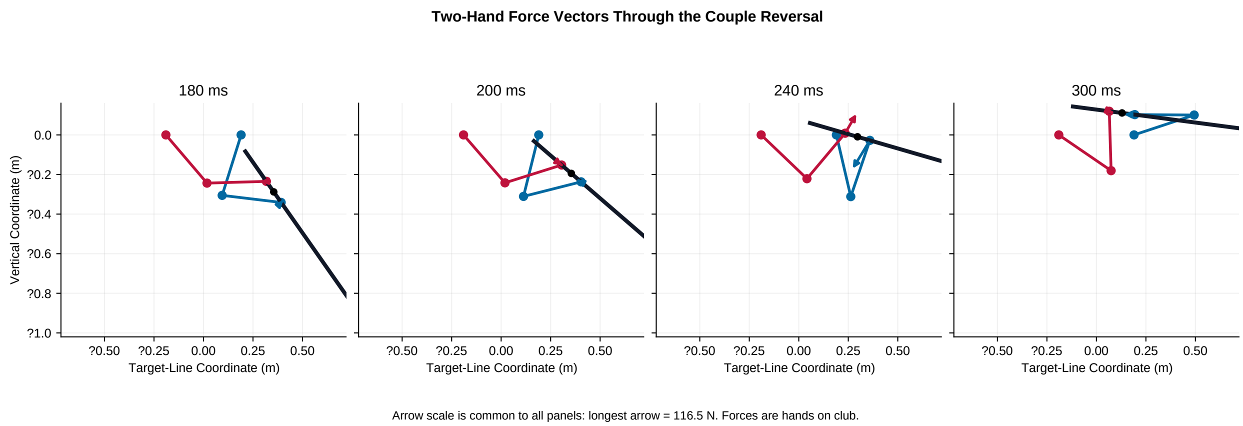

Figure 8.2: Two-Hand Force Vectors Through the Couple Reversal. Blue and red arrows are the right- and left-hand forces on the club; all panels use the same force scale. Black denotes the floating club, colored links denote the two arms, and the black point is the declared club center.

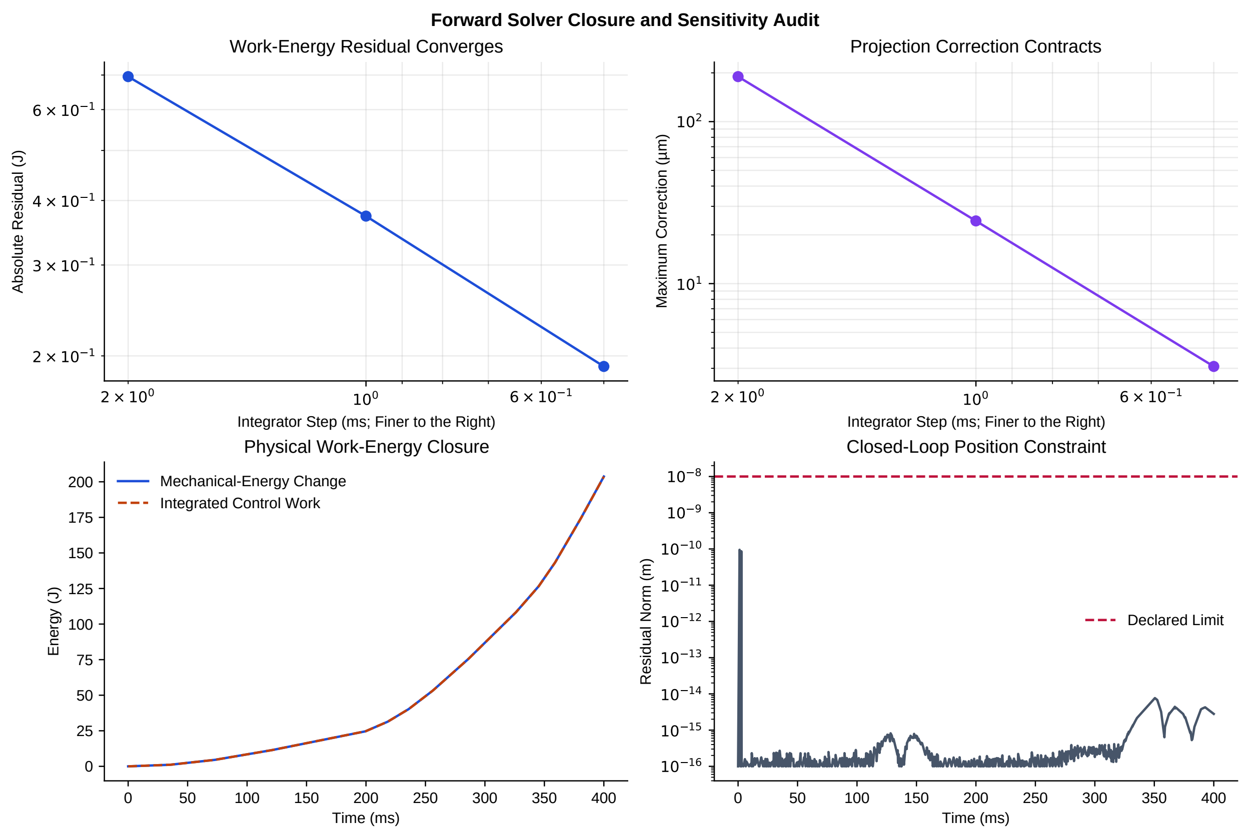

Figure 8.3: Forward Solver Closure and Sensitivity Audit. Work–energy residual and maximum projection correction contract under timestep refinement (top). Integrated control work overlays the mechanical-energy change, while the closed-loop position residual remains below its declared limit (bottom).

The work–energy residual decreases from 0.695 to 0.373 to 0.191 J as the step is refined from 2.0 to 1.0 to 0.5 ms. The maximum projection correction falls from 190 to 24.4 to 3.08 μm over the same sequence. Negative-couple onset is 0.1980, 0.1980, and 0.1985 s, a spread of only 0.5 ms. Tightening the projection tolerance by two orders of magnitude changes the 1.0 ms minimum couple by less than 5×10^{-8} N m and leaves onset unchanged. The reported reversal is therefore resolved relative to both timestep and projection tolerance in this model.

The accumulated projection-energy diagnostic is negative and converges toward zero with timestep refinement. It is not added to or subtracted from the physical work budget. Reporting it separately prevents numerical stabilization from being mistaken for passive energy transfer.

8.8 What the Forward Test Establishes

The forward calculation strengthens the archived pointwise result in one specific way: it demonstrates that an achieved state can carry a negative force-generated two-hand couple forward for a finite interval after all applied joint torques are removed. The negative control shows that the effect requires grip geometry, and the convergence studies show that it is not a timestep or projection-tolerance artifact at the declared resolution.

It does **not** establish that the negative couple is wholly passive in a human swing. The baseline state was created by prior applied torques; after the cut, gravity, velocity-dependent interaction terms, and constraint reactions remain. "Passive" here means zero *applied model command after the branch*, not absence of force, momentum, gravity, stored energy, or biological activation. Nor does the model include a moving torso, out-of-plane rotations, compliant wrists, a distributed shaft, hand-contact compliance, or ground reaction pathways.

For strategy, the result supports a conditional mechanical statement rather than a coaching prescription: club acceleration can coexist with a negative hand-force couple when the current state and grip geometry make the coupled reaction favorable. Consequently, maximizing positive wrist torque at every instant is not a necessary condition for increasing distal speed. Whether a specific golfer should reduce, reverse, or maintain wrist torque requires the higher-order moving-base, flexible-shaft, three-dimensional, and human-data tests that follow in the completion program.

Chapter 9

Gravity, Momentum, Damping, and Shaft-Flex Contributions

9.1 The Attribution Question

The preceding chapters established two different passive mechanisms. In the double pendulum, interaction forces can accelerate the distal segment at zero instantaneous wrist torque. In the archived two-hand model, separated contact forces can create a negative midpoint couple while the commanded free torques are zero. Neither result identifies the share attributable to gravity, velocity-dependent dynamics, dissipation, or shaft deformation. This chapter separates those terms in one declared, reproducible surrogate and asks four narrow questions:

1. Which terms create the instantaneous distal angular acceleration?
2. Which terms supply external work, dissipate energy, or merely redistribute it?
3. How much does one lumped shaft-flex mode change the matched delivery?
4. Do those conclusions persist when stiffness, damping, torque-cut time, impact window, and integration step are varied?

The analysis is deliberately a mechanism study. It is not a calibrated model of a named shaft, a reconstruction of a human swing, or a validation of the archived Simscape model. The implementation and machine-readable results are available in `flexible_shaft_study.py` and `shaft_contribution_study.json`.

9.2 A Matched Flexible and Rigid Pair

The model is a planar, three-coordinate, point-mass chain. The first coordinate is the arm angle, the second is the wrist-relative club angle, and the third, ϕ_2, is a lumped torsional shaft-deflection coordinate between proximal shaft mass and clubhead mass. The generalized state is

$$\mathbf{q} = (\theta_1, \phi_1, \phi_2)^{\mathsf{T}}. \tag{9.1}$$

The flexible and rigid cases have the same lengths, masses, gravity, joint damping, initial condition, and prescribed shoulder/wrist torque history. The rigid case is the exact coordinate reduction $\phi_2 = \dot{\phi}_2 = 0$, not a separately tuned model. This matched construction isolates the consequence of admitting one elastic degree of freedom.

The reference parameters are intentionally transparent: an 0.75 m arm, 0.45 m proximal shaft, 0.55 m distal shaft, 7.5 kg arm point mass, 0.15 kg proximal shaft point mass, and 0.20 kg distal head point mass. Projected

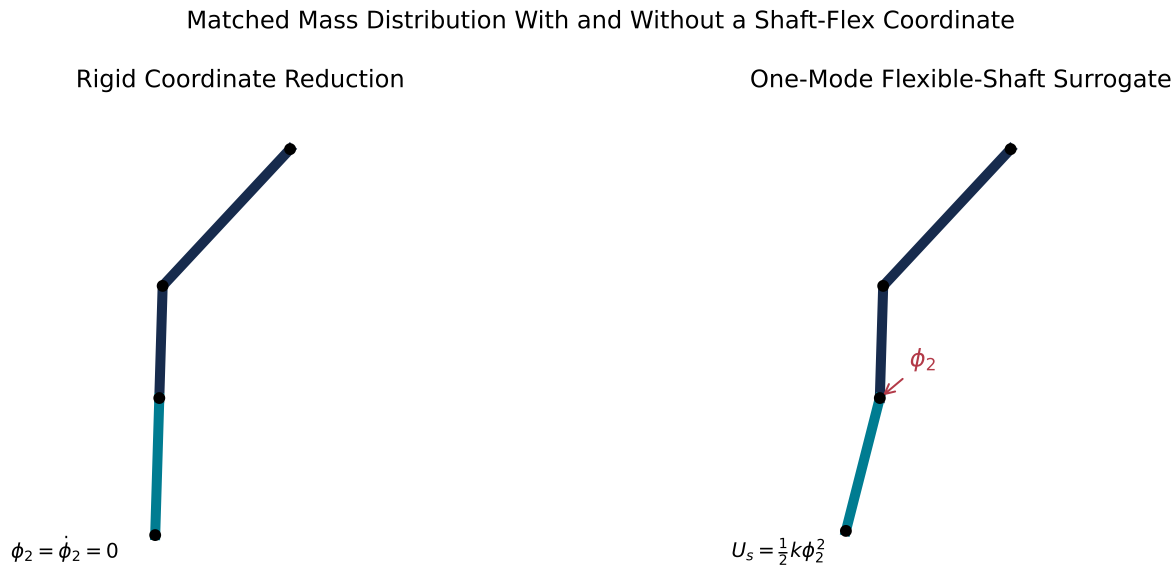

Figure 9.1: Matched Mass Distribution With and Without a Shaft-Flex Coordinate

gravity is 8.033 m/s². The linear shaft mode uses $k_s = 80$ N · m/rad and $c_s = 0.6$ N · m · s/rad. These values define the demonstration; they are not presented as a fitted equipment specification.

This point-mass idealization is more restrictive than a distributed Euler– Bernoulli or Timoshenko beam. It omits bending-plane coupling, torsion, frequency-dependent material loss, clubhead inertia, grip compliance, and three-dimensional handle motion. Large deflections in the sensitivity grid therefore identify stress tests of the surrogate, not credible predictions of a real shaft.

9.3 Equation-Level Contribution Accounting

The equations are written as

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} = \tau_{\text{ctrl}} - \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) - \mathbf{G}(\mathbf{q}) + \tau_{\text{joint damp}} + \tau_s + \tau_{s,d}, \tag{9.2}$$

where $\mathbf{C}$ contains the velocity-dependent inertial terms. The linear elastic and viscous shaft actions are

$$\tau_s = -k_s\phi_2, \qquad \tau_{s,d} = -c_s\dot{\phi}_2. \tag{9.3}$$

For every right-hand-side term $\mathbf{f}_j$, the corresponding generalized acceleration contribution is

$$\ddot{\mathbf{q}}_j = \mathbf{M}^{-1}\mathbf{f}_j, \qquad \ddot{\mathbf{q}} = \sum_j \ddot{\mathbf{q}}_j. \tag{9.4}$$

This is an exact algebraic decomposition at each state. It avoids assigning physical meaning to a single off-diagonal mass-matrix entry without accounting for the full coupled solve. In the tested trace, reconstructed and directly solved accelerations agree to floating-point precision.

Figure 9.2 shows the contributions to the absolute distal angular acceleration. The shaft-elastic term can be locally large because the reduced distal inertia is small. That statement concerns acceleration, not energy creation. A large instantaneous acceleration contribution can reverse sign later and perform nearly zero net work.

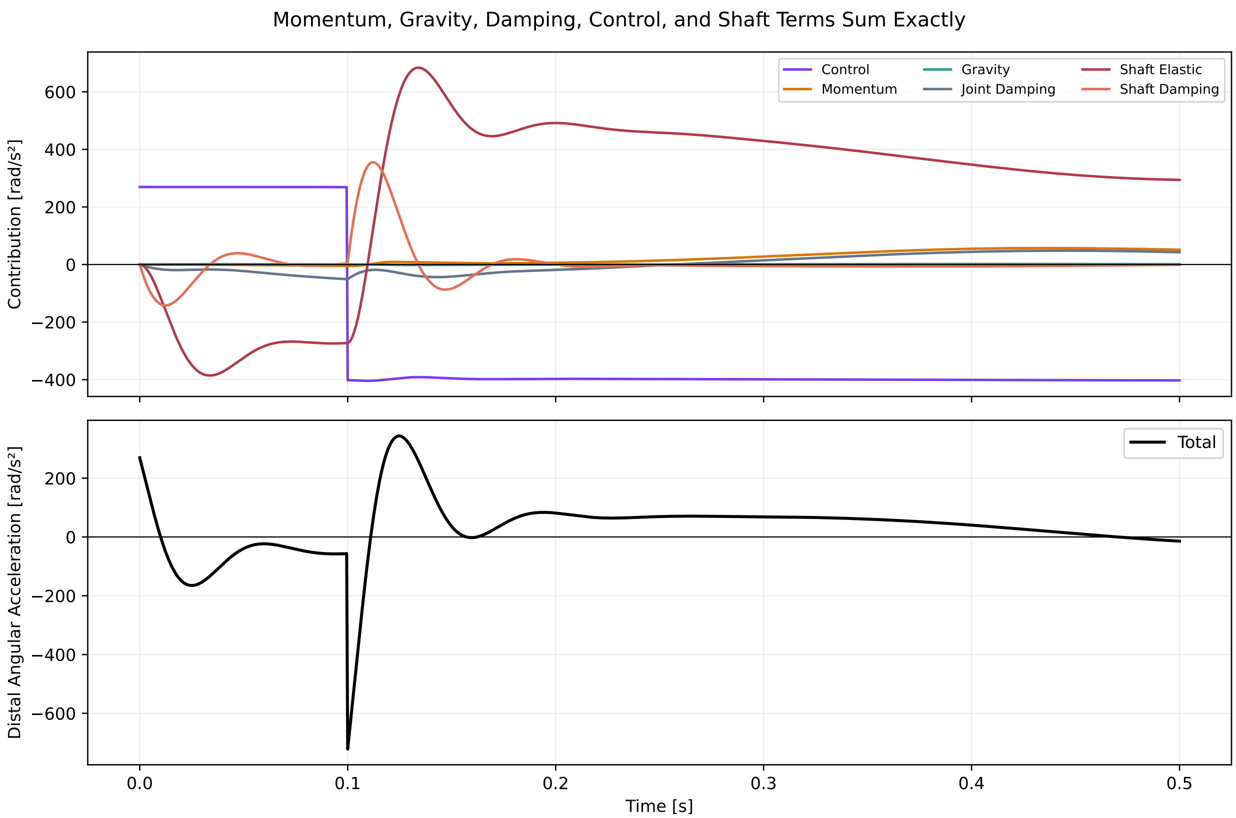

Figure 9.2: Momentum, Gravity, Damping, Control, and Shaft Terms Sum Exactly

9.3.1 Momentum Is Not an External Energy Source

For diagnostic symmetry, the analysis records the generalized quantity $(-\mathbf{C})^\mathsf{T}\dot{\mathbf{q}}$ and its time integral. In the reference trace that signed integral is 8.52 J. It must not be interpreted as 8.52 J of external work. The Coriolis/centrifugal vector arises from the coordinate representation of internal inertial coupling; it redistributes kinetic energy among generalized coordinates while the complete kinetic-energy derivative retains the required mass-matrix terms. Only boundary actuation, conservative potential changes, and declared dissipative processes enter the total energy balance below.

The same caution applies to labels such as "centrifugal power." A coordinate projection is useful for attribution, but it is not a new reservoir. The mechanism is exchange through coupled geometry and constraint forces.

9.4 Shaft Interface Force, Moment, and Power

The internal shaft interface transmits a planar force $\mathbf{F}_s$ and a moment $M_s = \tau_s + \tau_{s,d}$. The power delivered across the interface is the sum of force power and moment power, evaluated with consistent action, velocity, and sign conventions. The internal force performs equal and opposite work on the adjacent subsystems only after both translational and rotational terms are retained.

At the reference delivery crossing, the model predicts 56.34 N of interface force, -3.58° of shaft flex, +5.00 N · m of elastic moment, and -0.083 N · m of viscous moment. The positive elastic moment is especially important when this chapter is compared with Section 7: it cannot explain the archived -19.63 N · m pointwise two-hand midpoint couple. The two quantities belong to different models, reference points, and physical mechanisms.

9.5 Closed Work–Energy Accounting

The stored shaft energy is

$$U_s = \tfrac{1}{2}k_s\phi_2^2, \tag{9.5}$$

and the complete mechanical energy is

$$E = T + V_g + U_s. \tag{9.6}$$

Because gravity and the shaft spring are included in E, the external and dissipative balance is

$$\frac{dE}{dt} = \tau_{\text{ctrl}}^\mathsf{T}\dot{\mathbf{q}} + \tau_{\text{joint damp}}^\mathsf{T}\dot{\mathbf{q}} - c_s\dot{\phi}_2^2. \tag{9.7}$$

The shaft spring is absent from the right side because it stores and returns energy. Its signed generalized-power integral equals the negative change in strain energy, apart from numerical integration error; it is not a loss term.

The reference run receives 164.38 J of control work. Joint damping removes 9.21 J and shaft damping removes 0.97 J. Peak shaft strain energy is 0.720 J, compared with 227.88 J peak kinetic energy. The maximum accumulated closure error is 0.0322 J, approximately 0.02% of control work. Away from the piecewise-command switch, the root-mean-square differential residual is 0.0092 W. This closure is a numerical consistency result, not validation of the assumed physical parameters.

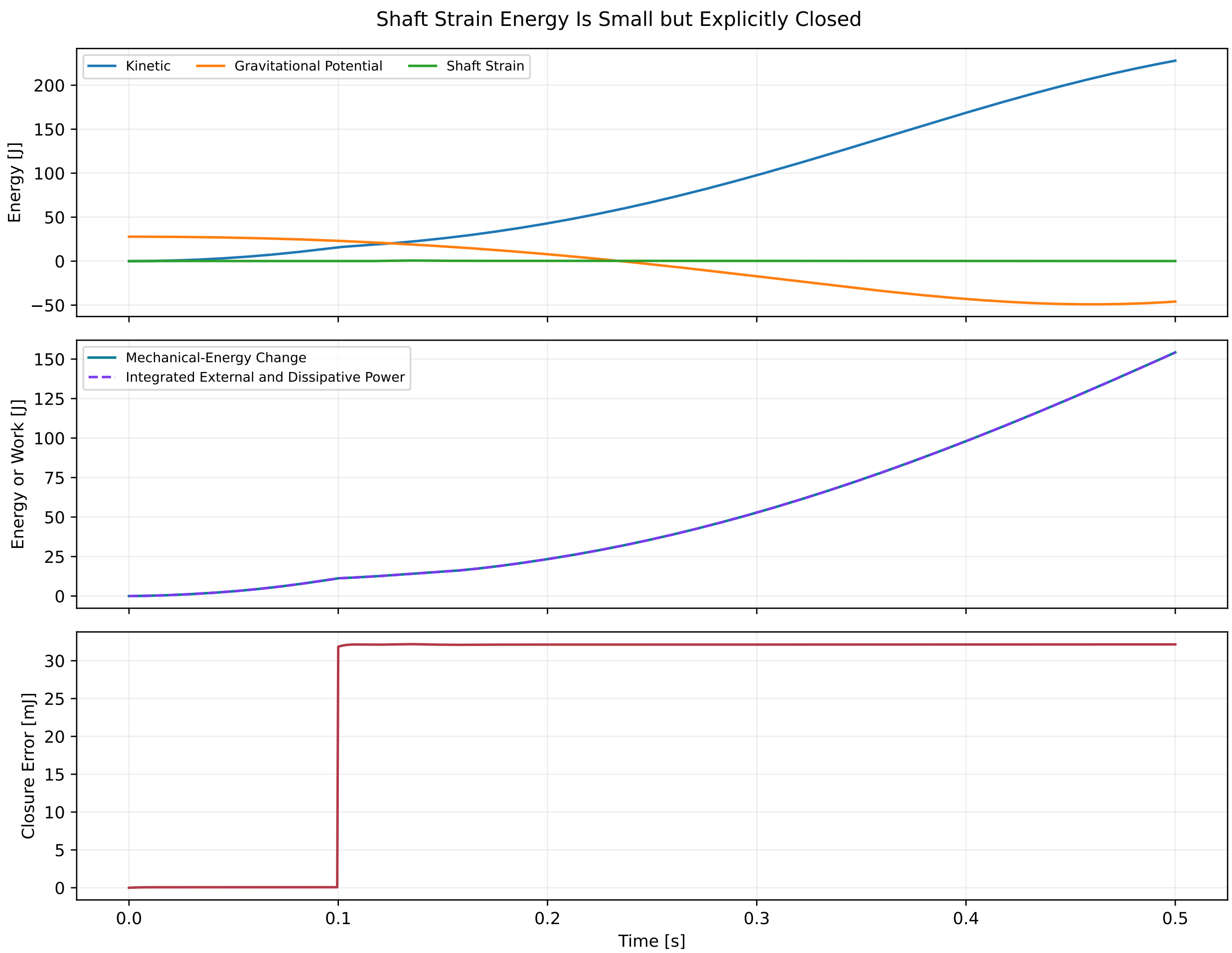

Figure 9.3: Shaft Strain Energy Is Small but Explicitly Closed

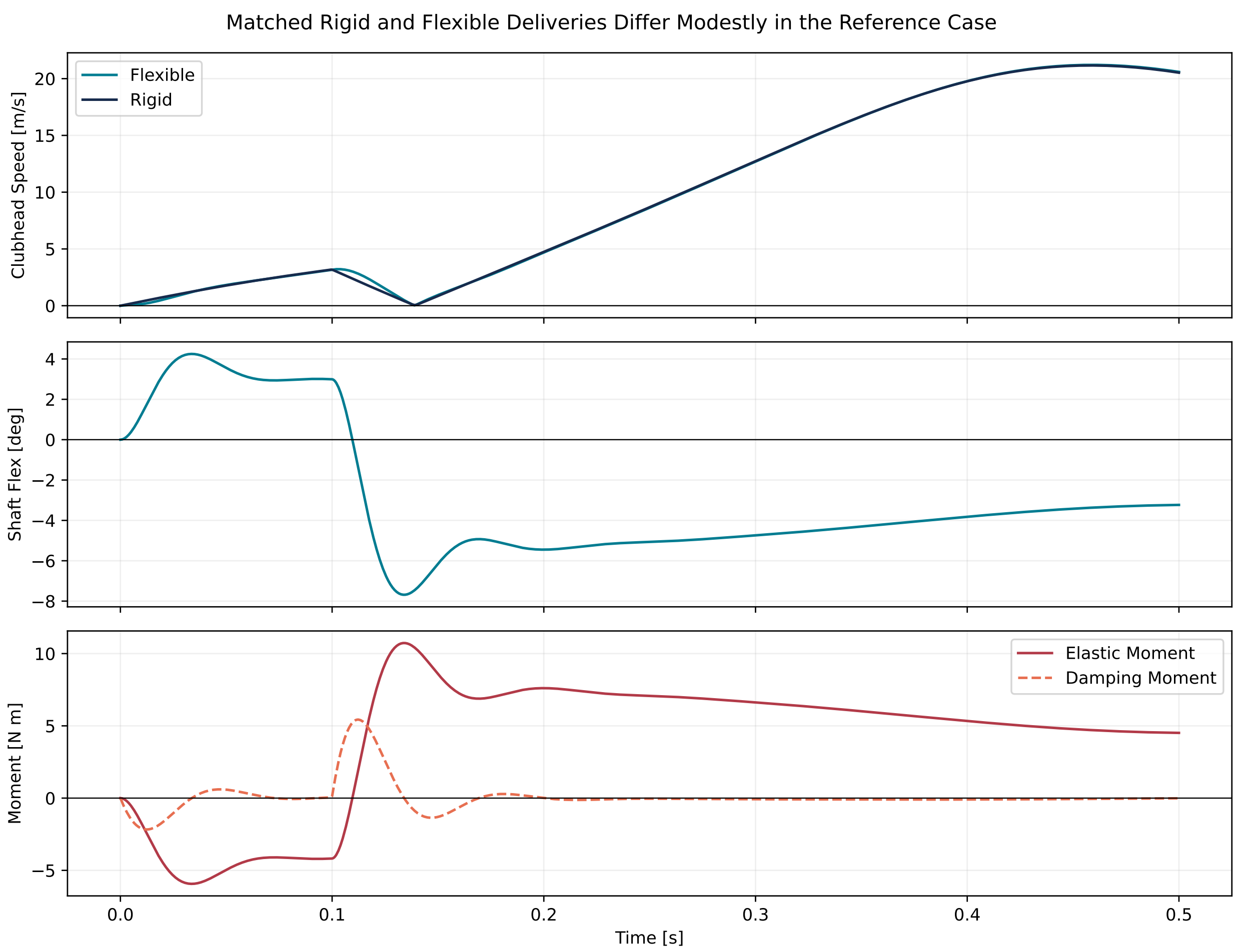

Figure 9.4: Matched Rigid and Flexible Deliveries Differ Modestly in the Reference Case

9.6 Reference Delivery: Flexibility Changes Timing Modestly

The flexible reference reaches the declared delivery crossing at 0.42725 s with 20.799 m/s clubhead speed. The matched rigid reduction reaches it 2.73 ms earlier at 20.691 m/s. The flexible-minus-rigid difference is therefore +0.108 m/s, or approximately 0.52% of the rigid value. Peak speeds are 21.213 and 21.159 m/s, respectively.

This result supports a restrained conclusion. The admitted elastic coordinate changes local forces, angular accelerations, pose, and delivery timing, but the reference speed effect is small. It does not support a general assertion that shaft recoil is either negligible or a primary source of clubhead speed. Different stiffness, damping, actuation, geometry, and endpoint definitions can change the result.

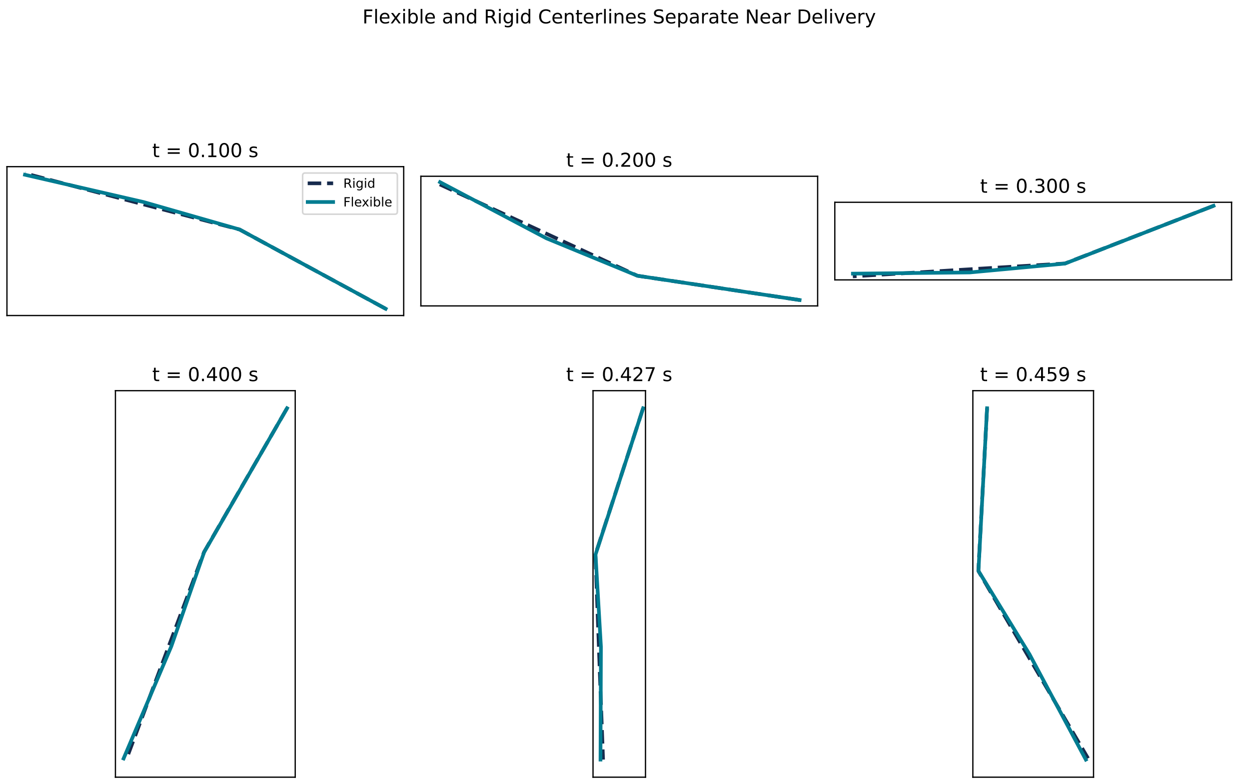

Figure 9.5: Flexible and Rigid Centerlines Separate Near Delivery

The centerline overlays in Figure 9.5 show why endpoint timing matters. Small angular differences accumulate into a visible distal-position change as the chain extends. Comparing speeds at a common clock time, a common configuration crossing, or the maximum within a time window answers different questions.

9.7 Physics Ablations

Removing projected gravity lowers reference delivery speed from 20.799 to 16.035 m/s and delays the crossing to 0.458 s. Removing the declared joint damping raises it to 22.969 m/s and delays the crossing to 0.435 s. Removing shaft damping produces 20.508 m/s at 0.428 s, increases impact flex to -5.66°, and raises peak stored strain energy from 0.720 to 2.805 J.

These are controlled ablations, not estimates of causal shares in a human swing. Turning off one term changes the subsequent state and hence every other state-dependent term. The comparison nevertheless establishes an important scale result for this model: gravity and joint damping alter the delivered speed more than the matched reference flex-versus-rigid change.

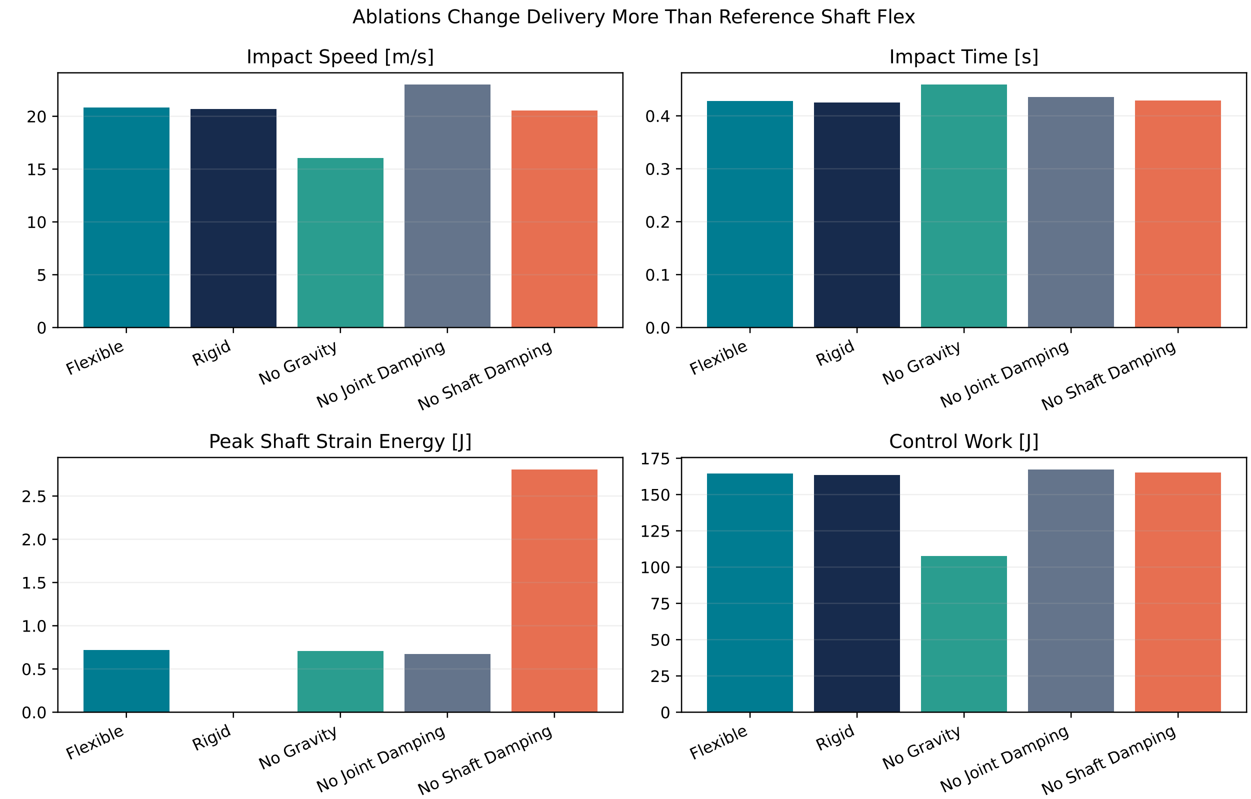

Figure 9.6: Ablations Change Delivery More Than Reference Shaft Flex

9.8 Robustness Across Stiffness, Damping, and Torque-Cut Time

The sensitivity study evaluates 120 combinations of six stiffnesses (10–320 $N \cdot m/rad$), four damping values (0–1.2 $N \cdot m \cdot s/rad$), and five command histories (no cut or cuts at 0.12, 0.18, 0.24, and 0.30 s). An extended 0.75 s horizon is used so that every declared case reaches its first delivery crossing.

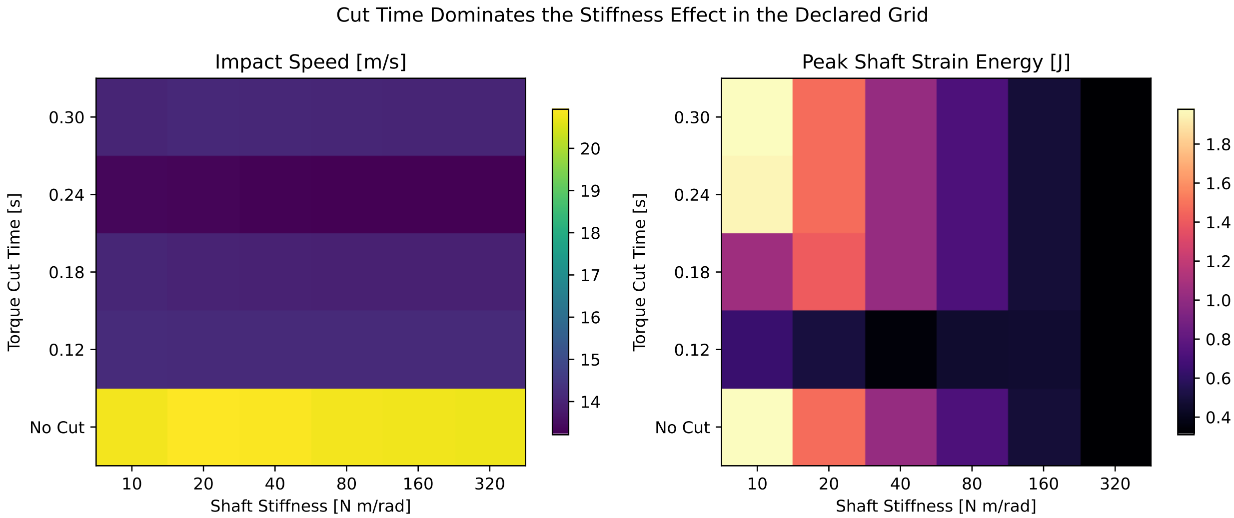

Figure 9.7: Torque-Cut Time Dominates the Stiffness Effect in the Declared Grid

Across the full grid, delivery speed spans 12.99–22.65 m/s and crossing time spans 0.425–0.719 s. Peak strain energy spans 0.161–18.05 J. The chosen torque-cut time produces the strongest organization visible in the speed map; stiffness more clearly organizes stored strain energy. This demonstrates why equipment-only comparisons are incomplete when the actuation history is not matched.

The full grid also includes impact flex from -42.9° to +11.1°. Those extreme values violate the small-deflection premise implicit in a one-mode linear torsional surrogate. They are retained as transparent stress-test results, not interpreted as physically realizable shaft shapes. A distributed-beam model with calibrated material and boundary properties is required before making equipment-design claims.

9.9 Impact Window and Numerical Resolution

The left panel of Figure 9.8 reports speed at the first configuration crossing and maxima within ±5, ±10, and ±20 ms. For the reference case, the values are 20.799, 20.915, 21.016, and 21.156 m/s. The window is therefore part of the estimand, not a plotting preference.

Halving the reference 0.5 ms step to 0.25 ms changes crossing speed by 5.8 mm/s and crossing time by 41 microseconds. Doubling it to 1 ms changes speed by 11.8 mm/s and time by 82 microseconds. The maximum energy-closure error scales from 0.0161 J at 0.25 ms to 0.0647 J at 1 ms, consistent with a converging fixed-step calculation around a piecewise command.

9.10 Implications for Late-Downswing Strategy

Three strategy-relevant conclusions survive the declared comparisons.

First, transfer is geometric and state dependent. Momentum coupling can create large distal acceleration without being an external energy source. Its useful effect depends on link orientation, angular velocities, and the direction of the constraint force relative to distal velocity.

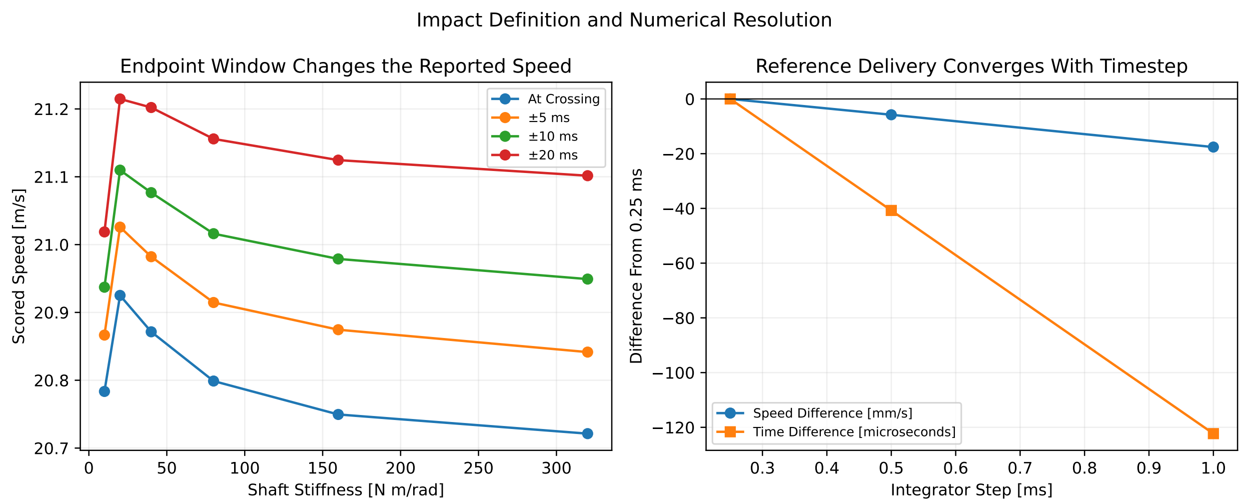

Figure 9.8: Impact Definition and Numerical Resolution

Second, elastic storage is a timing mechanism. A spring can temporarily retain energy, alter configuration, and return energy later, but the net benefit is not the peak strain-energy value. The relevant comparison is a matched endpoint under matched actuation, with damping and timing included.

Third, "negative torque" requires an identified object and reference point. In the WSCG two-hand audit, the late negative midpoint couple is a force-generated grip wrench. In this flexible surrogate, the reference shaft elastic moment at delivery is positive. Conflating these signs would erase the distinction between handle-contact geometry and shaft recoil.

Accordingly, the model suggests a testable strategy principle rather than a prescription: create the proximal motion and relative geometry that permit a late, velocity-aligned distal transfer while avoiding unnecessary dissipative work. Whether a golfer should actively restrain, release, or oppose a joint at a particular instant cannot be inferred from segment-speed order or torque sign alone. It requires joint power, interface wrench, state history, and a declared outcome window.

9.11 Falsification and Next Model Requirements

The present explanation would be weakened if a higher-fidelity model with matched mass distribution and actuation showed any of the following:

- the contribution sum fails after independent equation implementation;
- work–energy closure requires an unmodeled energy source;
- the flex-versus-rigid result reverses under small, physically calibrated parameter changes rather than only under broad stress tests;
- measured handle forces and shaft deformation cannot reproduce the predicted sign and timing of interface power; or
- a distributed shaft and two-hand closed-loop model attributes the archived negative couple primarily to an applied free torque rather than separated contact forces.

The next step is therefore not to add realism indiscriminately. It is to add degrees of freedom in an auditable sequence: moving hub, closed two-arm loop, distributed shaft, three-dimensional rotation, and finally data-constrained parameters. At each step, the rigid/flexible reduction, force/moment balance, work–energy closure, and endpoint definition should remain explicit.

Chapter 10

Coupled Base Motion and Club Compliance

10.1 Why the Mechanisms Must Be Coupled

The preceding forward two-hand experiment establishes that separated contact forces can create a late negative club couple after all commanded joint torques are set to zero. The shaft study separately establishes that elastic storage, viscous loss, and interaction forces can be closed in a forward model. Those results do not by themselves establish that the negative-couple mechanism survives when the shoulder base moves and the club flexes. Prescribing a mobile hub after solving a fixed-base trajectory cannot answer that question because base acceleration then changes reactions without being changed by them. Likewise, attaching a flexible point-mass chain to a single arm does not test whether two-hand differential forces survive the added compliance.

This chapter therefore evaluates one coupled system in which base translation, arm motion, grip reactions, proximal-club rotation, and shaft deflection are all simultaneous forward-dynamics outputs. The implementation, complete traces, and portable evidence record are available in `moving_base_flexible_club.py`, `run_moving_base_flexible_study.py`, and `moving_base_flexible_study.json`.

The question is deliberately narrow: does the registered planar mechanism remain mechanically possible after these two omitted degrees of freedom are admitted? The experiment does not calibrate a golfer, identify muscle forces, or predict the effect of a named shaft.

10.2 Coordinates and Geometry

The generalized coordinate vector is

$$\mathbf{q} = (\theta_R, \phi_R, \theta_L, \phi_L, x_b, y_b, x_g, y_g, \alpha, \beta)^{\mathsf{T}}, \tag{10.1}$$

where (x_b, y_b) translates a finite-mass shoulder base, (x_g, y_g) is the proximal club's grip-center position, α is its absolute angle, and β is the distal-club angle relative to α. Each arm uses an absolute upper-arm angle θ and a relative elbow angle ϕ. The distal angle is therefore $\alpha + \beta$.

For the planar direction and tangent operators

$$\mathbf{e}(\gamma) = \begin{bmatrix} \sin\gamma \\ -\cos\gamma \end{bmatrix}, \qquad \mathbf{t}(\gamma) = \frac{\partial \mathbf{e}}{\partial \gamma} = \begin{bmatrix} \cos\gamma \\ \sin\gamma \end{bmatrix}, \tag{10.2}$$

the right-hand position, for example, is

$$\mathbf{r}_{h,R} = \mathbf{r}_b + \mathbf{s}_R + L_u \mathbf{e}(\theta_R) + L_f \mathbf{e}(\theta_R + \phi_R), \tag{10.3}$$

and the corresponding grip point is

$$\mathbf{r}_{g,R} = \mathbf{r}_g + d_R \mathbf{e}(\alpha). \tag{10.4}$$

The four holonomic constraints are

$$\Phi(\mathbf{q}) = \begin{bmatrix} \mathbf{r}_{h,R} - \mathbf{r}_{g,R} \\ \mathbf{r}_{h,L} - \mathbf{r}_{g,L} \end{bmatrix} = \mathbf{0}, \qquad \mathbf{J} = \frac{\partial \Phi}{\partial \mathbf{q}}. \tag{10.5}$$

This construction leaves six feasible degrees of freedom at regular configurations. The two contact-force vectors are solved Lagrange multipliers; they are not allocated after the motion is known.

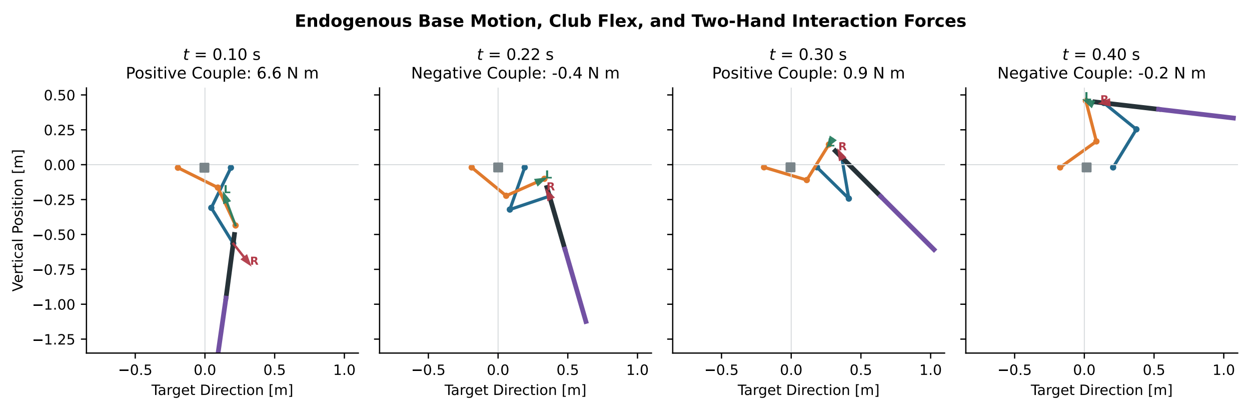

Figure 10.1: Endogenous Base Motion, Club Flex, and Two-Hand Interaction Forces

Figure 10.1 shows four states from the same forward trajectory. The square is the translating base, the black and purple segments are the proximal and distal club, and the arrows are the independently solved forces on the club. The force-generated couple becomes negative when the weighted tangential projections of the two arrows reverse; neither force norm nor shaft curvature alone determines that sign.

10.3 Inertia, Convective Terms, and the KKT Solve

The mass matrix is assembled from each center-of-mass Jacobian $\mathbf{J}_i$ and angular-velocity map $\mathbf{H}_i$:

$$\mathbf{M}(\mathbf{q}) = \sum_i m_i \mathbf{J}_i^\top \mathbf{J}_i + \sum_i I_i \mathbf{H}_i^\top \mathbf{H}_i. \tag{10.6}$$

This includes translation–rotation coupling between the base and both arms, and between the grip center and both club segments. The convective vector is formed from the exact centripetal accelerations of each center of mass. For a segment contribution $L\mathbf{e}(\gamma)$, that acceleration is $-L\dot{\gamma}^2\mathbf{e}(\gamma)$. This direct Jacobian construction avoids a finite-difference mass-matrix derivative in the primary dynamics.

The base is connected to the world by an isotropic linear spring and damper; the shaft mode has its own torsional spring and damper:

$$\mathbf{Q}_b = -k_b \mathbf{r}_b - c_b \dot{\mathbf{r}}_b, \qquad Q_\beta = -k_s \beta - c_s \dot{\beta}. \tag{10.7}$$

These values define a transparent mechanism experiment. The reference values are $m_b = 35$ kg, $k_b = 24{,}000$ N/m, $c_b = 500$ N $\cdot$ s/m, $k_s = 80$ N $\cdot$ m/rad, and $c_s = 0.6$ N $\cdot$ m $\cdot$ s/rad. They are not fitted torso or shaft properties.

At every step, the acceleration and contact multipliers solve

$$\begin{bmatrix} \mathbf{M} & -\mathbf{J}^{\mathsf{T}} \\ \mathbf{J} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{Q}_{\text{control}} + \mathbf{Q}_{\text{damp}} - \mathbf{h} - \nabla V \\ -\dot{\mathbf{J}}\dot{\mathbf{q}} \end{bmatrix}. \tag{10.8}$$

The solver fails closed if the mass matrix is not positive definite, the four-row constraint Jacobian loses rank, or either KKT residual exceeds its declared tolerance. Velocity-Verlet integration is followed by mass-metric configuration and velocity projection. Projection corrections and their energy changes remain explicit numerical diagnostics rather than hidden stabilization work.

10.4 Separating Grip and Shaft Moments

The force-generated couple about the grip center is

$$M_F = (\mathbf{r}_{g,R} - \mathbf{r}_g) \times \mathbf{F}_R + (\mathbf{r}_{g,L} - \mathbf{r}_g) \times \mathbf{F}_L. \tag{10.9}$$

Direct wrist torque is reported separately as $M_W = \tau_{w,R} + \tau_{w,L}$. The internal shaft moment is also separate:

$$M_s = -k_s\beta - c_s\dot{\beta}. \tag{10.10}$$

M_s is internal to the whole club, whereas $M_F + M_W$ is the external grip couple on it. Combining them into a single curve would obscure both the system boundary and the mechanism.

The two-point contact power closes against the wrench transported to the grip center:

$$\mathbf{F}_R \cdot \mathbf{v}_R + \mathbf{F}_L \cdot \mathbf{v}_L = (\mathbf{F}_R + \mathbf{F}_L) \cdot \mathbf{v}_g + M_F\dot{\alpha}. \tag{10.11}$$

The maximum discrepancy is 1.42×10^{-13} W in the reference trace. This identity is important because a negative couple need not imply negative net contact power: resultant force power can offset or exceed couple power.

10.5 Whole-System Energy Balance

The mechanical energy contains kinetic energy of the base, four arm segments, and two club segments; gravitational potential; base-spring energy; and shaft strain energy:

$$E = T + V_g + \tfrac{1}{2}k_b\|\mathbf{r}_b\|^2 + \tfrac{1}{2}k_s\beta^2. \tag{10.12}$$

Ideal constraints perform zero net work on the complete system. The remaining balance is

$$\frac{dE}{dt} = \mathbf{Q}_{\text{control}}^{\mathsf{T}}\dot{\mathbf{q}} - c_b\|\dot{\mathbf{r}}_b\|^2 - c_s\dot{\beta}^2 - c_j\sum_{k=1}^{4}\dot{q}_k^2. \tag{10.13}$$

Over 0.4 s, the reference run gains 91.521 J of mechanical energy from 111.133 J of applied work and -19.624 J of declared dissipation. The absolute work–energy residual is 0.0130 J. It falls monotonically from 0.0483 J at a 2 ms step to 0.0254 J at 1 ms and 0.0130 J at 0.5 ms. The maximum position projection falls from 69.2 µm to 8.87 µm and then 1.12 µm over the same refinement. Constraint rank remains four; the maximum position, velocity, KKT, and acceleration-constraint residuals are respectively 7.47×10^{-11} m, 2.01×10^{-15} m/s, 2.30×10^{-13}, and 1.52×10^{-13}.

10.6 Coupled Reference Response

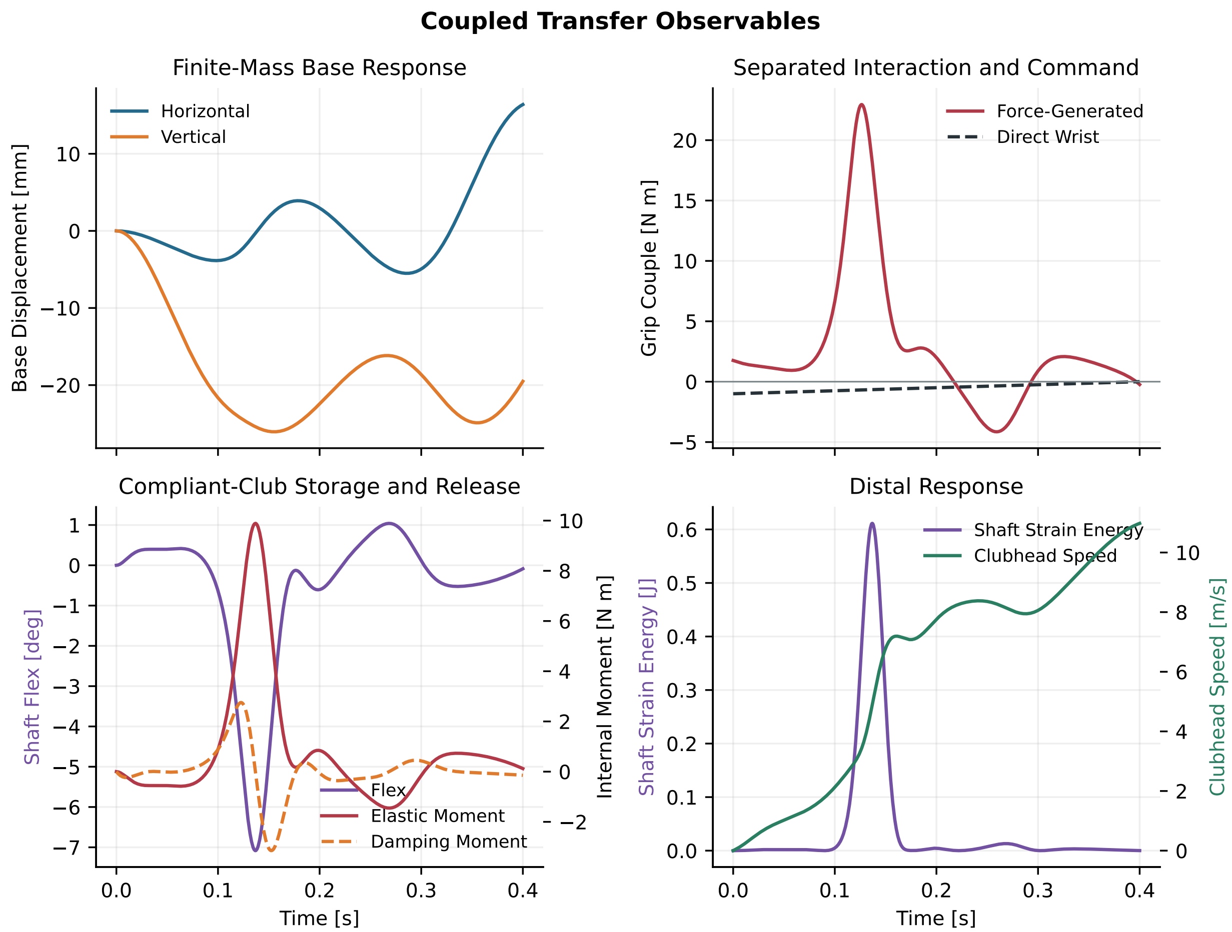

Figure 10.2: Coupled Transfer Observables

The base moves by at most 26.9 mm and the shaft coordinate reaches 7.08° in absolute value, so neither added coordinate is dormant. Peak shaft strain energy is 0.611 J and peak clubhead speed is 10.98 m/s. These values are model outputs under the declared loads, not equipment or performance predictions.

The force-generated grip couple spans -4.15 to +22.95 N · m and first crosses negative at 0.2175 s. By contrast, the direct wrist torque spans -1 to 0 N · m. The negative force-couple interval therefore cannot be relabeled as the direct wrist command. The shaft elastic moment reaches 9.89 N · m in absolute value and the damping moment reaches 3.15 N · m, but both remain separately identified at their internal interface.

10.7 Same-State Intervention and Geometric Negative Control

At exactly 0.200 s, the commanded trajectory is branched without changing any of its ten coordinates or velocities. All six joint commands are then set to zero for 50 ms. The branch begins with a force-generated couple of -0.675 N · m, reaches -2.17 N · m, and remains negative for the full recorded interval. Its work–energy residual is 0.00266 J; the energy loss is accounted for by the declared damping terms.

The force couple changes discontinuously at the intervention even though the coordinates and velocities do not. This is expected: contact multipliers are algebraic dynamics outputs, so removing generalized force at the same state changes the reactions required to satisfy the acceleration constraints.

This intervention does not establish that muscles are passive. It establishes that, after the coupled state has been created, continued applied torque is not mathematically necessary for the modeled negative grip couple over that interval. The contact reactions still depend on inertia, gravity, base restoring forces, shaft elasticity, and damping.

The geometric negative control sets both grip offsets to zero while retaining two independent hand constraints. Equation Equation 10.9 then predicts $M_F = 0$ for any finite contact forces. The executed trace returns zero to machine precision. This rules out a solver sign convention or contact-force magnitude alone as the source of the reported couple.

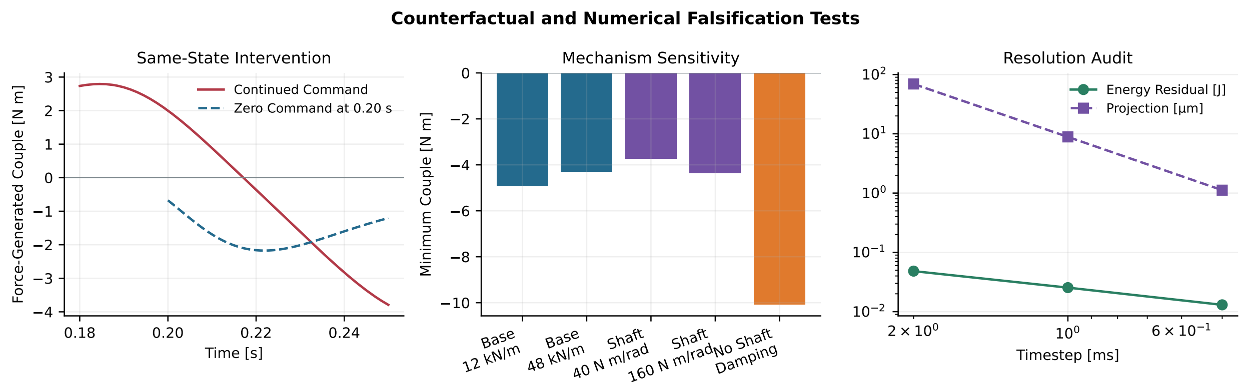

Figure 10.3: Counterfactual and Numerical Falsification Tests

10.8 Mechanism Sensitivity

The registered sensitivity cases vary one parameter while retaining the same initial configuration and open-loop command. Halving base stiffness to 12 kN/m increases maximum base displacement to 49.6 mm and moves the first negative crossing to 0.211 s; doubling it to 48 kN/m reduces displacement to 15.9 mm and yields a 0.217 s crossing. Both retain a negative minimum couple.

Changing shaft stiffness from 40 to 160 $\mathrm{N \cdot m/rad}$ changes peak flex from 11.17° to 3.94° and peak strain energy from 0.761 to 0.379 J. The negative-couple minimum remains between -3.72 and -4.37 $\mathrm{N \cdot m}$. Removing shaft damping increases flex, stored energy, and peak clubhead speed and advances the first negative crossing to 0.160 s; it does not eliminate the sign reversal.

These cases demonstrate local transport of the mechanism, not global robustness. The parameter range is too small to represent population or equipment uncertainty, and the open-loop command is not reoptimized after a parameter change. The appropriate next step is a registered ensemble with identifiability and uncertainty bounds, developed after the spatial model so that planar parameter compensation is not mistaken for physical certainty.

10.9 What This Tier Supports and What It Does Not

Three findings now survive the move from a rigid fixed-base model to this coupled tier:

1. separated hand-force moment arms can generate a late negative club couple;
2. the couple can persist for a finite interval after a same-state command removal; and
3. force/couple power, whole-system energy, constraints, and numerical projection can be closed simultaneously while base motion and shaft flex evolve endogenously.

The result would be contradicted at this tier if the sign reversal disappeared, the zero-command branch became immediately nonnegative, coincident grip points retained a force couple, base or flex motion had to be prescribed, wrench power failed its identity, or the finding changed under timestep refinement. None of those registered failure conditions occurs in the executed study.

The model still omits base rotation, legs and ground reactions, anatomical joints, out-of-plane motion, distributed beam modes, contact compliance, activation dynamics, and measured parameter distributions. It also cannot identify biological effort from constraint forces. Consequently, this chapter supports a planar mechanism-feasibility statement only. The next fidelity gate is a spatial full-body experiment that preserves the same wrench, intervention, energy, and discrepancy observables in two independent dynamics engines.

Chapter 11

Arm–Wrist Torque Allocation and Transmission Preload

11.1 The Hypothesis Requires Two Separate Tests

A club can receive a moment from direct wrist moments, from the moment of two separated hand forces about the club reference point, or from both. Those mechanisms are mechanically distinguishable even when their net club moment is identical. They are not, however, uniquely recoverable from club kinematics alone. The central comparison in this section therefore holds the club-level task fixed and varies only the modeled actuator allocation.

The word *slack* presents a different problem. It can refer to a contact gap, series-elastic extension, low tangent stiffness, delayed force development, or simply low activation. Those conditions are not interchangeable. The second experiment therefore declares one operational definition: a rotational dead zone followed by a linear series stiffness and first-order torque development. It asks what that element predicts during a change in actuator roles. It does not claim that the wrist, grip, or shoulder girdle is literally a backlash joint.

This separation is essential. If two complete swings are assigned different kinematics, a difference in club speed cannot identify whether allocation, preload, geometry, or the changed trajectory caused the result. Here the allocation comparison is same-state and task-matched; the transmission comparison uses the same post-transition net torque target. The implementation and evidence are available in `torque_allocation_preload.py`, `torque_allocation_preload_study.json`, and the corresponding `NPZ trace archive`.

11.2 Physiological Evidence and Its Limits

Scapular electromyography in 15 competitive male golfers reported trail-side levator scapulae, rhomboid, and trapezius activity during takeaway; lead-side trapezius and rhomboid activity during the forward swing; and high trail-side serratus anterior activity during the forward swing (Jobe, Perry, and Pink 1995). This is compatible with phase-dependent shoulder-girdle coordination. It does not establish the direction of either hand force, a bilateral grip couple, or a unique mapping from scapular activation to club moment.

The wrist evidence is likewise informative but non-identifying. In 15 subelite male golfers, extensor carpi ulnaris activity differed by side and swing phase, yet downswing activation was not significantly related to the reported clubhead kinematics (Robinson et al. 2023). A three-dimensional kinetic study found that wrist action and hand trajectory were important to the simulated swing, but its individual-subject results do not resolve a universal actuator allocation (Nesbit 2005a). A systematic review covering 92 golf-kinematics studies found substantial methodological heterogeneity and contradictory results (Bourgain et al. 2022). These sources motivate measurement; they do not prove either extreme strategy.

Direct bilateral measurement is necessary because closed-chain inverse dynamics does not uniquely allocate forces between two hands. Instrumented grip studies have demonstrated that this ambiguity can be reduced with strain gauges or a six-axis grip sensor (Koike 2016; Choi and Park 2020). The present model therefore reports hand-force requirements as predictions to test, not as measured biological loads.

11.3 Matched-Task Allocation

At a fixed configuration $\mathbf{q}$ and velocity $\dot{\mathbf{q}}$, the KKT system from Section 8 is affine in generalized control. Define the control-only club angular acceleration as

$$\ddot{\alpha}_C(\mathbf{u}) = \ddot{\alpha}(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{u}) - \ddot{\alpha}(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{0}). \tag{11.1}$$

Two actuator subspaces are evaluated. The *proximal* subspace contains the four shoulder and elbow generalized torques and sets both direct wrist moments to zero. The *wrist* subspace contains the two direct wrist moments and sets the four proximal torques to zero. Within each subspace, the minimum Euclidean-norm control is selected subject to the same target $I_C\ddot{\alpha}_C = 8$ N m. This is a reproducible mathematical allocation; the proximal subspace must not be relabeled as a measured scapular strategy.

Convex mixtures of the two task-normalized controls generate an allocation fraction from zero (proximal-only) to one (wrist-only). Because the KKT solve is linear in same-state control, every mixture retains the task. Across 19 club angles and 21 allocation fractions, the maximum task error is 5.33×10^{-15} N m. The direct wrist moment plus the moment generated by the two-hand force couple closes to the same net moment within 4.44×10^{-15} N m.

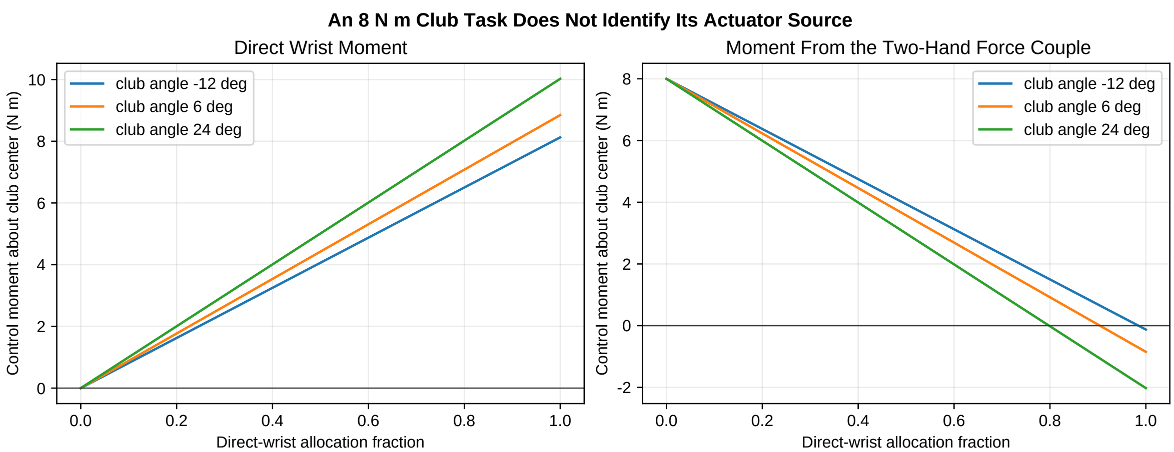

Figure 11.1: The Same Net Club-Moment Task Can Be Divided Between Direct Wrist Moment and a Two-Hand Force Couple

The task match does not make the internal demands equivalent. Across the declared geometry, RMS modeled hand force spans 7.58–91.51 N and the generalized-torque norm spans 5.74–25.45 N m. For these particular unweighted metrics, intermediate-to-wrist-dominant allocations minimize hand force, while wrist-dominant allocations minimize generalized-torque norm. The minimizing fraction changes with club geometry.

This result rejects two shortcuts. First, a measured net club moment does not identify whether the arms or wrists supplied it. Second, a single scalar `effort` cannot select the best allocation without declared weights for hand force, joint torque, work, stability, discomfort, accuracy, and robustness. The model supports an allocation surface, not universal technique advice.

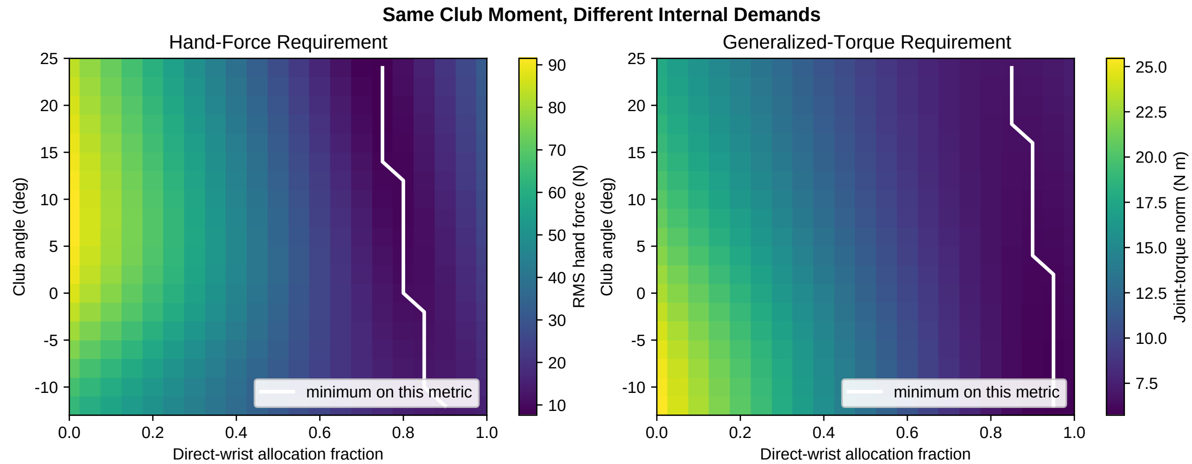

Figure 11.2: Modeled Demand Minima Vary With Geometry and Objective

11.4 A Falsifiable Definition of Slack

For each actuator channel i, desired torque is converted to an equilibrium deflection x_i^* through a dead zone δ_i and stiffness k_i:

$$x_i^* = \operatorname{sgn}(\tau_i^*) \left(\delta_i + \frac{|\tau_i^*|}{k_i} \right), \qquad \dot{x}_i = \frac{x_i^* - x_i}{T_i}, \tag{11.2}$$

$$\tau_i(x_i) = k_i \operatorname{sgn}(x_i) \max(|x_i| - \delta_i, 0). \tag{11.3}$$

The declared numerical experiment uses $k = 600$ N m rad^{-1}, $\delta = 0.012$ rad, and $T = 18$ ms for both abstract channels. These are synthetic sensitivity parameters, not identified tissue or grip properties. An interval with $|x_i| \leq \delta_i$ is the only condition called `zero transmission` here.

Two matched net-torque programs are compared. In the *persistent-direction* program, the arm channel remains positive and the wrist channel remains negative across the transition, with net torque changing from 6 to 10 N m. In the *role-reversal* program, the arm channel changes from -4 to +16 N m while the wrist channel changes from +10 to -6 N m, preserving the same pre- and post-transition net targets. Each is evaluated from its pre-transition loaded equilibrium and from a relaxed zero-deflection state.

From the loaded state, the persistent-direction program has no zero-transmission interval and a net torque-error impulse of 0.0717 N m s. The role-reversal program spends 11.5 ms and 22.0 ms in zero transmission for the arm and wrist channels, respectively, and its error impulse is 0.1954 N m s. A relaxed start produces an error impulse of 0.2011 N m s for either program. Thus preload improves continuity for a fixed desired program, while a sign reversal can discard that advantage under this particular dead-zone model.

The interpretation is conditional. Biological series elasticity normally stores and returns energy rather than acting as backlash, and short-range muscle stiffness can increase with activation (Araz et al. 2023; De Groote, Allen, and Ting 2017). Series compliance can also delay force rise (Luo, Cooke, and Pate 1994; Mayfield, Cresswell, and Lichtwark 2016), while co-contraction can improve task-relevant stiffness at an energetic cost (Berret et al. 2024). Consequently, neither zero compliance nor maximum co-contraction is presumed optimal. Useful compliance can filter impact, store energy, and permit coordination; unwanted gaps or low tangent stiffness can impair rapid torque transmission. The distinction must be measured.

11.5 Critical Comparison of the Two Extremes

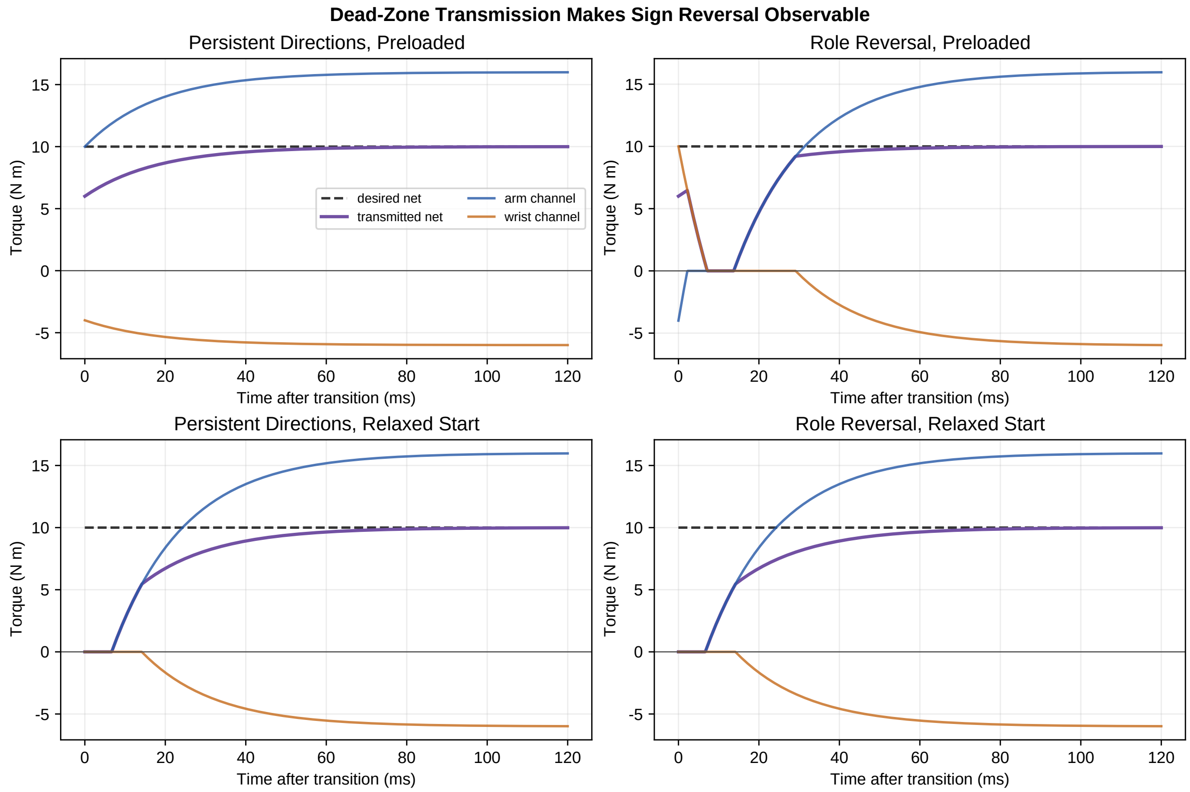

Figure 11.3: Transmission Continuity for Persistent-Direction and Role-Reversal Programs

Question	Persistent Arm Drive With Wrist Resistance	Wrist-to-Arm Role Reversal
Club-level task	Can match the same net moment	Can match the same net moment
Transmission continuity in the declared dead-zone model	Favored when each channel remains loaded in the same direction	Penalized when both channels cross zero
Possible coordination benefit	Reuses a directionally persistent proximal command	Separates preparation and delivery roles
Principal modeled cost	May require larger hand forces or proximal generalized torques, depending on geometry	May require reversal delay and transient torque error
Physiological evidence	Scapular phase activity is compatible but non-identifying	Wrist phase activity is compatible but non-identifying
Main falsifier	Measured bilateral forces show no predicted couple or preload continuity	Measured reversal occurs without force/stiffness gap and yields equal or better robustness

The persistent-direction strategy therefore has a clear conditional advantage: if the relevant biological transmission behaves like the declared dead-zone element, retaining load direction can preserve immediate torque transmission. It is not established as universally superior. The wrist-led preparation may reduce proximal loading, exploit elastic storage, or improve accuracy in a participant-specific system; the current model does not price those outcomes.

11.6 Measurements That Can Refute the Interpretation

The experimental protocol in Section 17 should add the following preregistered observables:

1. bilateral six-axis grip wrenches and grip-pressure distribution, transformed to a declared club reference point;
2. wrist moments from a closed-chain inverse-dynamics solve constrained by the measured internal grip wrench;
3. ultrasound or validated musculotendon-length proxies where feasible, plus perturbation-based or system-identification estimates of tangent stiffness;
4. surface EMG for forearm, shoulder, and scapular muscles, interpreted as activation timing rather than joint torque;
5. transition-aligned force-development delay, zero-force duration, torque-error impulse, club angular acceleration, shaft strain, and launch outcomes; and
6. participant-held-out comparisons of the two extremes and intermediate allocations, with no universal optimum asserted from group means alone.

The interpretation is falsified if the same club task does not show the predicted allocation-dependent hand forces, if inferred channel reversal does not produce a measurable transmission delay under a documented gap or low stiffness, or if any claimed performance advantage disappears on participant holdout. Observing scapular or wrist EMG timing alone is insufficient.

Chapter 12

Distributed-Shaft Reference and Modal Reduction

12.1 Structural Question and Evidence Boundary

The coupled model in Section 10 admits one linear torsional flex coordinate. That coordinate is sufficient to test whether stored elastic energy can coexist with the two-hand negative-couple mechanism, but it cannot represent a distributed bending field or decide when higher shaft modes matter. This section therefore compares a one-mode reduction with a higher-order Euler–Bernoulli reference assembled by the repository's shared finite-element shaft implementation. The exact analysis and evidence are in `shaft_beam_reference.py`, `run_shaft_beam_reference.py`, and `shaft_beam_reference.json`.

This is a structural verification experiment, not an equipment calibration. The modal observations used for identification are generated from a declared synthetic truth. No measured shaft frequencies, EI profile, clubhead inertia, or human loading data enter the result.

12.2 Distributed Beam and Tip Inertia

Each tapered beam element carries transverse displacement and section rotation at both nodes. Its standard consistent mass and bending-stiffness matrices are assembled into M_b and K_b, the butt displacement and rotation are clamped, and declared clubhead translation and rotary inertia are added at the free-end degrees of freedom. The generalized eigenproblem is

$$\mathbf{K}_b\phi_i = \omega_i^2\mathbf{M}_b\phi_i, \qquad \phi_i^{\mathsf{T}}\mathbf{M}_b\phi_j = \delta_{ij}. \tag{12.1}$$

The reference shaft is 1.0 m long with a 15.0 mm butt diameter, 8.5 mm tip diameter, 1.0 mm wall, 1,600 kg/m^3 density, 112 GPa elastic modulus, 0.205 kg tip mass, and 4.8×10^{-4} kg m^2 tip rotary inertia. These values define the synthetic test article only. Twenty-four elements form the executed reference; comparison against 48 elements changes the first three frequencies by at most 0.0896%. The converged first three frequencies are 5.240, 62.931, and 137.909 Hz.

12.3 Synthetic Identification and Uncertainty

Two parameters—the elastic modulus and clubhead mass—are identified from the first two synthetic modal frequencies. Optimization occurs in log-parameter space to enforce positivity. With an assumed independent frequency uncertainty of 0.05 Hz, local covariance is propagated from the residual Jacobian. The fit recovers

112.000 GPa and 0.2050 kg with a maximum modal residual of 9.94×10^{-9} Hz. The corresponding assumed-noise 95% intervals are 111.651–112.350 GPa and 0.1970–0.2133 kg.

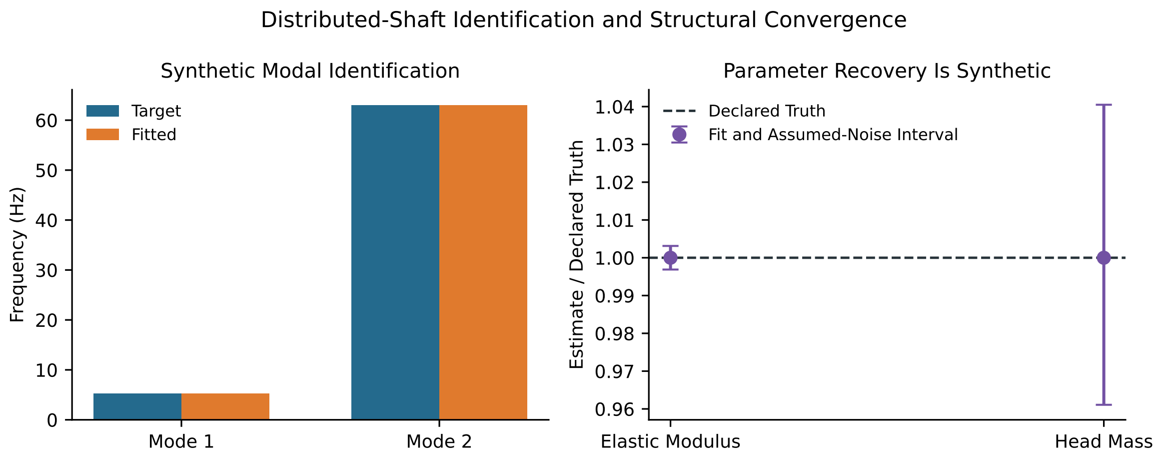

Figure 12.1: Synthetic Identification and Beam Convergence

Figure 12.1 verifies that the inference path can recover an identifiable synthetic truth. It does not show that real equipment follows the uniform-modulus, axisymmetric, Euler–Bernoulli assumptions. Practical equipment identification still requires measured modal frequencies, boundary conditions, mass properties, and their uncertainties.

12.4 Matched Reduced and Higher-Order Responses

The mass-normalized modal coordinates satisfy

$$\ddot{\eta}_i + 2\zeta\omega_i\dot{\eta}_i + \omega_i^2\eta_i = \phi_i^{\mathsf{T}}\mathbf{f}(t), \tag{12.2}$$

with a damping ratio of 0.018. The reduced model retains only the first mode. The higher-order reference retains six modes. Both receive exactly the same load history and use fixed-step fourth-order Runge–Kutta integration at 12.5 μs. A slow 80 ms tip-force pulse probes first-mode behavior; a short 4 ms tip-force-and-moment pulse deliberately introduces higher-frequency content.

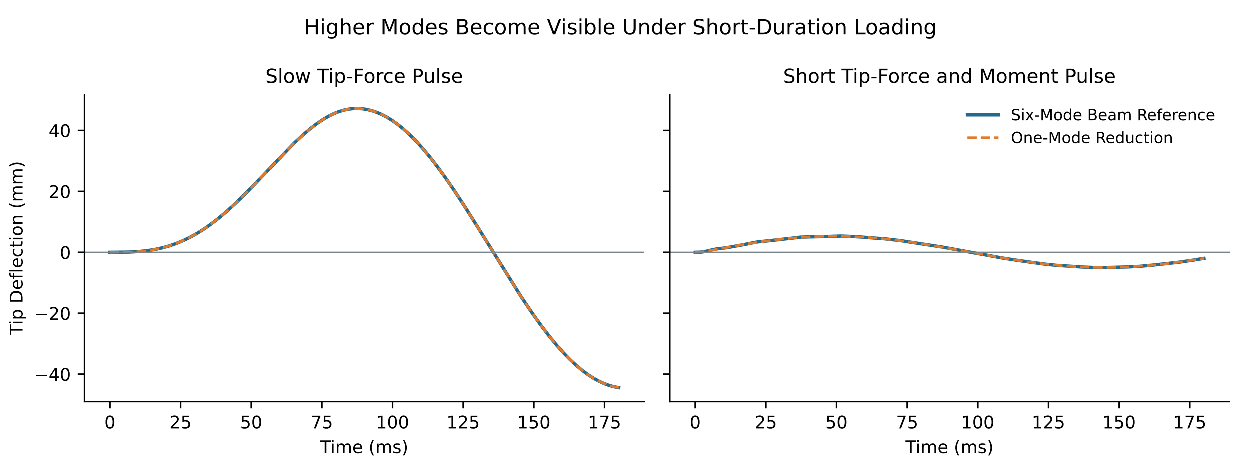

Figure 12.2: Reduced and Higher-Order Tip Responses

Under the slow pulse, the tip-deflection RMS discrepancy is 0.000687 mm. Under the short pulse it grows to 0.0413 mm, sixty times larger, even though the peak values remain close (5.287 mm reduced and 5.339 mm reference). Peak-only agreement therefore conceals a structural discrepancy in the response history. The one-mode model is a useful low-frequency surrogate, not a general replacement for the distributed beam.

12.5 Work–Energy Closure and Remaining Coupling Gate

For either modal truncation,

$$E = \tfrac{1}{2}\sum_i(\dot{\eta}_i^2 + \omega_i^2\eta_i^2), \qquad \dot{E} = P_{\text{tip}} - \sum_i 2\zeta\omega_i\dot{\eta}_i^2. \tag{12.3}$$

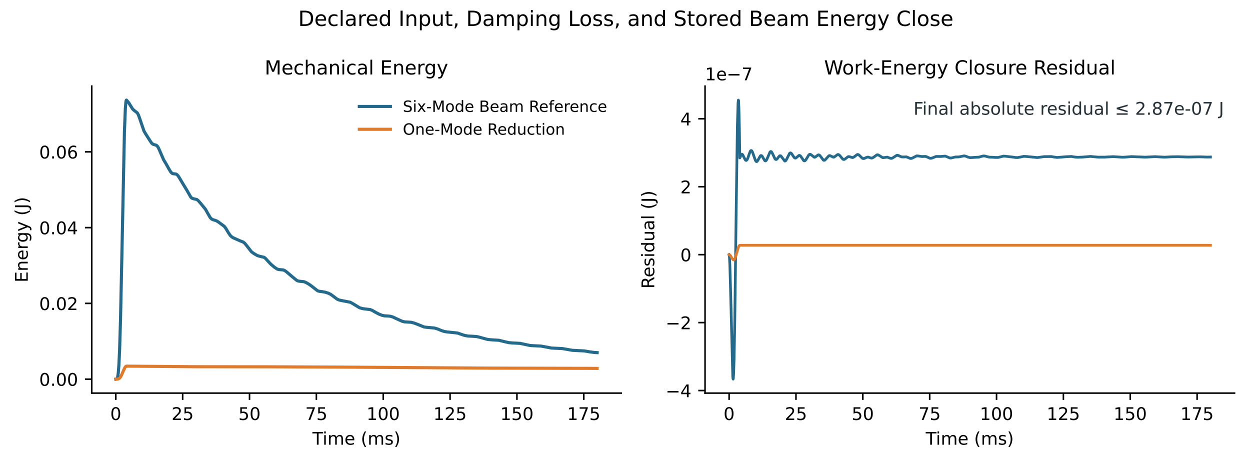

Figure 12.3: Beam Input, Damping, and Energy Closure

The maximum absolute integrated work–energy residual is 2.87×10^{-7} J for the six-mode reference and 2.77×10^{-8} J for the one-mode reduction. This closes the isolated beam comparison and makes damping loss explicit.

The isolated experiment narrows, but does not itself close, the structural coupling gap. Section 13 next inserts the transported modes into the moving-base, two-hand KKT solve and repeats the contact, constraint, and whole-system energy interventions. Equipment-specific language remains prohibited until measured modal, damping, inertia, and boundary-condition data are supplied.

Chapter 13

Forward Coupling of a Distributed Modal Shaft

13.1 Question, Scope, and Claim Boundary

The isolated beam experiment in Section 12 established that a one-mode reduction can conceal excitation-dependent response history. It did not establish whether the two-hand force-couple mechanism survives when the distributed shaft participates in the same forward dynamics as the moving base, arms, grips, and clubhead. This section closes that specific numerical gate. Finite-element-derived bending modes are inserted directly into the planar constrained solve from Section 10; the grip forces, modal coordinates, base motion, and clubhead motion are simultaneous outputs of one KKT system.

The implementation, deterministic study, and machine-readable result are available in `moving_base_modal_shaft.py`, `run_moving_base_modal_shaft_study.py`, and `moving_base_modal_shaft_study.json`. The corresponding trace archive is `moving_base_modal_shaft_study.npz`.

This remains a synthetic mechanism experiment. The shaft bending rigidity is four times the isolated structural reference (implemented as 448 GPa with the reference geometry) so that the declared drive remains inside a conservative small-deflection screen. That multiplier is a numerical stress-test choice, not a material, commercial-shaft, or equipment-calibration claim. No measured EI profile, damping, grip compliance, hand force, body segment, or launch data enter the calculation.

13.2 Coupled Coordinates and Distributed Inertia

The generalized coordinate vector is

$$\mathbf{q} = [\theta_R, \phi_R, \theta_L, \phi_L, x_B, y_B, x_G, y_G, \alpha, \eta_1, \dots, \eta_n]^\mathsf{T}, \tag{13.1}$$

where the first four angles describe the two planar arms, (x_B, y_B) is the finite-mass translating base, (x_G, y_G, α) describes the club root, and η_i are mass-normalized shaft bending coordinates. At shaft station s,

$$\mathbf{r}(s, \mathbf{q}) = \mathbf{r}_G + s\,\mathbf{e}(\alpha) + \left[\sum_{i=1}^{n} \Phi_i(s)\eta_i\right] \mathbf{n}(\alpha). \tag{13.2}$$

The finite-element mode shapes from Equation 12.1 are evaluated by Hermite interpolation at Gaussian quadrature stations. Distributed shaft mass, clubhead translation mass, and clubhead rotary inertia then contribute through their exact point Jacobians,

$$\mathbf{M}_s(\mathbf{q}) = \int_0^L \rho A(s)\mathbf{J}(s,\mathbf{q})^\top \mathbf{J}(s,\mathbf{q})\,ds + m_h \mathbf{J}_h^\top \mathbf{J}_h + I_h \mathbf{J}_\omega^\top \mathbf{J}_\omega. \tag{13.3}$$

This construction retains rigid–modal inertial coupling. It is not a beam response calculated after a rigid-club trajectory has already been prescribed. The transported quadrature basis reproduces the first three finite-element frequencies to a maximum relative discrepancy of 1.09×10^{-5} and the modal mass matrix differs from identity by at most 2.22×10^{-16}. For the declared four-times-EI case, those frequencies are 10.480, 125.891, and 276.066 Hz.

13.3 Constraint Forces, Couples, and Power

Both hand endpoints remain constrained to separated grip points. With four holonomic constraints $\mathbf{c}(\mathbf{q}) = \mathbf{0}$, the acceleration and reaction multipliers are obtained without a rank-deficient fallback:

$$\begin{bmatrix} \mathbf{M}(\mathbf{q}) & -\mathbf{J}_c^\top \\ \mathbf{J}_c & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{Q} - \mathbf{h}(\mathbf{q},\dot{\mathbf{q}}) \\ -\dot{\mathbf{J}}_c \dot{\mathbf{q}} \end{bmatrix}. \tag{13.4}$$

The two solved forces on the club are $\mathbf{F}_R$ and $\mathbf{F}_L$. Their force-generated grip couple about the club-root reference is

$$M_F = (\mathbf{r}_R - \mathbf{r}_G) \times \mathbf{F}_R + (\mathbf{r}_L - \mathbf{r}_G) \times \mathbf{F}_L. \tag{13.5}$$

This quantity remains separate from the directly commanded wrist moment and from the internal elastic generalized forces. The contact-power identity is checked at every sample in both equivalent forms,

$$\mathbf{F}_R \cdot \mathbf{v}_R + \mathbf{F}_L \cdot \mathbf{v}_L = (\mathbf{F}_R + \mathbf{F}_L) \cdot \mathbf{v}_G + M_F \dot{\alpha}. \tag{13.6}$$

The maximum discrepancy is 1.71×10^{-13} W in the baseline and 1.14×10^{-13} W after the intervention. Position, velocity, acceleration, and KKT residuals are also recorded rather than inferred from visual agreement.

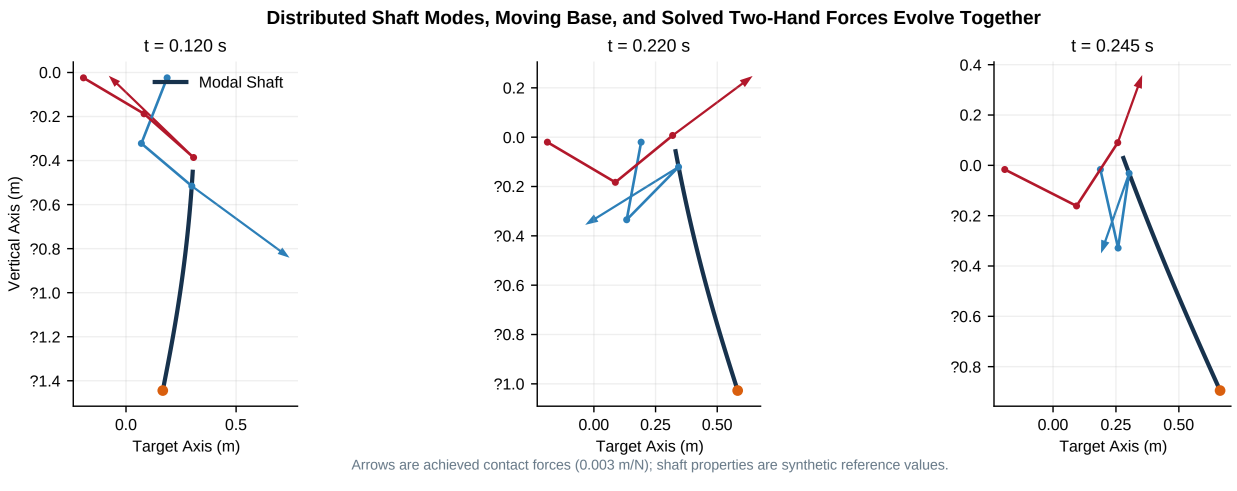

Figure 13.1: Forward Geometry, Distributed Shaft, and Achieved Grip Forces

Figure 13.1 shows three states from the integrated trajectory. The arrows are the achieved constraint forces, not specified loads. Their directions change with the arm–grip geometry while the modal centerline, base, and clubhead evolve in the same solve. Arrow length uses one common display scale; force interpretation still requires the registered reference point and power variables.

13.4 Model-Use Screen and Baseline Closure

Linear Euler–Bernoulli kinematics are inappropriate if the numerical case is allowed to bend without a declared range-of-use guard. This study therefore preregisters the screening rule

$$\max_t |w(L,t)|/L < 0.05. \tag{13.7}$$

The observed maximum is 0.03378, so the declared baseline passes. The 5% value is a conservative model-use screen for this resource, not a universal theorem about beam validity. Torsion, shear deformation, geometric nonlinearity, impact, and material anisotropy remain absent even when the screen passes.

At a 0.25 ms step, the base moves by at most 26.65 mm, the modal tip deflection reaches 33.78 mm, and the first three modal coordinates all depart from zero. Peak shaft strain energy is 0.533 J and peak clubhead speed is 7.000 m/s. The force-generated couple spans -18.65 to 31.06 N m. Mechanical energy changes by 40.285 J while integrated applied and dissipative work is 40.279 J, leaving a 0.00574 J absolute residual. These are internal consistency observations for the synthetic case, not estimates of golfer or equipment performance.

13.5 Excitation-Dependent Mode Truncation

One-, three-, and six-mode systems are integrated from identical initial states under two resolved inputs. A smooth drive represents the low-frequency case; a 2 ms wrist-moment pulse deliberately excites higher frequencies. The six-mode result is the within-model comparison reference, not experimental truth.

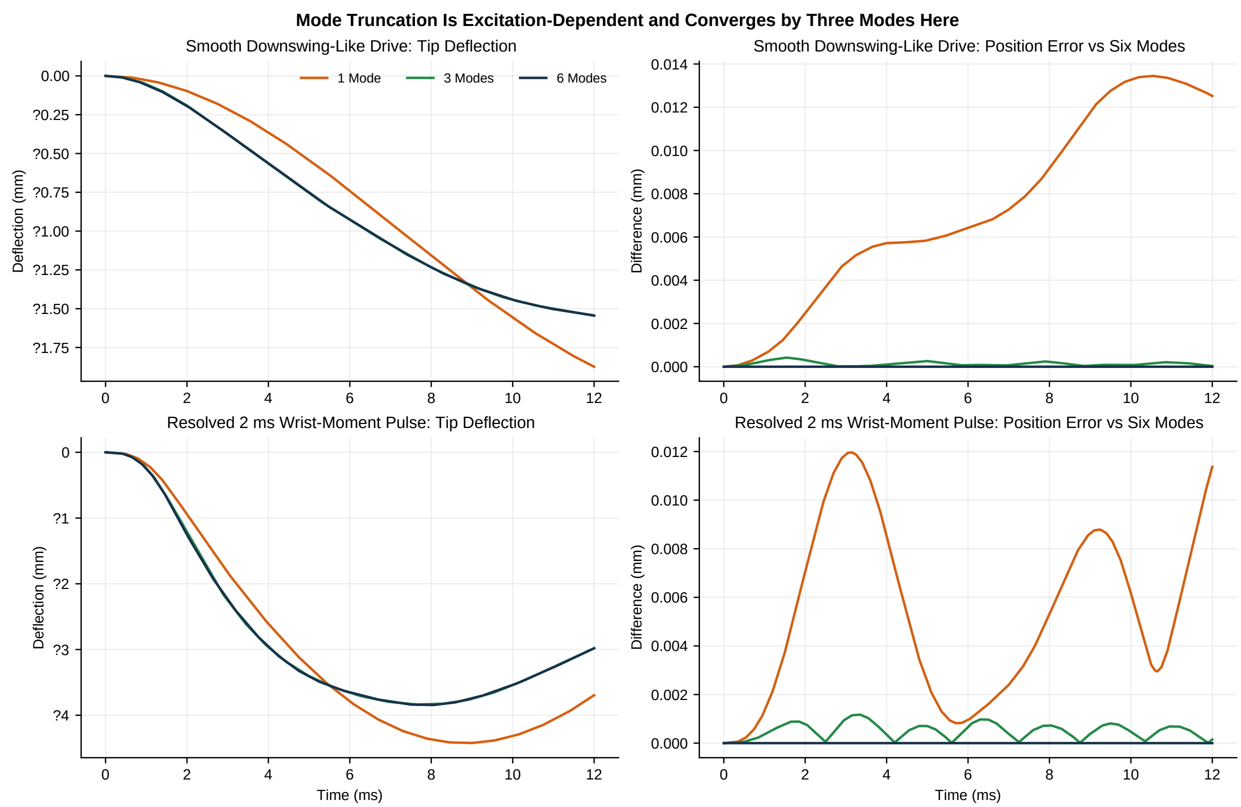

Figure 13.2: Mode-Truncation Response Under Smooth and Short Inputs

For the smooth drive, the one-mode tip-deflection RMS discrepancy relative to six modes is 0.0483 mm and its maximum clubhead-position discrepancy is 3.42 μm. Retaining three modes reduces those values to 0.000406 mm and 0.106 μm. Under the short pulse, the one-mode deflection discrepancy grows to 0.267 mm, 5.53 times its smooth-input value, while the three-mode discrepancy remains 0.00108 mm and the clubhead-position discrepancy remains 0.196 μm. Figure 13.2 therefore supports a conditional statement: three modes are converged for the two declared excitations and timestep, whereas a single-mode approximation is input-dependent. It does not establish that three modes suffice for impact or arbitrary loading.

13.6 Same-State Zero-Command Intervention

At 0.220 s, the complete state $(\mathbf{q}, \dot{\mathbf{q}})$ is copied exactly and all shoulder, elbow, and wrist commands are set to zero. The branch is then integrated for 30 ms. Because the branch shares its cut state bit for bit, any subsequent difference from a zero-inertia or relabeled-load construction cannot be attributed to a changed initial condition.

The force-generated couple is already negative at the cut and remains negative for the complete 30 ms branch, reaching -14.94 N m. This is a statement about the continued constrained motion of the declared model after command removal; it is not evidence that biological muscles are inactive. The branch continues to contain inertia, gravity, base restoring forces, joint damping, modal elasticity, and shaft damping.

Two exact geometry controls bound the interpretation. Moving both grip moment arms to the same reference point makes the force-generated couple identically zero while retaining the achieved force records. Reversing the registered moment arms reverses the couple with zero residual. Thus the negative sign is not a force-magnitude label: it depends on the cross products in Equation 13.5.

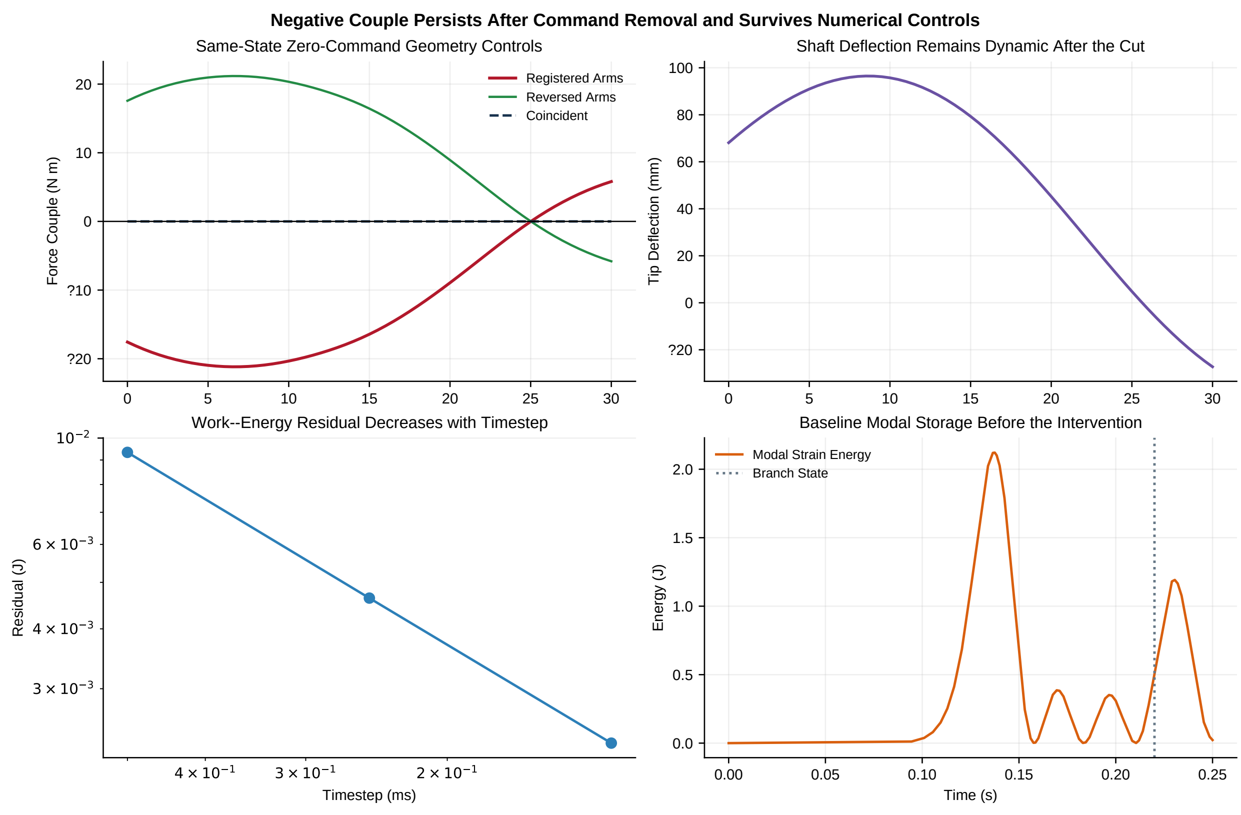

Figure 13.3: Same-State Killswitch, Geometry Controls, and Numerical Closure

The intervention is repeated at 0.25, 0.125, and 0.0625 ms. The absolute work–energy residual decreases monotonically from 0.00683 to 0.00340 to 0.00169 J, while the negative interval remains 30 ms and the minimum couple converges from -14.94 to -14.97 N m. This separates a persistent mechanism result from a fixed-step artifact without claiming zero numerical error.

13.7 Falsifiers and Remaining Inference Gaps

The coupled distributed-shaft result would fail its declared claim if any of the following occurred: transported modes did not reproduce the finite-element frequencies; higher modes remained numerically inert under the resolved pulse; the negative interval disappeared with timestep refinement; the coincident or reversed moment-arm controls failed; the 5% deflection screen failed; or the constraint, contact-power, and work–energy residuals ceased to close.

None of those failures occurs in this synthetic planar case. The supported claim is consequently narrow: a finite-element-derived modal shaft can be coupled into the moving-base two-hand forward solve while preserving the registered negative-couple intervention and numerical identities. Equipment calibration, anatomical arms, base rotation, three-dimensional shaft bending and torsion, compliant tissue, impact, muscle coordination, optimal control, human strategy, and coaching benefit remain untested. Those omissions define the next model tiers; they are not conclusions supplied by this one.

Chapter 14

Spatial Full-Body Common-State Dynamics

The preceding forward models establish a planar mechanism under increasing coupling, but a planar result cannot answer whether the same wrench geometry survives out-of-plane motion or whether it is an artifact of one dynamics implementation. This chapter executes a spatial comparison at identical achieved states. It uses one immutable reduced full-body model definition in two independent formulations: native MuJoCo inverse dynamics and a Lagrange–Christoffel implementation assembled directly from body Jacobians (Todorov, Erez, and Tassa 2012; Featherstone 2008).

This is a deliberately bounded test. It is a real three-dimensional dynamics calculation, not a rotation of planar vectors. The body and club move about all three world axes; the club has three translational and three rotational coordinates; and the hand forces include out-of-plane components. The two formulations evaluate exactly the same position, velocity, acceleration, inertias, gravity, and applied action–reaction loads. That common-state design isolates dynamics implementation from trajectory divergence.

The hand loads are prescribed. Therefore the experiment can test spatial wrench transport, geometry interventions, and inverse-dynamics equivalence. It cannot establish that the loads would emerge passively from compliant two-hand contact. That distinction controls every claim below.

14.1 Why a Common Model Is Necessary

Names such as "full body" or "humanoid" do not establish model equivalence. The available native engine artifacts differ in joint count, floating-base representation, lower-body topology, club attachment, and treatment of the second hand. Comparing their force histories directly would mix at least four effects:

1. numerical implementation;
2. coordinate and frame conventions;
3. structural model differences; and
4. distinct initial conditions or inputs.

No tolerance can make that comparison causal. The present experiment instead defines one study-specific tree and generates the MuJoCo representation from the same in-memory specification consumed by the independent analytical formulation. A SHA-256 digest of the canonical joint, inertia, and attachment record is stored for both implementations. The two recorded hashes must be identical before an outcome is evaluated.

This common model is a reduced full-body model rather than a subject-specific anatomical reconstruction. It contains pelvis and lower-body mass, a spatial torso, bilateral three-axis shoulders, elbows, wrists, and a separately moving club. Twenty generalized coordinates comprise 14 body coordinates and six club coordinates. Thirty-two spherical inertia elements include explicit small carrier inertias for otherwise massless compound-joint frames. Those carrier masses are present in both formulations and are part of the model hash; they are not an engine-specific numerical patch.

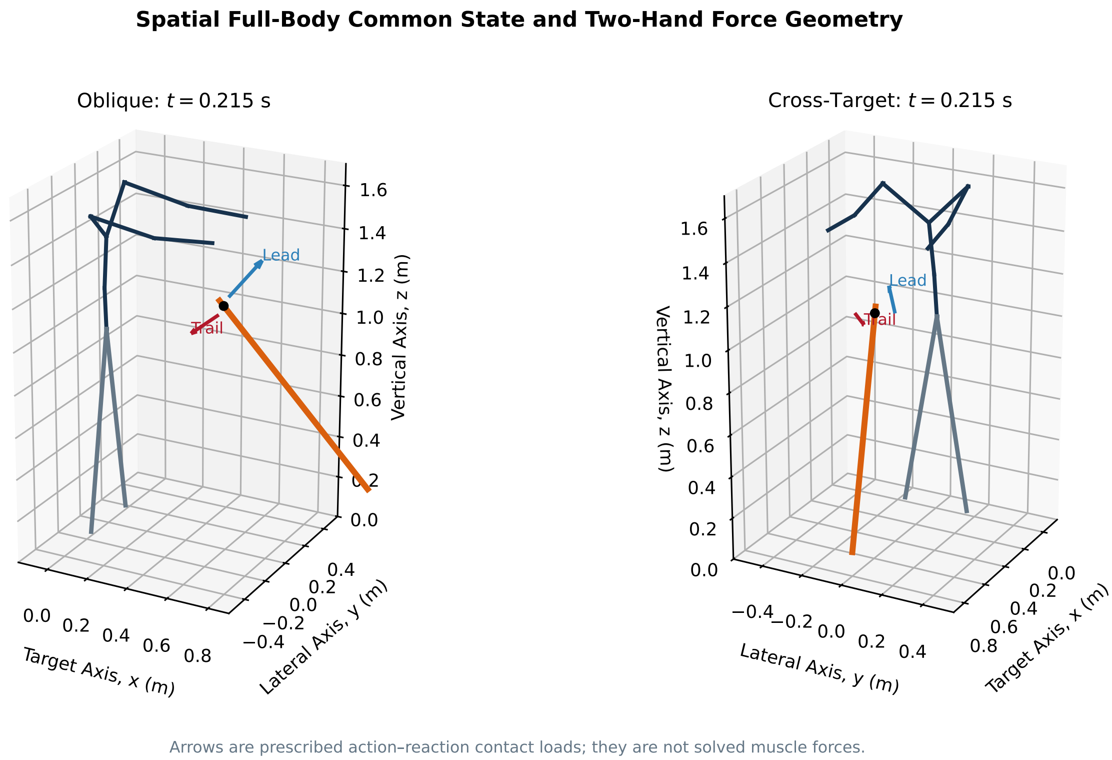

Figure 14.1: Spatial Full-Body Common State and Two-Hand Force Geometry

Figure 14.1 shows the achieved state at 0.215 s from two views. The lateral club displacement spans 35.0 mm over the registered window, and the model axes have rank three. The blue and red arrows are the lead- and trail-hand forces on the club. They are prescribed contact loads, not estimates of muscle force or measured human hand force.

14.2 Shared Spatial Dynamics Definition

Let the generalized state be

$$\mathbf{q} = \begin{bmatrix} \mathbf{q}_B & \mathbf{r}_C & \eta_C \end{bmatrix}^{\mathsf{T}}, \tag{14.1}$$

where $\mathbf{q}_B \in \mathbb{R}^{14}$ contains the reduced body coordinates, $\mathbf{r}_C \in \mathbb{R}^3$ is club-frame translation, and $\eta_C \in \mathbb{R}^3$ is the ordered club rotation coordinate. Every joint is a scalar revolute or prismatic coordinate. This avoids hiding a quaternion/Euler remapping inside the engine comparison.

For body b with mass m_b, world rotation $\mathbf{R}_b$, isotropic local inertia $\mathbf{I}_b$, linear COM Jacobian $\mathbf{J}_{v,b}$, and angular Jacobian $\mathbf{J}_{\omega,b}$, the analytical mass matrix is

$$\mathbf{M}(\mathbf{q}) = \sum_b \left(m_b \mathbf{J}_{v,b}^{\mathsf{T}} \mathbf{J}_{v,b} + \mathbf{J}_{\omega,b}^{\mathsf{T}} \mathbf{R}_b \mathbf{I}_b \mathbf{R}_b^{\mathsf{T}} \mathbf{J}_{\omega,b} \right). \tag{14.2}$$

The independent bias calculation uses

$$\mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) = \dot{\mathbf{M}}\dot{\mathbf{q}} - \frac{1}{2}\nabla_{\mathbf{q}}\left(\dot{\mathbf{q}}^{\top}\mathbf{M}\dot{\mathbf{q}}\right) + \nabla_{\mathbf{q}}V, \tag{14.3}$$

with centered finite differences of the mass matrix and an analytical gravity gradient. MuJoCo independently compiles the same tree and evaluates its native inverse dynamics. Both outputs are normalized to the same required-action convention,

$$\tau_{\text{required}} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{h} - \mathbf{Q}_{\text{contact}}. \tag{14.4}$$

This normalization matters. MuJoCo's inverse-dynamics output reports inertial plus bias generalized force; applied generalized loads are not subtracted from that field automatically. The first implementation audit exposed a 26% apparent discrepancy when the two sides used different external-load conventions. Applying Equation 14.4 explicitly reduced the maximum relative discrepancy to 2.14×10^{-11}. The preregistered 5% relative and 0.75 generalized-force absolute bounds were not changed.

14.3 Two-Hand Wrench and Action–Reaction Contract

At the declared club reference O, the lead and trail forces produce

$$\mathbf{F}_O = \mathbf{F}_L + \mathbf{F}_T, \qquad \mathbf{M}_O = (\mathbf{r}_L - \mathbf{r}_O) \times \mathbf{F}_L + (\mathbf{r}_T - \mathbf{r}_O) \times \mathbf{F}_T. \tag{14.5}$$

The complete wrench is $\mathcal{W}_C = [\mathbf{F}_O^{\top}, \mathbf{M}_O^{\top}]^{\top}$. The equal and opposite body-side wrench is $\mathcal{W}_B = -\mathcal{W}_C$. For a compatible contact twist $\mathcal{V} = [\mathbf{v}_O^{\top}, \omega^{\top}]^{\top}$,

$$\mathcal{W}_C^{\top}\mathcal{V} + \mathcal{W}_B^{\top}\mathcal{V} = 0. \tag{14.6}$$

The recorded maximum residual in Equation 14.6 is zero to stored precision. This checks the common virtual-work convention. It does not assert no work at a slipping or compliant biological contact, because the two sides would then have different point velocities and contact storage or dissipation would require its own state.

The force-generated scalar club couple is the moment projection on the local club axis $\hat{\mathbf{e}}_C$,

$$M_F = \hat{\mathbf{e}}_C^{\top}\mathbf{M}_O. \tag{14.7}$$

No direct club torque command is applied. This keeps direct couple and force-generated couple separate, but prescribed force is still an external input. "Zero direct torque" is therefore not synonymous with "passively generated contact force."

14.4 Registered Interventions and Tolerances

Three interventions are evaluated at the same achieved state and with the same force histories:

- **Registered Geometry:** maintain the declared 0.18 m signed separation of lead and trail contact points;
- **Reversed Moment Arm:** reverse the signed contact separation without changing either force vector; and
- **Coincident Hands:** set both force application points to the same reference, which analytically requires the force-generated couple to be zero.

The preferred result is not used to select a tolerance. Before inspecting the event-window outcome, the numerical region was fixed at 0.75 absolute generalized-force units, 5% relative error, and 4 ms intervention-grid alignment error. Because both formulations receive one registered load history, this last quantity checks input alignment; it is not a second event detector. The analytical derivative was evaluated with 2×10^{-6} and 10^{-6} coordinate steps. Their maximum generalized-force difference is 1.47×10^{-9}, far below the predeclared absolute region.

The experiment is falsified at this tier if any of the following occurs:

1. the two implementations use different model hashes;
2. the same-state inverse-dynamics residual exceeds either registered bound;
3. complete action–reaction wrench power does not close;
4. reversing the moment arm fails to reverse the force-generated couple;
5. coincident force application points retain a finite force couple; or
6. the purported spatial trajectory has no out-of-plane motion.

14.5 Executed Results

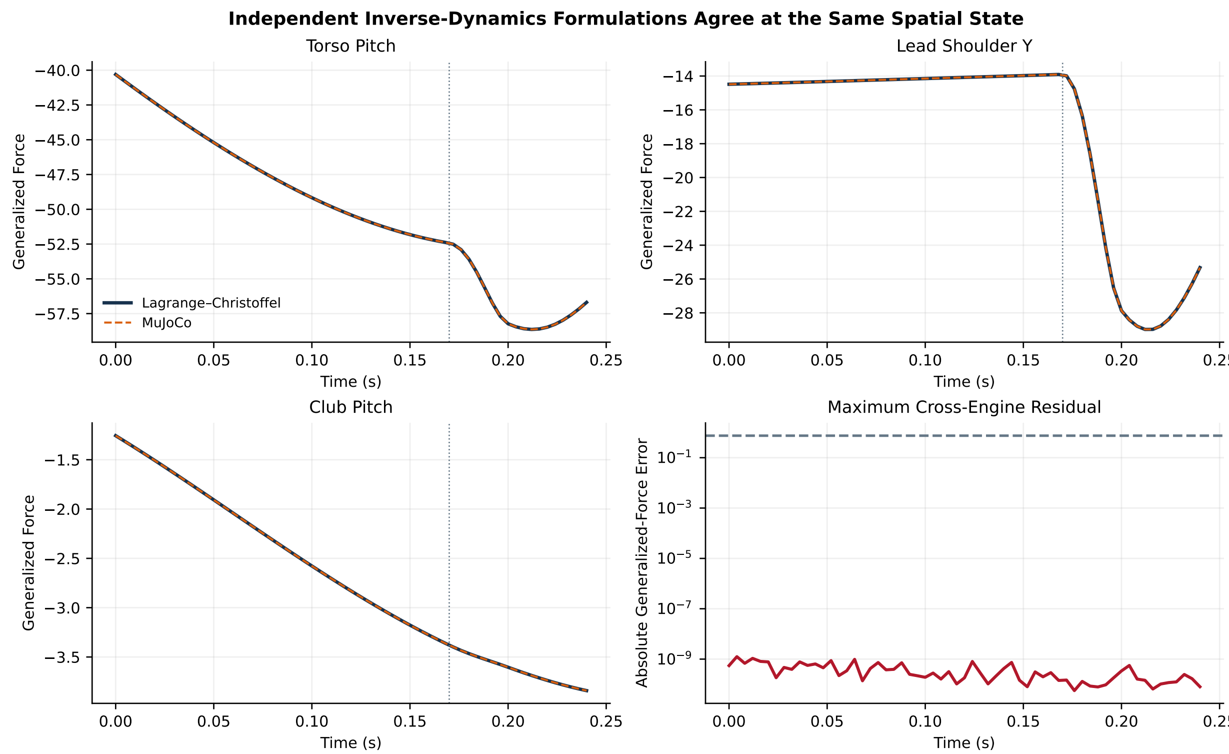

Figure 14.2: Independent Inverse-Dynamics Formulations Agree at the Same Spatial State

Figure 14.2 overlays three representative generalized actions. The curves are visually indistinguishable. Across all 20 coordinates and 61 event-aligned states, the maximum absolute discrepancy is 1.25×10^{-9} and the maximum relative discrepancy is 2.14×10^{-11}. The registered intervention-grid alignment error is zero by construction and is reported separately from the dynamics residual. This result supports implementation equivalence for the declared common model and same-state inverse-dynamics observable.

It does not show that arbitrary MuJoCo and analytical humanoid models agree. The equivalence depends on the identical model digest, joint order, spherical inertias, gravity vector, state, and generalized external-load convention. Changing any of those inputs creates a new experiment and requires a new tolerance record.

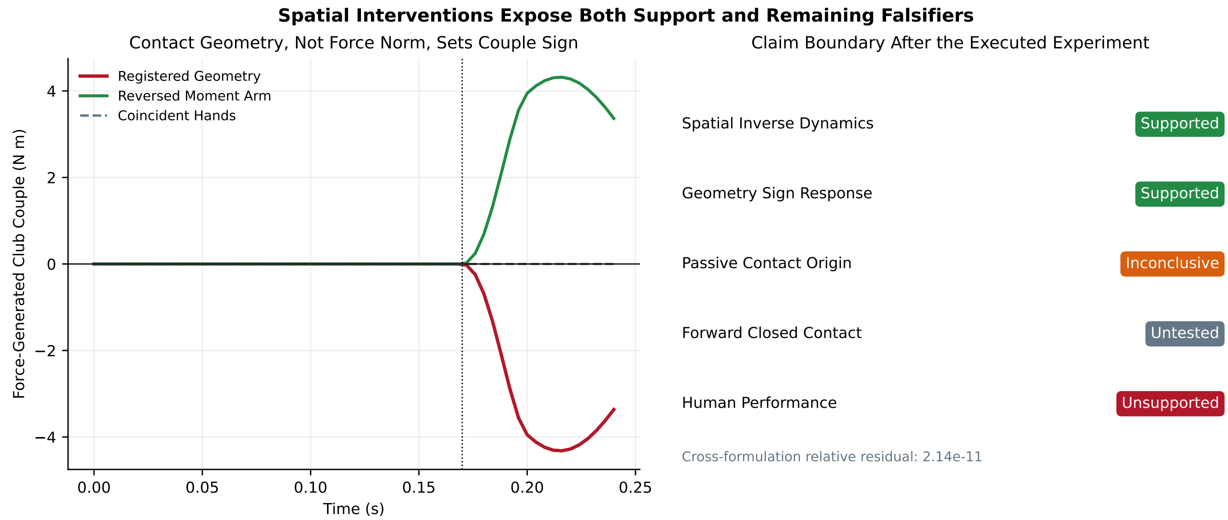

Figure 14.3: Spatial Interventions Expose Both Support and Remaining Falsifiers

The registered geometry reaches a minimum force-generated couple of $-4.32\ \mathrm{N} \cdot \mathrm{m}$ and first becomes negative at 0.172 s. Reversing only the signed moment arm yields an equal positive response; the maximum sign-reversal residual is $8.88 \times 10^{-16}\ \mathrm{N} \cdot \mathrm{m}$. Collapsing the contact separation makes the force-generated couple exactly zero while retaining the total force history. These results reject force magnitude alone as an explanation of the couple sign in this model.

The intervention is intentionally stronger than a visual pose comparison. It holds force vectors and achieved state fixed while changing only the geometric lever arm. Consequently, the sign response is attributable to Equation 14.5 within the declared model. It is not attributable to a change in force norm, event timing, solver state, or reference-frame rotation.

14.6 Falsifiability Gained and Claims Still Open

The executed experiment advances the evidence boundary in three ways.

First, the common spatial wrench is now coupled to a nonplanar reduced full-body inverse-dynamics calculation. The earlier three-dimensional result was only a frame-transformation audit. Here the mass matrix, bias forces, gravity, external generalized loads, and body Jacobians change with a genuine spatial state.

Second, the same model is evaluated through independent algorithms. Agreement therefore tests implementation transport rather than coordinate relabeling. The discovered external-load convention error demonstrates why that test has scientific value: two individually plausible calculations disagreed until the reported estimand was made identical.

Third, the geometry intervention remains exact in the spatial tier. It supports the statement that separated force application points can set club-couple sign independently of total force magnitude.

Four stronger statements remain open:

- **Passive Contact Origin Is Inconclusive.** The hand forces are inputs. A forward compliant two-hand solve must determine whether equivalent loads emerge after a distal command killswitch.
- **Forward Cross-Engine Contact Is Untested at This Tier.** Same-state inverse dynamics removes trajectory divergence by design. The independently integrated reduced contact experiment in Section 15 addresses that separate estimand without promoting this prescribed-load result.
- **Anatomical Magnitude Is Unsupported.** The reduced tree and spherical inertias are not subject-specific anatomy, muscle, or tissue mechanics.

- **Human Strategy Is Unsupported.** No measured trajectory or force record is held out for validation, and no coaching prescription follows from this mechanism experiment.

The next chapter tests the reduced contact mechanism under independent forward integration. Identifiability and uncertainty are then treated before control claims. Parameter compensation can otherwise make a structurally inadequate model reproduce one output while assigning the wrong contribution to contact, base motion, passive tissue, or actuation.

Chapter 15

Independently Executed Spatial Forward Contact

The common-state experiment in Section 14 asks whether two dynamics formulations return the same generalized action at identical achieved states. It deliberately prescribes the hand forces and removes trajectory divergence. The experiment in this chapter asks the next, stronger question:

> Can a negative force-generated couple arise from the achieved state of a forward compliant two-hand system, persist after its grounded driver is removed, and survive independent integration by two actual multibody engines?

The answer is affirmative for one declared reduced model. MuJoCo and Pinocchio independently integrate the same 13-coordinate contract using their native forward-dynamics algorithms (Todorov, Erez, and Tassa 2012; Carpentier et al. 2019). Neither engine receives a prescribed contact-force history. Each contact force is calculated from the engine's own achieved relative hand–club displacement and velocity. The common contact law, reference driver, timestep, and output convention are held fixed so that engine implementation is the principal changed factor.

This result closes the *reduced forward-contact* gate, not the anatomical or human gate. The hand bodies are finite-mass translational carriages, not arms; the grounded driver is not muscle; and the grip parameters are declared rather than identified from measurements. Those boundaries are properties of the experiment, not qualifications added after observing its outcome.

15.1 Model Estimand and Deliberate Reduction

The model contains two finite-mass hand carriages and one free rigid club. The hand positions are $\mathbf{h}_L, \mathbf{h}_T \in \mathbb{R}^3$. The club state is a world position $\mathbf{x}_C$, orientation $\mathbf{R}_C \in SO(3)$, linear velocity $\mathbf{v}_C$, and angular velocity ω_C. Its lead and trail grip offsets, expressed in the club frame, are

$$\rho_L = \begin{bmatrix}-0.080 & 0.015 & 0\end{bmatrix}^{\mathsf{T}} \text{ m}, \qquad \rho_T = \begin{bmatrix}0.080 & -0.015 & 0\end{bmatrix}^{\mathsf{T}} \text{ m}. \tag{15.1}$$

The achieved club-side contact positions and velocities are therefore

$$\mathbf{p}_i = \mathbf{x}_C + \mathbf{R}_C \rho_i, \qquad \dot{\mathbf{p}}_i = \mathbf{v}_C + \omega_C \times (\mathbf{R}_C \rho_i). \tag{15.2}$$

The reduction is purposeful. A full arm introduces joint topology, anthropometric estimation, muscle redundancy, and tissue compliance at the same time as contact and engine divergence. The carriage model isolates a narrower estimand: whether finite hand-side inertia and compliant two-point closure can generate the proposed spatial wrench without direct club actuation. A positive result makes an anatomical test worthwhile; it does not substitute for one.

The common record fixes each hand mass at 0.55 kg, club mass at 0.32 kg, principal club inertias at $[0.0012, 0.025, 0.025]$ kg m^2, gravity at -9.81 m/s^2 on the vertical axis, a 0.240 s interval, and a 0.25 ms timestep. The complete JSON-serializable record is hashed before either engine is constructed. Both realized adapters report the same SHA-256 model digest.

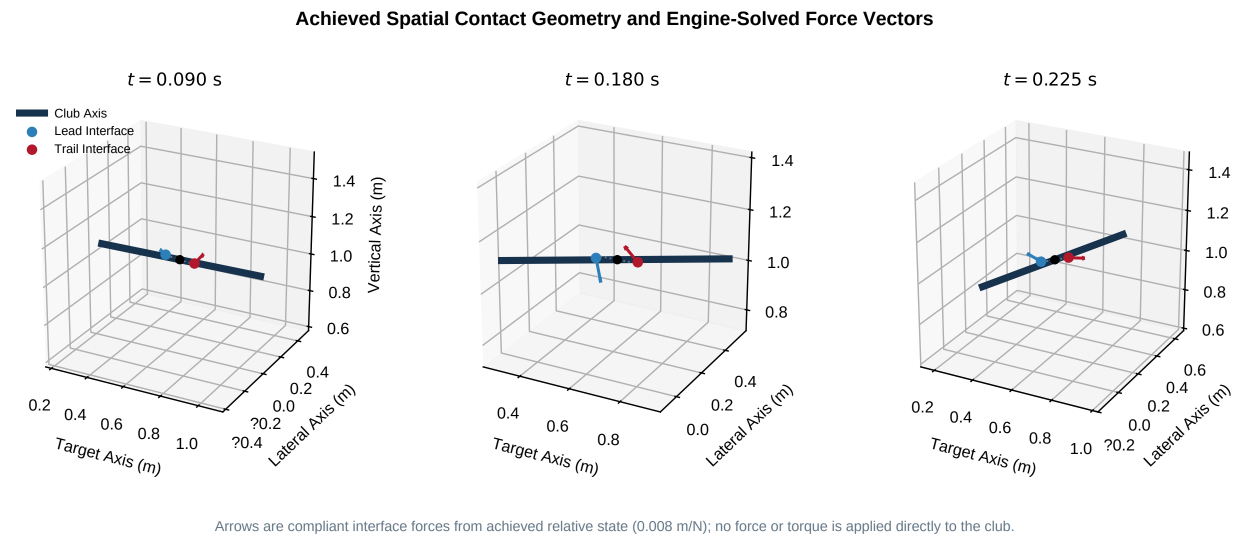

Figure 15.1: Achieved Spatial Contact Geometry and Engine-Solved Force Vectors

Figure 15.1 shows three achieved MuJoCo states. The blue and red arrows are not prescribed inputs. They are the compliant interface forces obtained from the achieved engine state at each sample. The two engines start from identical hand positions, club position, orientation, and zero velocity.

15.2 Paired Contact and Grounded Driver Contracts

Each interface uses a paired Kelvin–Voigt law,

$$\mathbf{F}_i = k_c(\mathbf{h}_i - \mathbf{p}_i) + c_c(\dot{\mathbf{h}}_i - \dot{\mathbf{p}}_i), \tag{15.3}$$

with $k_c = 1800$ N/m and $c_c = 18$ N s/m. The club receives $+\mathbf{F}_i$ at $\mathbf{p}_i$ and the hand receives $-\mathbf{F}_i$ at $\mathbf{h}_i$. The force-sum residual is therefore an exact runtime invariant rather than an outcome tolerance. The contact storage and dissipation terms are

$$U_{c,i} = \frac{1}{2}k_c\|\mathbf{h}_i - \mathbf{p}_i\|^2, \qquad P_{d,i} = -c_c\|\dot{\mathbf{h}}_i - \dot{\mathbf{p}}_i\|^2. \tag{15.4}$$

A world-referenced trajectory moves the finite hand carriages through a grounded spring–damper driver,

$$\mathbf{G}_i = k_d(\mathbf{h}_i^\star - \mathbf{h}_i) + c_d(\dot{\mathbf{h}}_i^\star - \dot{\mathbf{h}}_i), \tag{15.5}$$

where $k_d = 420$ N/m and $c_d = 24$ N s/m. The declared target contains a downswing-like rotation, translation, changing swing-plane tilt, and a bounded bilateral out-of-plane component. It is a manufactured excitation designed to exercise three-dimensional coupling. It is not a measured hand path.

No force or torque is applied directly to the club. The only club-side generalized load is the transported contact wrench

$$\mathcal{W}_C = \begin{bmatrix} \sum_i \mathbf{F}_i \\ \sum_i (\mathbf{p}_i - \mathbf{x}_C) \times \mathbf{F}_i \end{bmatrix}. \tag{15.6}$$

The equal and opposite driver wrench about the world origin is retained as a reduced ground-pathway reaction proxy. Because the carriages omit legs, pelvis, trunk, and anatomical arms, that proxy is not a prediction of measured ground-reaction force. It answers the bookkeeping question "where does the external reference-driver action close?" and prevents that pathway from being silently omitted.

15.3 Independent Engine Realizations

The common experiment executes in two native libraries, both version 3.8.0 in the archived run:

- **MuJoCo.** Two three-axis slide bodies and one free club body are compiled from generated MJCF. World-frame hand and point forces are mapped through `mj_applyFT`; `mj_forward` computes continuous-time acceleration.
- **Pinocchio.** Two `JointModelTranslation` branches and one `JointModelFreeFlyer` branch are assembled programmatically. Equal and opposite forces are transformed into each joint's local spatial-force convention; the articulated-body algorithm computes acceleration.

Both engines use the same semi-implicit position/velocity update after their native acceleration evaluation. This is a controlled numerical choice, not an assertion that the engines have identical internal integrators. It removes an uninteresting integration-method difference while retaining independent mass, bias, gravity, kinematics, spatial-force mapping, and forward-dynamics implementations. The Pinocchio adapter uses a local-world-aligned velocity at the moving club origin; using a world-origin spatial velocity instead creates an artificial contact-velocity discrepancy. That frame distinction is now covered by the forward trajectory gate.

The comparison evaluates complete trajectories, not just matched states. The declared engineering regions are 3 mm RMS and 9 mm maximum club-position difference, 0.035 rad maximum orientation difference, 10% relative RMS complete-wrench difference, and 8% normalized energy discrepancy. They are algorithmic acceptance regions, not statistical confidence intervals. Exact initial-state equality, action–reaction closure, analytical geometry controls, and timestep refinement provide independent checks that a passing trajectory is meaningful.

15.4 Forward Cross-Engine Results

Figure 15.2 shows that the two independently integrated trajectories remain close. In the continuing-driver branch:

- club-position RMS difference is 12.75 µm and maximum difference is 37.95 µm;
- maximum orientation difference is 8.94×10^{-8} rad;
- relative RMS complete-wrench difference is 0.256%; and
- normalized energy discrepancy is 0.0235%.

In the driver-killswitch branch, the corresponding values are 16.96 µm, 77.89 µm, 9.42×10^{-8} rad, 0.274%, and 0.0461%. Every observed value is well inside its declared region. The result is stronger than the preceding common-state inverse-dynamics comparison because numerical error can now accumulate and feed back into the contact law. It remains narrower than a cross-engine anatomical comparison because the engines realize one reduced common model.

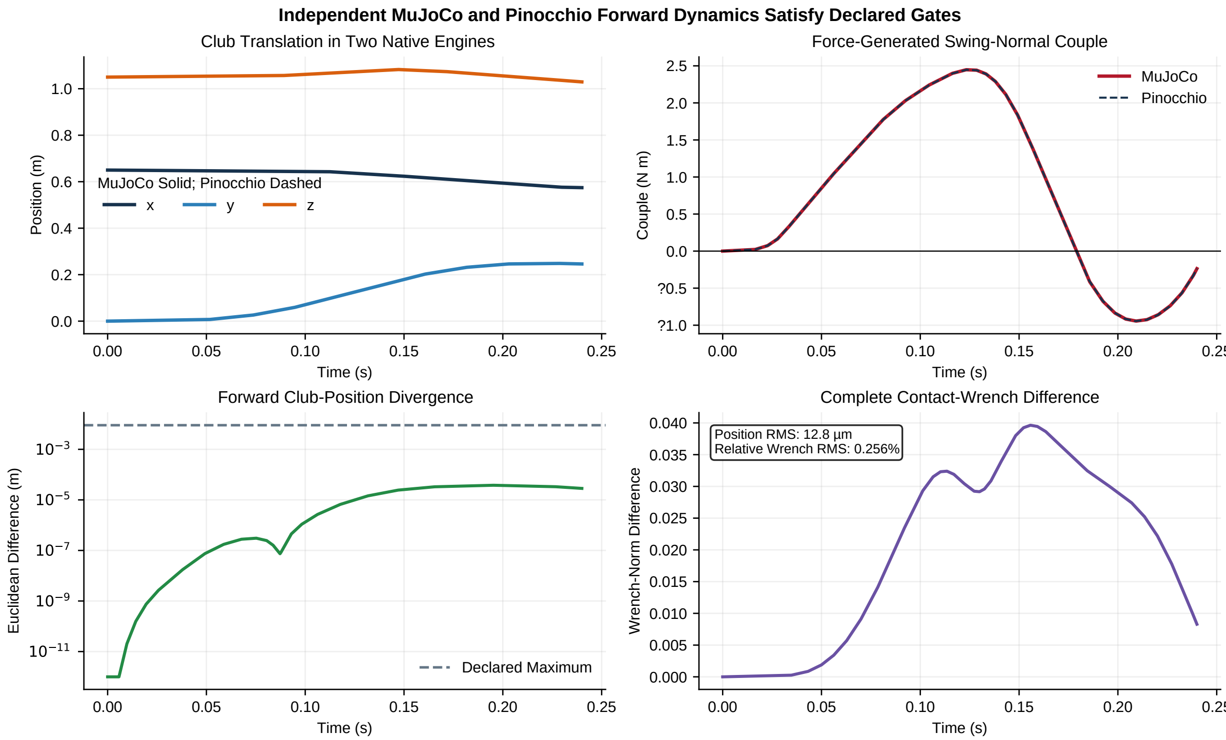

Figure 15.2: Independent MuJoCo and Pinocchio Forward Dynamics Satisfy Declared Gates

15.5 Exact Same-State Driver Killswitch

The baseline and intervention branches are bit-identical through $t_k = 0.180$ s. At that sample, the intervention sets $\mathbf{G}_L = \mathbf{G}_T = \mathbf{0}$ for the remainder of the rollout. Contact, gravity, hand inertia, club inertia, and interface storage/dissipation remain active. This intervention removes the complete grounded driver rather than merely setting a named distal torque to zero.

After the killswitch, both engines retain a negative force-generated swing-normal couple for 37.5 ms and reach approximately -0.409 N m. The branch also retains 7.55 degrees of club-axis out-of-plane evolution, a peak club-long-axis rotation rate of 13.59 rad/s, and a peak force-generated long-axis couple of 0.242 N m. These observables are separated deliberately: the negative quantity is a projection of contact moment on the instantaneous swing normal, while long-axis couple and rotation report torsional evolution. Calling either projection “the club torque” without naming its axis would be ambiguous.

The ground-pathway proxy reaches 56.85 N in resultant force and 62.61 N m in moment before the branch. It becomes exactly zero with the driver at the killswitch, while the two compliant interfaces continue to exchange force and moment internally. This separates prior external input from subsequent interaction dynamics. It does not make the pre-killswitch motion unforced or biologically passive.

15.6 Geometry Controls and Work–Energy Closure

Two same-force geometry controls are evaluated without advancing a new trajectory:

1. transport the achieved forces through the club reference, making both moment arms zero; and
2. negate both achieved moment arms while retaining force vectors and state.

The coincident-grip couple is exactly zero at every stored sample. Reversing the arms reverses the swing-normal couple with a maximum residual below 9.2×10^{-16} N m. These controls rule out total force magnitude,

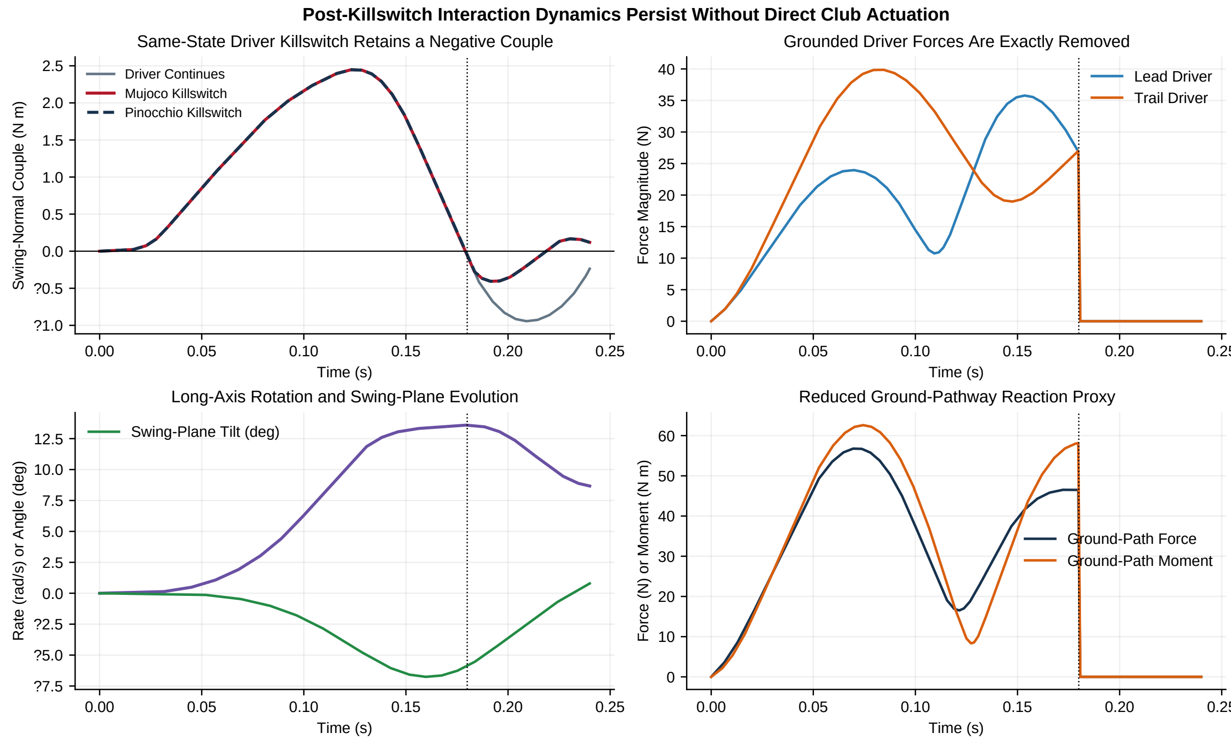

Figure 15.3: Post-Killswitch Interaction Dynamics Persist Without Direct Club Actuation

integration history, and engine state as explanations for the sign response. Within the declared wrench equation, moment-arm geometry is necessary.

The work–energy residual includes engine-native rigid-body kinetic and potential energy, interface storage, driver power, and contact dissipation. For the MuJoCo killswitch branch its maximum absolute value decreases from 0.0611 to 0.0305 to 0.0153 J as the timestep decreases from 0.50 to 0.25 to 0.125 ms. The monotone approximately first-order reduction is consistent with the shared semi-implicit step. At 0.25 ms, the residual is about 0.55% of the 5.52 J total-energy range. Pinocchio's corresponding maximum is 0.0349 J. The residual is reported rather than normalized away because it is a direct falsifier of the discretized experiment.

15.7 What the Experiment Falsifies

The committed record fails closed if any of these conditions occurs:

1. either imported library lacks its native model and forward-dynamics API;
2. the generated adapters report different common-model digests;
3. baseline or killswitch trajectories exceed a declared cross-engine region;
4. contact action–reaction force does not close at each evaluation;
5. the coincident-grip couple is nonzero;
6. reversed moment arms do not reverse couple sign;
7. baseline and killswitch states differ before the branch sample;
8. the post-killswitch negative interval is shorter than 30 ms; or
9. the work–energy residual fails to decrease under timestep refinement.

The experiment would therefore reject the stated reduced mechanism if a second native solver does not transport it or if it depends on hidden direct club actuation. Passing those tests does not establish that humans use the mechanism, that the declared stiffness represents tissue, or that a player should seek a particular force pattern.

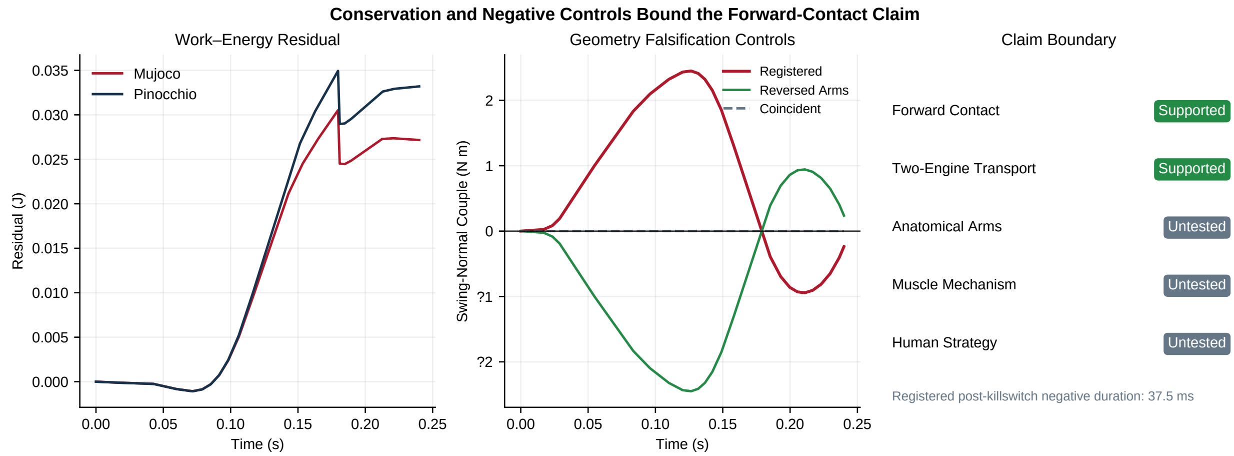

Figure 15.4: Conservation and Negative Controls Bound the Forward-Contact Claim

15.8 Reproduction and Platform Boundary

The evidence is stored in `data/spatial_forward_contact_study.json` and `data/spatial_forward_contact_study.npz`. Reproduction requires the repository's MuJoCo and Pinocchio optional engine dependencies. The native Pinocchio wheel used here executes in the Linux/WSL environment; a package named `pinocchio` on the default Windows Python index can resolve to an unrelated stub and is explicitly rejected by the adapter.

```
python3 -m scripts.research.proximal_distal_energy.run_spatial_forward_contact_study
python3 -m scripts.research.proximal_distal_energy.make_spatial_forward_contact_figures
python3 -m pytest tests/research/test_spatial_forward_contact.py -q
```

The next transport step is not "more confidence" in the carriage model. It is replacement of the carriages with subject-scaled articulated arms and a distributed shaft while preserving the same engine identity, wrench, killswitch, energy, event, and claim-boundary contracts.

Chapter 16

Coupled Uncertainty, Identifiability, and Control

16.1 Questions and Evidential Boundary

The preceding studies isolate feasible transfer mechanisms. They do not imply that a single parameter set is representative, that internal hand forces can be recovered from a net wrench, or that the fastest nominal command is robust. This chapter therefore asks three narrower questions in the coupled moving-base/flexible-club model:

1. Which uncertain inputs are associated with the largest changes in declared delivery observables?
2. Which internal quantities and parameters are identifiable from the chosen observations?
3. Do preselected delayed-actuation programs retain their advantages on an independently generated held-out ensemble?

The answers are model-screening results. Parameter ranges are engineering envelopes, not fitted population distributions; the actuator is a bounded command surrogate, not muscle physiology; and delivery is evaluated at a fixed registered time rather than at a simulated ball collision.

16.2 Registered Coupled Model and Design

Every rollout uses the ten-coordinate forward KKT model of Section 10. Base translation, both two-link arms, four grip constraints, two solved hand reactions, and one compliant shaft mode evolve in the same solve. This matters: uncertainty is propagated through the mechanism and its constraint reactions rather than through a disconnected response surface.

A deterministic Latin hypercube distributes samples across each declared range (McKay, Beckman, and Conover 1979). The global screen contains 24 points (seed 8426). Control programs are compared on a separate six-point training ensemble (seed 8449) and a six-point held-out ensemble (seed 8450). All runs use 0.24 s duration and a 4 ms step. The design varies 12 quantities simultaneously:

Parameter	Range	Interpretation
Anthropometric scale	0.95–1.05	Common arm-length scale
Limb-mass scale	0.90–1.10	Common arm-mass scale
Inertia-distribution scale	0.85–1.15	Segment inertia about its center
Base-mass scale	0.85–1.15	Translating-base inertia
Base-stiffness scale	0.75–1.25	Translational restoring stiffness

Parameter	Range	Interpretation
Joint-damping scale	0.60–1.40	Passive rotary damping
Grip-separation scale	0.85–1.15	Two-hand geometric lever arm
Shaft-stiffness scale	0.70–1.30	Lumped flex stiffness
Shaft-damping scale	0.60–1.40	Lumped flex damping
Activation delay	0.015–0.055 s	Pure command delay
Activation time constant	0.020–0.055 s	First-order command lag
Impedance scale	0.70–1.30	Linear joint-impedance proxy

These ranges were declared before selecting a preferred program. They bound a computational experiment; they do not encode prevalence or correlations in a golfer population.

16.3 Actuator and Outcome Contract

For commanded joint torque $u(t)$, the applied drive uses a delayed target and first-order state a,

$$\dot{a} = \frac{u(t - t_d) - a}{T_a},$$

followed by a torque-rate limit, asymmetric torque–velocity saturation, and a linear impedance term. The delivered torque is therefore history dependent and authority limited. No state is labeled as neural activation, muscle force, tendon force, reflex response, or co-contraction.

The registered outputs are delivery speed at 0.24 s; a planar face/path proxy; peak individual solved hand force; integrated squared delivered joint torque (the *effort proxy*); minimum force-generated grip couple; and peak shaft flex. The effort proxy has units associated with squared torque integrated over time; it is not metabolic or muscular cost. The face/path quantity is a planar orientation proxy, not a three-dimensional clubface measurement.

16.4 Global Sensitivity and Identifiability

Partial rank correlation coefficients (PRCCs) screen monotonic multivariable associations after ranking inputs and outputs and residualizing each variable against the remaining inputs (Marino et al. 2008). PRCC is useful for prioritization, but it is not causal attribution, a variance decomposition, or proof that the response is globally monotonic. With 24 samples for 12 inputs, the estimates are deliberately reported as a compact screening tier rather than precise population sensitivities.

Across the global ensemble, the 5th/median/95th percentiles are 4.005/5.188/ 5.840 m/s for delivery speed, 5.45/8.98/12.14 degrees for the face/path proxy, 140.7/162.3/200.0 N for peak individual hand force, and 66.0/82.0/94.4 for the effort proxy. Peak shaft flex spans 3.36/4.17/5.50 degrees. The signed minimum force couple spans -0.00478/-0.00035/approximately 0 N m in this particular global program; its small magnitude warns against transporting the much larger negative-couple values from different branch protocols without matching state and command.

The largest absolute PRCC is activation delay for delivery speed and effort, activation time constant for face/path error, limb mass for hand force, grip separation for minimum force couple, and shaft stiffness for peak flex. These rankings are hypotheses for larger designs, not settled parameter importance.

Identifiability is assessed separately from sensitivity (Raue et al. 2009). At a single planar instant, the two hand forces contain four scalar components, but their transported net force and scalar couple provide only three independent equations. The wrench map therefore has rank 3 and nullity 1: individual hand forces are structurally non-identifiable from the net planar wrench alone. Independent contact measurements or an additional constitutive assumption are required.

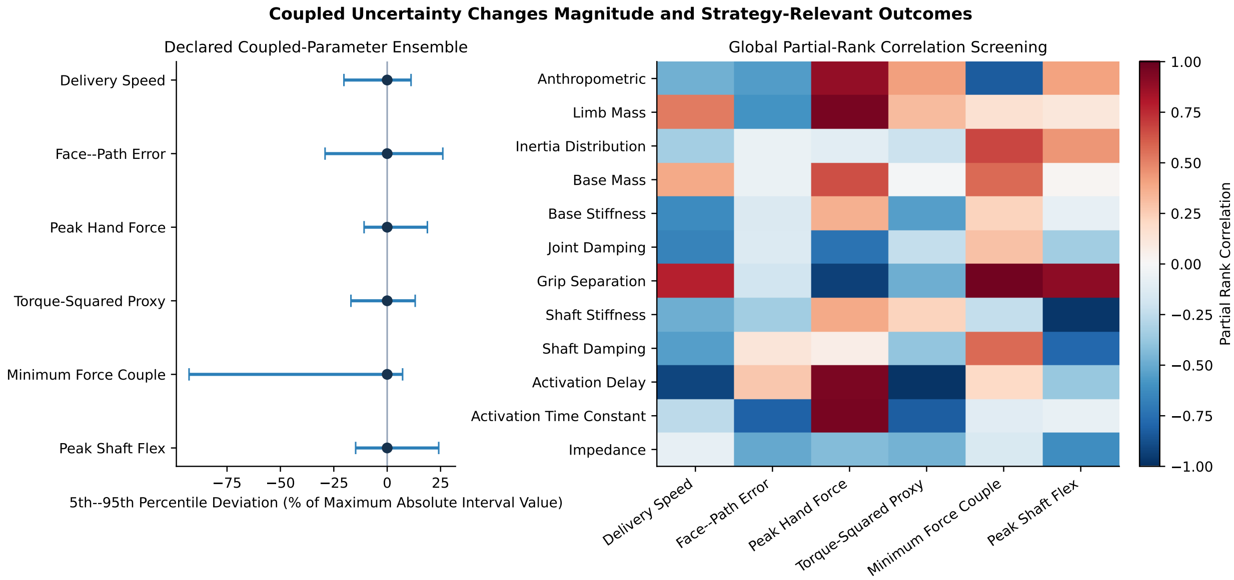

Figure 16.1: Uncertainty Intervals and PRCC Screening. Left: 5th–95th percentile ranges, normalized only for visual comparison. Right: partial rank correlations for the six registered outputs.

For the 12 uncertain parameters, a standardized sensitivity matrix is formed from the six summary observables. Its six singular values are 1.785, 1.146, 1.039, 0.863, 0.379, and 0.239. The effective rank at the registered 5% rule is six, so the full parameter vector has a nullity lower bound of six under this summary observation set. This is a local practical-identifiability screen, not a profile-likelihood analysis and not proof that every six-parameter subset is estimable.

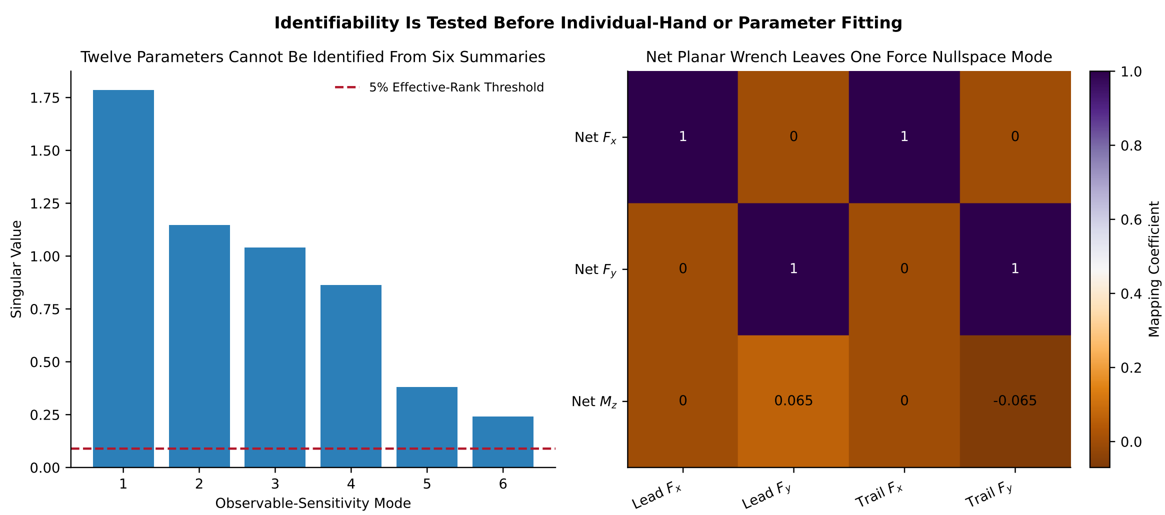

Figure 16.2: Identifiability Audit. The net planar wrench cannot uniquely allocate both hand forces, and six summary observables cannot identify the full 12-parameter vector.

16.5 Preselected Programs and Held-Out Evaluation

Eight programs were fixed before comparing results: passive wrist, early drive, late drive, restrain-then-drive, higher impedance, later release, lower drive, and early restrain. Five objectives are retained rather than collapsed prematurely: maximize the 10th-percentile delivery speed; minimize mean face/path error; minimize the 90th-percentile individual-hand force; minimize the effort proxy; and minimize delivery-speed dispersion.

Seven programs are nondominated on the training ensemble; all eight are nondominated on held-out samples. That instability is substantive evidence against a universal optimum: even the poor-speed early-drive program retains a held-out tradeoff in another objective. A balanced equal-range Chebyshev rule selects later release, a speed-priority rule selects early restrain, and a load-priority rule selects higher impedance. These are objective-specific selections, not interchangeable recommendations.

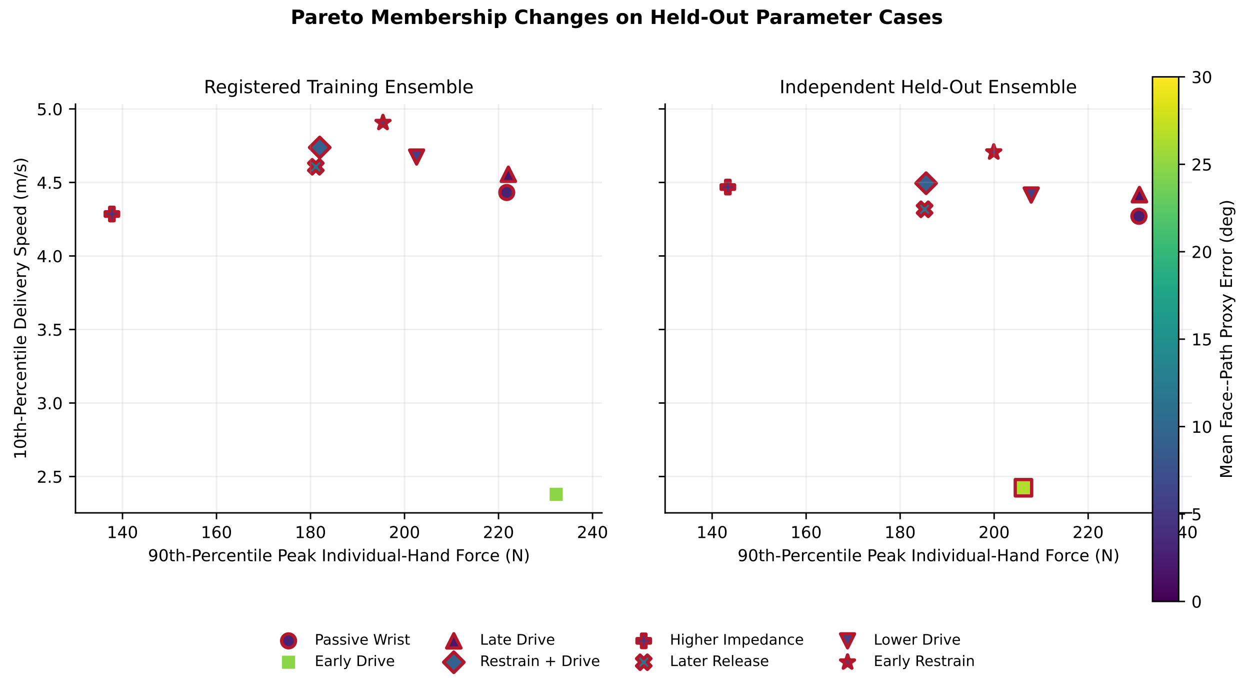

Figure 16.3: Training and Independently Held-Out Pareto Comparisons. Membership changes across ensembles, so no program is promoted as universally optimal.

16.6 Bounded Test of Delayed Opposing Command

Hypothesis H4 asks whether a delayed actuator can benefit from an anticipatory opposing command. In the held-out ensemble, early restrain raises the 10th-percentile delivery speed to 4.706 m/s compared with 4.417 m/s for late drive, but worsens mean face/path error from 1.58 to 5.44 degrees. Thus H4 is supported only as a conditional lower-tail speed result for these bounded model schedules. It does not demonstrate human preactivation, neural timing, or a generally superior strategy.

Numerical closure remains materially below the reported effects across all rollouts: the maximum position-constraint residual is 9.99×10^{-11} m, velocity-constraint residual 2.40×10^{-15} m/s, KKT residual 1.90×10^{-13}, and contact-power residual 6.40×10^{-14} W.

16.7 Claims, Falsifiers, and Next Measurements

This phase supports four bounded conclusions: coupled uncertainty can change delivery materially; the chosen summaries cannot identify all 12 parameters; net wrench cannot identify individual planar hand forces; and command selection depends on the objective and evaluation ensemble. It does **not** support a universal program, physiological effort inference, human preactivation, or a coaching prescription.

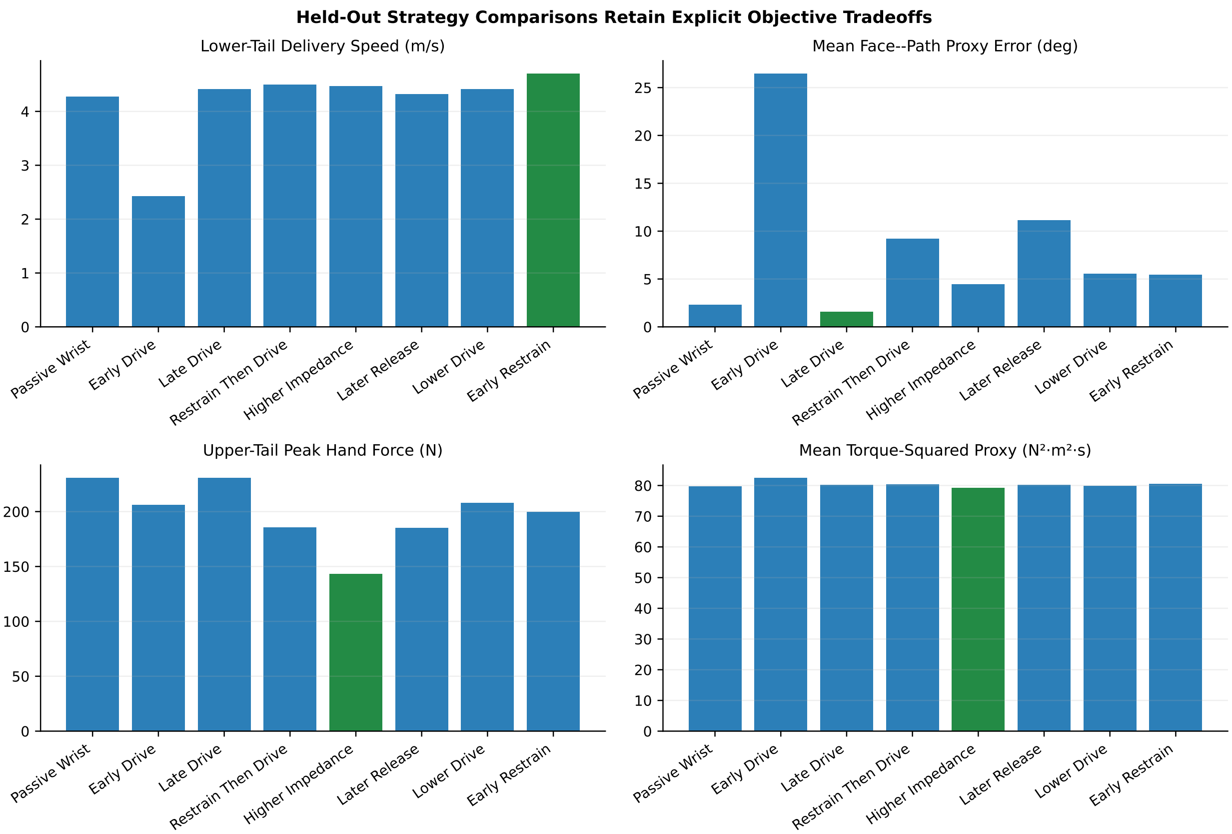

Figure 16.4: Held-Out Strategy Tradeoffs Across Speed, Planar Delivery Error, Hand Load, Effort Proxy, and Dispersion.

The results would be challenged by a larger registered ensemble that reverses the H4 lower-tail speed ordering, a direct two-hand measurement that uniquely resolves the wrench null mode without extra assumptions, or an independently implemented model in which the declared tradeoffs disappear under the same observables and actuator bounds. The next experimental protocol should measure both hand wrenches, club kinematics, ground reaction, and synchronization uncertainty; estimating all internal forces from club motion alone is not an acceptable substitute.

Chapter 17

Experimental Falsification Protocol

17.1 Why Readiness Is Not Evidence

Human measurements are the decisive next tier, but an empty data slot must not be filled with model output or an opportunistic dataset lacking the required observables. Protocol `proximal-distal-human-falsification-v1` is frozen before human outcomes are accessed. Its validator qualifies acquisition metadata, modality coverage, provenance, participant splits, and inference boundaries. The committed input is synthetic and the result is explicitly `synthetic_dry_run_only`; all human predictions remain untested.

17.2 Registered Measurements

H1–H3 require synchronized full-body kinematics, bilateral hand wrenches, club kinematics, and ground reaction. H4 additionally requires launch-monitor delivery and surface EMG or an equivalent governed activation measurement. The arm–wrist allocation hypotheses in Section 11 also require grip-pressure distribution, a closed-chain inverse-dynamics estimate conditioned on the measured internal grip wrench, and a preregistered estimate of force-development delay or tangent stiffness. Ultrasound-derived musculotendon length is preferred where feasible. EMG remains an activation timing measure and is not converted directly to joint torque. The protocol fixes SI units, minimum rates, and maximum synchronization uncertainty before outcome access. Hand wrench and ground reaction are sampled at at least 1000 Hz, club motion at at least 500 Hz, and the primary non-launch streams are synchronized within 1–2 ms.

Every stream carries a SHA-256 digest, coordinate frame, reference point, sampling rate, units, synchronization uncertainty, calibration record, and acquisition identity. Governed source data stay outside the public repository. Public evidence contains approved pseudonyms and derived results, never names, contacts, birth dates, medical identifiers, or identities inferred from filename, directory, session, row order, club, or device.

17.3 Fixed Outcomes and Falsifiers

Prediction	Primary Estimand	Falsifier or Inconclusive Condition
EXP-H1	Participant-level late distal drift work	Hierarchical interval enters the negligible region, or reconstruction/residual gates fail
EXP-H2	Event-aligned force–velocity projection and transported wrench power	Sign order is smaller than timing uncertainty or fails reference/frame controls

Prediction	Primary Estimand	Falsifier or Inconclusive Condition
EXP-H3	Negative equivalent club-couple interval after active/passive inventory	Sign is explained by measured active/passive terms or fails filtering/residual sensitivity
EXP-H4	EMG onset–reversal interval and held-out delivery association	Anticipatory interval or held-out association disappears under registered sensitivity
EXP-H5	Bilateral force-couple and direct-wrist allocation under a matched club task	Internal-force pattern does not vary with the assigned allocation or fails wrench closure
EXP-H6	Transition-aligned zero-force duration, tangent stiffness, and torque-error impulse	Inferred channel reversal has no measurable transmission gap, or preload does not improve same-program continuity

EXP-H3 can establish an observational interval, not muscular inactivity. EXP-H4 can establish timing and held-out association, not that a neural strategy caused the outcome.

17.4 Calibration, Filtering, and Residual Gates

The laboratory frame is right-handed and SI-valued. The club-frame transform, wrench reference, functional joint centers, segment inertias, and ground reference are retained. The primary low-pass cutoff is selected before held-out outcomes are opened; residual-analysis and plus/minus 20% cutoff variants are mandatory sensitivities. Event-specific filter tuning is prohibited.

Primary-window force and wrench values are never imputed. Kinematic gaps are limited to 10 ms and reported. Force residual greater than 10% body weight or moment residual greater than 5% body-weight-times-height fails the primary inverse-dynamics gate. Event uncertainty combines device synchronization, resampling, and event localization. A predicted order within that interval is inconclusive.

17.5 Participant-Level Holdout and Hierarchy

The split unit is the participant, never the swing. At least 25% of eligible participants are assigned to the held-out set by governed random allocation before fitting or threshold selection. Repeated swings are nested within participants; participant contrasts, not swing rows, are the primary units. Hierarchical partial pooling reports both within- and between-participant uncertainty. Skill is analyzed only if it is a governed declared variable and is never inferred from performance, session, filename, row order, or club.

17.6 Negative Controls

The experiment retains four controls. Rotating complete wrench/twist pairs or transporting them together to a second reference must preserve power. Shuffled timing within a prespecified noncausal window tests spurious event alignment. Replacing bilateral loads with net wrench must recover the known individual-hand null mode. Model-only sign-reversing geometry interventions remain labeled as model counterfactuals rather than human interventions.

Each outcome is classified as supported, contradicted, inconclusive, or untested. Missing modality, residual failure, ambiguous timing, or unapproved provenance cannot be silently excluded after outcomes are known.

17.7 Executed Dry Run and Open Gate

The synthetic record contains two pseudonymous fixture participants separated at participant level and all six modality descriptors. The validator confirms units, hashes, sampling, synchronization, split uniqueness, and absence of identity fields. Deliberate identity, unit, hash, synchronization, split, and governance violations fail closed in tests.

No governed human dataset was available in this phase. Accordingly, EXP-H1 through EXP-H4 remain `untested_no_governed_human_data`. Evaluation begins only after an ethics/oversight reference, consent and reuse basis, private data authority, synchronized streams, calibration records, frozen participant split, and analysis-release authorization pass the same validator. The protocol removes analytical discretion; it does not remove the need for measurements.

Chapter 18

Open-Resource Release and Reviewer Qualification

18.1 Release Contract

The scientific product is distributed as an open research resource rather than as evidence for a software platform. Ten presets identify reproducible entry points: analytical double pendulum, forward two-hand, coupled moving-base/flexible-club, isolated synthetic distributed shaft, forward distributed modal shaft, matched-task arm–wrist allocation and transmission, reduced spatial common state, reduced two-engine spatial forward contact, uncertainty/control, and experimental readiness. Preset names preserve the model ladder and never imply that a higher unexecuted tier is available.

The release manifest hashes the report source and PDF, chapter source, linked bibliography, evidence JSON/NPZ/CSV, publication PDF/SVG figures, analysis scripts, evidence schema, data dictionary, citation metadata, and reviewer guide. A separate `CHECKSUMS.sha256` is sorted by path for standard tooling. Validation is read-only and fails on missing or changed bytes, unsafe paths, and malformed records. Updating expected hashes requires a separate explicit write command after analysis, render, and visual gates pass.

18.2 Reviewer Path

The reviewer workbench begins with the claim–evidence–alternative–falsifier matrix, then groups static SVG/PDF figures and downloadable data by model tier. It distinguishes analytical attribution, branched forward counterfactuals, common-state inverse dynamics, uncertainty/control screening, and experimental readiness. This is a deliberately static, link-stable review surface; it does not recompute results in a browser or introduce a second scientific code path.

The data dictionary defines common units, state and wrench fields, power/work terms, residuals, provenance, ensemble partitions, and interpretation boundaries. Citation metadata points to the article and requests the exact release commit. Code and repository-authored documentation use the repository's MIT license; linked papers, external archives, and the registered WSCG source retain their own boundaries.

18.3 Qualification Commands

From the repository root:

```
python -m scripts.research.proximal_distal_energy.qualify_open_release list-presets
python -m scripts.research.proximal_distal_energy.qualify_open_release validate
```

The first command exposes canonical deterministic entry points. The second validates the pinned bundle but never changes it. Maintainers regenerate the manifest only through the explicit `write` action after all source data, figures, article source, rendered PDF, and documentation are final.

18.4 Qualified and Open Claims

The release records planar interaction dynamics and negative-couple feasibility as supported at their declared tiers, geometry response as supported through reduced spatial common-state inverse dynamics, and modal reduction behavior as supported on a synthetic structural shaft case. It records forward distributed-shaft coupling as supported only for the declared synthetic planar tier. It records post-killswitch contact persistence as supported only in the reduced spatial carriage model, human predictions as untested, and a universal controller as unsupported.

Five release-level completion gates remain visible: subject-scaled articulated spatial contact with calibrated grip and distributed shaft; equipment-calibrated beam and measured grip properties; measured tissue-level preload and slack identification; governed held-out human evaluation; and external archive deposit with a persistent identifier. Archive deposition changes external state and has not been performed. A reproducible local bundle is not described as a permanently archived release until that deposit is verified.

Chapter 19

A Common-Observable Ladder Toward Higher-Order Models

19.1 Why a Model Ladder Needs Fixed Observables

Adding degrees of freedom does not automatically strengthen a mechanism claim. It can instead change the question by changing the reference point, frame, contact definition, or reported power. A useful fidelity ladder must preserve the observable while relaxing one structural assumption at a time. This chapter therefore carries one common interaction record through a three-coordinate chain, a prescribed mobile-hub inverse-dynamics comparison, forward rigid- and flexible-club two-hand systems, and arbitrary three-dimensional frame rotations. The reduced full-body common-state tier in Section 14 then adds nonplanar body and club inverse dynamics in two independent formulations.

The common record contains:

- model tier and time;
- Cartesian frame and reference point;
- force and equivalent couple acting on the declared distal subsystem; and
- linear and angular velocity of that same point in that same frame.

Force power, couple power, and total wrench power are derived rather than stored as unrelated fields. This makes dimensional and sign errors visible and permits exact frame and reference-transport tests. The implementation is available in `mechanism_ladder.py`, and the recorded analysis is generated by `run_mechanism_ladder_study.py`.

Figure 19.1 separates the lower-order mechanism tests from the spatial dynamics boundary. The rotated 3-D tier is a coordinate-frame audit, not an out-of-plane swing simulation; the later common-state tier supplies that nonplanar step. The historical mobile-hub tier prescribes hub motion and calculates the required reaction change. By contrast, Section 10 now evolves base translation, two-hand reactions, and club flex together. These distinctions are part of the result.

19.2 The Common Wrench–Power Contract

At reference point A, the interaction record is the wrench $(\mathbf{F}, \mathbf{M}_A)$ and twist $(\mathbf{v}_A, \omega)$. Its power is

$$P_A = \mathbf{F} \cdot \mathbf{v}_A + \mathbf{M}_A \cdot \omega. \tag{19.1}$$

If the reference moves to point B, with $\mathbf{r}_{BA} = \mathbf{r}_B - \mathbf{r}_A$, then

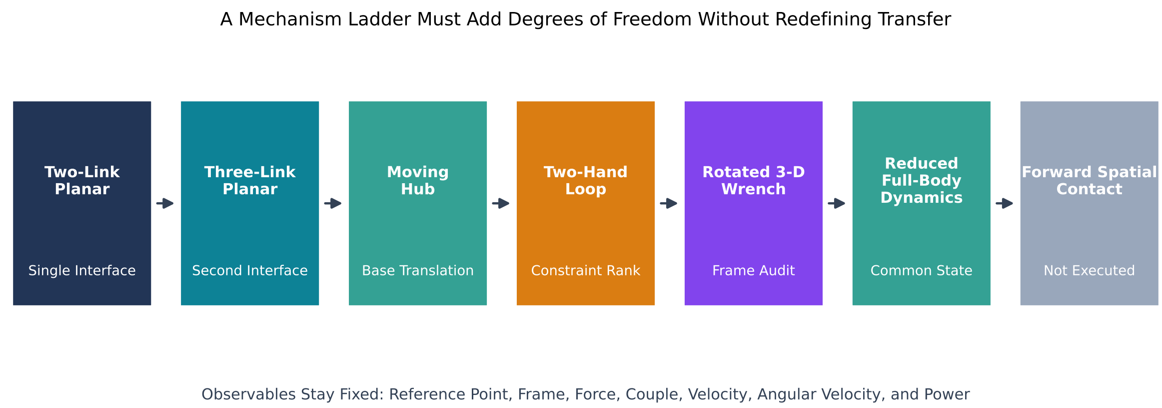

Figure 19.1: A Mechanism Ladder Must Add Degrees of Freedom Without Redefining Transfer

$$\mathbf{M}_B = \mathbf{M}_A - \mathbf{r}_{BA} \times \mathbf{F}, \tag{19.2}$$

and rigid-body point kinematics require

$$\mathbf{v}_B = \mathbf{v}_A + \omega \times \mathbf{r}_{BA}. \tag{19.3}$$

Substitution into Equation 19.1 gives $P_B = P_A$. Transporting the moment without transporting the velocity, or vice versa, creates an artificial power difference. This is the power counterpart of the wrench-transport rule introduced in Section 7.4.1.

Under a proper frame rotation $\mathbf{R} \in SO(3)$, all Cartesian vectors transform by the same $\mathbf{R}$. Orthogonality then preserves vector norms and dot products. The audit applies 51 different proper rotations and a fixed reference translation to sampled three-link wrenches. Maximum force-norm, couple-norm, rotation-power, and transport-power residuals are respectively 7.1×10^{-15} N, 8.9×10^{-16} N $\cdot$ m, 2.4×10^{-13} W, and 5.7×10^{-14} W.

These residuals establish implementation consistency. They do not show that a planar model contains the physics of a three-dimensional swing. Their purpose is to ensure that any later discrepancy is physical or numerical rather than a silent frame-definition change.

19.3 Three Coordinates and Two Internal Interfaces

The executed three-link tier reuses the reference trace from Section 9. Its bodies are the arm point mass, proximal shaft point mass, and distal head point mass. Thus it adds a second internal interface and a third coordinate, but it does not yet add an independently actuated torso. The distal interface record is evaluated at the second joint:

$$P_{23} = \mathbf{F}_{23} \cdot \mathbf{v}_{J_2} + M_{23}\omega_3. \tag{19.4}$$

At the interpolated delivery crossing, $t = 0.427252$ s. The nearest stored 0.5 ms sample has 56.37 N interface force, $+4.91$ N $\cdot$ m interface couple, 32.09 W force power, 74.60 W couple power, and 106.69 W total wrench power. The precise values are model outputs, not expected human ranges.

Figure 19.3 shows that the qualitative lesson from the double pendulum survives: neither force magnitude nor moment sign uniquely determines transfer. Total interface power changes sign while the force magnitude

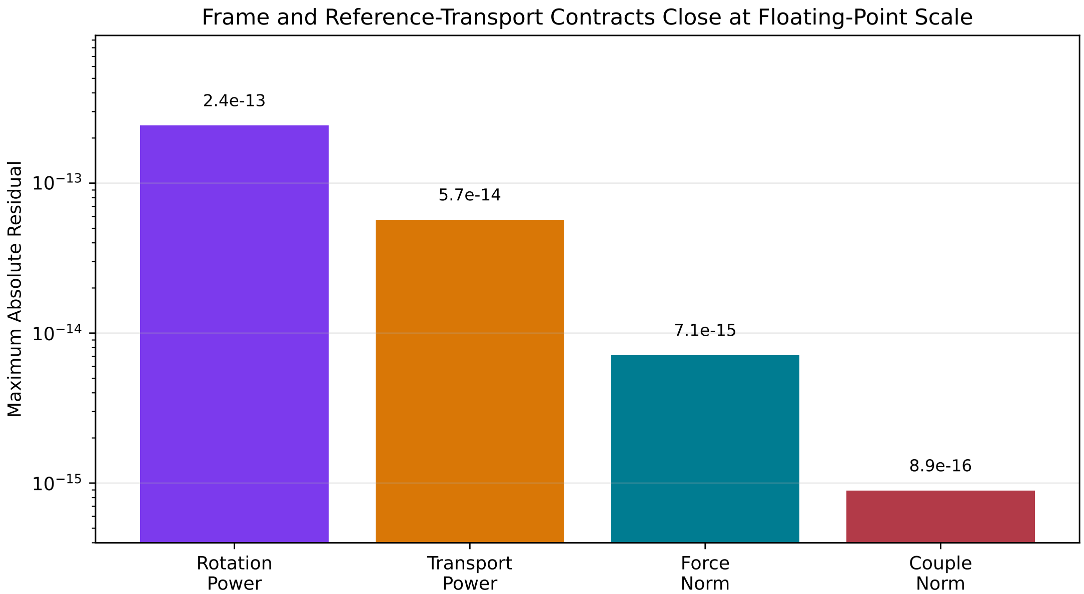

Figure 19.2: Frame and Reference-Transport Contracts Close at Floating-Point Scale

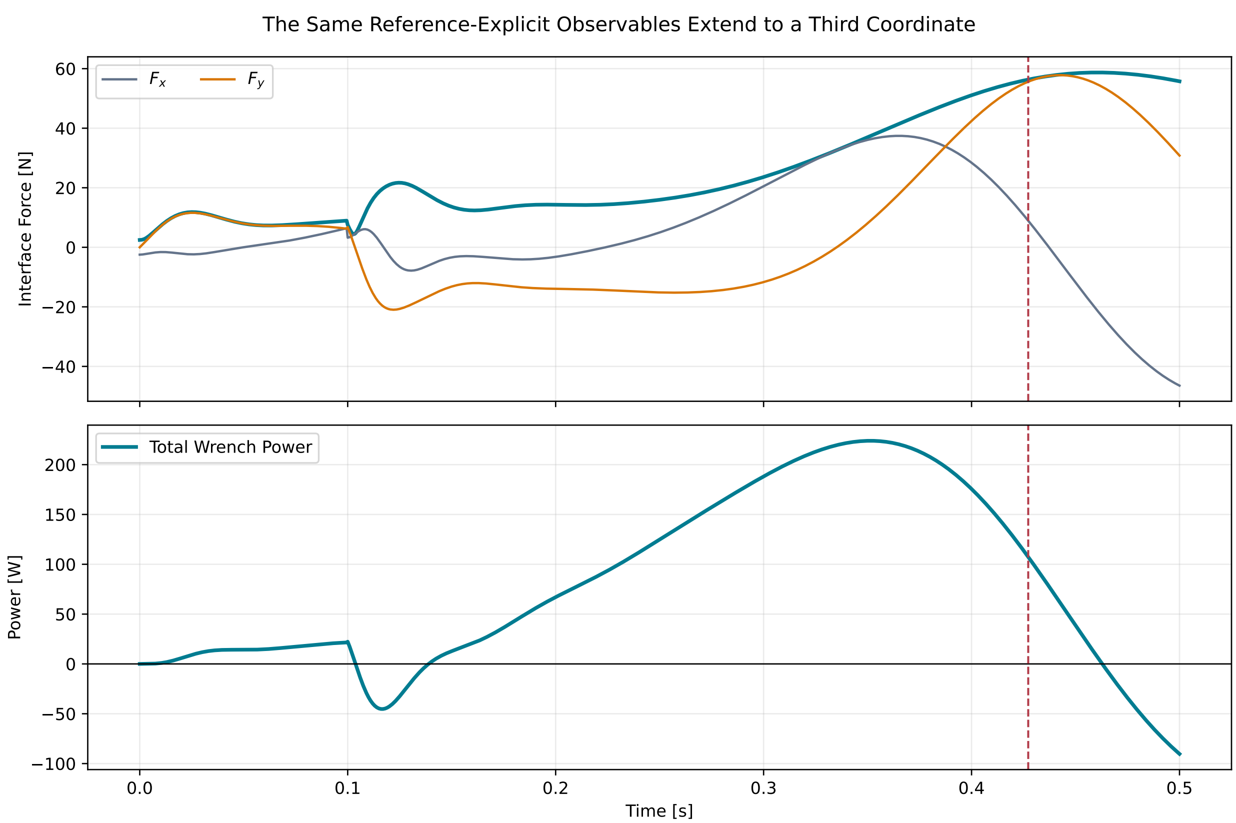

Figure 19.3: The Same Reference-Explicit Observables Extend to a Third Coordinate

remains positive. Conversely, force and couple power can reinforce one another or oppose one another depending on velocity alignment.

The result does not prove invariance to every added link. A torso–arm–club model with an actuated first interface can alter both state history and distal reaction. What survives here is the accounting structure: force, couple, velocity, and angular velocity remain sufficient to state interface power without naming a particular generalized coordinate.

19.4 Prescribed Mobile-Hub Inverse Dynamics

A moving hub changes two parts of the interaction calculation. Its acceleration changes the force required to accelerate the supported mass, and its velocity changes the point at which force power is evaluated. For a prescribed hub acceleration $\mathbf{a}_H$ and distal supported mass m_d, the force increment in the executed comparison is

$$\Delta\mathbf{F}_H = m_d\mathbf{a}_H. \tag{19.5}$$

The corresponding interface velocity is $\mathbf{v}_{J,H} = \mathbf{v}_{J,0} + \mathbf{v}_H$. The zero-amplitude case reduces exactly to the fixed-hub trace: maximum force and power differences are both zero to stored precision.

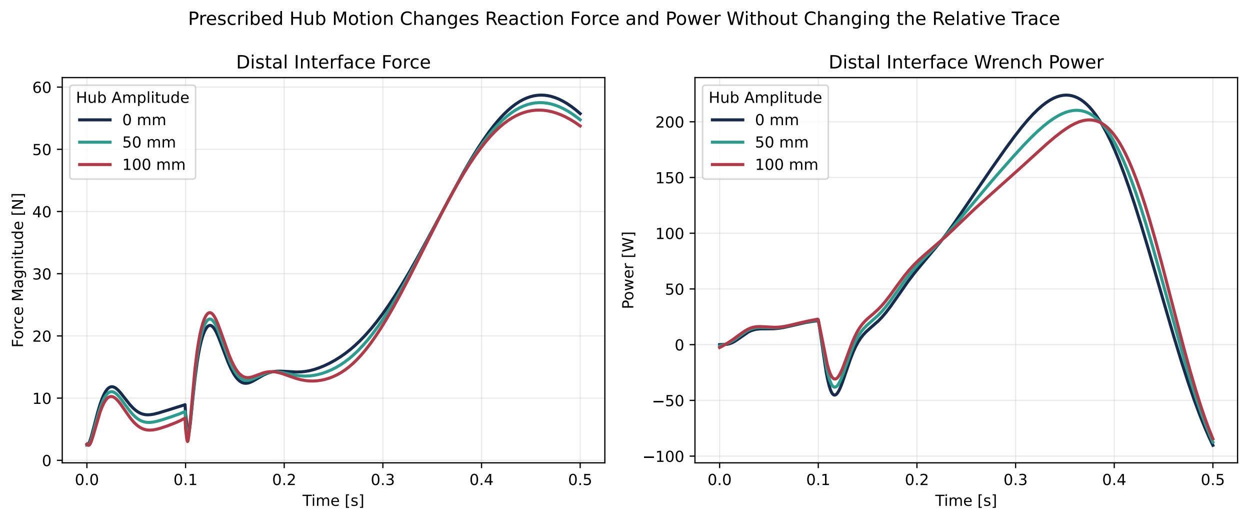

Figure 19.4: Prescribed Hub Motion Changes Reaction Force and Power Without Changing the Relative Trace

The declared hub path is a two-frequency planar translation at 1.25 Hz, tested at amplitudes 0, 25, 50, and 100 mm. At 100 mm, the maximum distal-force shift is 2.67 N and maximum instantaneous power change is 38.09 W. The signed power difference integrated over the half-second trace is -0.174 J. These values are specific to the prescribed path and the 0.20 kg supported distal mass.

The comparison is useful precisely because it is not a new forward swing. The relative trajectory is held fixed, so Equation 19.5 exposes how base motion changes the inverse-dynamics reaction. A forward torso model would also change the relative coordinates and controls. Prior hub-path studies show that such motion can materially affect model performance and inferred work (Nesbit and McGinnis 2009; Nesbit and McGinnis 2014); the present calculation supplies a transparent bridge rather than substituting for that higher-order analysis.

19.5 Forward Coupled Moving-Base Tier

The prescribed-hub audit isolates one inverse-dynamics term, while the coupled experiment in Section 10 removes its central restriction. Its finite-mass base responds to arm and club reactions; both arms remain closed to the floating grip; and shaft flex responds to the same trajectory. The model emits the same reference-explicit force, couple, point velocity, angular velocity, and power record used by the lower tiers.

This tier preserves three registered qualitative responses. A late force-generated negative couple occurs with finite hand separation, survives a 50 ms same-state zero-command branch, and collapses exactly when both contact moment arms are set to zero. Simultaneously, the contact power identity and whole-system work–energy balance close under timestep refinement. Thus the earlier prescribed-hub result remains a useful analytical audit, but it is no longer the highest executed moving-base result.

The forward tier is still planar and its base has translation but no rotation. Its one torsional flex mode is not a distributed beam. Transport to spatial dynamics is therefore evaluated separately in Section 14 rather than inferred from the planar result.

19.6 Closed-Loop Two-Hand Constraint Geometry

The two-hand wrench audit in Section 7 starts from recorded contact forces. A closed-loop model adds a prior question: which generalized velocities are compatible with maintaining both contacts? The executed planar geometry uses five coordinates,

$$\mathbf{q}_c = (\theta_L, \theta_T, x_G, y_G, \psi_G)^\mathsf{T}, \tag{19.6}$$

for lead- and trail-arm angles and grip pose. Two Cartesian constraints per hand give

$$\dot{\mathbf{c}} = \mathbf{J}_c(\mathbf{q}_c)\dot{\mathbf{q}}_c = \mathbf{0}, \tag{19.7}$$

with a 4×5 constraint Jacobian. At every one of 201 sampled geometries, $\mathbf{J}_c$ has rank four and a one-dimensional nullspace. The maximum residual obtained by multiplying the Jacobian by its computed nullspace vector is 6.8×10^{-16}.

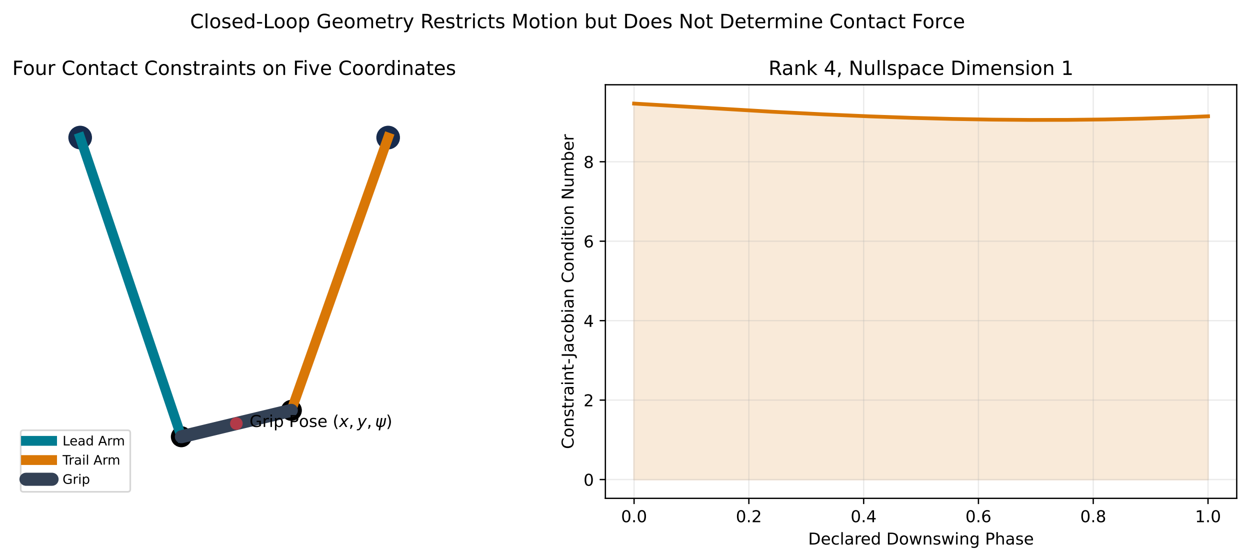

Figure 19.5: Closed-Loop Geometry Restricts Motion but Does Not Determine Contact Force

The condition number remains between 9.05 and 9.46 in the declared sweep. That shows stable local velocity compatibility for this geometry. It does not determine the Lagrange multipliers or hand forces. Those require mass, acceleration, actuation, and constraint-stabilization equations. In particular, constraint rank cannot establish the sign of the equivalent couple.

This distinction connects the two evidence types cleanly:

- the constraint Jacobian defines admissible closed-loop motion;
- the dynamic solve determines contact reactions consistent with that motion;
- the equivalent wrench reduces those reactions about a declared point; and
- wrench power determines whether that action supplies or removes club energy.

The archived WSCG force tables remain the dynamic evidence for the negative two-hand couple. The present Jacobian audit proves the geometry contract needed for a later forward closed-loop reproduction without pretending to have run that reproduction.

19.7 Three-Dimensional Frame Audit

Planar vectors can be embedded in three dimensions and expressed in arbitrary proper frames. This is a necessary test because golf-model engines differ in axis conventions, quaternion ordering, angular-velocity frames, and wrench reference points. A correct adapter must preserve physical invariants even when every component changes (Featherstone 2008).

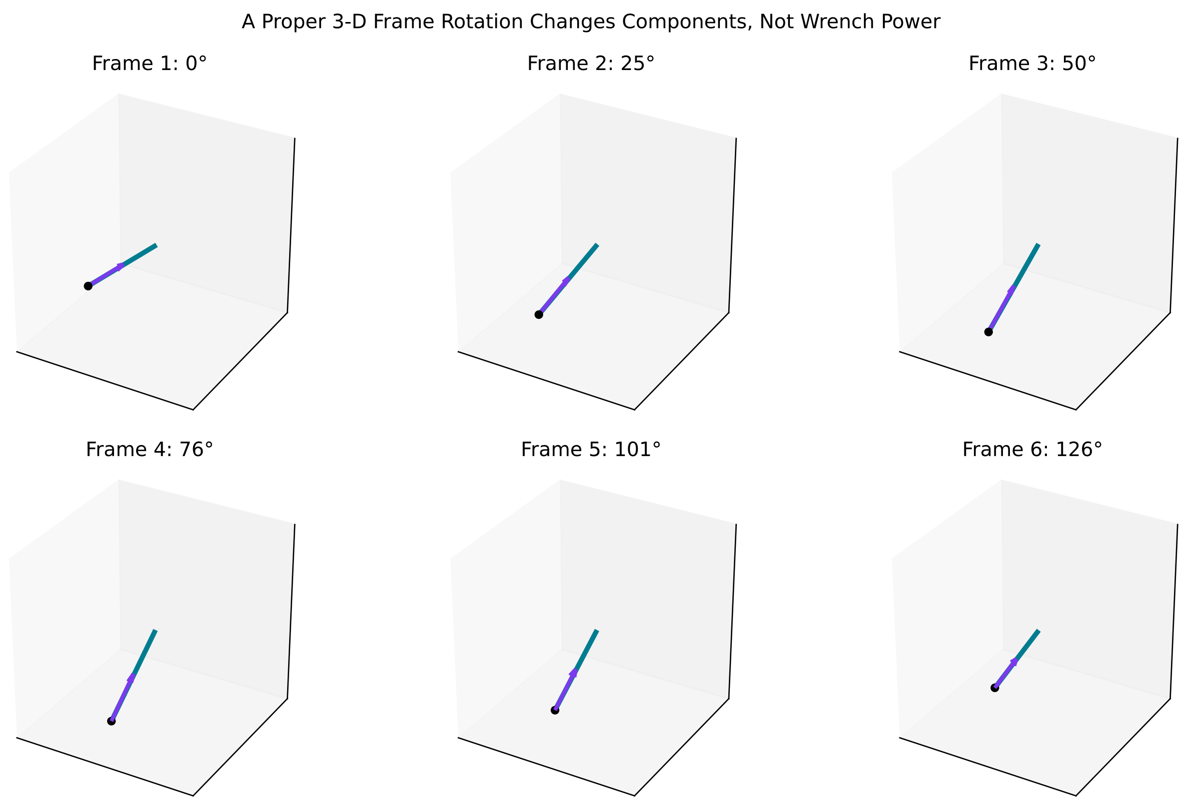

Figure 19.6: A Proper 3-D Frame Rotation Changes Components, Not Wrench Power

Figure 19.6 shows the delivery wrench in six rotated frames. The arrow components and apparent pose change, but force norm, couple norm, and total power do not. The transformation is performed on the complete wrench and twist, not on force alone.

This executed audit is three-dimensional algebra applied to a planar trajectory. It tests frame handling and source-to-engine adapters. It does not include out-of-plane angular velocity, long-axis club rotation, aerodynamic forces, three-dimensional joint motion, or full-body contact dynamics. Calling it a "3-D swing validation" would be incorrect.

19.8 Reduced Full-Body Common-State Inverse Dynamics

The spatial experiment in Section 14 uses 20 generalized coordinates, a separately moving six-coordinate club, bilateral arms, torso, pelvis, and lower-body mass. It evaluates 61 identical achieved states through MuJoCo and an independent Lagrange–Christoffel formulation. The common model hash, gravity, external loads, state, and required-action convention are identical.

The maximum relative generalized-action discrepancy is 2.14×10^{-11}. Reversing only the signed hand moment arm reverses the -4.32 N $\cdot$ m force-generated couple, and coincident application points remove that couple exactly. Unlike the rotated-wrench audit, this tier contains out-of-plane motion and state-dependent spatial mass and bias terms.

The result remains common-state inverse dynamics with prescribed hand loads. It does not solve closed contact, execute a zero-command forward branch, or identify passive biological load generation. The reduced forward-contact tier in Section 15 addresses the first two limitations without promoting either model to anatomical or biological evidence.

19.9 Reduced Spatial Forward Contact

The next tier replaces prescribed hand loads with two compliant point interfaces between finite-mass hand carriages and a free rigid club. MuJoCo and Pinocchio independently evaluate achieved kinematics, spatial-force mapping, mass, bias, gravity, and native forward dynamics. The two adapters share a hashed parameter record, contact law, semi-implicit update, and output schema.

In the exact same-state driver-killswitch branch, the complete grounded driver is removed at 0.180 s. The force-generated swing-normal couple remains negative for 37.5 ms in both engines and reaches approximately -0.409 N m. Coincident grips remove the couple exactly; reversed moment arms reverse its sign below 9.2×10^{-16} N m residual. Cross-engine trajectory, wrench, orientation, and energy gates pass, and the work–energy residual decreases monotonically as timestep is halved.

This executes independent forward spatial contact at a reduced mechanism tier. It does not add anatomical arms, subject-specific inertias, muscles, measured grip compliance, a distributed shaft, or human observations. The evidence boundary therefore moves from "forward contact unexecuted" to "reduced forward contact supported; articulated human transport untested."

19.10 What Survives, What Changes, and What Remains Open

The discrepancy matrix in Figure 19.7 summarizes the evidence boundary. Nine statements survive the executed tiers:

1. interaction force may remain nonzero under zero instantaneous distal command;
2. force magnitude does not determine power;
3. reference-point transport requires paired moment and velocity transport;
4. moving-base acceleration changes required reaction through supported mass; and
5. closed-loop constraint rank determines admissible velocity dimension, not contact-force magnitude or couple sign;
6. nonplanar generalized action is reproducible across the two declared inverse-dynamics formulations; and
7. reversing spatial contact moment-arm geometry reverses the force-generated club couple while coincident points remove it;

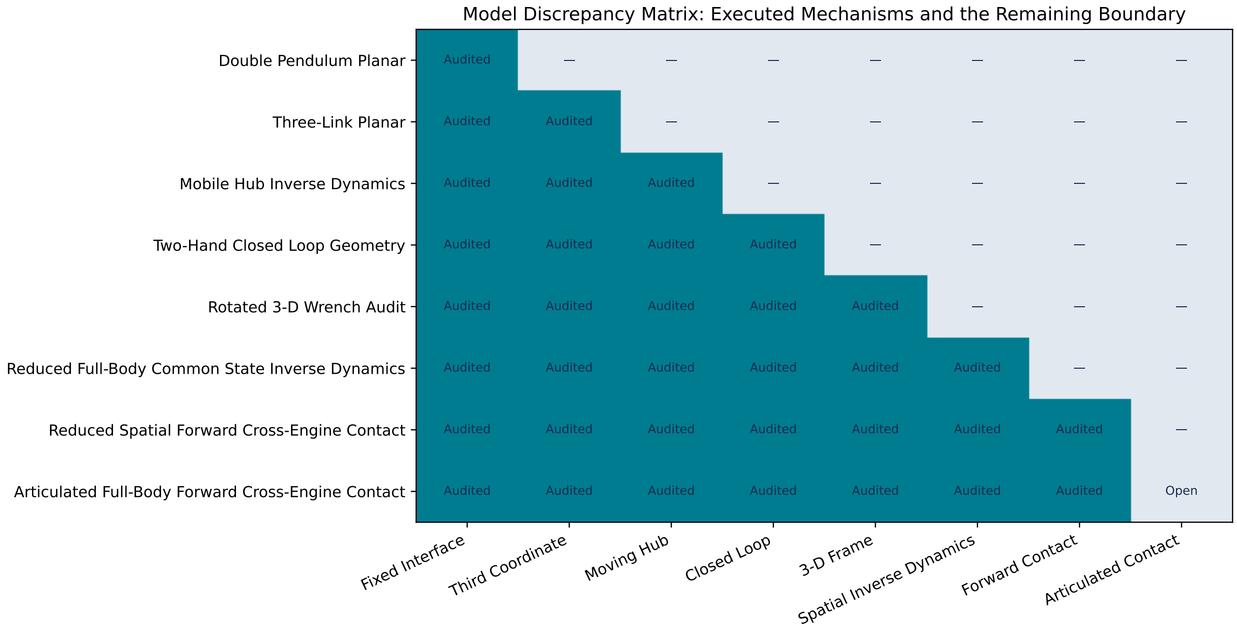

Figure 19.7: Model Discrepancy Matrix: Executed Mechanisms and the Remaining Boundary

8. the negative force-generated couple persists after complete grounded-driver removal in the declared reduced spatial compliant-contact model; and
9. that forward-contact result transports through native MuJoCo and Pinocchio dynamics within declared trajectory, wrench, orientation, and energy regions.

Other claims do not yet survive because they have not been tested. The present analysis does not show that the reduced result survives anatomical arms, that its magnitude is invariant to a distributed shaft, or that independently authored native humanoid models produce matched force histories. Those are separate structural and empirical questions.

19.10.1 Tier-by-Tier Interpretation

Planar Double Pendulum. This tier supplies the smallest model in which a proximal constraint force can do work on a distal body. Its strength is exact interpretability: radial and tangential projections, matched-state counterfactuals, and interface work can be checked directly. Its weakness is structural. A fixed hub and single wrist collapse torso motion, two-hand closure, and shaft deformation into omitted or prescribed effects. The tier can establish a mechanism but cannot determine its magnitude in a golfer.

Three-Coordinate Planar Chain. Adding the third coordinate introduces a second interface and an internal storage mode. The common schema remains valid, and total interface power remains the sum of force and couple power. The tier demonstrates that a locally large elastic acceleration contribution need not be a large net energy source. Because the added coordinate represents lumped shaft flex rather than an independently actuated torso, conclusions about proximal segment strategy remain provisional.

Prescribed Mobile Hub. Translating the base demonstrates that "proximal" motion changes both the reaction necessary to accelerate distal mass and the velocity used in force power. Holding the relative trace fixed isolates that inverse-dynamics effect. It also prevents a stronger claim: in a forward model, hub motion and joint motion co-evolve, so the calculated 38.09 W maximum power shift is not an expected benefit of a particular body translation.

Two-Hand Closed Loop. Constraint rank and nullspace dimension formalize the kinematic closure that a single-wrist pendulum omits. The audit shows one locally feasible velocity coordinate for the declared four-constraint, five-coordinate geometry. It does not solve for constraint multipliers. The negative equivalent-couple evidence therefore still comes from the archived force solution, while the Jacobian supplies the geometry contract a forward reproduction must satisfy.

Rotated Three-Dimensional Wrench. The proper-rotation audit is a required adapter test. It catches axis swaps, improper reflections, inconsistent moment transport, and mixed angular-velocity frames before an engine comparison is interpreted physically. Passing it establishes coordinate consistency only. Out-of-plane dynamics require nonplanar states and forces, not merely three stored vector components.

Reduced Full-Body Common-State Dynamics. This tier aligns the model digest, joint order, state, gravity, external loads, and inverse-dynamics estimand. It therefore isolates implementation transport and executes genuine out-of-plane dynamics. Its agreement is not anatomical validation because the model is reduced and its hand loads are prescribed.

Reduced Spatial Forward Contact. This tier adds independently solved two-hand compliance, state divergence, a same-state driver killswitch, long-axis rotation, swing-plane evolution, and reduced ground-pathway bookkeeping. It transports the declared mechanism through two native engines. Its hand carriages and declared contact law prevent anatomical, tissue, muscle, or human inference.

Articulated Full-Body Forward Contact. This tier remains open because it must replace the carriages with subject-scaled articulated arms, retain independently solved two-hand contact, and preserve event alignment without changing the observable. Contact calibration, stabilization, actuation, and anatomical uncertainty must be registered before the preferred outcome is inspected.

19.10.2 Cross-Engine Gate Definition

The executed common-state comparison did not begin with visually similar motion. It required identical model hashes and checked generalized action, wrench power, geometry interventions, and event timing. A forward comparison must retain those checks and additionally pass, at minimum:

- frame and reference-point metadata equality;
- force and moment balance residuals below declared absolute and relative tolerances;
- constraint rank and nullspace agreement away from singular events;
- wrench-power agreement after reference transport;
- matched gravity, damping, actuator, and contact settings; and
- an event-aligned comparison that distinguishes interpolation error from trajectory divergence.

Engine capability is not evidence that the comparison was run. The machine-readable discrepancy table separately marks reduced full-body common-state inverse dynamics and reduced two-engine forward contact as `executed`; articulated full-body forward contact remains `not_executed`. This split prevents a carriage-model result from silently becoming an anatomical claim.

19.11 Strategy Implications Across the Ladder

The higher-order audits reinforce a conditional strategy statement. Creating clubhead speed is not equivalent to maximizing a single joint force, delaying release by a fixed time, or producing a particular torque sign. Useful transfer requires the complete interaction wrench to align with the distal twist during the intended delivery interval.

A moving hub can alter both reaction and force power even when relative joint motion is held fixed. A two-hand loop can restrict admissible motion while still allowing a large opposed-force mode. A third coordinate can store energy and change timing without being a net source. A frame rotation can reverse or mix plotted components without changing physical power.

The practical model-building consequence is direct: optimize and report an event-window objective with force, couple, power, work, and constraint diagnostics together. A strategy that looks favorable only in one coordinate, reference point, or instantaneous sample has not yet demonstrated robust transfer.

19.12 Limitations and Next Falsification Tests

The ladder remains deliberately incomplete:

- the three-link trace remains an arm–proximal-shaft–distal-head surrogate;
- the original hub comparison remains prescribed, although the newer coupled tier evolves a finite-mass translating base;
- the forward two-hand tiers use ideal planar point constraints and declared mechanism-study parameters rather than anatomical or contact calibration;
- the coupled club has one lumped torsional mode rather than a distributed spatial beam;
- the reduced full-body tier uses prescribed hand loads and spherical inertia elements rather than solved contact and subject-specific anatomy; and
- the forward spatial tier uses two translational hand carriages, declared Kelvin–Voigt contacts, and a rigid club rather than anatomical arms, measured tissue, or the distributed shaft.

The next decisive model test replaces the carriages with subject-scaled articulated arms and couples the distributed shaft while emitting the same common schema and preserving the exact same-state command-removal branch. Failure of force/moment balance, power invariance, contact-rank agreement, or matched-event parity at that tier would falsify transport of the reduced passive-persistence claim.

Chapter 20

Empirical Evidence in Golfers

20.1 The Work and Power Budget of the Swing

Nesbit's full-body kinematic-kinetic study and the companion work-power analysis of Nesbit and Serrano remain the most complete published energy budgets of the swing (Nesbit 2005b; Nesbit and Serrano 2005). Using subject-specific full-body models of four golfers of widely differing skill, they report that the trunk complex (lumbar and thoracic spine plus hips) contributes 68.7–72.2% of total swing work, the shoulder and arm joints 24.3–28%, and the legs only 3.3–3.8% (Nesbit and Serrano 2005). Two further observations bear directly on the timing question. First, power generation and clubhead speed decreased systematically with increasing handicap. Second, the phase analysis locates the *generation* of energy predominantly in the earlier downswing (torso-dominated) and the *conversion* into clubhead speed late (shoulder/arm and club interface) — generation and transfer are temporally separated, which is a precondition for the retain-early/release-late strategy to even be possible.

The ground is the ultimate source of the external impulse. Han and colleagues characterized golfer-ground interaction in 63 skilled golfers and found meaningful associations between ground-interaction force and moment parameters and maximum clubhead speed (Han et al. 2019). Recent work has pushed this to an explicitly energetic formulation: in thirty golfers (fifteen professionals, fifteen high-level amateurs), foot-ground interaction variables alone explained 35.5% of clubhead-speed variance, rising to 75.4% when trunk sequencing and an *impulse-based energy-transfer efficiency* measure (effectiveness of mechanical impulse transmission from proximal segments to the club) were added; mediation analysis attributed the dominant indirect pathway to transfer efficiency rather than to sequencing per se (Rachnavy et al. 2026). That distinction — transfer *efficiency* mattering where sequence *order* alone does not — recurs throughout the skill-level literature below.

20.2 Segmental Kinetic-Energy Sequencing

Two studies measured the quantity this document cares most about — segment *kinetic energy*, not merely angular speed — and both found the P→D structure at the energetic level. Anderson, Wright and Stefanyshyn computed segmental kinetic energies in the golf swing and documented the sequenced build-up and outward migration of energy along the chain (Anderson, Wright, and Stefanyshyn 2006). Kenny, McCloy, Wallace and Otto reproduced segmental kinetic-energy sequencing in a large-scale computer-simulated golf swing, confirming the ordered energetic handoff in a full forward-dynamic setting (Kenny et al. 2008). These studies matter here for two reasons. First, they validate the segment-energy time course as a practical observable (the same observable computed for the model in Section 24). Second, they show the energetic sequence is not an artifact of angular-velocity bookkeeping: energy itself arrives late in the club, after the proximal segments have peaked.

20.3 Kinematic-Sequence Evidence and Skill-Level Differences

The descriptive P→D findings are robust in skilled populations. Tinmark and colleagues, studying elite golfers across full and partial shots, found a significant proximal-to-distal temporal ordering of peak speeds (pelvis → torso → hands/club) with successive amplification of peak speed at every level of effort, in both men and women (Tinmark et al. 2010). Cheetham and colleagues compared amateur recreational players with touring professionals on transition- and downswing-phase kinematic sequence parameters and found professionals showed higher peak-speed magnitudes and more consistent sequencing and timing characteristics (Cheetham et al. 2008). Zheng and colleagues' three-dimensional comparison of professional and amateur swings found systematic temporal and spatial differences, with amateurs deviating progressively more from professional norms as handicap increased (Zheng et al. 2008). Meister and colleagues provide elite rotational benchmarks (pelvis-thorax separation and its timing) against which amateurs can be compared (Meister et al. 2011), and Lindsay, Mantrop and Vandervoort review the broader catalogue of skill-level biomechanical differences (Lindsay, Mantrop, and Vandervoort 2008).

Torso-pelvis interaction variables predict performance. Myers and colleagues found upper-torso and pelvis rotation (and their separation) associated with ball speed in golfers across skill levels (Myers et al. 2008); Chu, Sell and Lephart identified a small set of biomechanical variables — among them trunk and pelvis kinematics at specific downswing events — explaining a large share of driving-performance variance (Chu, Sell, and Lephart 2010); Joyce and colleagues related three-dimensional trunk kinematics to clubhead speed across clubs (Joyce et al. 2013). On the widely coached "X-factor" specifically, Kwon and colleagues showed the computed values depend materially on methodology and that, in skilled golfers, X-factor parameters were not straightforwardly associated with clubhead velocity (Kwon et al. 2013) — another instance of a popular proxy dissolving under measurement scrutiny.

Three refinements prevent an overly tidy reading:

1. **Sequence order is common but not universal — even among the best.** Skilled golfers whose peak ordering deviates from the canonical pelvis-thorax-arm-club pattern are documented, and recent work examines how alternative sequence patterns among elite players relate to clubhead speed (Lee, Ehrlich, and Cain 2026). On-plane analyses of the "axle-chain" system in 66 skilled golfers found clubhead-speed associations concentrated in specific angular-motion characteristics rather than in gross ordering (Madrid et al. 2020).

2. **Within-player timing differences do not straightforwardly explain shot quality.** Comparing well-timed and mistimed shots *within* highly skilled players, Neal and colleagues found no significant differences in the timing lags between peak angular speeds of contiguous segments (Neal et al. 2007) — the felt sense of "timing" lives somewhere other than the inter-peak lags of a five-segment model. Movement variability is itself structured and functional in golf (Langdown, Bridge, and Li 2012; Glazier 2011), so single-trial sequence snapshots carry limited information.

3. **Method sensitivity.** As Section 2.5 noted, the identified sequence depends on component-selection methodology (Marsan et al. 2019), so cross-study comparisons of ordering statistics carry real uncertainty. Historical measurements with specialized instrumentation reached compatible conclusions about trunk-pelvis motion but with their own method dependence (McTeigue et al. 1994).

What survives these refinements: *skilled golfers generate larger proximal speeds, separate transition from release more distinctly, and transmit impulse and energy to the club more efficiently; merely exhibiting a P→D peak order is neither unique to skill nor sufficient for it* (Cheetham et al. 2008; Zheng et al. 2008; Rachnavy et al. 2026; Lee, Ehrlich, and Cain 2026). Coaching perception has partly internalized this: when surveyed systematically, coaches' key swing parameters overlap with, but do not coincide with, the parameters the biomechanical literature supports (Smith et al. 2015).

20.4 Where the Transfer Occurs

Synthesizing Section 20.1, Section 20.2 and Section 20.3 with the mechanics of Section 2: in a skilled swing, the dominant energy *generation* occurs in the trunk-hip complex during the first two-thirds of the downswing

(Nesbit and Serrano 2005); the dominant *transfer to the club* occurs in the final phase, through the wrist interface, by a mechanism that is substantially parametric/geometric (wrist-cock release under centrifugal loading, hub-path tightening) with a modest late active wrist contribution (White 2006; Miura 2001; Sprigings and Neal 2000; Nesbit and McGinnis 2009). Less-skilled players truncate exactly this separation: early release ("casting") moves energy into the club before proximal speed has peaked, producing the lower distal peak speeds and transfer ratios observed in amateur cohorts (Cheetham et al. 2008; Zheng et al. 2008; Milburn 1982). Chapter Section 24 reproduces this entire pattern — including the casting failure mode — inside a fully instrumented model where every energy pathway is measurable exactly.

Chapter 21

The Timing Question and the Counterfactual Framework

21.1 Two Strategies, Stated Precisely

Let $E_p(t)$ be the kinetic energy of the proximal subsystem (trunk and arms, with the club riding folded) over a downswing $t \in [t_0, t_{\text{imp}}]$, and define the transfer functional $P_{\text{w}}(t)$ as the inter-segment power crossing the wrist interface — joint-force power plus moment power, in the sense of Section 2.5.2. The two strategies are:

- **S1 (transfer early):** make $P_{\text{w}}(t)$ large as soon after transition as possible; the club begins acquiring its final energy early.
- **S2 (retain early, release late):** keep $P_{\text{w}}(t)$ small early (the wrist transmits centripetal force but little power) while proximal torques maximize E_p; then allow or drive a large P_{w} concentrated in the final phase.

The double-pendulum, delayed-release and optimal-control literatures reviewed in Section 2 uniformly favor S2 within their model classes (Milburn 1982; Pickering and Vickers 1999; Sprigings and Neal 2000; Sprigings and MacKenzie 2002; Sharp 2009; White 2006). The mechanistic reasons compose into a coherent account:

1. **Inertia management.** With the wrist cocked, the system's moment of inertia about the hub is small, so proximal torque buys angular velocity cheaply; early release raises the inertia while proximal torque is still accelerating the system (White 2006; T. P. Jorgensen 1999).
2. **Actuator physiology.** Torque-velocity limits mean the wrist's small muscles add meaningful power only while wrist angular velocity is modest; early wrist torque squanders that window and can even require *negative* (restraining) torque later (Sprigings and Neal 2000).
3. **Parametric transfer needs developed centrifugal loading.** The passive outward energy flow at release, and the inward-pull gain, scale with the centrifugal field established by proximal rotation — weak early, strong late (White 2006; Miura 2001).
4. **Interaction torques do the late work for free.** Motion-dependent torques accelerate the distal segment and decelerate the proximal one automatically once release begins; the golfer's job is to *time* this cascade, not to muscle it (Putnam 1993; Herring and Chapman 1992).

Empirically, S2-like signatures separate skill groups (Section 20.3), while the within-expert null result on inter-peak lags (Neal et al. 2007) suggests that once S2 is grossly achieved, residual performance variance lives in finer variables (release profile shape, hub-path geometry, impulse-transfer efficiency) — exactly what the counterfactual framework below is designed to expose.

21.2 What Would Falsify S2?

To keep the question scientific rather than rhetorical, we state what evidence would count against the retain-early/release-late hypothesis:

- an optimal-control solution, in a validated 3D model with physiological actuator limits, whose optimal wrist-interface power is front-loaded;
- measured expert swings in which early wrist-interface power (normalized) is *positively* associated with clubhead speed across players;
- counterfactual (ZTCF/ZVCF) analyses showing that in faster swings the control term dominates club acceleration *early* rather than late;
- **or, in the direct manipulation of Section 23: torque programs that drive the wrist early outperforming programs that retain early and drive late, under matched proximal effort.**

The last test is executed in Section 24.

21.3 ZTCF and ZVCF: Definitions

Two counterfactual accelerations are defined *pointwise* at each measured state $(q, \dot{q}, \tau)$ of a swing:

$$\ddot{q}_{\text{ZTCF}} \;=\; M(q)^{-1}\,(-\,b(q,\dot{q})), \tag{21.1}$$

$$\ddot{q}_{\text{ZVCF}} \;=\; M(q)^{-1}\,(\tau \;-\; b(q,\mathbf{0})) \;=\; M(q)^{-1}\,(\tau \;-\; g(q)), \tag{21.2}$$

where the simplification in Equation 21.2 holds because the Coriolis term (quadratic in velocity) and viscous dissipation (linear in velocity) vanish at $\dot{q} = \mathbf{0}$. The ZTCF (Equation 21.1) answers *"what would the system do right now if every actuator switched off?"* — it is the **drift** field of the affine control system, containing gravity plus all motion-dependent interaction dynamics. The ZVCF (Equation 21.2) answers *"what would the currently applied torques, together with gravity, produce from a standstill in this pose?"* — it removes every motion-dependent term but is *not* the pure control contribution, since gravity remains in Equation 21.2; the pure torque term is $M^{-1}\tau$ in the split below. Because Equation 2.1 is affine in τ, the actual acceleration superposes exactly:

$$\ddot{q} \;=\; \underbrace{M^{-1}(-b(q,\dot{q}))}_{\text{drift = ZTCF}} \;+\; \underbrace{M^{-1}\tau}_{\text{control}}, \tag{21.3}$$

an identity enforced to numerical tolerance by the accompanying tests (Section 29).

21.4 Two Operational Variants

Two distinct operationalizations are available, and the difference matters for interpretation.

Pointwise (instantaneous) counterfactuals. The engine-agnostic Python implementation (`simulation_backends/ztcf_zvcf.py`) evaluates Equation 21.1 and Equation 21.2 at each sample of an already-measured trajectory — explicitly *not* a forward-integrated rollout. Each sample answers a local "what if"; this is the right tool for drift/control *time profiles* along a real swing, and it is the tool used in Section 24.

Re-simulation counterfactuals. The original MATLAB/Simscape pipeline (`run_ztcf_simulation.m`) re-simulates the full model with a torque "killswitch" applied at time t, for a sweep of t values, tabulating the counterfactual state reached with torques off. The pipeline forms the difference table DELTA = BASE−ZTCF (`process_data_tables.m`), attributing to active control everything the passive system would not have done. A companion ZVCF model applies the measured torques to the same pose with velocities zeroed

(SCRIPT_ZVCF_GENERATOR.m); the ZVCF model is run without gravity so that, when its samples are set against the ZTCF (which retains gravity), the gravitational contribution is not double-counted. This decomposition is meaningful *pointwise*, at matched states, in the sense of Equation 21.3: the ZVCF script records one sample per zero-velocity pose, and separately integrated counterfactual rollouts do not superpose at the trajectory level in a nonlinear multibody system, because M and b evolve differently once the counterfactual state diverges. The MATLAB pipeline additionally computes, for every joint and every table (BASE, ZTCF, DELTA), the linear work $\int \mathbf{F} \cdot \mathbf{v}\, dt$, angular work $\int \tau \cdot \omega\, dt$, total work and power, and the fractional contributions ZTCF/BASE and DELTA/BASE (calculate_work_impulse.m, calculate_total_work_power.m) — precisely the Robertson-Winter-style joint-interface accounting of Section 2.5.2, including a kinematic-sequence plot computed *separately for the BASE, ZTCF and DELTA components* (SCRIPT_324_PLOT_ZTCF_KinematicSequence.m).

21.5 The Counterfactual Form of the Timing Hypothesis

The ZTCF/ZVCF language makes the S2 hypothesis crisp. Along a measured downswing, project the drift and control accelerations onto the club's speed-producing direction and integrate the associated powers. S2 predicts, for high-performing swings:

- **Early downswing:** the *control* term should dominate energy input at proximal joints, while at the wrist the control contribution to club angular acceleration should be small or negative (restraining lag against the developing centrifugal field). The club's ZTCF profile should show the drift field already "wants" to open the wrist — control is *holding energy back*.
- **Late downswing:** the *drift* term (parametric transfer, centrifugal release) should dominate club acceleration, with a brief late window of positive wrist control power (Sprigings and Neal 2000). The active share of club-side work should concentrate before release; the passive share should account for most club energy gain after release (White 2006).

This pattern was initially motivated by torque-removal analyses. Section 24 provides a reproducible quantitative test in a two-segment model; conversely, had the early-drive programs won the sweep, S2 would have been falsified in exactly the terms of 3. Bounded-actuator, impact-definition, and model-parameter sensitivity checks are reported here; re-simulation parity, 3D full-body models, and measured swings remain future work.

Chapter 22

Computational Framework

The numerical work uses an open-source, backend-neutral implementation of a planar two-segment golf model. Model parameters are represented once and rendered to the analytical equations of motion and optional simulation backends. The present results use the deterministic analytical ODE backend; they do not depend on a proprietary solver or graphical interface.

22.1 Analysis Components

The workflow separates four responsibilities:

1. **Dynamics.** A validated mass matrix and bias-force provider advances the two generalized coordinates under specified shoulder and wrist torques.
2. **Counterfactual decomposition.** At each sampled state, acceleration is split into passive drift and applied-control components. These pointwise quantities are not presented as alternative forward trajectories.
3. **Energy accounting.** Segment kinetic and potential energies, actuator work, and wrist-interface force and moment power are evaluated with a Robertson–Winter balance check (Robertson and Winter 1980).
4. **Reproducibility.** Fixed-step integration, explicit parameter records, machine-readable outputs, and regression tests connect every reported number to a declared computation.

The source implementation and versioned data are linked once in the code and data availability statement (Section 27). Repository names are omitted from the scientific argument because software provenance is not evidence for the physical claims.

22.2 Validation Boundary

The implementation includes unit tests for drift-plus-control reconstruction, work–energy consistency, deterministic replay, impact detection, and headline ordering. The reduced full-body common-state tier now compares MuJoCo with an independent Lagrange–Christoffel implementation generated from one hashed model. That agreement reduces implementation risk for the declared inverse-dynamics estimand; it does not validate the primary planar model or remove structural limitations such as prescribed spatial hand loads, simplified inertias, missing forward contact, or simplified actuators.

Chapter 23

Computational Methods

This chapter specifies the simulation experiments whose results appear in Section 24. Everything described here is implemented in three committed modules under `scripts/research/proximal_distal_energy/` (`swing_model.py`, `interaction_forces.py`, `torque_programs.py`, `run_experiments.py`, and `run_interaction_force_st` with figures from the paired figure scripts), and every output carries a provenance record (git SHA, integrator settings, parameter values). The reproduction commands are in Section 28.

23.1 Model

The simulated plant is a backend-neutral two-segment golf model (Section 22): an arm segment rotating about a fixed hub, hinged at the wrist to a club segment consisting of a distributed shaft plus a clubhead point mass at the tip, moving in an inclined swing plane with projected gravity. Baseline parameters are (`GolfModelParams.default()`; full table in Section 30): arm length 0.75 m and mass 7.5 kg; club length 1.0 m with 0.15 kg shaft and 0.20 kg clubhead; plane inclination 35°; viscous damping 0.4 and 0.25 $\mathrm{N \cdot m \cdot s/rad}$ at shoulder and wrist. Angles follow the declared convention: θ_1 is the arm angle from the downward vertical in the swing plane, θ_2 the wrist angle of the club relative to the arm, and the club's absolute angle is $\theta_{\text{club}} = \theta_1 + \theta_2$.

Rollouts use the ODE reference backend (analytic dynamics, fixed-step RK4) with $\Delta t = 1$ ms over a 0.9 s horizon. Every rollout starts from the same top-of-backswing state, $\theta_1(0) = -2.2$ rad, $\theta_2(0) = -1.57$ rad (a 90° wrist cock), $\dot{q}(0) = 0$ — an arm carried well past horizontal with the club folded over the shoulder, comparable to the classical two-lever setups (T. P. Jorgensen 1999; Pickering and Vickers 1999).

23.2 Torque Programs

All programs share a constant shoulder torque τ_s active from $t = 0$ (two levels: 60 and 100 $\mathrm{N \cdot m}$), so the sweep isolates the wrist channel. The wrist torque follows one of three profiles, parameterized by an onset time t_{on}:

- **passive**: $\tau_w(t) = 0$ throughout (free hinge) — the White limit (White 2006);
- **drive-only**: $\tau_w(t) = 0$ for $t < t_{\text{on}}$, then $\tau_w = +15$ $\mathrm{N \cdot m}$ (opening);
- **restrain-then-drive**: $\tau_w(t) = -R$ (cock-retaining) for $t < t_{\text{on}}$ with $R \in \{5, 10\}$ $\mathrm{N \cdot m}$, then $\tau_w = +15$ $\mathrm{N \cdot m}$.

The onset grid is $t_{\text{on}} \in \{0, 0.025, \dots, 0.35\}$ s (15 values). With both shoulder-torque levels this yields 92 distinct programs; `drive-only` at $t_{\text{on}} = 0$ is the S1 ("transfer early") pole of the design and `restrain-then-drive` with late onset is the S2 pole. Torque magnitudes sit inside the ranges used by comparable two- and three-segment studies (Sprigings and Neal 2000; Pickering and Vickers 1999), scaled to this model's inertias.

E1b (actuator-bound sensitivity). Because constant torques overstate what muscle can deliver at speed, a companion sweep (`e1b_bounded_torque.py`) replaces both concentric drives with a linear torque-velocity bound, $\tau(\omega) = \tau_{\text{iso}}\,\text{clip}(1 - \omega/\omega_{\text{max}}, 0, 1)$ (shoulder: $\tau_{\text{iso}} = 100$ N · m, $\omega_{\text{max}} = 20$ rad/s against arm speed; wrist: 20 N · m, 30 rad/s against opening speed; restraint constant, since eccentric torque does not taper with shortening velocity). State-dependent torques require closed-loop integration, done with the backend's own `mass_matrix`/`bias_forces` primitives under the identical RK4 settings.

23.3 Impact Definition and Validity

Impact is the first upward crossing of $\theta_{\text{club}} = 0$ (club pointing straight down in the swing plane), linearly interpolated between samples; the score of a program is the clubhead (distal tip) speed at that instant. Because the shoulder torque is constant, a program that keeps the club folded through the bottom of the swing can postpone the crossing into a second rotation while the shoulder keeps pumping energy — an artifact, not a swing. A trial is therefore **invalid** unless the arm angle at the crossing satisfies $\theta_1 \leq 2.0$ rad (a first-pass delivery); 29 of the 92 programs were excluded by this rule, all of them heavy-restraint or very-late-onset programs. This constraint is conservative *against* the S2 hypothesis: it discards only trials that would otherwise have inflated the restraint arm of the sweep.

23.4 Recorded Quantities

For every program: the full state trajectory, applied controls, impact time and clubhead speed. For four representative programs at $\tau_s = 60$ N · m (passive; drive-only at $t_{\text{on}} = 0$; the best drive-only; the best restrain-then-drive), additionally:

Segment energies. Arm kinetic energy $\frac{1}{2}I_1^{\text{hub}}\dot\theta_1^2$; club kinetic energy $\frac{1}{2}m_2\|\mathbf{v}_{c2}\|^2 + \frac{1}{2}I_2^{\text{com}}(\dot\theta_1 + \dot\theta_2)^2$ with the club centre-of-mass velocity $\mathbf{v}_{c2}$ computed analytically from the state.

Wrist-interface power accounting (E4). Following Section 2.5.2: the joint-force power $P_f = \mathbf{F} \cdot \mathbf{v}_{\text{wrist}}$, with the joint reaction force recovered from Newton's second law for the club segment, $\mathbf{F} = m_2\mathbf{a}_{c2} - m_2\mathbf{g}$ (centre-of-mass acceleration by central differences of the analytic velocity); the net wrist moment power on the club $(\tau_w - b_w\dot\theta_2)(\dot\theta_1 + \dot\theta_2)$; the wrist actuator power $\tau_w\dot\theta_2$; and the residual check that the club's total mechanical energy rate equals joint-force power plus moment power (reported as a figure in Section 24; gravity is not added separately because potential energy is included in the club's mechanical energy).

Pointwise counterfactual split (E2). At every sample, the `drift_and_control_split` (Equation 21.3) evaluated with the applied torques, yielding drift and control components of both joint accelerations; the club's absolute angular-acceleration split is the row sum.

Exact interaction-force audit (E5). Analytic club-centre-of-mass acceleration replaces finite differentiation and separates the wrist reaction force into proximal and distal tangential and centripetal terms plus gravity reaction (Section 3.3). Each vector is projected onto the analytic wrist velocity to obtain a signed power pathway. A late-downswing matched-state experiment then integrates the commanded and zero-torque futures for 120 ms, explicitly distinguishing a trajectory-level killswitch from pointwise ZTCF (Section 3.7). Newton balance, club moment balance, force and power reconstruction, matched initial state, and initial ZTCF equivalence are unit-tested.

Matched-state counterfactual ensemble (E6). Eight cut times, including three samples bracketing the 0.100 s wrist-command switch, are crossed with four forward horizons and three fixed timesteps. Commanded and zero-torque futures share the exact interpolated source state. Divergence is recorded in generalized position and velocity, wrist force, force power, integrated force work, and terminal clubhead speed. The 80 ms subset is repeated with gravity disabled and with damping disabled. The 0.5 ms grid is the timestep reference, and the registered WSCG chart's stored DELTA series are independently reconstructed as BASE minus counterfactual after time alignment (Section 6).

Phase budgets. With the downswing divided at $t_{\text{imp}}/2$: club kinetic-energy gain, wrist actuator work, shoulder work, and integrated joint-force transfer, per half.

Robustness analyses (E1c and E1d). E1c re-scores the same 92 trajectories using fixed arm positions, peak horizontal clubhead velocity, peak first-pass speed, a matched arm angle, and five validity bounds. E1d repeats the four-strategy comparison after changing one model input at a time: arm length (±10%), arm mass (±15%), club length (±10%), clubhead mass (±20%), swing-plane inclination (25° and 45° versus 35°), and both damping coefficients (0.5× and 1.5×). For each E1d case, late-drive and restrain-then-drive onset are re-optimized on the original onset grid. These ranges test local structural sensitivity; they are not population percentiles or confidence intervals.

23.5 Design Rationale and Scope

This deliberately simple model is a planar 2-DOF chain with a fixed hub and rigid shaft. That choice is the point, not a compromise: the classical timing results were established in exactly this model class (Section 2.3), so the analysis first reproduces the class's behavior under explicit instrumentation — where every energy pathway is exactly measurable and the counterfactual contract is tested — before scaling to models where the same questions are harder to instrument. What the 2-DOF results can and cannot establish is taken up explicitly in Section 25; the extension pathway (re-simulation parity, higher-fidelity models, and motion capture) is specified in Section 26.

Chapter 24

Results

All numbers in this chapter are computed by the committed experiment suite (Section 23) from the recorded outputs in `data/` (`e1_sweep.json`, `representative_traces.npz`, `results_summary.json`); figures are generated by `make_figures.py`. Speeds are quoted to 0.1 m/s and times to 1 ms.

24.1 E1 — Timing of the Distal Handoff

Figure 24.1 shows clubhead speed at impact as a function of wrist-drive onset time for both shoulder-torque levels. Table 24.1 lists the representative programs.

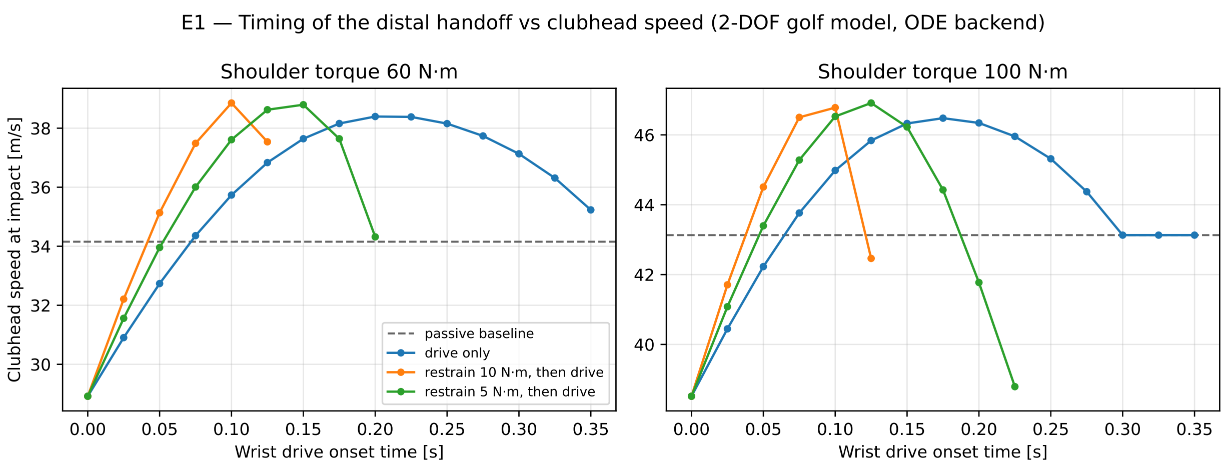

Figure 24.1: E1 sweep: clubhead speed at impact versus wrist-drive onset time, for drive-only and restrain-then-drive profiles at two shoulder-torque levels. Dashed line: fully passive wrist baseline.

Table 24.1: Representative programs at $\tau_s = 60$ N $\cdot$ m.

Program ($\tau_s = 60$ N $\cdot$ m)	t_{on} [s]	t_{imp} [s]	Clubhead speed [m/s]	vs passive
Passive wrist	—	0.370	34.2	—
Early drive (S1 pole)	0.000	0.327	28.9	**−15.3%**
Best drive-only	0.200	0.352	38.4	**+12.4%**

Program ($\tau_s = 60$ N · m)	t_{on} [s]	t_{imp} [s]	Clubhead speed [m/s]	vs passive
Best restrain-then-drive ($R = 10$)	0.100	0.349	38.9	**+13.8%**

Three findings, consistent across both torque levels:

1. **Driving the wrist from the top is the worst valid strategy.** At $\tau_s = 60$ N · m the early-drive program reaches 28.9 m/s versus 34.2 m/s for a completely passive wrist — active early distal effort *destroys* 15% of the speed a free hinge would have delivered. The early-drive program is the worst valid trial in the entire 92-program sweep. At $\tau_s = 100$ N · m the pattern repeats (early drive is the minimum of its profile curve; passive baseline 43.1 m/s).
2. **The same wrist torque, applied late, is worth +12%.** Clubhead speed rises monotonically with onset time until an interior optimum ($t_{\text{on}} = 0.200$ s at 60 N · m, 0.175 s at 100 N · m — roughly the middle of the downswing, with drive active through the final ~40%), reaching 38.4 and 46.5 m/s respectively.
3. **Retain-early-then-drive is best of all.** Programs that apply a *negative* (cock-retaining) wrist torque early and the identical positive drive late beat the best drive-only programs at both torque levels: 38.9 m/s ($R = 10$ N · m until 0.100 s) at $\tau_s = 60$, and 46.9 m/s ($R = 5$ N · m until 0.125 s) at $\tau_s = 100$. The margin over the best drive-only program is modest (+0.5 m/s at both levels), consistent with most of the retention benefit being achievable passively; but the ordering is strict, and heavy or over-long restraint fails the first-pass validity rule

   (2) rather than winning — the benefit is a *timed* retention, not maximal retention.

24.2 E1b — Sensitivity to Actuator Torque-Velocity Bounds

A standing objection to constant-torque sweeps is that real muscle torque falls with shortening velocity, shrinking the late-drive window (Sprigings and Neal 2000). E1b re-runs the sweep with a linear torque-velocity bound on both concentric drives (shoulder: 100 N · m isometric, zero at 20 rad/s of arm speed; wrist: 20 N · m isometric, zero at 30 rad/s of opening speed; restraint left constant, as eccentric torque does not taper with shortening velocity), integrated closed-loop with the same RK4 settings (Figure 24.2). The ordering survives intact: early drive 28.8 m/s < passive 30.8 < best late drive 34.6 (onset 0.175 s) < best restrain-then-drive 34.8 (R = 10 N · m until 0.100 s). Effect sizes compress — late drive is worth +12.2% over passive and the restraint margin narrows to +0.2 m/s — precisely the compression bounded-actuator theory predicts, while the optimal onsets move slightly *earlier* (0.175–0.100 s versus 0.200–0.100 s), since a velocity-limited wrist needs a slightly longer window to deliver its work. The S2 ordering is therefore not an artifact of unbounded actuators.

24.3 Kinematic Sequence

Figure 24.3 shows arm and club angular-speed traces. All four programs produce the qualitative P→D pattern — arm speed peaks, then club speed peaks higher — as expected from interaction dynamics alone (Section 2.2). The programs differ exactly where the skill literature says they should (Section 20.3): the early-drive swing opens the club immediately, its arm peak is depressed, and its club peak is the lowest of the four; the retain-then-drive swing holds the club folded longest, achieves the highest arm speed before release, and converts it into the highest club peak. Impact in the early-drive swing occurs with the arm still 0.36 rad *short* of vertical — the clubhead has overtaken the hands before the arm arrives, the model's version of "casting" / premature release — whereas the two best programs deliver impact with the arm 0.7–0.8 rad *past* vertical (hands ahead of the clubhead, forward shaft lean), matching the geometry of skilled impact.

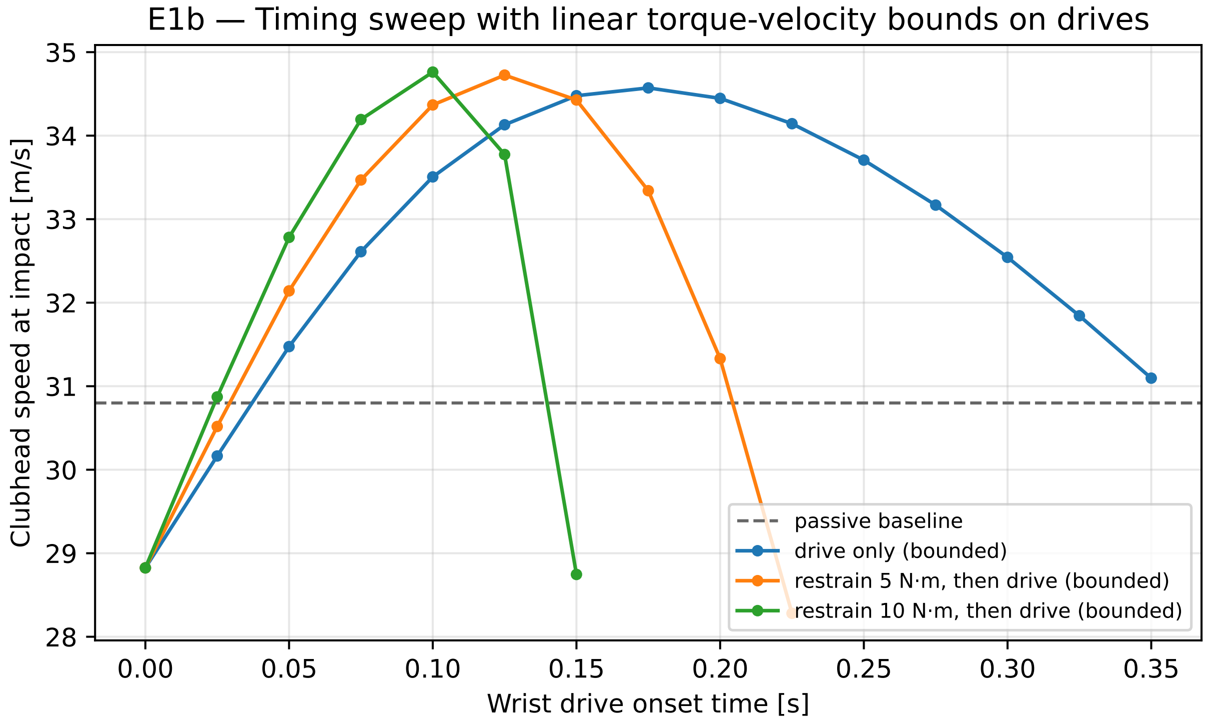

Figure 24.2: E1b: the timing sweep repeated with linear torque-velocity bounds on the shoulder and wrist drives.

24.4 Segment Energies

Figure 24.5 shows arm and club kinetic-energy time courses, and Figure 24.6 the early-half/late-half budget. The energetic story is unambiguous:

- **Early half.** The winning programs put the *least* energy into the club early: 5.0 J (best restrain) and 7.3 J (best drive-only) versus 22.9 J for the early-drive program (passive: 7.8 J). Meanwhile they bank more energy in the arm: early-half shoulder work is 54.7 J (best restrain) and 51.9 J (best drive) versus 33.9 J (early drive) — the early-driven club back-reacts on the arm and *steals its runway*.
- **Late half.** The ordering inverts: late club kinetic-energy gain is 180.5 J (best restrain) and 174.2 J (best drive) versus 84.4 J (early drive) and 136.4 J (passive).
- **Active retention is measurable.** The best restrain program's early-half wrist actuator work is *negative* (−1.9 J) and its early-half joint-force transfer is likewise slightly negative (−1.0 J): the wrist interface is momentarily a net energy *withdrawer* from the club while the proximal system loads — the literal, quantitative meaning of "retaining energy proximally".

24.5 E4 — Wrist-Interface Power Accounting

Figure 24.7 (left) shows the Robertson-Winter interface powers for the best restrain-then-drive swing. The joint-force power — energy carried into the club *through the joint force*, the pathway that requires no wrist torque at all — is small and briefly negative early, then surges in the final ~0.1 s to dominate the club's energy supply (late-half integral 131.7 J versus 25.7 J of late wrist actuator work). The wrist moment channel contributes a brief late pulse, exactly the "latter stages" window identified by Sprigings and Neal (Sprigings and Neal 2000). The right panel verifies the accounting: the club's mechanical-energy rate tracks the sum of joint-force and moment power across the swing (residual at the level of the finite-difference approximation),

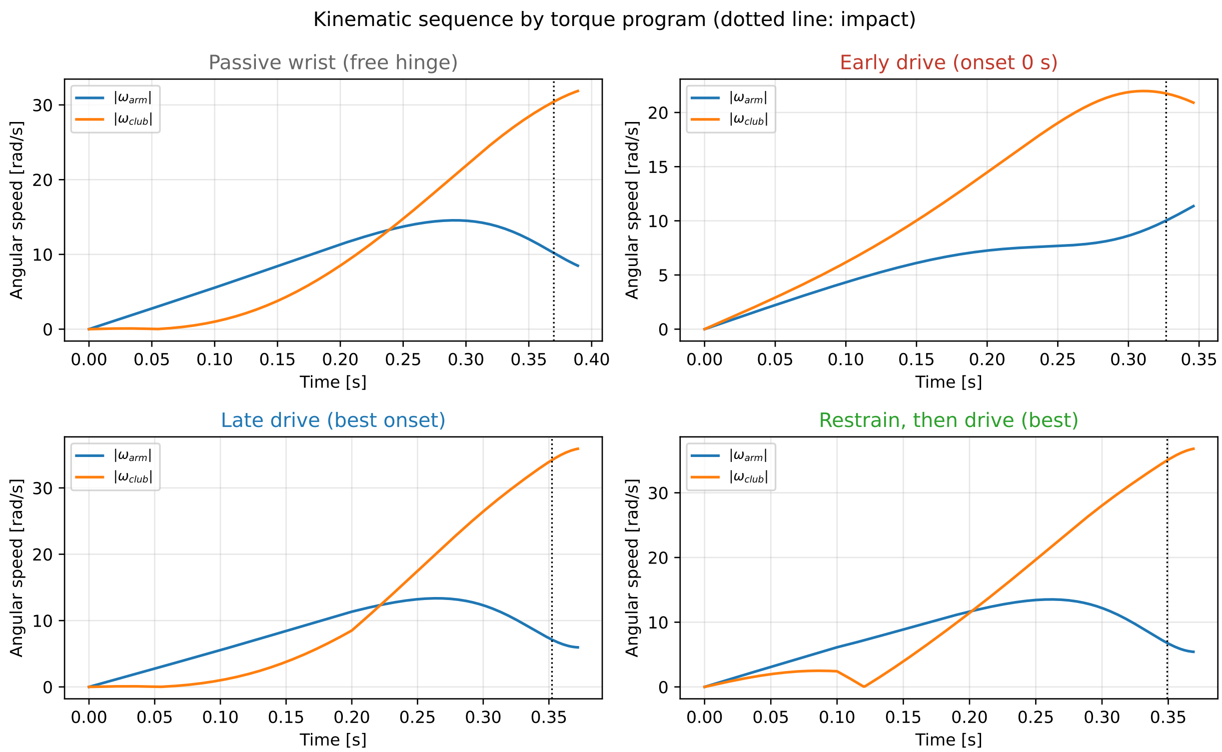

Figure 24.3: Kinematic sequence (arm and club angular speeds) for the four representative programs at $\tau_s = 60$ N $\cdot$ m. Dotted vertical line: impact.

confirming the implementation of the segmental balance of Section 2.5.2.

24.6 E2 — Counterfactual Drift/Control Split

Figure 24.8 shows the pointwise drift/control decomposition (Equation 21.3) of the club's absolute angular acceleration for the S1 pole (early drive) and the S2 winner (restrain-then-drive). In the early-drive swing, the control term supplies a large share of club acceleration from the first instant — the S1 signature — and the drift term never develops the late surge seen elsewhere. In the restrain-then-drive swing the picture reverses: early, the control term *opposes* the drift term at the club (restraint holding back a drift field that already wants to release); late, the drift term — the ZTCF, i.e. what the accumulated motion would do with every actuator switched off — dominates the club's acceleration entirely, with the control term a small additive assist. This is, instant by instant, the counterfactual signature that Section 21.5 predicted for high-performing swings: **early control retains; late drift delivers.**

24.7 E1c and E1d — Robustness to Scoring and Model Parameters

Re-scoring all trajectories under five alternative families of impact criteria preserved the qualitative strategy ordering. This rules out the specific concern that the result was created by the club-vertical impact definition, but it does not test unmodeled impact physics.

The one-at-a-time parameter study also preserved the ordering in all 13 cases. Across those cases, clubhead-speed ranges were 26.4–31.3 m/s for early drive, 32.9–35.6 m/s for the passive wrist, 36.5–40.6 m/s for the best late drive, and 36.8–41.3 m/s for the best restrain-then-drive program. The best drive onset shifted from 0.175 to 0.225 s, while the best restrained onset shifted from 0.100 to 0.150 s.

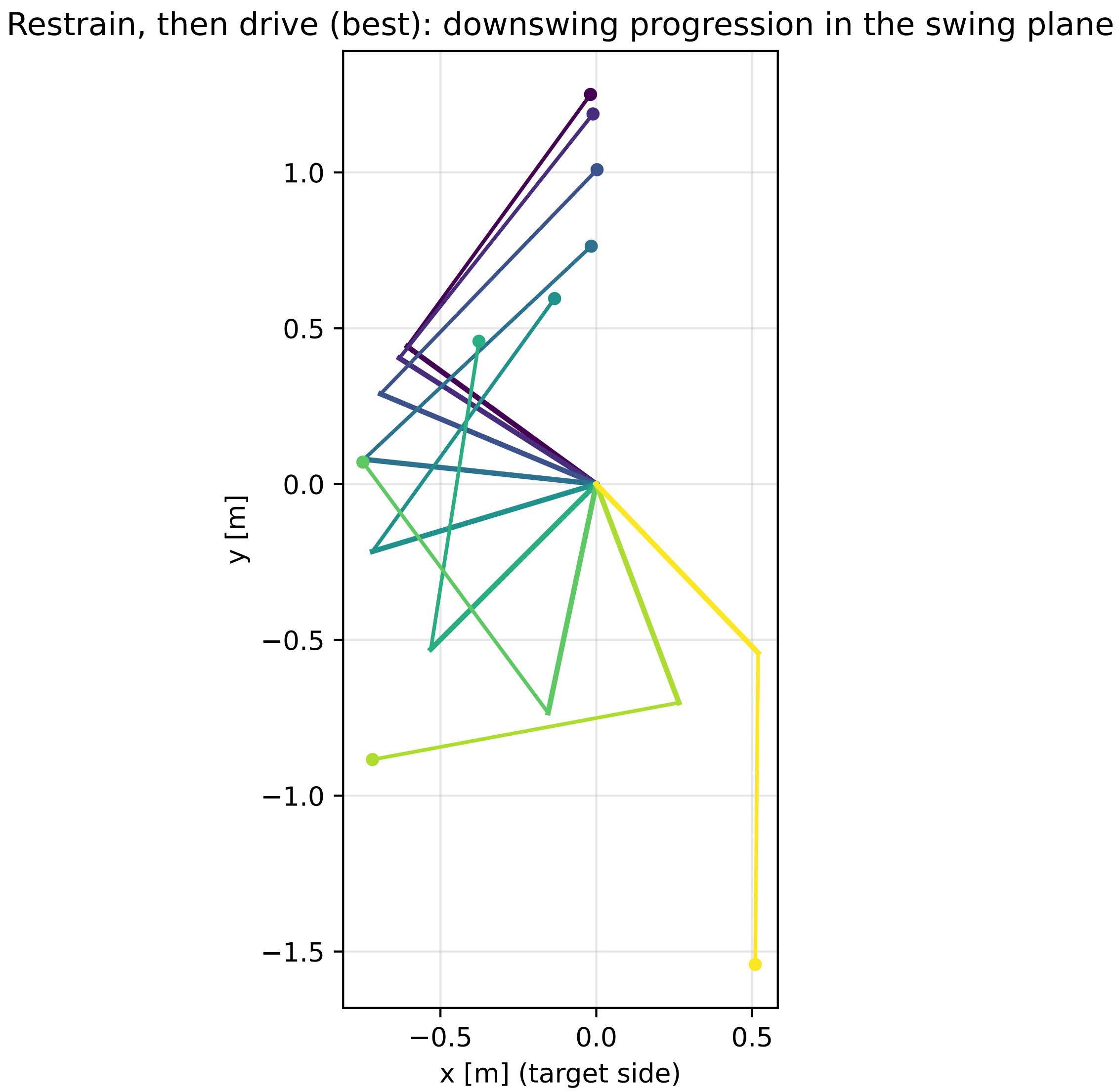

Figure 24.4: Stick-figure montage of the best restrain-then-drive swing in the swing plane (downswing progression at equal time steps).

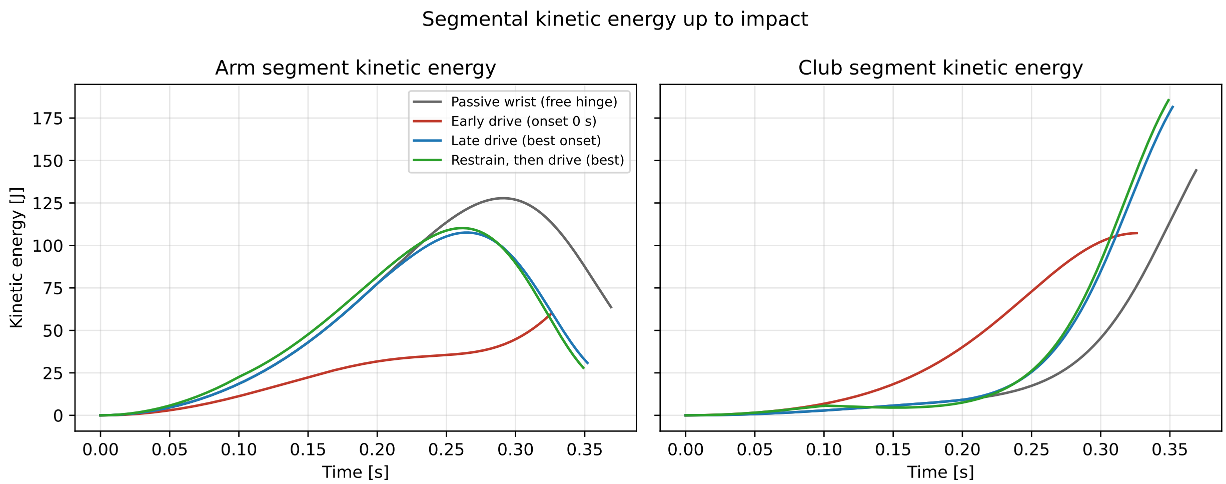

Figure 24.5: Segmental kinetic energies up to impact for the four representative programs.

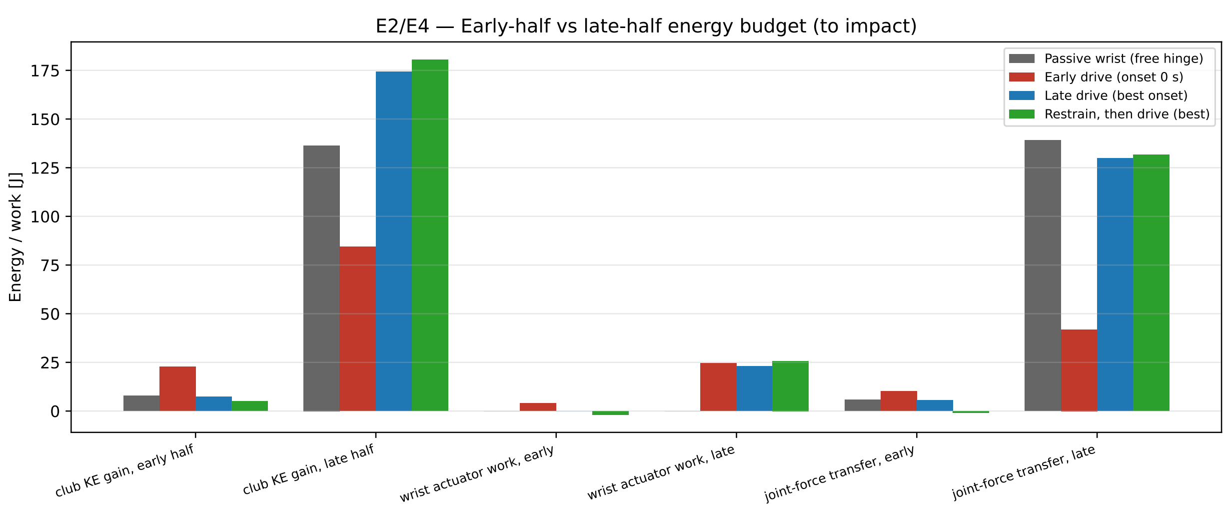

Figure 24.6: Early-half versus late-half energy budget (club kinetic-energy gain, wrist actuator work, integrated joint-force transfer) for the representative programs.

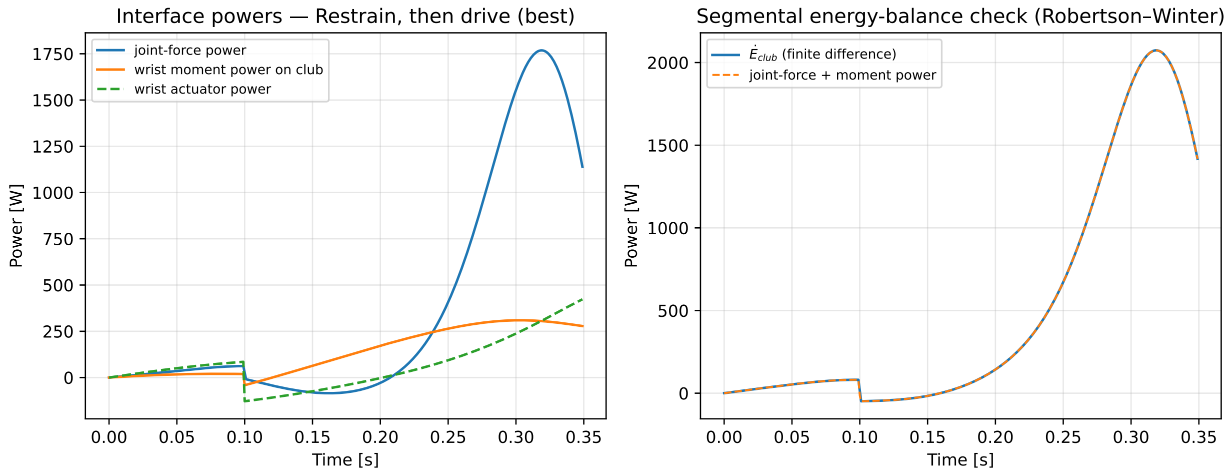

Figure 24.7: Left: wrist-interface powers for the best restrain-then-drive swing. Right: segmental energy-balance check — club mechanical-energy rate versus the sum of interface powers.

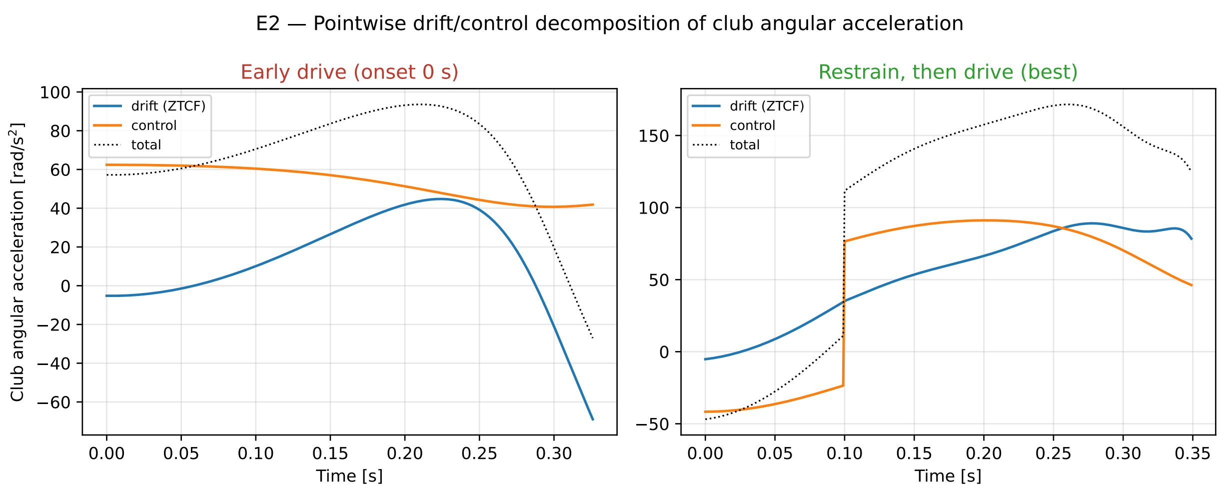

Figure 24.8: Pointwise drift (ZTCF) versus control decomposition of club absolute angular acceleration, for the early-drive (left) and best restrain-then-drive (right) programs.

The stable ordering supports a mechanism-level interpretation within the local parameter neighborhood. The shifting optimum argues against treating one onset time as a general prescription. Because parameters were varied individually, the study does not characterize interaction effects or a population distribution.

24.8 Summary of Findings

Within this model class, under matched proximal effort and a consistently defined first-pass impact: (i) early distal drive is lower than the passive condition (-15% in the baseline case); (ii) identical distal effort applied late is worth $+12\%$; (iii) timed active retention followed by late drive is best at both effort levels, with its early-phase wrist work measurably negative; (iv) the late club energy supply is dominated by the passive joint-force/parametric pathway, not the wrist actuator; (v) the counterfactual split shows control acting early as restraint and drift delivering late — the S2 pattern of Section 21.5 in full; and (vi) the entire ordering survives linear torque-velocity bounds, alternative scoring definitions, and the declared one-at-a-time parameter perturbations, although effect sizes and optimal onset times change (Section 24.2, Section 24.7).

Chapter 25

Discussion

25.1 Agreement With the Literature

Every headline result of Section 24 has an antecedent in the literature of Section 2, and the quantitative agreement is close enough to be meaningful rather than merely directional. The harm of early release reproduces the delayed-release results of Pickering and Vickers and of Sprigings and MacKenzie (Pickering and Vickers 1999; Sprigings and MacKenzie 2002) and Milburn's original observation that early uncocking truncates arm acceleration (Milburn 1982). The +12% value of late wrist drive brackets Sprigings and Neal's 4% gain from a physiologically bounded late wrist torque (Sprigings and Neal 2000) — our unbounded constant 15 $\mathrm{N} \cdot \mathrm{m}$ torque and longer active window make a larger gain expected. The dominance of the passive late energy pathway (joint-force transfer of 131.7 J versus 25.7 J of late wrist work in the best swing) is White's parametric transfer made explicit in Robertson-Winter accounting (White 2006; Robertson and Winter 1980). The casting geometry of the early-drive swing — clubhead overtaking the hands before the arm reaches vertical — is the model's rendering of the amateur release failure documented across skill-level studies (Cheetham et al. 2008; Zheng et al. 2008), and the impact geometry of the winning programs (hands ahead, arm 0.7–0.8 rad past vertical) matches the skilled pattern. Finally, the drift/control time signature — control as early restraint, drift as late delivery — is the counterfactual pattern proposed in Section 21.5, now exhibited by this model.

25.2 Why Retention Wins: A Mechanistic Reading

The energy budgets localize the causal structure. Early wrist drive fails through *two* simultaneous channels: it raises the system's moment of inertia about the hub while the shoulder is still accelerating (so the same shoulder torque buys less arm speed — the early-drive program banks 21 J less early shoulder work than the retain program despite identical τ_s), and it spends the wrist's work budget at a time when the joint-force/parametric channel cannot yet amplify it (White 2006). Timed restraint wins through the mirror mechanisms: it holds the compact configuration slightly longer than a free hinge would (the drift field already wants to open the wrist — the restraint is fighting centrifugal release, not gravity), banks that margin as arm speed, and then releases into a fully developed centrifugal field where the passive channel converts arm energy to club energy at the maximum rate. The negative early wrist work (−1.9 J) is small in absolute terms but diagnostic: retention is an *energetic withdrawal*, not merely an absence of drive — precisely the "restraining torque" role that appears in Jorgensen's fitted expert swing and in Sprigings-Neal-type actuator solutions (T. P. Jorgensen 1999; Sprigings and Neal 2000; Herring and Chapman 1992).

Two honest qualifications temper the S2 reading. First, the margin of active restraint over the best *passive-then-drive* program is small (+0.5 m/s at both effort levels): in this model most of the retention benefit is available from simply *not driving* the wrist, with active holding adding a final increment. Second, retention is only valuable *timed*: heavier or longer restraint programs failed the first-pass validity rule rather than winning — over-retention postpones delivery past the useful window. "Retain early, release late" is therefore not "retain maximally"; it is *retain until the drift field can pay, then stop retaining.*

25.3 What the 2-DOF Model Cannot Establish

The results chapter demonstrates the S2 pattern in the model class where the classical timing literature lives; it does not yet demonstrate it for a golfer. The specific gaps, each with a tracked remediation:

- **Spatial Body and Distributed-Club Dynamics.** The hub-path and inward-pull mechanisms (Miura 2001; Nesbit and McGinnis 2009; Nesbit and McGinnis 2014), shaft dynamics (MacKenzie and Sprigings 2009; Betzler et al. 2012; Osis and Stefanyshyn 2012), and long-axis rotations (Marshall and Elliott 2000) are outside the primary 2-DOF model. The forward two-hand and coupled moving-base/flexible-club tiers address planar closure, translation, and one lumped flex mode; base rotation, distributed beams, and the reduced full-body common-state tier now addresses nonplanar inverse dynamics. The reduced MuJoCo/Pinocchio forward-contact tier adds compliant spatial closure, long-axis rotation, swing-plane evolution, and a reduced ground-pathway proxy. Subject-scaled articulated arms, base–ground dynamics, and a coupled distributed beam remain appropriate staged extensions.
- **Simplified Actuator Models.** E1b shows the S2 ordering survives a linear torque-velocity bound on the drives with compressed margins (Section 24.2), but activation-rate limits, history dependence, and eccentric-side detail remain unmodeled. The coupled study adds pure delay, first-order activation, rate limits, torque–velocity bounds, and an impedance proxy, but these remain command-level surrogates; full optimal-control treatments with physiological actuators are the appropriate refinement (Sprigings and Neal 2000; Sharp 2009; McNally and McPhee 2018).
- **Coupled Sensitivity Is a Screening Tier.** Alternative impact definitions and single-parameter perturbations preserved the ordering (Section 24.7). The 12-input Latin-hypercube study now exposes coupled effects, parameter non-identifiability, and held-out strategy tradeoffs. Its ranges are engineering envelopes and its sample is small; measurement uncertainty and population distributions remain unmodeled.
- **Counterfactuals Remain Model-Tier Specific.** The Python pathway now includes both pointwise ZTCF profiles and forward same-state killswitches in the double-pendulum and constrained two-hand tiers. The coupled extension now preserves the finite negative-couple branch with endogenous base translation and club flex. The spatial common-state tier transports the geometry response across independent inverse-dynamics formulations. The reduced spatial forward tier then preserves a negative contact couple after complete driver removal in both native engines. Its hand carriages and declared contact law do not establish persistence in anatomical arms or human dynamics.
- **No Human Data Yet.** The decisive observational test — wrist interface power time courses across skill levels from motion capture
  (3) — awaits an appropriately governed motion-capture study.

25.4 Implications, Cautiously Stated

For modeling practice, the analysis shows that the timing question is tractable with counterfactual decomposition, and that energy-level accounting (not peak ordering) is the observable that separates strategies — supporting the methodological argument of Section 2.5 and the transfer-efficiency findings in recent human data (Rachnavy et al. 2026). For coaching interpretation, the results align with instruction that discourages "hitting from the top" and encourages retained lag with late release (Cochran and Stobbs 1968; T. P. Jorgensen 1999) — but a 2-DOF torque-program result is evidence about mechanism, not a prescription for any individual player, whose constraints (mobility, strength, consistency, injury history (Cole and Grimshaw 2016; Hume, Keogh, and Reid 2005)) sit outside this model. Any coaching translation should also respect that within-player timing variability in experts is real and partly functional (Neal et al. 2007; Langdown, Bridge, and Li 2012; Glazier 2011).

Chapter 26

Conclusions and Research Priorities

26.1 Conclusions

The literature does not support equating proximal-to-distal peak order with a requirement to transfer mechanical energy outward as early as possible. Interaction dynamics, optimal-control studies, and segmental energy measurements instead motivate a timing hypothesis: retain a compact proximal configuration while proximal speed develops, then accelerate the distal handoff later in the downswing.

The present simulations provide model-derived support for that mechanism. Under matched shoulder torque in a planar two-link model, early wrist drive reduced clubhead speed relative to a passive wrist, whereas optimized late drive increased it. A small early restraining torque followed by late drive was best among the tested programs. Energy and counterfactual accounting located the modeled mechanism in early arm-energy retention and late joint-force transfer rather than in early active wrist work.

This conclusion is conditional. The result survived torque-velocity bounds, alternative scoring rules, and 13 one-at-a-time parameter cases, but the optimal onset shifted across cases. The study therefore supports a robust ordering within a local model neighborhood, not a universal timing value or an individual coaching prescription. Planar motion, simplified actuators, and absence of human data remain material limitations.

The forward constrained two-hand extension resolves one previously open mechanistic question. Starting from the exact same 0.200 s state, the force-generated club couple remains negative for 50 ms after all applied joint torques are removed. The effect disappears exactly when both grip moment arms are removed and remains stable under timestep and projection refinement. This supports finite passive-couple persistence in the declared planar model, not a claim about muscle inactivity or human technique.

The coupled moving-base/flexible-club extension resolves the next structural question at the planar tier. Base translation, shaft flex, and both grip forces evolve in one forward constrained solve. The negative force-generated couple survives the same-state zero-command intervention, while coincident grip points remove it exactly; constraint, contact-power, and work–energy residuals close under timestep refinement. This transports the mechanism beyond a fixed base and rigid club without promoting it to a spatial or physiological result.

The distributed-shaft comparison closes a narrower structural gate. A one-mode reduction and six-mode Euler–Bernoulli reference share the same assembled mass, stiffness, damping, tip inertia, and loads. The reduced model closely follows slow loading but misses higher-mode history under a short force-and-moment pulse, even when their peak deflections are similar. Synthetic modal identification, mesh convergence, and work–energy closure pass. This does not calibrate equipment or show that the result survives coupling the beam into the two-hand constrained solve.

The reduced full-body common-state extension resolves implementation transport for a narrower spatial estimand. MuJoCo and an independent Lagrange–Christoffel formulation agree across 20 coordinates and 61 nonplanar states to 2.14×10^{-11} relative error. Reversing the hand moment arm reverses the force-generated couple and coincident points remove it exactly. Because the forces are prescribed and the state is shared, this supports spatial inverse-dynamics and geometry transport but leaves passive forward contact inconclusive at that tier.

The reduced spatial forward-contact extension resolves that narrower mechanism gate. Native MuJoCo and Pinocchio independently integrate two finite-mass hand carriages, paired compliant interfaces, and a free rigid club without direct club actuation. Their baseline and same-state killswitch trajectories satisfy the declared position, orientation, wrench, and energy regions. Removing the complete grounded driver leaves a negative force-generated swing-normal couple for 37.5 ms in both engines; coincident and reversed-moment-arm controls close exactly, and the work–energy residual decreases under timestep refinement. This supports reduced spatial contact persistence, not anatomical arms, muscles, calibrated grip tissue, equipment, or human strategy.

The coupled uncertainty/control phase rejects two additional shortcuts. A net planar wrench cannot uniquely identify both hand forces, and six summary observables cannot identify the complete 12-parameter vector. Separate training and held-out ensembles leave objective-dependent tradeoffs among all eight command programs. Early restraint improves held-out lower-tail speed relative to late drive but increases the planar face/path error, so the result supports a conditional model mechanism rather than a universal or physiological strategy.

The experimental phase is preregistered but not empirically executed. Its machine-readable protocol fixes six synchronized modalities, participant-level holdout, identity-safe provenance, filtering and residual sensitivities, and four prediction-specific falsifiers. The synthetic dry run passes all intake and fail-closed controls, but it cannot advance H1–H4. Governed human data are still required.

The open-resource layer now supplies model-tier presets, a hash-pinned release manifest, standard checksums, a data dictionary, citation metadata, and a claim-first reviewer workbench. Validation fails closed on artifact drift. The bundle remains a qualified local release candidate rather than a persistent archive: articulated subject-scaled contact, equipment-calibrated beam coupling, governed human evaluation, and external identifier deposit remain open.

26.2 Research Priorities

The next work should reduce the largest inferential risks in dependency order:

1. **Articulated Spatial Forward-Contact Counterfactual.** Replace the executed hand-carriage reference with subject-scaled articulated arms and calibrated grip interfaces while preserving the two-engine same-state killswitch, force, couple, power, work, pathway, and closure observables.
2. **Calibrated Uncertainty and Constrained Control.** Extend the executed multivariable screen with population-informed distributions, measurement error, profile-likelihood or Bayesian calibration, larger held-out ensembles, and a constrained optimal-control comparator.
3. **Spatial Model-Fidelity Ladder.** Add base rotation, a three-dimensional club, ground pathways, and couple the executed distributed/modal shaft reference into the forward constrained solve without changing the registered interface observables. Each rung should carry conservation checks and cross-engine tolerance gates.
4. **Execute the Frozen Human-Data Protocol.** Acquire governed synchronized measurements that pass the fixed inclusion, residual, missingness, participant-holdout, and provenance gates. Sequence order alone must not be treated as evidence of energy transfer.
5. **Arm–Wrist Allocation and Preload Falsification.** Execute the bilateral grip-wrench, pressure, activation, stiffness, and participant-holdout tests registered in Section 11. Treat the dead-zone mechanism as rejected if its predicted gap is absent under documented low tangent stiffness or if any performance difference disappears on holdout.
6. **Open-Resource Release.** Maintain deterministic commands, machine-readable results, data dictionaries, checksums, citation metadata, and explicit license boundaries. Archive validated releases with a persistent identifier.

The implementation roadmap and acceptance criteria are tracked in the open research epic. The matched-task allocation and transmission program is tracked separately in its falsification epic.

Code and Data Availability

Analysis code, tests, figures, machine-readable outputs, and parameter records are available in the open-source repository. The repository name identifies provenance only; none of the scientific claims depend on adoption of the broader software project. The principal summary is recorded in `data/results_summary.json`, and the parameter-sensitivity results are recorded in `data/e1d_parameter_sensitivity.json`. The spatial common-state evidence is recorded in `data/spatial_full_body_study.json`. The two-engine spatial forward-contact evidence is recorded in `data/spatial_forward_contact_study.json`. The uncertainty, identifiability, and control evidence is recorded in `data/uncertainty_control_study.json`. The distributed-shaft structural evidence is recorded in `data/shaft_beam_reference.json`. The forward distributed-modal shaft and arm–wrist allocation evidence are recorded in `data/moving_base_modal_shaft_study.json` and `data/torque_allocation_preload_study.json`. The frozen experimental protocol and synthetic readiness classification are recorded in `data/experimental_protocol_v1.json` and `data/experimental_protocol_readiness.json`.

Chapter 27

Software and Data Availability

The open-source implementation separates the analytical model, torque programs, counterfactual evaluation, energy accounting, robustness analyses, and figure generation. The scientific entry points are collected under `scripts/research/proximal_distal_energy/`, with shared model parameters and dynamics under `simulation_backends/`. This is a provenance statement, not a capability claim about the surrounding software.

The current evidence boundary includes these known limitations:

- A multi-phase matched-state torque-killswitch ensemble is complete for one declared torque program; other torque programs, model classes, and uncertainty distributions remain necessary before making general persistence claims.
- The archived two-hand ZTCF remains pointwise along the BASE state history. Separate forward rigid-club and coupled moving-base/flexible-club models now demonstrate finite zero-command persistence. A reduced full-body spatial common-state tier now tests nonplanar inverse dynamics, but passive spatial contact and human variants remain open validation steps.
- The primary model has two planar links and a fixed hub; torso, hub-path, long-axis rotation, and closed-loop arm effects are absent.
- The shaft extension is a three-coordinate point-mass surrogate with one linear torsional mode. It is not a calibrated distributed-beam or equipment model, and extreme low-stiffness deflections exceed its linear premise.
- The coupled actuator adds delay, first-order activation, rate and torque–velocity limits, and impedance, but it remains a nonphysiological command surrogate.
- Coupled uncertainty is screened over declared engineering envelopes; population distributions and measurement uncertainty are not estimated.
- No human motion or force data are analyzed; the result is mechanistic and model-derived, not a skill-group effect.
- The model ladder now executes reduced full-body common-state inverse dynamics in MuJoCo and an independent Lagrange–Christoffel formulation. The hand loads are prescribed, so independently solved forward two-hand contact is still unexecuted.

Chapter 28

Reproducibility Guide

From the repository root (Python 3.11+ with the project dependencies):

```
# Primary sweep, counterfactuals, and interface-power analysis
python3 -m scripts.research.proximal_distal_energy.run_experiments

# Registered WSCG source extraction and interaction-force analysis
python3 -m scripts.research.proximal_distal_energy.extract_wscg_charts
python3 -m scripts.research.proximal_distal_energy.run_interaction_force_study
python3 -m scripts.research.proximal_distal_energy.run_counterfactual_ensemble
python3 -m scripts.research.proximal_distal_energy.run_two_hand_wscg_analysis
python3 -m scripts.research.proximal_distal_energy.run_shaft_contribution_study
python3 -m scripts.research.proximal_distal_energy.run_mechanism_ladder_study
python3 -m scripts.research.proximal_distal_energy.run_forward_two_arm_study
python3 -m scripts.research.proximal_distal_energy.run_moving_base_flexible_study
python3 -m scripts.research.proximal_distal_energy.run_shaft_beam_reference
python3 -m scripts.research.proximal_distal_energy.run_spatial_full_body_study
python3 -m scripts.research.proximal_distal_energy.run_uncertainty_control_study
python3 -m scripts.research.proximal_distal_energy.run_experimental_protocol_dry_run
python3 -m scripts.research.proximal_distal_energy.qualify_open_release validate

# Robustness analyses
python3 -m scripts.research.proximal_distal_energy.e1b_bounded_torque
python3 -m scripts.research.proximal_distal_energy.e1c_impact_sensitivity
python3 -m scripts.research.proximal_distal_energy.e1d_parameter_sensitivity

# Figures and document
python3 -m scripts.research.proximal_distal_energy.make_figures
python3 -m scripts.research.proximal_distal_energy.make_interaction_force_figures
python3 -m scripts.research.proximal_distal_energy.make_counterfactual_figures
python3 -m scripts.research.proximal_distal_energy.make_two_hand_wscg_figures
python3 -m scripts.research.proximal_distal_energy.make_shaft_contribution_figures
python3 -m scripts.research.proximal_distal_energy.make_mechanism_ladder_figures
python3 -m scripts.research.proximal_distal_energy.make_forward_two_arm_figures
python3 -m scripts.research.proximal_distal_energy.make_moving_base_flexible_figures
python3 -m scripts.research.proximal_distal_energy.make_shaft_beam_reference_figures
python3 -m scripts.research.proximal_distal_energy.make_spatial_full_body_figures
python3 -m scripts.research.proximal_distal_energy.make_uncertainty_control_figures
cd docs/research/proximal_distal_energy_transfer
```

```
quarto render proximal_distal_energy_transfer.qmd --to pdf
cd ../../..
python3 -m scripts.research.proximal_distal_energy.optimize_article_pdf
```

Outputs land in `data/` (JSON and NPZ with integrator and parameter provenance) and `figures/` (PDF and SVG). The open-chain analyses use fixed-step RK4; the forward constrained two-hand analysis uses velocity Verlet with mass-metric projection. The analyses contain no stochastic elements. Re-runs should reproduce the recorded values up to floating-point environment differences. The final optimizer requires PyMuPDF, performs lossless object/stream compaction, and refuses an artifact that changes the page count, URI-link count, or outline-entry count.

Chapter 29

Validation Evidence

Tests exercise analytical double-pendulum ground truth, optional cross-backend agreement, drift-plus-control reconstruction, work–energy consistency, deterministic replay, impact edge cases, headline values, and the parameter-case contract. The wrist-interface balance also checks that the club mechanical-energy rate matches the summed interface powers to the finite-difference tolerance. These checks reduce numerical and implementation error; they do not validate the structural model assumptions.

Chapter 30

Model Parameters and Program Definitions

Table 30.1: Baseline model parameters and experiment constants.

Parameter	Value	Notes
Arm length l_1	0.75 m	baseline
Arm mass m_1	7.5 kg	both arms lumped
Arm COM ratio	0.45	$l_{c1} = 0.3375$ m
Arm inertia about COM	$0.3516\ \mathrm{kg \cdot m^2}$	about hub: $1.2059\ \mathrm{kg \cdot m^2}$
Club length l_2	1.0 m	grip to clubhead
Shaft mass	0.15 kg	COM ratio 0.43
Clubhead mass	0.20 kg	point mass at tip
Club composite mass m_2	0.35 kg	$l_{c2} = 0.7557$ m
Club inertia about COM	$0.0403\ \mathrm{kg \cdot m^2}$	about wrist: $0.2402\ \mathrm{kg \cdot m^2}$
Plane inclination	35°	projected gravity $8.033\ \mathrm{m/s^2}$
Damping (shoulder, wrist)	$0.4, 0.25\ \mathrm{N \cdot m \cdot s/rad}$	viscous
Integrator	RK4, $\Delta t = 1$ ms	analytical ODE backend
Initial state	$\theta_1 = -2.2$, $\theta_2 = -1.57$ rad, $\dot{q} = 0$	top of backswing
Shoulder torque τ_s	$60\ /\ 100\ \mathrm{N \cdot m}$	constant from $t = 0$
Wrist drive	$+15\ \mathrm{N \cdot m}$ from t_{on}	onset grid 0–0.35 s, step 0.025
Wrist restraint R	$5\ /\ 10\ \mathrm{N \cdot m}$ before t_{on}	restrain-then-drive only
Impact	first $\theta_1 + \theta_2 = 0$ crossing	valid iff $\theta_1 \leq 2.0$ rad

Recorded outputs:

- `data/e1_sweep.json` — one row per torque program plus provenance.
- `data/representative_traces.npz` — representative state, control, counterfactual, energy, speed, and interface-power traces.
- `data/results_summary.json` — representative impacts and phase budgets.
- `data/e1b_bounded_sweep.json` — torque-velocity-bounded sweep.
- `data/e1c_sensitivity.json` — alternative impact-definition results.
- `data/e1d_parameter_sensitivity.json` — 13 one-at-a-time parameter cases.
- `data/interaction_force_mechanisms.npz` — exact force, power, geometry, and matched-state kill-switch arrays.
- `data/interaction_force_summary.json` — headline force, work, and divergence metrics with counterfactual contracts.

- `data/wscg_2024_hand_force_series.csv` — hash-verified values from the source presentation's embedded chart cache.
- `data/wscg_2024_source_provenance.json` — source hashes, series inventory, and interpretation boundary.
- `data/counterfactual_ensemble.json` — 96 baseline cut/horizon/timestep rows, gravity and damping ablations, convergence metrics, and WSCG DELTA checks.
- `data/counterfactual_selected_traces.npz` — commanded and zero-torque state, force, and power traces for three representative cuts.
- `data/wscg_two_hand_raw/*.csv` — portable, column-labeled exports of the archived BASE, ZTCF, and DELTA two-hand tables.
- `data/two_hand_wscg_analysis.json` — source hashes, wrench-reconstruction residuals, reversal times, sensitivity results, and interpretation boundary.
- `data/two_hand_wscg_analysis.npz` — local force modes, moment and power decompositions, poses, and geometry-counterfactual arrays.
- `data/shaft_contribution_study.json` — matched rigid/flexible results, termwise acceleration and power summaries, ablations, 120 robustness cases, endpoint windows, timestep convergence, and interpretation boundaries.
- `data/shaft_contribution_traces.npz` — reference, rigid, and ablation state, speed, force, acceleration-contribution, power, work, and energy traces.
- `data/mechanism_ladder_study.json` — common-schema invariance residuals, prescribed mobile-hub cases, closed-loop constraint diagnostics, and the fail-closed model-discrepancy table.
- `data/mechanism_ladder_traces.npz` — three-link interface, mobile-hub, and closed-loop conditioning arrays used by the higher-order figures.
- `data/forward_two_arm_study.json` — forward model parameters, command, closure, killswitch ensemble, negative control, and numerical sensitivities.
- `data/forward_two_arm_study.npz` — baseline and representative branch states, contact forces, force modes, moments, power, energy, and solver residuals.
- `data/moving_base_flexible_study.{json,npz}` — coupled base/flex branch, negative control, parameter screen, convergence, states, forces, and energy.
- `data/shaft_beam_reference.{json,npz}` — synthetic modal identification, finite-element convergence, paired one/six-mode responses, and work–energy closure; not equipment calibration.
- `data/spatial_full_body_study.{json,npz}` — common-model hashes, prescribed wrench interventions, two-formulation inverse dynamics, and spatial states.
- `data/uncertainty_control_study.{json,npz}` — declared design, actuator contract, uncertainty intervals, PRCC screen, identifiability audit, training/held-out program comparison, closure, and rollout arrays.
- `data/experimental_protocol_v1.json` — frozen inclusion, measurement, processing, participant-split, prediction, and inference contract.
- `data/experimental_protocol_{dry_run,readiness}.json` — synthetic-only intake fixture and fail-closed readiness classification; no human observations.
- `release_manifest.json` and `CHECKSUMS.sha256` — exact release artifact inventory, model-tier presets, claim status, open gates, hashes, and sizes.

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